Principles of Management and Economics

VI Semester SEE Question Papers

Course codes: HS261TA, 21HS51B, 18HEM51, 18HEM61, 16HEM51 / 16HEM61, 16HEM51, 12HSM51, HS251TA

- Feb/Mar 2026 (HS251TA)
- Aug 2025 (HS261TA)
- Feb/Mar 2025 (HS251TA)
- Model Question Paper (2022 Scheme) (HS251TA)
- Apr/May 2024 (21HS51B)
- Apr 2024 (18HEM51)
- Jun 2023 Makeup (18HEM51)
- Jun 2023 Makeup (18HEM61)
- Mar/Apr 2023 (18HEM51)
- Grade Improvement Nov 2021 (18HEM61)
- Jan/Feb 2021 Supplementary (16HEM51)
- Jan/Feb 2021 (18HEM51)
- Dec 2020/Jan 2021 (18HEM51)
- Fast Track Oct 2020 (16HEM51 / 16HEM61)
- Nov/Dec 2019 (16HEM51)
- Nov/Dec 2019 Makeup (16HEM51)
- Nov/Dec 2018 (16HEM51)
- Nov/Dec 2017 (12HSM51)
- Nov/Dec 2016 (12HSM51)

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RV COLLEGE OF ENGINEERING®

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V Semester B.E. Regular / Supplementary Examination February / March 2026

Common to BT / AIML / CD / CS / CY / IS

PRINCIPLES OF MANAGEMENT AND ECONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	List and briefly describe any two functions of management under POSDCORB.	02	1	1
	1.2	Mention any two levels of management with suitable examples.	02	1	2
	1.3	What is the purpose of organizing?	02	2	2
	1.4	Differentiate between line authority and staff authority.	02	2	2
	1.5	Ravi works in a company with high job security and a good salary. However, he feels stuck and wants opportunities to be creative and to grow. Which need is currently motivating Ravi?	02	3	1
	1.6	A CEO leads a company through major change by motivating employees with a long-term vision and encouraging innovation. Which leadership style is this and why?	02	3	2
	1.7	Distinguish between real flow and money flow.	02	2	4
	1.8	What is a public good?	02	3	2
	1.9	State the four components of GDP using the expenditure method.	02	2	4
	1.10	Why do economists build the model?	02	2	4

PART-B

2	a	A manufacturing company is not meeting its production targets. Explain how different levels of management can help solve the problem.	08	3	1
	b	Define and explain the importance of the principle of 'Unity of Command' and 'Esprit de Corps' given by Henry Fayol.	08	1	4
3	a	*TechNova Solutions* is a mid-sized software development company experiencing rapid growth due to increased demand for its AI-based enterprise products. The CEO notices several emerging challenges: • Teams often duplicate similar work. • Employees are unclear about whom to report to during cross-functional projects. • Decision-making is slow because most approvals must go through senior management. • Managers complain about being overloaded with direct reports. • Some departments follow well-defined procedures, while others operate informally, causing inconsistencies. Analyze TechNova's current organizational issues and propose a redesigned organizational structure.	08	4	2

	b	A new coaching institute plans to launch skill-development programs. The management wants clarity in long-term and short-term direction. Apply the concepts of strategic, tactical, and operational goals to this institute and create one example of each.	08	3	2
		OR			
4	a	A start-up faces declining customer interest due to new technology introduced by competitors. Propose a strategic management plan for the start-up and justify each step.	08	3	2
	b	CityCare Hospital is a medium-sized healthcare institution with multiple departments such as Emergency, ICU, General Medicine, Radiology, and Administration. Recently, the hospital has faced several operational issues: • Nurses in the Emergency Department receive instructions from both the Chief Medical Officer and visiting specialist doctors, causing confusion. • Radiology technicians are unsure whether to follow directives from the Head of Radiology or from doctors in other departments who demand immediate reports. • Some junior doctors bypass their department heads and directly approach the hospital director for approvals. • Conflicts arise when two managers give contradictory instructions to the same staff member. • Employees feel stressed because they are not clear about who has final authority in decision-making. Using the concept of chain of command, analyze the problems CityCare Hospital is facing.	08	4	2
5	a	Critically evaluate Maslow's theory of motivation.	08	3	3
	b	Illustrate the five leadership styles in Blake and Mouton's Managerial Grid.	08	2	4
		OR			
6	a	Explain McGregor's Theory X and Theory Y assumptions in detail.	08	1	3
	b	Explain Blake and Mouton's Managerial Grid Theory of leadership.	08	1	2
7	a	Explain with suitable examples how a mixed economy combines features of market and command economies. Give suitable examples.	08	3	4
	b	Explain circular flow of economics with a neat sketch.	08	2	4
		OR			
8	a	What do you understand by elasticity of demand? Explain its various importance and applications.	08	2	4
	b	What are the characteristic features of an oligopolistic industry? How does it differ from monopolistic competition?	08	2	4
9	a	Explain Keynesian model of Macroeconomics and how it is different from classic model in macroeconomics.	08	2	1
	b	Critically assess the role of parliamentary discussion in budget approval.	08	3	1
		OR			
10	a	Explain the functioning and importance of the Keynesian cross model.	08	2	4
	b	A major oil-importing economy faces a sudden increase in global crude oil prices by 40%. Firms' production costs increase sharply. Analyse the impact on output and price level using the AS–AD model. Suggest one policy the government can adopt.	08	3	4

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Regular Examinations August-2025

PRINCIPLES OF MANAGEMENT AND ECONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	At google, many projects are addressed by small, focused employee teams that quickly form and complete work. This approach exemplifies $a(n)$ _____ organization structure.	01	1	1
	1.2	The initial aim of the Hawthorne Studies was to examine the effect of various lighting levels on _____.	01	1	1
	1.3	A technology company aiming to grow its market share over the next five years will develop a _____ plant that includes goals such as launching products in new international markets, investing in cutting-edge R&D, and increasing digital marketing efforts to boost brand visibility.	01	1	2
	1.4	A consumer electronics company adopts a strategy by emphasizing product innovation and sleek design, allowing it to differentiate itself from competitors. This is an example of a _____ strategy.	01	1	2
	1.5	_____ needs are mainly physiological / biological and unlearned.	01	1	3
	1.6	_____ needs can be met by challenging jobs, creative tasks, advancement opportunities, and achievement in work.	01	1	3
	1.7	John Maynard Keynes, one of the greatest economists of the twentieth century, pointed out that economics is not just a subject area but also a _____.	01	1	4
	1.8	The _____ is the percentage change in the quantity demanded of a good or service divided by the percentage change in the price.	01	1	4
	1.9	_____ is a fundamental economic indicator that measures the total market value of all final goods and services produced within a country over a specific time period, typically a year or a quarter.	01	1	5
	1.10	A travel agent who loses their job because consumers now book trips online using travel websites and apps experiences _____ unemployment.	01	1	5
	1.11	Identify the three common characteristics that define an organization.	02	1	1
	1.12	What does SMART stand for in goal setting, and give one example of a quantitative objective?	02	1	2
	1.13	What are the three core needs identified in Alderfer's ERG Theory?	02	1	3
	1.14	How does product differentiate enable firms in a monopolistic competition market structure to influence prices? Give an example to support your answer.	02	1	4
	1.15	A coffee shop increases the price of a cup of coffee from \$3 *to* \$3.30. As a result, the number of cups sold per day drops from 200 to 180. Calculate the price elasticity of demand for coffee at this shop. Based on your answer, is the demand elastic, inelastic, or unit elastic?	02	2	5

PART-B

2	a	Discuss Henry Mintzberg's model of managerial roles. Evaluate the relevance of this model in the context of contemporary management practices.	08	4	1

	b	Evaluate the Systems Approach to Management and its relevance in modern organizational settings. Strengthen your arguments with an example.	08	4	1
3	a	Discuss the key differences between strategies and tactical planning. Illustrate with an example how can a company might implement both strategic and tactical planning to achieve its objectives.	08	3	2
	b	Define the BCG Matrix and explain its purpose in strategic planning. Discuss how the BCG Matrix assists organizations in investment decisions.	08	3	2
		OR			
4	a	Define any three types of departmentalization. Analyze the advantages and disadvantages of each type.	08	2	2
	b	Analyze centralization and decentralization in the context of organizational structure, providing real-world examples to illustrate your points.	08	3	2
5	a	What are the categories of motivation according to the thermists? Explain.	08	2	3
	b	Analyze how Maslow's Hierarchy of needs can enhance management practices in organizations, with real-life example.	08	4	3
		OR			
6	a	Contrast McGregor's Theory X and Theory Y, highlighting their assumptions about human nature, and their implications for management practice and organizational behavior.	08	3	3
	b	Critically assess Hersey and Blanchard's approach to situational leadership.	08	3	3
7	a	Distinguish between microeconomics and macroeconomics and explain how both perspectives are useful in understanding the economy.	08	3	4
	b	Illustrate and explain the role of the circular flow diagram in understanding the economic system.	08	3	4
		OR			
8	a	Define and distinguish between price elasticity of demand and supply. What do these measures reveal about market behavior?	08	3	4
	b	How do oligopolistic markets differ from other market structures? Explain with reference to characteristics, formation, and firm behavior.	08	2	4
9	a	Define a price index and explain how it is used to measure inflation. Illustrate your answer with examples.	08	3	5
	b	Outline the major exchange rate mechanisms used by various countries and briefly explain their features.	08	2	5
		OR			
10	a	How does the Keynesian Cross Model work? Detail its main components, and the role played by the multiplier.	08	2	5
	b	Explain the key stages involved in the national budgeting process in India. How does this process ensure transparency and accountability in public finance? Discuss.	08	2	5

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RV COLLEGE OF ENGINEERING®

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V Semester B.E. Regular Examinations Feb/Mar – 2025

Common to AIML / CSE / ISE / ECE / EEE / EI / ET

PRINCIPLES OF MANAGEMENT AND ECONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Which step in *POSDCORB* involves delegating tasks and giving structured instructions?	01	2	2
	1.2	_____ is the primary scientist associated with the foundation of Administrative Theory.	01	1	1
	1.3	Differentiate between macroeconomics and microeconomics.	02	2	4
	1.4	_____ quadrant of the BCG Matrix represents high market share and high market growth.	01	1	3
	1.5	Define inflation.	01	2	5
	1.6	Define formalization.	02	1	2
	1.7	_____ have provided the foundation for our current theories of motivation, leadership, group behaviour and development, and numerous other behavioural approaches.	01	1	1
	1.8	Adam Smith called price mechanism as _____.	01	1	1
	1.9	The market state that satisfy all the essential features of a perfect competitive market except identity of product is known as _____.	01	2	2
	1.10	According to Fayol's 14 principles of management, 'esprit de corps' refers to _____ and _____.	02	1	3
	1.11	Mention the 4 different measures of *GDP*.	02	3	4
	1.12	Define economic planning.	02	2	4
	1.13	In *AS–AD* model the *AD* curve represents the relationship between the _____ in the economy and the _____.	02	2	3
	1.14	_____ is father of Economics.	01	1	1

PART-B

2	a	Illustrate different management function in detail.	08	2	2
	b	Briefly describe Fayol's 14 principles of management in detail.	08	2	1
3	a	Define Planning. Illustrate the different types of plans in detail.	08	2	2
	b	Bring out the differences between Centralization and decentralization.	08	2	2
		OR			
4	a	Bring out the different types of Organizational strategies.	08	2	2
	b	Illustrate different quadrants of *BCG* Matrix.	08	3	3
5	a	Briefly describe *D.C.* McClelland Achievement Motivation Theory.	08	2	3
	b	Illustrate Douglas McGregor theory x and theory y in detail.	08	2	3
		OR			

6	a	Illustrate Robert Blake and Jane Mouton developed Leadership Grid theory.	08	3	3
	b	Differentiate between Transactional and Transformational Leadership.	08	3	3
7	a	Briefly describe circular flow model of economics with help of diagram.	08	2	4
	b	Illustrate different types of elasticity demand in detail.	08	3	5
OR					
8	a	Differentiate between inductive and deductive reasoning.	08	2	4
	b	Illustrate the features of Capitalistic Economy.	08	2	5
9	a	Taking an example explain Price Elasticity of Demand and Price Elasticity of Supply.	08	3	4
	b	Briefly describe why do Oligopolies exist in today's competitive world.	08	3	4
OR					
10	a	Illustrate key components of classical growth theory within the framework of macroeconomic models.	08	2	5
	b	Briefly describe *IS–LM* model along with equations.	08	2	5

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RV COLLEGE OF ENGINEERING®

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V Semester B.E. Examinations – Model Question Paper (2022 Scheme)

DEPARTMENT OF INDUSTRIAL ENGINEERING AND MANAGEMENT

PRINCIPLES OF MANAGEMENT AND ECONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	The power to hold workers accountable for their actions and to make decisions about the use of organizational resources is known as _____.	01	2	4
	1.2	_____ was one of the first to recognize that organizations could be viewed from the perspective of individual and group behavior.	01	2	1
	1.3	Management Science came into existence in a _____ context.	01	2	2
	1.4	In the behavioral management approach what is the primary focus of the manager is _____.	01	2	2
	1.5	In a mechanistic organization, communication is mostly _____.	01	2	2
	1.6	_____ Structures include cross-hierarchical and cross-functional teams and a flat structure.	01	2	3
	1.7	Grouping activities on the basis of customer flow is _____ departmentalization.	01	2	4
	1.8	Procter & Gamble has separate departments for Tide, Pampers, Charmin, and Pringles. This is an example of departmentalization by _____.	01	2	4
	1.9	In Maslow's hierarchy of needs, _____ is the highest level of need?	01	3	4
	1.10	The offering of employee health insurance addresses which of Maslow's needs?	01	2	5
	1.11	Individuals become leaders based on their _____.	01	2	5
	1.12	The ability to influence people through granting or withholding benefits that are of interest to them is called _____.	01	2	3
	1.13	Mr. X is the only person in the company with expertise in the E-Commerce field. This source of power is called _____.	01	2	2
	1.14	The demand for a good increases when the price of a substitute _____ and also increases when the price of a complement _____.	01	2	2
	1.15	The price elasticity of demand is 5.0 if a 10 percent increase in the price results in a _____ decrease in the quantity demanded.	01	2	2
	1.16	Adam Smith called price mechanism as _____.	01	2	3
	1.17	An economy that interacts with other economies is known as _____.	01	2	4
	1.18	_____ growth is the percentage increase in the real GDP.	01	2	4
	1.19	The price elasticity of demand measures the _____ in quantity demanded due to a change in price.	01	3	4

	1.20	A shift in the supply curve to the right indicates an _____ in supply.	01	2	5

PART-B

2	a	Discuss Hawthorne effect in today's workplace. How can managers apply these concepts to improve work performance?	08	3	1
	b	Elaborate on the contributions of Henry Fayol to Management theory and justify the relevance of this theory for today's organizations.	08	2	1
3	a	Though decentralization is a welcome idea for efficient administration of any organization, but it comes with its own drawbacks. Discuss.	08	3	2
	b	In today's business environment, more than any preceding era, the only constant is change. Successful organizations effectively manage change, continuously adapting their bureaucracies, strategies, systems, products, and cultures to survive the shocks and prosper from the forces that decimate the competition. With this context explain in detail the Strategic Management process.	08	3	2
		OR			
4	a	The Taj West End Hotel has divisions such as reception, outdoor catering, conference hall maintenance and sales. With a schematic representation, illustrate the type of organization structure with advantages and disadvantages.	08	2	2
	b	Describe the situations where the decision making can be Decentralized. Explain any two cases in detail.	08	2	2
5	a	Theory X follows lower order needs and Theory Y follows higher order needs. Elucidate using Maslow's hierarchy of needs theory.	08	3	3
	b	Discuss the findings of Ohio state studies in reference to this following quotation by Mahatma Gandhi "I suppose leadership at one time meant muscles; but today it means getting along with people".	08	3	3
		OR			
6	a	Explain the three key linkages in expectancy theory and their role in motivation, which affect a worker's level of performance with an example?	08	2	3
	b	Exhibit the relationship between task behavior and relationship behavior using Blake & Mouton's managerial grid.	08	3	3
7	a	Discuss the role of firms in the circular flow model. How do they interact with households? How does the financial sector contribute to the circular flow of income and expenditure?	08	2	4
	b	Explain the relationship between elasticity and pricing strategies. How do firms use the concept of price elasticity to determine pricing policies in both competitive and monopolistic markets?	08	3	4
		OR			
8	a	Describe the income approach to measuring GDP. What are the different types of income included in this approach. Also discuss the role of exports and imports in GDP calculations. How do they impact the overall GDP of a country?	08	3	4
	b	Describe the Keynesian cross model. Explain how it demonstrates the relationship between aggregate demand and national income. Discuss the concept of equilibrium output in this model and how government spending can influence equilibrium.	08	2	4
9	a	Would you expect supply to play a more significant role in determining the price of a basic necessity like food or a luxury like perfume? Explain.	08	3	5
	b	Illustrate the concept of a shortage and a surplus in a market. What are the implications of each for prices and quantities exchanged?	08	2	5

		OR			
10	a	Compare and contrast elastic, inelastic, and unitary price elasticity's of demand and supply. Provide examples for each.	08	3	5
	b	Will the firms in an oligopoly act more like a monopoly or more like competitors? Briefly explain.	08	2	5

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Regular Examinations April/May-2024

COMMON TO AS/BT/CH/CV/IEM/ME

PRINCIPLES OF MANAGEMENT AND ECONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

PART-A

			M	BT	CO
1	1.1	Why efficiency and effectiveness are important to management?	02	1	1
	1.2	How do systems theory and the contingency approach make managers better at what they do?	02	2	2
	1.3	Define organizational structure and organizational design.	02	1	1
	1.4	State two important reasons for an appropriate span of management.	02	2	3
	1.5	How do reward packages, such as pensions and share option schemes, help motivate the workforce?	02	2	3
	1.6	Mention the contemporary concerns affecting leadership.	02	2	4
	1.7	List any two uses of Microeconomics.	02	1	4
	1.8	What do you understand by the term Monopolistic competition?	02	1	3
	1.9	Mention any two equivalent ways of measuring GDP using the circular flow model.	02	2	4
	1.10	What are the different ways of classifying unemployment according to economists?	02	1	2

PART-B

2	a	Researchers at Harvard Business School found that the most important managerial behaviors involve two fundamental things: enabling people to move forward in their work and treating them decently as human beings. In the context of the above statement, explain in detail Mintzberg's managerial roles.	08	2	2
	b	Henry Fayol described the practice of Management as a distinct function. Explain fundamental principles of management that could be applied to various organizational situations.	08	2	2
3	a	Imagine a tech start-up that has recently launched three products: ground-breaking AI software (high growth, low market share), a mobile app for productivity (moderate growth, moderate market share), and a legacy software product (low growth, high market share). Apply the BCG matrix to advise the start-up on how to allocate resources strategically.	08	2	2
	b	Tata Group of India has business in chemicals, communications and IT, consumer products, energy, engineering, materials, and services. With a schematic representation, explain the type of organizational structure with its advantages and disadvantages.	08	2	2

		OR			
4	a	With more than 2.1 million employees worldwide working in various departments, functional areas, and stores, Walmart Stores Inc., uses strategic management to help coordinate and focus employees' efforts on what's important as determined by its goals. In this context, explain in detail the strategic management process involved.	08	4	2
	b	A manufacturing firm produces standardized products in a sable market environment. It employs a hierarchical structure with clearly defined roles and strict procedures. However, the company is facing challenges adapting to changing consumer preferences and technological advancements. Analyze this scenario in terms of mechanistic and organic structures.	08	4	2
5	a	At the workplace, the workers put inputs into the job, such as education, experience, effort, energy, and expect to get some outcomes such as salary, reward, promotion, verbal recognition, and interesting and challenging work each in equal amounts. Explain in detail the importance of Adam's Equity theory in the given scenario.	08	5	3
	b	Steve Jobs achieved unwavering loyalty and commitment from Apple's technical staff in the early 1980s by articulating a vision of personal computers that would dramatically change the way people lived. Explain the leadership traits the above situation is relevant to where leader has to develop according to the readiness of the follower.	08	4	5
		OR			
6	a	"The corporate world is filled with stories of leaders who failed to achieve greatness because they failed to understand the context they were working in." Explain how the Fiedler Model would support the given statement.	08	3	4
	b	The manger wants two of his employees to switch their vacations from the summer to the spring so that the job needs will be filled suitably during the summer. Explain how Vroom's expectancy model is applied in the given situation.	08	3	4
7	a	Companies often build models of their new products, which are more rough and unfinished than the final product will be but can still demonstrate how the new product will work. Explain how economists apply the scientific method in the given scenario.	08	2	4
	b	Explain the main issue in the modern debate between the New Keynesians and the New Classical economists. If wages and prices are sticky in the short run – then what can we expect and what role does government play, according to the Keynesians?	08	2	3
		OR			
8	a	A rapidly developing country has hired you to present a specific strategy for keeping their inflation and unemployment rates low while maintaining a healthy GDP growth rate. Explain in detail the components of GDP that can be used to accomplish this.	08	2	2
	b	A country decides to liberalize its capital account and open up to international capital flows. Apply the IS-LM model to examine the effects of increased capital mobility on domestic interest rates, investment, and exchange rates.	08	2	2

9	a	A new version of a popular smartphone is released with advanced features, making older models less desirable. Analyze how this technological obsolescence affects the equilibrium price and quantity of the older smartphone models in the market.	08	3	3
	b	In a market dominated by a few large firms, one firm consistently sets the price for its products, and others follow suit. Analyze the rationale behind this price leadership strategy in an oligopolistic market. What factors might determine which firm takes on the role of price leader?	08	3	3
		OR			
10	a	Discuss how the price elasticity of supply for labour varies across different types of jobs, such as highly-skilled professions versus low-skilled manual labour. What factors determine the responsiveness of workers to changes in wages?	08	3	2
	b	A clothing retailer has built a strong brand reputation for its high-quality garments and unique designs. Analyze how brand loyalty influences the retailer's market power and pricing decisions in a monopolistically competitive market.	08	3	2

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations April-2024

COMMON TO EI / CH / IEM / ME / AS / CV

INTRODUCTION TO MANAGEMENT AND ECONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

			M	CO	BT
1	1.1	List any two important skills required by manager.	02	1	2
	1.2	Mention the relevance of systems approach.	02	1	2
	1.3	Differentiate between Authority and Responsibility.	02	1	2
	1.4	What is formalization?	02	1	1
	1.5	What are the three components of expectancy theory?	02	2	2
	1.6	Mention the importance of managerial grid.	02	3	2
	1.7	Mention any two reasons for economic problems.	02	4	2
	1.8	Distinguish between positive economics and normative economics.	02	4	2
	1.9	Define recession and expansion.	02	4	1
	1.10	What is Keynesian economics?	02	4	2

PART-B

2	a	Is Taylor's theory of scientific management still useful in today's business? Explain with the help of case example.	08	1	2
	b	Highlight the important features of systems approach with the help of suitable examples.	08	1	2
3	a	Differentiate between organic and mechanistic organization structure with suitable examples.	08	1	2
	b	Assume that after your final year, you are going to pursue your higher studies abroad. Chalk out systematic plan how you will start your preparation for the same.	08	2	3
		OR			
4	a	Identify the various factors that influence the amount of centralization.	08	2	3
	b	List and explain six key elements involved in an organizational design.	08	2	3

5	a	Apply theory X and theory Y to your student life. What characteristics student will possess if the above mentioned theories applied to him?	08	2	3
	b	Design a questionnaire to understand leadership qualities of a student studying in an engineering institution. Questionnaire should consist of atleast eight questions apart from personal information.	08	3	2
		OR			
6	a	Identify one example each for transactional leader and transformational leader from real life examples. Justify your answer with identifying traits present in them.	08	3	3
	b	Design a questionnaire to understand motivational level of an employee working in a software industry. Questionnaire should consist of atleast eight questions apart from personal information.	08	3	3
7	a	Identify the various components included and excluded from measuring our Gross Domestic Product.	08	4	2
	b	Identify the various factors which will help to shift in the supply curves to left or right with the help suitable graphs.	08	4	3
8	a	Examine the differences between Keynesian model and classical model of macroeconomics.	08	4	3
	b	With suitable examples, explain the key features of Mundell-Fleming Model.	08	4	2

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations June – 2023 (Makeup)

COMMON TO EI / CH / IEM / ME / AS / CV

INTRODUCTION TO MANAGEMENT AND ECONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART – A

1	1.1	Differentiate between the two management terms efficiency and effectiveness.	02
	1.2	How do systems theory and contingency approach make managers better at what they do?	02
	1.3	Differentiate between goals and plans.	02
	1.4	Contrast mechanistic and organic structures	02
	1.5	Explain the three key elements of motivation.	02
	1.6	Define leader and leadership.	02
	1.7	Differentiate between exchange rate and effective exchange rate.	02
	1.8	What is Consumer Price Index (CPI)? How is it calculated?	02
	1.9	Mention / list any two roles of government in a capitalist society system used for economic management of a country.	02
	1.10	In the labour market, what causes a shift for labour in demand curve and supply curve for labor?	02

PART – B

2	a	Explain the salient features of scientific management and general administrative theory.	10
	b	"An organization should be viewed as an unified, purposeful system composed of interrelated parts". Elucidate with appropriate examples.	06
3	a	With a flow chart, explain in detail, the eight steps in the decision making process.	08
	b	Explain any four types of goals that an organization might have.	08
		OR	
4	a	Explain with flowcharts the five common forms of departmentalization.	08

	b	Pfizer, is a global pharmaceutical company, looking for ways to help the employees to be more efficient and effective. An internal study conducted at Pfizer, revealed that approximately $20-40\%$ of time the employees were spending in doing support work rather than doing actual work. Hence, Pfizer by use of "magic button" (online form) outsourced the support work. By this approach Pfizer saved $66500\ hours$. i) What structural implications – good and bad this approach have? ii) Explain in terms of different organizational design structures.	08
5	a	Contrast McGregor's Theory X and Theory Y with Herberg's two-factor theory.	08
	b	Explain the situational leadership theory giving its salient features.	08
		OR	
6	a	How do equity theory and expectancy theory explain employee motivation? Discuss.	08
	b	$3M\ CEO$ George Buckley continues leadership excellence based on six leadership attributes, the ability to chart the course, energize and inspire others, demonstrate ethics, integrity and compliance, deliver results, raise the bar & innovate resourcefully. Mr. Buckley believes that leaders should build respect and trust to those who work for the company. Discuss how attributes might be developed.	08
7	a	Highlight the major themes concerning decision making and markets with respect to micro economics.	08
	b	With the help of a neat diagram, illustrate the function and scales of economic entities in a typical economy.	08
8	a	Define GDP. Give a comprehensive explanation of the circular flow model.	08
	b	With the help of a neat sketch, explain the concept of investment-savings, liquidity money $(IS-LM)$ model used in macroeconomics.	08

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Makeup Examinations June – 2023

INTRODUCTION TO MANAGEMENT AND ECONOMICS

(COMMON TO CV/EI/CH/ IM/ME/AS)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

		PART – A	
1	1.1	Differentiate between the two management terms efficiency and effectiveness.	02
	1.2	How do systems theory and contingency approach make managers better at what they do?	02
	1.3	Differentiate between goals and plans.	02
	1.4	Contrast mechanistic and organic structures	02
	1.5	Explain the three key elements of motivation.	02
	1.6	Define leader and leadership.	02
	1.7	Differentiate between exchange rate and effective exchange rate.	02
	1.8	What is Consumer Price Index (CPI)? How is it calculated?	02
	1.9	Mention / list any two roles of government in a capitalist society system used for economic management of a country.	02
	1.10	In the labour market, what causes a shift for labour in demand curve and supply curve?	02
		PART – B	
2	a	Explain the salient features of scientific management and general administrative theory.	08
	b	"An organization should be viewed as an unified, purposeful system composed of interrelated parts". Elucidate with appropriate examples.	08
3	a	With a flow chart, explain in detail, the eight steps in the decision making process.	08
	b	Explain any four types of goals that an organization might have.	08
		OR	
4	a	Explain with flowcharts the five common forms of departmentalization.	08

	b	Pfizer, is a global pharmaceutical company, looking for ways to help the employees to be more efficient and effective. An internal study conducted at Pfizer, revealed that approximately $20 - 40\%$ of time the employees rather than doing actual work, were spending in doing support work. Hence, Pfizer by use of "magic button" (online form) outsourced the support work. By this approach Pfizer saved $66500\ hours$. i) What structural implications – good and bad this approach have? ii) Explain in terms of different organizational design structures.	08
5	a	Contrast McGregor's Theory X and Theory Y with Herberg's two-factor theory.	08
	b	Explain the situational leadership theory giving its salient features.	08
		OR	
6	a	How do equity theory and expectancy theory explain employee motivation? Discuss.	08
	b	3M CEO George Buckley continues leadership excellence based on six leadership attributes, the ability to chart the course, energize and inspire others, demonstrate ethics, integrity and compliance, deliver results, raise the bar & innovate resourcefully. Mr. Buckley believes that leaders should build respect and trust to those who work for the company. Discuss how attributes might be developed.	08
7	a	Highlight the major themes concerning decision making and markets with respect to micro economies.	08
	b	With the help of a neat diagram, illustrate the function and scales of economic entities in a typical economy.	08
8	a	Define GDP. Give a comprehensive explanation of the circular flow model.	08
	b	With the help of a neat sketch, explain the concept of investment-savings, liquidity money (IS-LM) model.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)
V Semester B. E. Examinations Mar/Apr-2023
COMMON TO EI / CH / IEM / ME / AS / CV

INTRODUCTION TO MANAGEMENT AND ECONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	A manager applies various theories of Management in his/her unique personalised way. What nature of management does this statement indicate?	01
	1.2	The festive season was about to come. Mrs. Rubi, Head of Production department, called for a meeting of all the subordinates and communicated the objectives. Now she is establishing authority-responsibility relationship between superior and subordinates to avoid any delay. Identify the function of management in the above case.	01
	1.3	Sridhar's father is working as a section in-charge in a government office. Identify the level of management at which he is working?	01
	1.4	In the Wealth of Nations, Adam Smith described the breakdown of jobs into narrow and repetitive tasks and called this as _____.	01
	1.5	In a classroom discussion, Kalesh argued that planning is a function to be performed only at top-level management. But Suresh is of the opinion that planning is required at all levels of management. a) Do you support the argument of Kalesh or Suresh? b) Justify your answer.	02
	1.6	Why a mission statement is so important in the strategic management process?	02
	1.7	Supermarket is an organization in which only the managers are the leaders. Identify the type of organisation.	01
	1.8	Uranus Limited is a company dealing in metal products. The work is mainly divided into functions including production, purchase, marketing, accounts and personnel. Identify the type of organizational structure followed by the organization.	01
	1.9	Chao has a lot of internal drive. His overall goal in life is to "be all that he can be" in all aspects of his life. Which level of Maslow's hierarchy of needs is reflected in Chao's attitude?	01
	1.10	Melissa is a good role model for her team mates. She wants to make sure her team does well and wins the conference championship. Her desire to attain this goal is described as _____ motivation.	01

	1.11	Which leadership style tends to centralize authority and make unilateral decisions?	01
	1.12	Classify the following into intermediate products and final products: a) Furniture purchased by a school. b) Chalk, duster, etc, purchased by a school.	02
	1.13	Give two examples each for micro and macroeconomics.	02
	1.14	When there is high inflation in the economy, how will it affect the supply of money in the economy?	01
	1.15	What is the role of a Central Bank in the following exchange rate? a) Fixed exchange b) Floating exchange	02

PART-B

2	a	Aapka Vidyalaya' believes in the holistic development of students and encourages team building through a mix of curricular, co-curricular and sports activities. On its Founder's Day, a stage performance had to be put up. A committee of ten prefects was constituted to plan different aspects of the function. They all decided to use recycled paper for decoration. There was a spirit of unity and harmony and all the members supported each other. With mutual trust and a sense of belonging, the programme was systematically planned and executed. Kartik, one of the prefects, realised that the group had unknowingly applied one of the principles of management while planning and executing the programme. He was so inspired by the success of this function that he asked his father to apply the same principle in his business. His father replied he was already using this principle. i) Identify the principle of management applied for the success of the programme. ii) State any two features of management highlighted in the above paragraph. iii) Identify any two values which 'Aapka Vidylaya' communicated to society.	08
	b	Sanchit, after completing his entrepreneurship course from Sweden returned to India and started a coffee shop 'Aroma Coffee Can' in a famous mall in New Delhi. The specialty of the coffee shop was the special aroma of coffee and a wide variety of flavours to choose from. Somehow, the business was neither profitable nor popular. Sanchit was keen to find out the reason. He appointed Sameera an *MBA* from a reputed college, as a Manager to find out the causes for the same. Sameera took feedback from the clients and found out that though they loved the special unique aroma of coffee but were not happy with the long waiting time being taken to process the order. She analysed and found out that there were many unnecessary obstructions in between which could be eliminated. She fixed a standard time for processing the order. She also realized that there were some flavours whose demand was not enough. So, she also decided to stop the sale of such flavours. As a result within a short period Sameera was able to attract the customers. Identify and explain any two techniques of scientific management used by Sameera to solve the problem.	08

3	a	Agile Ltd.' is a well known automobile manufacturing company in India. The company plans to increase the sale of its sedan cars by 20% in the next quarter. In order to achieve the desired target, the marketing team of the company considers the impact of policy of the government towards diesel vehicles and the level of competition in this segment of cars. They explore the various available options like offering more discount to dealers and customers, providing more customer friendly finance options, lucky draws on test drives, increasing advertising, offering more of free accessories on the purchase of the car, etc. A thorough analysis of the various available options is done keeping in view the relative viability of each option. The company decides to pursue the option of offering more discount to dealers and customers in order to boost the sale of sedan cars. In order to implement the plan, they determine the various discount packages and communicate the same to their product dealers. To make the prospective consumers aware about the new available benefits, advertisements are made through various sources of print and electronic media. The market analysis of the company keep a close watch on the revenue from the sedan cars to study the effect of new initiatives by the company to promote its sales. In context of the above case: i) Name the function of management described in the above paragraph. ii) Identify and explain the various steps involved in process, the function of management as identified in part (a) by quoting lines from the paragraph.	08
	b	Conceptually, strategy of a firm consists of two inseparable parts: business strategy and corporate strategy. i) Distinguish between business strategy and corporate strategy. ii) Identify the key elements considered to develop and formulate such master strategy.	08
		OR	
4	a	Steelo Ltd. decided to set-up its steel manufacturing factory in the backward area of Orissa where very less job opportunities were available. People of that area welcomed this effort of Steelo Ltd. To attract people to work in its factory, it also decided to provide many other facilities like a school, hospital, market etc. in the factory premises. Steelo Ltd. started earning huge profits. Another competing company asked its production manager Aslam to investigate the reasons of earning huge profits by Steelo Ltd. Aslam found that in both the companies, there was systematic coordination among the various activities to achieve the organizational goals. Every employee knew who was responsible and accountable to whom. The only difference was that in his organization, communication took place only through the scalar chain whereas Steelo Ltd. was allowing the flow of communication in all the directions as per the requirement which led to faster spread of information as well as quick feedback.	

		i) Identify the type of organization which permits Steelo Ltd. the flow of communication in all the directions. ii) State another advantage of the type of organization identified in part (i) above. iii) State any two values which Steelo Ltd. wanted to communicate to society.	08
	b	Atul joins as a Regional Sales head in the export division of a $FMCG$ (fast moving consumer goods) company. In a departmental meeting, he asks one of his subordinates, Manik, to take charge of the company's new office in Dubai. He allocates the work to him and grants the necessary authority. However, within a month by seeking regular feedback on the extent of work accomplished from Manik, Atul realizes that Manik is not doing the work as per his expectations. So he takes away the authority delegated to him and re delegates the work to Prakrit. Due to the time wasted in this switch over, the work at the Dubai office suffers tremendously and the company is not able to meet its desired goals. In context of the above case: i) Why is Manik supposed to give regular feedback about work to Atul? ii) Can the authority granted to a subordinate be taken back and re delegated to another person? iii) Can Manik be held responsible for not meeting the work related expectations of Atul? Give a suitable reason in support of your answer. iv) Differentiate between authority, responsibility and accountability on the basis of origin and flow.	08
5	a	Amit Verma joined ABC Technologies in January after having worked at HKY Computers where he had worked as a Software developer. Amit felt that ABC offered better career prospects, as it was growing much faster than HKY which was a relatively smaller company. Amit joined as a Senior Software Developer at ABC Technologies with a good pay hike. He joined Archita Mehta's five-member team. While she was efficient at what she did and extremely intelligent, she had neither the time nor the leaning to groom her team members. Time and again, Amit found himself thinking of Ram Kapoor, his old boss, and of how he had been such a positive influence. Archita, on the other hand, even without actively doing anything, had managed to significantly lower his motivation levels. i) What could be the reasons for Amit was feeling disillusioned? Answer the questions using Maslow's hierarchy of needs? ii) What should Amit do to overcome his tension? What can a team leader do to ensure high levels of motivation among his/her team members?	08
	b	Teachers are often viewed as leaders in academic context along with bosses/manager in professional, politicians/elected officials in civic, and parents in personal contexts. For each of these leaders and contexts. i) Explain in brief the important leadership styles that each should possess.	

		ii) Discuss some of the influences in each context that may affect the leader and his or her leadership traits.	08
		OR	
6	a	Employees weigh what they put into a job situation (input) against what they get from it (outcome) and then compare their input-outcome ratio with the input-outcome ratio of others. If they perceive equity, they feel that their situation is fair, and that justice prevails. If the ratios are unequal, inequity exists and they will view themselves as either under-rewarded or over-rewarded. Explain in detail the importance of Adam's Equity theory in the given scenario.	08
	b	Steve is a CEO of a major healthcare organization. He is typically calm, cool, and collected and likes to take the time to evaluate all of his options before making a decision. When Steve went to work on Monday, he was faced with survivors being brought in from three different traumas (a plane crash, a high-rise fire, and a multi-vehicle accident). He was forced to become more authoritative and to quickly delegate tasks to some of his more senior staff, such as setting up a mobile trauma unit in one of the parking lots. Explain in detail the relationship between task behavior and relationship behavior using Blake & Mouton's managerial grid in the given situation.	08
7	a	A sanitizer manufacturing company wants to become a market leader. For this purpose the manager follows an activity with certain logical steps. The first step suggested by him is to increase profits by at least 30% in the next quarter. Explain how economists apply the scientific method in the given scenario.	08
	b	The Fast Food companies like the McDonald and Burger King who sells the burger in the market are the most common type of example of monopolistic competition. The two companies mentioned above sell an almost similar type of products but are not the substitute of each other. Now which product a particular consumer likes the most and of which company totally depends on him. Apart from the burger, other products are also sold by these companies like French fries, soft drinks, etc. All these products of the mentioned companies are of similar nature but there is no congruency between the products sold by the two as each one has a slightly different shape and taste. i) How a manufacturer does balance the Demand and Supply for his product in the market. ii) Bring out the meaning of Elasticity, Monopoly, Monopolistic and Oligopoly in the parlance of market driven economy.	08
8	a	A rapidly developing country has hired you to present a specific strategy for keeping their inflation and unemployment rates low while maintaining a healthy rate of GDP growth. Explain in detail the components of GDP that can be used to accomplish this.	08
	b	Assume that the government of a big open economy decides to increase its public spending that influence domestic income and interest rate given that we face perfect capital mobility under the flexible exchange rate regime. Explain in what way the Mundell Fleming model has an impact on this.	08

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RV COLLEGE OF ENGINEERING®

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VI Semester B.E. Grade Improvement Examinations Nov – 2021

FOUNDATIONS OF MANAGEMENT AND ECONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

Answer any FIVE full questions out of TEN. Each carries 20 marks.

		PART-A	M	BT	CO
1	1.1	List the functions of management.	01	–	–
	1.2	What is the meaning of the term 'Quantitative approach'?	01	–	–
	1.3	Name the types of goals and plans.	01	–	–
	1.4	Write the meaning of the corporate strategy.	01	–	–
	1.5	What is the meaning of the term work specification?	02	–	–
	1.6	Meaning of the chain and command is _____.	01	–	–
	1.7	List the early theories of motivation.	01	–	–
	1.8	Write the steps of Maslow's theory.	01	–	–
	1.9	Name the behavioral theories.	01	–	–
	1.10	Explain the term Herzberg's two factor theory.	01	–	–
	1.11	What is the meaning of managerial grid?	01	–	–
	1.12	Measuring of contingency theory of leadership is _____.	01	–	–
	1.13	Importance of economics is _____.	01	–	–
	1.14	Equilibrium in markets for goods and services _____.	01	–	–
	1.15	Price elasticity means _____.	01	–	–
	1.16	Differentiate between micro and macroeconomics.	01	–	–
	1.17	Define supply and demand.	01	–	–
	1.18	Illustrate the neo-classical synthesis.	01	–	–
	1.19	Define growth theory.	01	–	–

PART-B

2	a	Explain the management functions, roles and skills.	08	–	–
	b	Differentiate between scientific management and administrative theory.	06	–	–
	c	With one case study, explain contemporary approach.	06	–	–
3	a	Rajiv family's small business which employs 25 semi-skilled workers. During the first week on the job, his dad called him in and said, 'Rajiv, I had a chance to observe you working with the men and women for the past two days and although I hate to, I feel I must say something. You are just too nice to people. I know that this human behavior stuff has been taught to you at the institute, but it just does not work here. I remember when we discussed the Hawthorne studies when I was in the institute, everybody there got excited, but believe me, there is more to managing people than just being nice to them.' Answer the following question. (i) Explain briefly the Hawthorne studies that were explained by Rajiv to his father. (ii) Do you think Rajiv's father understood and interpreted the Hawthorne studies correctly? – Comment (iii) If you were Rajiv, how would you explain to your father the new perspective that is needed for a business to be successful?	10	–	–

	b	Principles of Taylor and Fayol are mutually complementary. One believed that the management should share the gains with the workers, while the other suggested that employee's compensation should depend on the earning capacity of the company and should give them a reasonable standard of living. (i) Identify and explain the principles of Taylor and Fayol referred to in the above statements. (ii) Excluding the principles identified in (i) explain briefly the other principles of Fayol to the contribution of management.	10	–	–
4	a	With a case study, explain the types of goals and plans.	10	–	–
	b	Explain corporate and competitive strategies.	10	–	–
5	a	Explain the overview of designing organizational structure.	10	–	–
	b	Explain the following terms. i. Chain of command ii. Span of control iii. Formalization iv. Mechanistics and organic structure.	10	–	–
6	a	With a flowchart, explain the hierarchy of needs of theory.	10	–	–
	b	With one case study, explain Herzberg's two factor theory.	10	–	–
7	a	Explain the following terms. i. Blake and Mouton's managerial grid ii. Contingency theories of leadership.	10	–	–
	b	Differentiate between transactional and transformational leadership.	10	–	–
8	a	Explain the importance of economics, micro-economics and macro-economics.	10	–	–
	b	With a case study, explain the changes in income and prices affecting consumption choices.	10	–	–
9	a	In an analysis of the market for paint, an economist discovers the facts listed below. State whether each of these changes will affect supply or demand, and in what direction. (i) There have recently been some important cost-saving inventions in the technology for making paint. (ii) Paint is lasting longer, so that property owners need not repaint as often. (iii) Because of severe hailstorms, many people need to repaint now. (iv) The hailstorms damaged several factories that make paint, forcing them to close down for several months.	10	–	–
	b	Describe briefly the circular flow economy model diagram with neat sketch.	10	–	–
10	a	With a flowchart or block diagram, explain the components of GDP (Gross Domestic Product).	08	–	–
	b	Differentiate the Classical model, Keynesian Cross model.	06	–	–
	c	Explain with an example, the exchange rate determination.	06	–	–

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B.E. Examinations Jan/Feb 2021

FOUNDATIONS OF MANAGEMENT AND ECONOMICS

Time: 02 Hours ***Maximum Marks: 50***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

		PART-A	**M**	**BT**	**CO**
1	1.1	Mention any two experiments conducted under the "Hawthorne studies".	02	–	–
	1.2	Draw the hierarchy of plans.	02	–	–
	1.3	Give any two examples of Dissatisfiers mentioned in "Herzberg's Two Factors Theory".	02	–	–
	1.4	List the various players involved in model of simple economy.	02	–	–
	1.5	Mention any two measure of *GDP*.	02	–	–
		PART-B			
2		Explain how the contingency approach is appropriate for studying management.	08	–	–
3		Differentiate between various types of plans with suitable examples.	08	–	–
		OR			
4		Highlight various features of the two models of organizational design.	08	–	–
5		"The strength of a tendency to act in a certain way depends on the strength of an expectation that the act will be followed by a given outcome and on the attractiveness of that outcome to the individual". Do you agree with above statement? Justify your answer suitably.	08	–	–
		OR			
6		"Successful leadership is achieved by selecting the right leadership style, which is contingent on the follower's readiness". Justify the above statement with the help of contingency leadership theory.	08	–	–
7		Analyze the various themes of microeconomics.	08	–	–
8		Illustrate the National Income determination and changes in the price level with the help of a suitable macroeconomics model.	08	–	–

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B.E. Examinations Jan/Feb 2021

INTRODUCTION TO MANAGEMENT & ECONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

		PART-A	M	BT	CO
1	1.1	Mohan a manager does not distribute the work amongst his subordinates either according to capability. Which principle of management is ignored here?	01	–	–
	1.2	_____ roles involve people and other duties that are ceremonial and symbolic in nature.	01	–	–
	1.3	The process of setting goals and choosing the means by which these goals will be achieved is known as _____.	01	–	–
	1.4	An established set of decisions to deal with specific activities which recur frequently within the organisation is known as _____.	01	–	–
	1.5	The principle of Chain of command is found in _____ type of organization.	01	–	–
	1.6	_____ management principle states that each individual should report to only one boss in order to avoid conflict and/ or confusion.	01	–	–
	1.7	If a general manager asks the sales manager to recruit some salesman on his behalf, it is an instance of _____.	01	–	–
	1.8	At what level of an organisation does a corporate manager operate?	01	–	–
	1.9	A person's need for power and authority falls under which category of need in *ERG* theory of motivation?	01	–	–
	1.10	_____ leadership style is characterized by contingent reward and management by exception.	01	–	–
	1.11	_____ is a set of forces that energize, direct and sustain behaviour.	01	–	–
	1.12	_____ proposes that if a higher-level need fails to be filled then a person may regress and seek to further fill lower-level needs instead.	01	–	–
	1.13	A decrease in demand causes the equilibrium price to _____.	01	–	–
	1.14	The law of demand states that the quantity of a good demanded varies _____ with its price.	01	–	–
	1.15	The income elasticity of demand is the percentage change in the quantity demanded divided by the percentage change in _____.	01	–	–
	1.16	The market for automobiles is an example of _____ oligopoly.	01	–	–
	1.17	The *LM* curve will shift down when the _____ demand declines.	01	–	–
	1.18	The total amount of income from economic activities across the country in a year is called _____.	01	–	–
	1.19	Gross national product will exceed gross domestic product if net factor payments from abroad are _____.	01	–	–
	1.20	Bob spends cash on rice, tuition fee, and flour etc. In this scenario which index is used to estimate the average of the entire expenses?	01	–	–

PART-B

2	a	Rajiv family's small business which employs 25 semi-skilled workers. During the first week on the job, his dad called him in and said, 'Rajiv, I had a chance to observe you working with the men and women for the past two days and although I hate to, I feel I must say something. You are just too nice to people. I know that this human behavior stuff has been taught to you at the institute, but it just does not work here. I remember when we discussed the Hawthorne studies when I was in the institute, everybody there got excited, but believe me, there is more to managing people than just being nice to them.' Answer the following question. (i) Explain briefly the Hawthorne studies that were explained by Rajiv to his father. (ii) Do you think Rajiv's father understood and interpreted the Hawthorne studies correctly? – Comment (iii) If you were Rajiv, how would you explain to your father the new perspective that is needed for a business to be successful?	08	–	–
	b	Principles of Taylor and Fayol are mutually complementary. One believed that the management should share the gains with the workers, while the other suggested that employee's compensation should depend on the earning capacity of the company and should give them a reasonable standard of living. (i) Identify and explain the principles of Fayol and Taylor referred to in the above statements. (ii) Excluding the principles identified in (i) explain briefly the other principles of Fayol to the contribution of management.	08	–	–
3	a	Pan Masala and Sons is a manufacturer of tobacco products. It decided to increase profits by at least 20% in the next quarter. It has many options: A. To increase the working hours of labourers without additional payment. B. To employ children from the local community. C. To sell its products outside the schools and colleges. It opted the (B) and (C) options to achieve the target. (i) Identify the function of management described in the above Para. (ii) State the steps of the process involved in the function of management identified in (i) quoting the lines from the above Para. (iii) State any two values neglected by the organization.	08	–	–
	b	Explain briefly the steps of Strategic Management Process with an example.	08	–	–
OR					

4	a	"Shan Spices Ltd." are the manufacturer of different food specific spices like Rajmaa Masala, Cholley Masala, Aaloo Parantha Masala etc. Mr. Raghav, the owner of the company has created different departments for purchase, production, marketing, finance and human resource. There are thirty employees working in the organization. Planning is of paramount importance to the company as Mr. Raghav believes that effective planning leads to achievement of organizational objectives. So in order to make employees focus on objectives, he issued instructions that during working hours only official matters will be discussed. He made certain rules and code of conduct for the employees to follow, according to which employees are not allowed to visit and talk to the employees of other departments except for official work. He emphasized on work performance which resulted in smooth functioning of the organization. (i) Identify and state the type of organization mentioned in the above Para. (ii) State one feature of the concept identified in part (a) as mentioned in the above Para. (iii) What was the purpose behind the formulation of rules for the employees that restricted their personal communication with the employees of other departments? (iv) State two values violated by Mr. Raghav.	08	–	–
	b	Identify whether the following examples are Organic or Mechanistic in Structure and explain briefly the salient features of it. (i) Consulting Firm (ii) Department of Motor Vehicles. (iii) Universities (iv) Facebook Corporation.	08	–	–
5	a	Atuliya has started his own consultancy firm under the name 'Happy Go Lucky' after working for five years in a company as a wedding planner. Considering the fact that he is into a labour intensive business and motivation is a complex process, Atulia wants to offer such rewards and benefits that will help to fulfill the various needs of the employees and will inspire them to give their best to the organisation. In the context of the above case: (i) Why is motivation considered to be a complex process? (ii) Name the various types of needs that exist in a hierarchy within every human being. Also, suggest any two suitable incentives for each of these that Atulia may offer to his employees.	08	–	–
	b	Human beings cannot be motivated only through monetary incentives. Non-monetary incentives are also necessary to motivate them. Explain any four such non-monetary incentives.	08	–	–
OR					

6	a	Umang Gupta is the Managing Director of Denver Ltd. The company had established a good name for itself and had been doing well. It was known for timely completion of orders. The Production Manager, Ms. Kanta was efficiently handling the processing of order and had a team of fourteen motivated employees working under her. Everything was going on well. Unfortunately she met with an accident. Umang knew that in the absence of Ms. Kanta, the company may not be able to meet the deadlines. He also knew that not meeting the deadlines may lead to customer dissatisfaction with the risk of loss of business and goodwill. So, he had a meeting with his employees in which accurate the speedy processing of orders was planned. Everybody agreed to work as team because the behaviour of Umang Gupta was positive towards the employees of the organisation. Hence everyone put in extra time and efforts and the targets were met on time. Not only this, Umang visited Ms. Kanta and advised her to take sufficient rest. (i) Identify the leadership style of Umang Gupta and draw a diagram depicting the style. (ii) State any two values highlighted by the behaviour of Umang Gupta. (iii) List some of the primary characteristics of leadership identified in (i).	08	–	–
	b	Describe briefly the various Types of Situational Leadership Styles as identified by Blanchard and Hersey. Explain with an example.	08	–	–
7	a	In an analysis of the market for paint, an economist discovers the facts listed below. State whether each of these changes will affect supply or demand, and in what direction. (i) There have recently been some important cost-saving inventions in the technology for making paint. (ii) Paint is lasting longer, so that property owners need not repaint as often. (iii) Because of severe hailstorms, many people need to repaint now. (iv) The hailstorms damaged several factories that make paint, forcing them to close down for several months.	08	–	–
	b	Describe briefly the circular flow economy model diagram with neat sketch.	08	–	–
8	a	Describe briefly the four different types of *GDP* and show how the *GDP* is calculated based on income.	08	–	–
	b	Explain briefly how the *AS–AD* model incorporates growth, unemployment, and inflation in economy.	08	–	–

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RV COLLEGE OF ENGINEERING®

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V Semester B.E. Dec-20/Jan-21 Examinations

DEPARTMENT OF INDUSTRIAL ENGINEERING AND MANAGEMENT

INTRODUCTION TO MANAGEMENT AND ECONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

		PART-A	M	BT	CO
1	1.1	The power to hold workers accountable for their actions and to make decisions about the use of organizational resources is known as _____.	01	–	–
	1.2	_____ was one of the first to recognize that organizations could be viewed from the perspective of individual and group behavior.	01	–	–
	1.3	Management Science came into existence in a _____ context.	01	–	–
	1.4	In the behavioral management approach what is the primary focus of the manager is _____.	01	–	–
	1.5	In a mechanistic organization, communication is mostly _____.	01	–	–
	1.6	_____ Structures include cross-hierarchical and cross-functional teams and a flat structure.	01	–	–
	1.7	Grouping activities on the basis of customer flow is _____ departmentalization.	01	–	–
	1.8	Procter & Gamble has separate departments for Tide, Pampers, Charmin, and Pringles. This is an example of departmentalization by _____.	01	–	–
	1.9	In Maslow's hierarchy of needs, _____ is the highest level of need?	01	–	–
	1.10	The offering of employee health insurance addresses which of Maslow's needs?	01	–	–
	1.11	Individuals become leaders based on their _____.	01	–	–
	1.12	The ability to influence people through granting or withholding benefits that are of interest to them is called _____.	01	–	–
	1.13	Mr. X is the only person in the company with expertise in the E-Commerce field. This source of power is called _____.	01	–	–
	1.14	The demand for a good increases when the price of a substitute _____ and also increases when the price of a complement _____.	01	–	–
	1.15	The price elasticity of demand is 5.0 if a 10 percent increase in the price results in a _____ decrease in the quantity demanded.	01	–	–
	1.16	Adam Smith called price mechanism as _____.	01	–	–
	1.17	An economy that interacts with other economies is known as _____.	01	–	–
	1.18	_____ growth is the percentage increase in the real GDP.	01	–	–
	1.19	Microeconomics and Macroeconomics are the study of _____ and _____ respectively.	02	–	–

PART-B

2	a	Discuss Hawthorne effect in today's workplace. How can managers apply these concepts to improve work performance?	08	–	–
	b	Elaborate on the contributions of Henry Fayol to Management theory and justify the relevance of this theory for today's organizations.	08	–	–
3	a	Though decentralization is a welcome idea for efficient administration of any organization, but it comes with it's own drawbacks. Discuss.	08	–	–
	b	In today's business environment, more than any preceding era, the only constant is change. Successful organizations effectively manage change, continuously adapting their bureaucracies, strategies, systems, products, and cultures to survive the shocks and prosper from the forces that decimate the competition. With this context explain in detail the Strategic Management process.	08	–	–
		OR			
4	a	The Taj West End Hotel has divisions such as reception, outdoor catering, conference hall maintenance and sales. With a schematic representation, illustrate the type of organization structure with advantages and disadvantages.	08	–	–
	b	Describe the situations where the decision making can be Decentralized. Explain any two cases in detail.	08	–	–
5	a	Theory X follows lower order needs and Theory Y follows higher order needs. Elucidate using Maslow's hierarchy of needs theory.	08	–	–
	b	Discuss the findings of Ohio state studies in reference to this following quotation by Mahatma Gandhi "I suppose leadership at one time meant muscles; but today it means getting along with people".	08	–	–
		OR			
6	a	Explain the three key linkages in expectancy theory and their role in motivation, which affect a worker's level of performance with an example?	08	–	–
	b	Exhibit the relationship between task behavior and relationship behavior using Blake & Mouton's managerial grid.	08	–	–
7	a	Would you expect supply to play a more significant role in determining the price of a basic necessity like food or a luxury like perfume? Explain.	08	–	–
	b	Will the firms in an oligopoly act more like a monopoly or more like competitors? Briefly explain.	08	–	–
8	a	With the help of a neat sketch explain the concept of IS–LM model used in macroeconomics.	08	–	–
	b	Whether the classical growth theory of macroeconomics is relevant in today's economic scenario.	08	–	–

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RV COLLEGE OF ENGINEERING®

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V / VI Semester B. E. Fast track Examinations Oct-2020

FOUNDATIONS OF MANAGEMENT AND ECONOMICS

Time: 02 Hours ***Maximum Marks: 50***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	What is meant by efficiency and effectiveness in the context of management?	01
	1.2	Distinguish between systems theory and contingency approaches.	01
	1.3	What is meant by Organizational Structure?	01
	1.4	State two important reasons for an appropriate Span of Management.	01
	1.5	Mention the goals of Organizational Behavior.	01
	1.6	Mention the contemporary concerns affecting leadership.	01
	1.7	List any four problems in achieving efficiency in Production and Distribution.	01
	1.8	What is meant by the term Market Mechanism?	01
	1.9	Mention any two equivalent ways of measuring GDP using circular flow model.	01
	1.10	What are the different ways of classifying unemployment according to economists?	01

PART-B

2	Dell Inc. is a company that understands the importance of management skills. It started an intensive five-day offsite skills training program for their employees as a way to improve its operations. One of Dell's directors of learning and development thought this was the best way to develop skills who can build that strong relationship amongst the employees. In the above context, explain different Managerial Skills.	08
3	*CEO* of Cisco Systems likes to listen to voice mails forwarded to him from dissatisfied customers. He says "E-mail would be more efficient but I want to hear the emotions, I want to hear the frustration, I want to hear the caller's level of comfort with the strategy we're employing. I can't get that through e-mail". This manager understands the importance of customers. In this context, explain in detail the steps involved in the process of strategic management.	08
	OR	

4	Food marketer Smart Balance Inc. helps people stay trim and lean with its heart-healthy products. The company outsources almost every other organizational function including manufacturing, product distribution, Marketing and Sales. With a schematic representation, illustrate the suitable type of organizational structure with the advantages and disadvantages.	08
5	"Today's managerial challenge is to inspire, motivate and enable knowledge of the employees to solve day-to-day problems that cannot be anticipated". Explain in detail the importance of Maslow's Hierarchy of needs in the given scenario.	08
	OR	
6	"If the group situation was highly unfavorable but was led by a relationship-oriented leader, the group's performance could be improved by replacing that person with a task-oriented leader". Explain how Fiedler model would support the given statement.	08
7	Companies often build models of their new products, which are more rough and unfinished than the final product will be, but can still demonstrate how the new product will work. Explain how Economists apply the steps of scientific method in the given scenario.	08
8	Assume that the Government of a big open economy decides to increase its public spending that influence domestic income and interest rate given that we face perfect capital mobility under the flexible exchange rate regime. Explain in what way the Mundell Fleming model has an impact on this.	08

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Nov/Dec-19

Common BT / CH / CV / ME / IM / EI

FOUNDATIONS OF MANAGEMENT AND ECONOMICS

Time: 02 Hours ***Maximum Marks: 50***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Why managers are important to organizations?	01
	1.2	Mention different functions of management.	01
	1.3	What is the nature and purpose of planning?	01
	1.4	Bring out any two differences between Mechanistic and Organic organizational structures.	01
	1.5	List any two current issues in motivation.	01
	1.6	Mention the contemporary concerns affecting leadership.	01
	1.7	List any two uses of Microeconomics.	01
	1.8	What do you understand by the term Market Mechanism?	01
	1.9	Mention any two equivalent ways of measuring GDP using circular flow model?	01
	1.10	What are the different ways of classifying unemployment according to economists?	01

PART-B

2	In today's demanding and dynamic workplace, employees who want to be valuable assets must constantly upgrade their skills, and developing management skills can be particularly beneficial in today's workplace. In view of the above statement, explain the managerial skills.	08
3	With more than 2.1 million employees worldwide working in various departments, functional areas and stores, Walmart stores Inc.., uses strategic management to help coordinate and focus employees efforts on what is important as determined by its goals. In this context, explain in detail the strategic management process involved? **OR**	08

4	Tate group of India has businesses in chemicals, communications and *IT*, consumer products, energy, engineering materials and services. With a schematic representation illustrate the suitable type of organizational structure for the above context. List any two advantages and disadvantages.	08
5	At the workplace the workers put inputs into the job, such as education, experience, effort, energy and expect to get some outcomes such as salary, reward, promotion, verbal recognition, and interesting and challenging work each in equal amounts. Explain in detail the relevance of Adam's equity theory in the given scenario.	08
	OR	
6	"The corporate world is filled with stories of leaders who failed to achieve greatness because they failed to understand the context they were working in ". Explain how Fiedler Model would support the given statement.	08
7	Comparison often build models of their new products, which are more rough and unfinished than the final product will be, but can still demonstrate how the new product will work. Explain how economists apply the scientific method in the given scenario.	08
8	Assume that the government of a big open economy decides to increase its public spending that influence domestic income and interest rate given that we face perfect capital mobility under the flexible exchange rate regime. Explain in what way the Mundell Fleming model has an impact on this.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B.E. Examinations Nov/Dec – 19 (Makeup)

Common BT / CH / CV / ME / IM / EI

FOUNDATIONS OF MANAGEMENT AND ECONOMICS

Time: 02 Hours ***Maximum Marks: 50***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

		PART-A	M	BT	CO
1	1.1	_____ management methods called for optimizing the way that tasks were performed and doing the job in one "best" way.	01	–	–
	1.2	_____ concluded that managers perform ten interrelated activities that relate to decision making, using information and inter-personal relationships.	01	–	–
	1.3	Identify the type of plan in which the time span of the plan may vary from 5 to 10 years or more.	01	–	–
	1.4	A _____ organization structure is the simplest form and has clear lines of authority and ease of decision making.	01	–	–
	1.5	Name the person who suggested that for all human motivation, the following needs can be organized in a hierarchical manner: a) Physiological b) Safety c) Social d) Esteem e) Self-actualization.	01	–	–
	1.6	_____ leadership theory suggests that management style adapt itself to changing circumstances.	01	–	–
	1.7	In a centrally planned economy, the central problems are solved by _____ authority.	01	–	–
	1.8	Write the difference between microeconomic and macroeconomics.	01	–	–
	1.9	A _____ market is the place where workers and employees interact with each other.	01	–	–
	1.10	_____ is the total market value of the goods and services produced by a country's economy during a specified period of time.	01	–	–

PART-B

2	Explain briefly systems approach to management.	08	–	–
3	Distinguish between strategic and tactical planning.	08	–	–
	OR			
4	Write any eight differences between centralization and decentralization.	08	–	–
5	Explain Herzberg's two factor theory.	08	–	–
	OR			
6	Discuss any eight qualities of transactional leadership.	08	–	–
7	Explain the importance and uses of microeconomics.	08	–	–

8	What is meant by exchange rate? Explain the procedure to determine exchange rates.	08	–	–

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-18

Common BT / CH / CV / ME / IM / EI

FOUNDATIONS OF MANAGEMENT AND ECONOMICS

Time: 02 Hours ***Maximum Marks: 50***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	An example for the classical approach to management theory is _____.	01
	1.2	Name the person who is considered as "Father of Human Relations Approach".	01
	1.3	Any permanent change in the behavior is the outcome of _____.	01
	1.4	Maximum number of subordinates under a supervisor is known as _____.	01
	1.5	Drawing a general impression about an individual on the basis of a single characteristic is _____.	01
	1.6	Mention the key goals of organizational behavior.	01
	1.7	Mention some important components of GDP.	01
	1.8	What are the utilities of studying microeconomics?	01
	1.9	Suggest an economic model which will help us to understand total spending in the economy and its effects on output and inflation.	01
	1.10	Which sector of the economy contributes to the most of the Indian GDP?	01

PART-B

2	Explain how the contingency approach is appropriate for studying management?	08
3	A hotel has departments such as front desk operations, sales, catering, housekeeping, laundry and maintenance. What kind of organizations structure do you suggest for this hotel and why? List some advantages and disadvantages.	08
	OR	
4	Identify different factors that influence the amount of centralization and decentralization.	08

5	An individual tends to act in a certain way based on the expectation that the act will be followed by a given outcome and on the attractiveness of that outcome to the individual. Identify and explain the theory which supports the above statement.	08
	OR	
6	"Successful leadership is achieved by selecting the right leadership style, which is contingent on the followers readiness". Justify the above statement with help of the contingency leadership theory.	08
7	Highlight major themes concerning decision making and markets with respect to microeconomics.	08
8	With the help of neat diagram, explain the concept of $IS-LM$ model used in microeconomics.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-17

Common to EE / EC / IM / EI / TE

MANAGEMENT AND ORGANIZATIONAL BEHAVIOR

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	List any two challenges of being a manager.	02
	1.2	List the roles and corresponding skills of a manager.	02
	1.3	Define single use plan and standing plan.	02
	1.4	Mention any two differences between programmed and non-programmed decisions.	02
	1.5	Mention any two important features of strategic management.	02
	1.6	Identify the major behavioural science disciplines that contribute to Organizational Behaviour.	02
	1.7	Mention learning theories and their relevance in shaping behaviour.	02
	1.8	List the factors determining group structures.	02
	1.9	How the fundamental attribution error and self-servicing bias can distort attributions?	02
	1.10	Define leader and leadership.	02

PART-B

2	a	Define management. Propose various functions of management suitably applied to a modern manufacturing organization.	08
	b	An organization should be viewed as a unified, purposeful system composed of interrelated parts. Elucidate.	08
		OR	
3	a	Henry Fayol insisted that management was a skill like any other one that could be taught once its underlying principles were understood. Elaborate.	08
	b	Exhibit the fact that relations theory arose from early attempts to systematically discover the social and psychological factors that created effective human relations.	08
4	a	"Corporate level strategy is to endure a fair balance amongst the business units that use up cash and those that supply cash." Describe the methodology in evaluating biz units in the above context using BCG matrix	08
	b	"As managers use computers and software tools more often, they'll be able to make more rational decisions" With reference to the statement, explain eight steps of decision making process.	08
		OR	

5	a	"A manufacturing/*IT* organization makes multiple products at various locations across the globe which caters to the different needs of the customers." On the above case, identify the appropriate organizational structure. Support your answer with justification for the same.	08
	b	Explain the contingency factors that favor the mechanistic model and the organic model of organizational design.	08
6	a	"Imagine that your assistant is ideal in all respects, but in one thing, he or she is hopeless at taking phone messages for you when you're not in the office." Illustrate components of attitude and its impact on a dissatisfied employee.	06
	b	"*MBIT* model could be used to manage personal change and Big Five model to manage professional change." Explain.	10
		OR	
7	a	Suppose that you are the branch manager of Kotak Mahindra. Your subordinate – the cashier, pleads for your permission to leave early to home citing that his stomach is upset sue to lunch in the canteen. Analyse his behaviour from the perspective of attribution theory.	10
	b	What is social learning? Describe the implications of social learning theory for managing people at work.	06
8	a	Define group and the stages of group development.	08
	b	Explain the relevance of major types of teams found in the organizations in today's networked business environment.	08
		OR	
9	a	"Most of us have to work for living, and a job is a central part of our lives". In view of the statement, bring out the contrast between two theories of Motivation: Maslow's hierarchy of needs and McGrogor's theory.	08
	b	Expectancy theory emphasizes on three components and their relationship. Elucidate on these three components and their relationships leading to motivation in employees work.	08
10	a	The leadership traits of Mahatma Gandhi inspired many leaders across the world, just like the former *US* president Barack Obama once stated "Throughout my life, including my work as a young man on behalf of the urban poor, I have always found inspiration in the life of Mahatma Gandhi and his simple and profound lesson to be the change we seek in the world". Explain different leadership traits that the above situation is relevant to.	06
	b	Elucidate and infer on University of Michigan studies and Blake & Mouton Managerial grid on leadership styles with respect to organizational context.	10
		OR	
11	a	"Every individual employee in an organization plays a role in controlling work activities". Do you agree with this statement, or do you think the process of controlling is something that only managers are responsible for? Explain.	08
	b	What do you mean by organizational performance? Discuss the various types of tools used to monitor and measure organizational performance.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-16

Common to EE / EC / IM / EI / TE

MANAGEMENT AND ORGANIZATIONAL BEHAVIOR

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	Why efficiency and effectiveness are important to management?	02
	1.2	How does systems theory and the contingency approach make managers better at what they do?	02
	1.3	Define organizational structure and organizational design.	02
	1.4	State two important reasons as to why an appropriate span of management is required.	02
	1.5	What do you understand by the term the Task-Technology Approach?	01
	1.6	Mention the key goals of organizational behavior.	02
	1.7	List the four dimensions of Emotional Intelligence.	02
	1.8	Identify the shortcuts individuals use in making judgments about others.	02
	1.9	How the fundamental attribution error and self-serving bias can distort attributions?	01
	1.10	Define group think and social loafing.	02
	1.11	List the major dimensions of transformational leadership.	02

PART-B

2	a	Define Management. Propose various functions of Management as applied to a modern manufacturing organization. Highlight the specific requirements of a typical manufacturing organization from the functional management perspectives.	08
	b	Identify and categorize the Mintzberg's roles of managers regardless of type of organization and levels in the organization. Explain each role briefly.	08
		OR	
3	a	Henry Fayol described the practice of management as a distinct function. Explain fundamental rules of management that could be applied to all organizational situations.	10
	b	Several of our current management ideas and practices can be directly traced from contributions of quantitative approach. Is it true or not? Justify with a suitable example.	06

4	a	What is a goal? Explain the steps in Goal setting.	06
	b	Illustrate eight steps of decision making process with a suitable example of any organization.	10
		OR	
5	a	A hotel has departments such as front desk operations, sales, catering, housekeeping, laundry and maintenance. With a schematic representation, illustrate the type of organization structure with the advantages and disadvantages.	08
	b	What is the purpose of Job Enrichment and Job Enlargement?	08
6	a	Illustrate the components of attitude and its impact on a dissatisfied employee.	06
	b	One day the Boss comes in and he is nervous, edgy and argumentative. The next day he is calm and relaxed. Does this behavior suggest that personality traits are not consistent? Explain the situation with a suitable personality model.	10
		OR	
7	a	An employee does an unsatisfactory job on an assigned project. Explain the attribution process that this person's manager will use to form judgments about this employee's job performance.	10
	b	What is social learning? Explain with examples any two processes that determine the effectiveness of this type of learning.	06
8	a	Discuss the stages of Group Development.	08
	b	Explain the major types of teams found in the organizations. Distinguish between teams in formal and informal organizational settings.	08
		OR	
9	a	"Theory *X* follows lower order needs and Theory *Y* follows higher order needs". Elucidate using Maslow's hierarchy of needs theory.	08
	b	Explain the three key linkages in expectancy theory and their role in motivation, which affect a worker's level of performance with an example.	08
10	a	What are the distinctive aspects of Leadership as a process? Examine the important outcomes of Ohio state studies over University of Michigan studies. Bring out the differences in the outcome.	10
	b	Exhibit the relationship between task behavior and relationship behavior using Blake & Mouton's managerial grid. When each style is justified?	06
		OR	
11	a	Define controlling. Explain in detail the three steps in the control process. What is the role played by Feedback and Feed forward mechanisms in an organizational context?	08
	b	What do you mean by organizational performance? Describe the three frequently used organizational performance measures.	08

Supply Chain Management

VI Semester SEE Question Papers

Course codes: IM362IA, 18IM63

- Aug 2025 (IM362IA)
- Sep 2023 (18IM63)
- Aug 2022 (18IM63)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations August-2025

Industrial Engineering and Management

SUPPLY CHAIN MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	_____ strategy define the set of customer needs a firm seeks to satisfy through its products and services.	01	1	1
	1.2	Id execution of a process is initiated in response to a customer order then that process is called _____ process.	01	1	1
	1.3	_____analysis evaluates the present value of any stream of future cash flows and allows management to compare two streams of cash flows in terms of their financial value.	01	1	2
	1.4	Write the objective function to minimize total supply chain cost for capacitated plant location model.	01	1	2
	1.5	If optimal lot size is 980 units, annual demand is 12000 units, then average flow time is _____ years.	01	2	3
	1.6	If desired cycle service level is given, the required safety inventory is computed using the formula _____.	01	1	3
	1.7	_____ is a SCM collaborative initiative that aims to improve the design of products through collaborations between manufacturers and suppliers.	01	1	4
	1.8	A mode with higher transportation costs can be justified if it results in significantly lower _____.	01	1	4
	1.9	Fluctuations in orders increases as they move up the supply chain from retailers to wholesalers to manufacturers to suppliers is called _____.	01	1	4
	1.10	In the context of 'Managerial Levers to Achieve Coordination', sharing point of sale data improves _____ accuracy.	01	1	4
	1.11	List the basic steps in achieving strategic fit.	02	1	1
	1.12	Graphically show the relationship between number of facilities and inventory costs, transportation costs, facilities cost.	02	1	2
	1.13	What are the managerial levers to reduce safety inventory without reducing product availability?	02	1	3
	1.14	Give an example to demonstrate Milk run concept.	02	2	4
	1.15	Illustrate the key participants involved in a Business-to-Consumer (B2C) e-commerce transaction with practical example.	02	2	4

PART-B

2	a	Discuss the cycle and push/pull views of a supply chain. Which of the two views is very useful when considering strategic and operational decisions?	10	3	1
	b	Justify this statement with an example: 'It is necessary to achieve a fit between competitive and functional strategies'.	06	4	1

3	a	With the help of block diagram, explain a framework for supply chain network design decisions.	10	2	2
	b	Draw the diagram of 'Manufacturer storage with direct shipping' type of design option for transportation. Identify it's any THREE performance characteristics connected to service.	06	2	2
		OR			
4	a	Critically analyze the impact of globalization on supply chain networks and discuss the key risks that arise in global supply chains.	10	4	2
	b	During network design, managers need a methodology that allows them to estimate the uncertainty in their forecast of demand and price and then incorporate this uncertainty in the decision-making process. In this context, explain such a methodology for evaluating network design decisions under uncertainty.	06	2	2
5	a	Daily demand for PCs at Dell is normally distributed, with a mean of 2500 and a standard deviation of 500. A key component in PC assembly is the hard drive. The hard drive supplier takes an average of 7 days to replenish inventory at Dell. Dell is targeting CSL of 90 percent (providing a fill rate close to 100 percent) for its hard drive inventory. Evaluate the safety inventory of hard drives that Dell must carry if the standard deviation of lead time is seven days. Further, Dell is working with the supplier to reduce the standard deviation to zero. Evaluate the reduction in safety inventory that Dell can expect as a result of this initiative. Compare above 2 cases of safety inventory evaluations and bring out the key insight about impact of lead time uncertainty on safety inventory.	10	4	3
	b	Identify managerial levers that reduce lot size and cycle inventory in supply chain without increasing cost.	06	2	3
		OR			
6	a	Elaborate the impact of demand and supply uncertainties on the safety inventory in an organization.	10	3	3
	b	Explain the role of Information Technology (IT) in effective inventory management.	06	2	3
7	a	Evaluate the strengths and weakness of different modes of transportation.	10	2	4
	b	Given the customer density and distance from warehouse how should a firm like W.W. Grainger should tailor its transportation network? Explain.	06	2	4
		OR			
8	a	What is the role of supply contracts? Elaborate on types of contracts and their impact on supplier performance and information distortion.	10	2	4
	b	What are some ways that a firm like Wal-Mart benefits from good sourcing decisions?	06	2	4
9	a	What do you mean by bullwhip effect? In the context of supply chain management, how do coordination mechanisms and the bullwhip effect impact performance outcomes? Discuss.	10	3	4

	b	Elaborate on the core processes of CRM and their importance in managing customer relationships.	06	2	4
		OR			
10	a	What do you understand by VMI? Compare it with continuous replenishment and explain how manufacturers benefit from VMI.	10	3	4
	b	How does collaboration through CPFR impact supply chain performance? Discuss.	06	2	4

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)
VI Semester B. E. Examinations September-2023
Industrial Engineering and Management

SUPPLY CHAIN MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Decisions regarding inventory policies to be followed are made during _____ phase of supply chain.	01
	1.2	_____ cycle occurs at the manufacturer/supplier interface and includes all processes to ensure that materials are available for manufacturing to occur according to schedule.	01
	1.3	Graphically show intracompany intrafunctional scope of strategic fit.	02
	1.4	_____ analysis evaluates the present value of any stream of future cash flows and allows managers to compare different cash flow streams in terms of their financial value.	01
	1.5	Highlight the classification of network design decisions.	02
	1.6	Average flow time is calculated using the expression _____.	01
	1.7	The discount factor used to obtain the present value of money in the next period where k represents the rate of return is _____.	01
	1.8	State the two reasons which leads quantity discounts to increase the supply chain surplus.	02
	1.9	The set of business processes required to purchase goods and services is called _____.	01
	1.10	What do you mean by vendor managed inventory system?	02
	1.11	Party that requires the movement of products between two points in the supply chain is called _____.	01
	1.12	Cooperative supply chain relationships developed to enhance the overall business performance of both parties is a definition of _____.	01
	1.13	Mention the salient features of second price or Vickrey auction.	01
	1.14	Citing an example clarify which are the parties involved in $B2C$ e-business.	02
	1.15	If the manager at ABC store manages inventory such that the store does not run out of inventory in 6 out of 10 replenishment cycles, then store achives a CSL of _____%.	01

PART B

2	a	Discuss the push/pull processes in the Dell supply chain and identify the push/pull boundary.	08
	b	Compare and contrast efficient and responsive supply chains with examples.	08
3	a	Suggest a suitable design of distribution network for commodity products. Discuss the key features of such a design.	06
	b	Steel Appliances(SA) manufactures high quality refrigerators and cooking ranges. SA has one assembly factory located near Bangalore from which the products are supplied to the entire country. Demand has grown rapidly and the CEO of SA has decided to set up another factory to serve its eastern markets. The supply chain manager is asked to find a suitable location for the new factory. Three plants located in Chennai, Pune and Noida will supply parts to the new factory, which will serve markets in Assam, Meghalaya, Manipur, Nagaland and Arunachal Pradesh. The coordinate location, the demand in each market, required supply from each plant and the cost for each supply source or market are given in the following table. Use gravity location model to locate the new plant, also calculate the total transportation cost.	10

$Sources/markets$	$Transportation\ cost\ Rs./Ton\ km$	$Quantity\ in\ Tons$	$Coordinates$	
			Xi	Yi
$Supply\ sources$				
$Chennai$	0.90	500	700	1200
$Pune$	0.95	300	250	600
$Noida$	0.85	700	225	825
$Markets$				
$Assam$	1.50	225	600	500
$Meghalaya$	1.50	150	1050	1200
$Manipur$	1.50	250	800	300
$Nagaland$	1.50	175	925	975
$Arunachal\ Pradesh$	1.50	300	1000	1080

OR

4	a	How has globalization made strategic fit even more important to a company's success?	05
	b	Motorola has gone from manufacturing all its cell phones in house to almost completely outsourcing the manufacturing. What are the pros and cons of the two approaches?	05
	c	Locate the following product on the implied uncertainty (demand) spectrum: Salt, Rice, tooth brush, small cars, $SUV, IPOD$, Mobile phone with blue-tooth technology, Text books, Coffee, credit card.	06
5	a	Explain the role of cycle and safety inventory in supply chain practices.	08
	b	Weekly demand for microwave ovens at Best Buy store is normally distributed with a mean of 2500 and a standard deviation of 500. It takes two weeks to supply a Best Buy order. Best Buy targeting a CSL of 90% and monitors its inventory continuously. How much safety inventory of microwave ovens should Best Buy carry?	08

		OR	
6	a	Consider the following marginal unit discount problem. $\begin{array}{\|c\|c\|}\hline \textbf{\textit{Order Quantity}} & \textbf{\textit{Unit price (in Rs.)}} \\ \hline 0-4000 & 15/- \\ \hline 4000-7000 & 14/- \\ \hline Over\ 7000 & 13/- \\ \hline \end{array}$ Order cost= $Rs.150/-$, Holding charges= 10%, Annual Demand= 6000 units. Evaluate the order quantity.	10
	b	Weekly demand for 12" frames at the Frame shop is normally distributed with a mean of 250 and a standard deviation of 150. The store manager continuously monitors inventory and currently orders 1,000 frames each time the inventory drops to 600 frames. The manufacturer currently takes two weeks to fill an order. How much safety inventory does the store carry?	06
7	a	Explain the factors affecting transportation decisions.	10
	b	What is tailored transportation? Explain its importance in distribution.	06
8	a	Describe supply chain co-ordination and the bull whip effect and their impact on supply chain performance.	10
	b	What factors lead to batching of orders within a supply chain? How does this affect co-ordination? What actions can minimize large batches and minimize co-ordination.	06

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Examinations August-2022

Industrial Engineering and Management

SUPPLY CHAIN MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Cisco's decisions regarding its choice of supply sources for components, contract, manufacturing, and location and capacity of its warehouses are all under the purview of SC _____ decisions.	01
	1.2	Draw the cost-responsiveness efficient frontier curve.	02
	1.3	_____ cycle occurs at the manufacturer/ supplier interface and includes all pressures to ensure that materials are available for manufacturing to occur according to schedule.	01
	1.4	The coordinate location of a manufacturer is (40, 60). The coordinate location of its customer is (20,10). The manufacturer is required to supply 200 compressors to this customer. The cost of transporting one compressor is Rs.100. Using the gravity location model, determine the distance between the manufacturer and supplier, and the total transportation cost.	02
	1.5	Identify any two macro-economic factors that affect or influence the network design decisions in a supply chain.	02
	1.6	List the risks that are to be considered during network design.	02
	1.7	If the manager at B& M manages inventory such that the store does not run out of inventory in 6 out 10 replenishment cycles, then stores achieves a CSL of _____%.	01
	1.8	What are the two factors which determine appropriate level of safety inventory?	02
	1.9	A _____ is a transportation method in which a truck either delivers product from a single supplier to multiple retailers or goes from multiple suppliers to a single retailer.	01
	1.10	Mention any two benefits of effective sourcing decisions.	02
	1.11	Identify the supply macro processes.	02
	1.12	List any four managerial levers to achieve coordination.	02

PART B

2	a	Sketch the framework for structuring the drivers of supply chain performance. Choose any one of the Indian organizations and discuss the structuring of its drivers.	08
	b	Sketch and explain the push/pull process view of the Dell supply chain.	08
3	a	Discuss in detail about the role, factors influencing with respect to facility location for a pharmaceutical Supply chain.	08
	b	Identify the uncertainties influencing supply chain network design.	08
		OR	
4	a	What is a decision tree? Elucidate the methodology involved in evaluating supply chain network design decisions using decision tree analysis.	08
	b	Discuss how you evaluate the performance of a distribution network along the two dimensions namely: customer needs and the cost.	08
5	a	Consider the following marginal unit discount problem. **Order Quantity** | **Unit price (in Rs.)** 0-4000 | 15/- 4000-7000 | 14/- Over 7000 | 13/- Order cost= Rs. 150/-, Holding charges=10%, Annual Demand=6000 units Evaluate the economic order quantity.	10
	b	Weekly demand for 12" frames at the Frame shop is normally distributed with a mean of 250 and a standard deviation of 150. The store manager continuously monitors inventory and currently orders 1,000 frames each time the inventory drops to 600 frames. The manufacturer currently takes two weeks to fill an order. How much safety inventory does the store carry?	06
		OR	
6	a	Explain the role of cycle and safety inventory in supply chain practices.	08
	b	Weekly demand for microwave ovens at Best Buy store is normally distributed with a mean of 2500 and a standard deviation of 500. It takes two weeks to supply a Best Buy order. Best Buy targeting a CSL of 90% and monitors its inventory continuously. The manufacturer currently takes two weeks to fill an order. How much safety inventory of microwave ovens should Best Buy carry?	08
7	a	Identify the trade-offs that managers should consider when making transportation decisions. Explain.	10
	b	Justify in the context of tailored transportation: "a firm can meet customer needs at a lower cost by using tailored transportation to provide the appropriate transportation choice based on customer characteristics."	06
8	a	Describe supply chain co-ordination and the bull whip effect and their impact on supply chain performance.	10
	b	What factors lead to batching of orders within a supply chain? How does this affect co-ordination? What actions can minimize large batches and minimize co-ordination.	06

Ergonomics

VI Semester SEE Question Papers

Course codes: IM363IA

- Aug 2025 (IM363IA)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations August-2025

Industrial Engineering and Management

ERGONOMICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Define Human Factors Engineering.	02	1	1
	1.2	State two goals of Human Factors Engineering.	02	1	1
	1.3	What is system thinking in the context of ergonomics?	02	2	1
	1.4	Mention any two uses of anthropometric data.	02	1	2
	1.5	What is the role of NIOSH in workplace ergonomics?	02	2	1
	1.6	List any two differences between discrete and continuous controls.	02	3	3
	1.7	What is meant by “Design for All”?	02	1	1
	1.8	Define selective attention with an example.	02	2	2
	1.9	Mention any two interaction styles in HCI.	02	1	1
	1.10	What is divided attention? Provide an example.	02	1	2

PART-B

2	a	Explain the process of Human Factors Engineering with a suitable example.	08	2	1
	b	Illustrate how understanding users and tasks enhances ergonomics design.	08	3	3
3	a	Explain the biomechanics of the human spine during lifting activities.	08	4	1
	b	Describe the principles for designing workstations for sitting and standing posture.	08	3	2
		OR			
4	a	Explain the principles of workspace layout using anthropometric principles.	08	3	1
	b	What are the common low back problems encountered in industrial lifting tasks?	08	2	3
5	a	Explain the types of controls and their ergonomic requirements.	08	2	2
	b	Describe the role of labels, icons, and integrative displays in control rooms.	08	2	2
		OR			
6	a	Discuss the fifteen principles of display design with examples.	08	2	1
	b	Discuss ergonomics design principles for auditory and visual displays.	08	3	2

7	a	Explain how cultural factors influence ergonomic design for diverse populations.	08	2	2
	b	Discuss the ergonomics design strategies for people with functional limitations.	08	3	2
		OR			
8	a	Explain the considerations for designing products for elderly users.	08	2	2
	b	Explain the scope and benefits of "Design for All" concept in the context of Ergonomic Design of products.	08	2	4
9	a	Discuss the effect of divided attention and time sharing in cognitive.	08	2	3
	b	Explain how selective attention and perception influence task performance.	08	2	3
		OR			
10	a	Discuss the fifteen principles of HCI design.	08	2	3
	b	Describe various interaction styles in HCI. Give suitable examples.	08	2	2

Human Resource Management (and Development / Analytics)

VI Semester SEE Question Papers

Course codes: IM364TA, 21IM67, 18IM6D2, 16IM6D4, 12IM6C4

- Aug 2025 (IM364TA)
- Sep/Oct 2024 (21IM67)
- Sep 2023 (18IM6D2)
- Aug 2022 (18IM6D2)
- Grade Improvement Nov 2020 (16IM6D4)
- Apr/May 2019 (16IM6D4)
- Apr/May 2017 (12IM6C4)
- Apr/May 2016 (12IM6C4)

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Regular Examinations August-2025

Industrial Engineering and Management

HUMAN RESOURCE MANAGEMENT & ANALYTICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Name any four duties of an effective HR Manager.	02	1	1
	1.2	Explain how job analysis supports effective recruitment and selection.	02	2	1
	1.3	List two types of employee interviews.	02	1	2
	1.4	What is an unstructured interview? Mention one disadvantage.	02	2	2
	1.5	What are the main objectives of employee training programs?	02	2	3
	1.6	Define performance management and state one benefit.	02	1	3
	1.7	What is an HR dashboard? How does it help HR decision - making?	02	2	4
	1.8	Why is aligning HR analytics with business objectives important?	02	2	4
	1.9	Define predictive analytics in HR.	02	1	4
	1.10	Mention any two tools used for HR predictive analytics.	02	1	4

PART-B

2	a	Describe the talent management process .How does it contribute to achieving strategic organizational goals?	08	2	1
	b	You have been assigned to develop a detailed job description for a newly created Customer Service Manager role in a rapidly growing e-commerce company. Discuss the various methods you would use to conduct job analysis for this position and justify why these methods are appropriate for this role and Industry.	08	4	1
3	a	Compare internal and external recruitment sources. Provide examples of each in different organizational contexts.	08	2	2
	b	A mid-sized IT services company is facing high employee turnover, especially among skilled software developers, and is experiencing significant skill shortages impacting project delivery timelines. Discuss how workforce planning and forecasting can help address these issues effectively. Suggest specific tools or techniques the company can use to forecast workforce needs and develop strategies to retain and attract talent.	08	4	2
		OR			
4	a	Evaluate the effectiveness of work samples and simulations as employee selection tools. For what types of jobs are these methods most useful?	08	3	2

	b	A growing telecommunications company is planning to replace its current unstructured interview process with structured interview approach for hiring Customer Service Representatives. Compare unstructured and structured interviews, highlighting the benefits of structured interviews in improving hiring outcomes. Additionally, outline the key steps you would take to design an effective structured interviews guide tailored specifically for Customer Service Representative roles in this company.	08	4	2
5	a	Discuss the stages involved in the employee training process and outline key considerations when designing an effective training program.	08	2	3
	b	A Fast-growing fintech startup is worried that its current onboarding process is not effectively supporting new employees, leading to low retention rates within the first year. Critically elevate this onboarding process, propose key metrics the company should track to measure onboarding success over time and recommend strategies to enhance employee retention and engagement through improved onboarding practices.	08	4	3
		OR			
6	a	Define performance management and differentiate it from performance appraisal. Discuss essential components of a performance management system that supports continuous development.	08	2	3
	b	A national retail chain has recently revamped its performance management system to include self-assessments, manager feedback, and peer evaluations. Evaluate the effectiveness of incorporating these multiple feedback source in driving employee development. Discuss potential challenges the organization might face when implementing this multi-rater system, and suggest ways to ensure it effectively supports ongoing talent development.	08	4	3
7	a	Explain the role of predictive analytics in HR. Illustrate how HR departments can use predictive analytics to forecast turnover, hiring needs, and employee performance.	08	3	4
	b	A growing manufacturing firm relies heavily on HR analytics that primarily analyze candidates' qualifications and work experience during recruitment. Critically assess the limitation of focusing only on these data points. Suggest other types of data or analytical methods that could provide deeper insights to improve hiring quality and reduce employee turnover. Explain how predictive analytics can be leveraged to enhance recruitment outcomes in this context.	08	4	4
		OR			
8	a	Describe statistical modeling and its application in HR research. How can HR professionals use statistical models to analyze trends and make decisions on employee performance or turnover?	08	3	4

	b	A global consumer goods company uses HR dashboards to track training participation and completion rates. However, senior management believes the dashboards fail to provide sufficient insights into how training impacts employee performance and business outcomes. Critically evaluate the current use of HR dashboards for monitoring training effectiveness. Identify common shortcoming of such dashboards and suggest ways to redesign them to deliver more comprehensive, actionable insights that link training efforts to organizational goals.	08	4	4
9	a	Define root cause analysis and explain its application in HR analytics. Provide an example of how it can uncover causes of high turnover or absenteeism.	08	3	4
	b	A multinational corporation has extensively datafied its HR processes, using extensive employee data to drive decisions and improve operational efficiency. While this has led to productivity gains, many employees feel reduced to mere data point, resulting in lower morale and engagement. Critically evaluate the impact of HR datafication on employee morale and engagement. Recommend strategies the organization can implement to balance the benefits of data-driven HR with a more human-centric, empathetic approach to managing its workforce.	08	4	4
		OR			
10	a	Explain various tools and techniques used in HR predictive analytics. Provide a real-world example where a specific tool improved recruitment or employee engagement.	08	3	4
	b	A mid-sized retail company is considering investing in predictive analytics tools to enhance its recruitment process but is concerned about the return on investment (ROI) and whether the costs will be justified. Critically evaluate the potential ROI of using predictive analytics in recruitment. Discuss how the company can measure the effectiveness of its predictive models in terms of cost savings, improved quality of hires, and reduction in employee turnover.	08	4	4

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations Sept/Oct– 2024

Industrial Engineering and Management

HUMAN RESOURCE MANAGEMENT

Time: 02 Hours ***Maximum Marks: 50***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

			M	BT	CO
1	1.1	What is the purpose of questionnaires in job analysis?	02	1	1
	1.2	How do line managers contribute to *HRM*?	02	2	2
	1.3	What is personnel planning?	02	2	3
	1.4	What does "talent management" primarily focus on?	02	1	4
	1.5	What is the purpose of a needs assessment in training?	02	2	5

PART-B

2	a	Elaborate on the core competencies which are essential for *HRM* professionals?	06	2	1
	b	What are the different methods used to collect job analysis information?	06	2	1
3	a	How do work samples and simulations differ in their approach to evaluating candidates, and when might each be most effectively used?	07	3	2
	b	What is the selection process in recruitment? And list the steps that are typically involved.	07	1	2
		OR			
4	a	What are the various methods organizations use to recruit candidates internally. How do they benefit the company?	07	3	3
	b	State the differences between Structured Interviews and Unstructured Interviews.	07	2	3
5	a	Enumerate the best practices for effectively implementing a training program?	07	2	4
	b	How do performance management and performance appraisal differ from each other?	07	2	4
		OR			
6	a	In what ways does performance management contribute to an organization's success?	07	3	5
	b	What is employee appraisal? What role does it play in talent management?	07	3	5

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations September-2023

Industrial Engineering and Management

HUMAN RESOURCE MANAGEMENT AND DEVELOPMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Distinguish between line authority and staff authority.	01
	1.2	What does equal employment opportunity co-ordinations specialization involve as an included activity?	01
	1.3	What is strategic human resource development?	01
	1.4	What is strategy map as a tool?	01
	1.5	What is meant by job analysis?	01
	1.6	Write a note on job enrichment.	01
	1.7	What is the central idea behind virtual job fairs?	01
	1.8	Mention the two goals of on-campus recruiting.	01
	1.9	What is the aspect that is demonstrated in the content validity of a recruitment test?	01
	1.10	What does 'intelligent tests' measure?	01
	1.11	Name any two of the "*BIG FIVE*" traits focused by Industrial psychologists.	01
	1.12	In one sentence, describe the concept of a Management Assessment Centers.	01
	1.13	Mention the main purpose of Employee Orientation.	01
	1.14	What is the utility of the On-the-Job training?	01
	1.15	What is Vestibule training?	01
	1.16	What is organizational development? (In one sentence).	01
	1.17	Why it is generally considered to take 360-degree feedback more for development purpose than for pay purpose? Justify.	01
	1.18	Clearly describe a Behavioral Anchored Rating Scale (*BARS*) as an appraisal tool.	01
	1.19	Give any one reason as to why employee engagement is important.	01
	1.20	What is the meaning of the term psychological contract?	01

PART B

2	a	Why is it important for companies today to make their human resources into a competitive advantage? Explain how *HR* can contribute to doing this. What are some examples of employers who do this?	08

	b	What is the difference between a strategy a version and a mission? Give an example of each. Which one of the three contributes to create competitive advantages? Why?	08
3	a	Why should managers think of staffing, training, appraising and paying employees as part of the talent management process? Briefly summarize.	08
	b	Do you think companies can really do without detailed job description? Why or why not? In a company with only 25 employees, is there less need for job description? Why or why not?	08
		OR	
4	a	What are the 5 main sources of job candidates? Mention any one pro and any one cons of each of the five sources.	08
	b	Why is it important to understand how to recruit a more diverse workforce? What are the five main things a recruiter needs to do to recruit and retain a more diverse workforce?	08
5	a	What is the difference between reliability and validity in the context of employee testing? Why is it important to verify reliability and validity before administring the test for prospective employees? In what respects reliability and validity are similar?	08
	b	What are the advantages that Google realized by using screening techniques called "crowd sourcing"? How does this process work? How does this technique fit into Google's fast growth strategy?	08
		OR	
6	a	Distinguish clearly between structured interviews versus unstructured interviews. Why are structured interviews generally considered to be superior?	08
	b	Briefly discuss and give examples of atleast 5 common interviewing mistakes. What recommendations would you give for avoiding these interviewing mistakes?	08
7	a	"A well thought out orientation program is essential for all new employees, whether they have experience or not". Explain the key ideas of an orientation program and why you agree or disagree with this statement.	08
	b	What are some typical on-the-job training techniques? What do you think are some of the main drawbacks of relying on informal on-the-job training for breaking new employees into their job?	08
8	a	Discuss the pros and cons of the following four performance appraisal methods (list any one pro and one con). i) Alternation ranking method ii) Paired comparison method iii) Forced comparison method iv) Critical incident method.	08
	b	Compare and contrast performance management and performance appraisal. How does the use of information technology tools support performance management? Discuss the utility of digital dashboards.	08

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Examinations August-2022

HUMAN RESOURSE MANAGEMENT AND DEVELOPMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	What is meant by the term Human capital ?	02
	1.2	Elaborate on the term "Functional strategy" in the context of human resource management strategy.	02
	1.3	Distinguish between "Job description" and "Job specification".	02
	1.4	What is meant by the term "Job posting".	02
	1.5	Explain the term "Negligent hiring".	02
	1.6	Bring out the purpose of Miniature job training and evaluation approach in the hiring of candidates.	02
	1.7	What is a competency model?	02
	1.8	Mention the essence of Job instruction training.	02
	1.9	"Supervisors often rate unfriendly employees lower on all traits". What is this potential appraisal problem termed as?	02
	1.10	An appraisal process culminates with an interview. What is the purpose of this interview?	02

PART B

2	a	Illustrate the HR management responsibilities of line and staff managers.	08
	b	Explain each of the seven steps in the strategic planning process.	08
3	a	With the help of an illustration, describe the uses of Job analysis information in executing overall strategic plans.	08
	b	There is no standard format for writing job description. However, the description contain several common sections. Considering the same, draft a job description for the post of "Sales representative".	08
		OR	

4	a	Name and describe the main internal resources of the candidates in Recruitment process.	08
	b	Discuss the relevance of Recruiting Yield Pyramid in organizational hiring process. Support your answer with an illustrative example.	08
5	a	"Validity tells you whether the test is measuring what you think its supposed to be measuring". In the light of above statement, write a note on the following: i. Test validity ii. Content validity iii. Construct validity iv. Criterion validity.	08
	b	What is management assessment center? Review any four typical simulated tasks in the above method of testing.	08
		OR	
6	a	Provide an overview on any four methods of administering the interviews.	08
	b	List and explain the main ways/steps of conduction to make the interview useful and effective	08
7	a	Briefly review the five steps involved in a successful training process.	08
	b	Review the four most commonly used training techniques in implementation of training program.	08
8	a	Discuss the following techniques of performance appraisal. i. Graphic rating scale. ii. Behaviorally anchored rating scales.	08
	b	Building comprehensive employee retention programs by organizations follow certain steps. Review the steps.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Grade Improvement Examinations Nov-2020

Industrial Engineering and Management

HUMAN RESOURCE MANAGEMENT & DEVELOPMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of statistical tables permitted.

PART A

1	1.1	List out the three important duties the human resource manager would carry out.	02
	1.2	What is meant by Strategic Human Resource Management and list any two important tools used for it.	02
	1.3	How the Job Specifications for Trained versus Untrained Personnel would differ?	02
	1.4	What are Trend Analysis and Ratio Analysis in the context of recruitment and selection process?	02
	1.5	List any two ways that would make the Interview more useful.	02
	1.6	Distinguish unstructured interview and situational interview.	02
	1.7	List the five steps in the training process.	02
	1.8	How the computer based training would be more effective in training process? List any two points.	02
	1.9	List out any four aspects to be kept in mind when conducting the interview.	02
	1.10	List the 3-steps in performance appraisal process.	02

PART B

2	a	As a new member of the Board of Directors for a local bank, Jack Nelson was being introduced to all the employees in the home office. When he was introduced to Ruth Johnson, he was curious about her work and asked her what the machine she was using did. Johnson replied that she really did not know what the machine was called or what it did. She explained that she had only been working there for 2 *months*. However, she did know precisely how to operate the machine. According to her supervisor, she was an excellent employee. At one of the branch offices, the supervisor in charge spoke to Nelson confidentially, telling him that something was wrong, but she didn't know what. For one thing, she explained, employee turnover was too high, and no sooner had one employee been put on the job than another one resigned. With customers to see and loans to be made, she continued, she had little time to work with the new employees as they came and went.	

		All branch supervisors hired their own employees without communication with the home office or other branches. When an opening developed, the supervisor tried to find a suitable employee to replace the worker who had quit. After touring the 22 branches and finding similar problems in many of them, Nelson wondered what the home office should do or what action he should take. The banking firm generally was regarded as being a well-run institution that had grown from 27 to 191 employees during the past 8 *years*. The more he thought about the matter, the more puzzled Nelson became. He couldn't quite put his finger on the problem, and he didn't know whether to report his findings to the president. Questions: i) What do you think is causing some of the problems in the bank's home office and branches? ii) Do you think setting up an HR unit in the main office would help? iii) What specific functions should an HR unit carry out? What HR functions would then be carried out by supervisors and other line managers? What role should the Internet play in the new HR organization?	08
	b	Explain the concept of competitive and functional strategies by considering an example of one from service sector, one from manufacturing sector and one from *IT* sector.	08
3	a	What is Talent Management? Describe in detail the six steps in doing a job analysis.	08
	b	Considering an example for a 'Manager' post in a nationalized bank, write the job descriptions, including summaries and job functions, using the Internet and traditional methods.	08
		OR	
4	a	Describe briefly the main internal and any five outside sources of candidates in recruiting for a job.	08
	b	Considering an example of college campus recruitment, explain briefly the steps in recruitment and selection process.	08
5	a	Explain briefly the five steps in validating a test. Consider an example of your own and explain briefly the validation of such test.	08
	b	Describe the eight tests you could use for employee selection and how you would use them?	08
		OR	
6	a	Explain briefly the two examples of work sample/simulation tests.	06
	b	Describe briefly, the different steps to design and conduct an effective interview.	10

7	a	A well-thought-out orientation program is essential for all new employees, whether they have experience or not. Explain why you agree or disagree with this statement.	10
	b	Enumerate on any six management development programs in an organization.	06
8	a	List the techniques for appraising performance of employees.	10
	b	Compare and contrast the concept of performance management and performance appraisal.	06

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

HUMAN RESOURCE MANAGEMENT AND DEVELOPMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	List the four objectives of HRM.	02
	1.2	List any four benefits of strategic management.	02
	1.3	Name the different methods of collecting job data. (At least four methods)	02
	1.4	List the internal and external factors governing recruitment process.	02
	1.5	Write a note on ability tests and personality tests used in recruitment and selection process.	02
	1.6	List the three objectives of interviews.	02
	1.7	List any four points regarding how training benefits the organization.	02
	1.8	List any four learning principles.	02
	1.9	List the four guidelines for setting performance goals.	02
	1.10	Write the features of Behaviourally anchord rating scales ($BARs$).	02

PART B

2	a	Using block diagram, elaborate line and staff organization. Mention any two advantages and disadvantages.	08
	b	List the importance of HR management in today's organizational context.	08
3	a	What is Job Analysis? Elaborate it in detail with an appropriate example.	08
	b	Explain any four demand forecasting techniques used for manpower planning.	08
		OR	
4	a	Illustrate the job design and job specification for purchase manager.	08
	b	What are the internal factors governing recruitment? Explain any four of them in detail.	08
5	a	With the help of neat sketch explain different steps involved in selection process.	08
	b	List and ascertain any three types of selection tests.	08
		OR	

6	a	Elaborate on any four types of Interview.	08
	b	Discuss: i) Back ground investigation ii) Work sample.	08
7	a	Ascertain at least eight benefits of training the employee.	08
	b	Differentiate "on the job training" and "off the job training" concepts of training an employee.	08
8	a	Explain 360 degree appraisal method? Assess at least two merits and two demerits.	08
	b	List the traditional methods of appraisal. Outline the features of any two methods of appraisal.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

HUMAN RESOURCE MANAGEMENT AND DEVELOPMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	What is the importance of *HRM* in an organization?	02
	1.2	List the objectives of Human Resource Management.	02
	1.3	Differentiate *HRM* policy and *HRM* procedure.	02
	1.4	What is Human Resource Planning?	02
	1.5	Differentiate between recruitment and selection.	02
	1.6	List advantageous of performance appraisal in organization.	02
	1.7	What are the benefits of 360 degree feedback?	02
	1.8	What is employee welfare?	02
	1.9	List any four employee welfare laws followed by industries in India.	02
	1.10	List any four non-statutory welfare benefits.	02

PART B

2	a	Briefly discuss the scope of Human Resource Management.	08
	b	List and briefly discuss the functions of Human Resource Management.	08
		OR	
3	a	Briefly discuss the steps involved in Human Resources Planning process	08
	b	Briefly discuss how the job analysis is being performed in *HRM*.	08
4	a	Discuss the factors to be considered for developing recruitment plan.	08
	b	List and discuss the steps involved in selection process of an employee.	08
		OR	
5	a	Illustrate the training stages in training process from a system perspective.	08
	b	What is training evaluation? What are the objectives and purpose of evaluation? and what is to be evaluated?	08

6	a	What is performance appraisal? List the objectives of performance appraisal.	08
	b	Write a note on performance appraisal tools and techniques and also list performance appraisal biases.	08
		OR	
7	a	Briefly discuss how 360 degree performance appraisal is being performed in industries?	08
	b	Briefly discuss the legal issues associated with performance appraisals.	08
8	a	Write a note on employee welfare measures. List the objectives of employee welfare measure scheme in an organization.	08
	b	List and discuss the different types of welfare activities carried out by industries.	08
		OR	
9	a	List different approaches to labor welfare in organizations and discuss any two of them.	08
	b	List statutory welfare benefits schemes or provisions provided by industries to employee health.	08
10	a	What is human resource accounting? Discuss the benefits and limitations of the same.	08
	b	What are the reasons that encouraged global organizations to develop international HR strategies? and list the differences between $IHRMN$ and domestic HRM.	08
		OR	
11	a	Discuss why is Management of Diversity important? What are the major issues in Management of Diversity in today's organizations?	08
	b	Discuss the role of human resource management in employee retention in organizations.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

HUMAN RESOURCE MANAGEMENT AND DEVELOPMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART-A

1	1.1	List out the functions of HRM.	02
	1.2	Compare HRM with HRD in two to three sentences.	02
	1.3	List out any five benefits of Manpower Planning.	02
	1.4	Distinguish between Job Description and Job Specification in two sentences.	02
	1.5	Summarize 'Competency Mapping' in two sentences.	02
	1.6	Recognize any four approaches to Executive Development.	02
	1.7	Identify any five employee welfare measures.	02
	1.8	Identify any four components for all round development.	02
	1.9	Explain HR Audit in two sentences.	02
	1.10	Identify any five rewards given to employees in manufacturing industries.	02

PART-B

2	a	List and explain the objectives of Human Resource Management and Development.	08
	b	Distinguish between Human Resource Management and Human Resource Development. Give suitable examples.	08
		OR	
3	a	Explain Man Power Planning and relate it to Human Resource Inventory.	08
	b	Discuss the methods of manpower forecasting.	08
4	a	Classify the various sources and techniques of recruitment.	08
	b	Examine the various tests used for scientific selection and suggest a method for selecting a 'lab attender'.	08
		OR	

5	a	Identify the training needs for 'lab attenders' and choose a method to train them and justify good choice.	08
	b	Formulate a training program for 'Lab Attenders' and briefly estimate a budget for the same making suitable assumptions. State explicitly the assumptions you have made.	08
6	a	Identify and briefly bring out the advantages and limitations of any two methods of Performance Appraisal.	08
	b	Develop a suitable performance appraisal system for 'lab attenders'.	08
		OR	
7	a	Examine the '360° method of Appraisal' and evaluate its applications to a 'Front office Bank Clerk'.	08
	b	Investigate the emerging trends of the *PA* process and predict the same for appraising 'Bank Clerks' during the next decade.	08
8	a	Classify the types of Employee Welfare Measures and develop a program to motivate the lab attenders to your college.	08
	b	Distinguish between Statutory and Non-Statutory Welfare programs.	08
		OR	
9	a	Recognize the approaches to employee welfare and identify a suitable approach to cover the non-teaching staff of your institution.	08
	b	Explain how employee welfare measures could be administered.	08
10	a	Describe 'Human Resources Audit' and list out its advantages.	08
	b	Discuss the characteristics of International Human Resources Management.	08
		OR	
11	a	Identify the merits and demerits of flexible working.	08
	b	Examine the work life balance among *IT* professionals and give your comments.	08

Elements of Financial Management

VI Semester SEE Question Papers

Course codes: IM266TEQ, 21IE6F8

- Aug 2025 (IM266TEQ)
- Sep/Oct 2024 (21IE6F8)

IM266TEQ

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations August-2025

Industrial Engineering and Management

ELEMENTS OF FINANCIAL MANAGEMENT (Institutional Elective)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of Interest factor table is permitted.

PART-A

			M	BT	CO
1	1.1	List the three broad areas of Financial Decision making in the organization.	02	1	1
	1.2	What is meant by the term Financial Intermediary? Give an example.	02	2	1
	1.3	List the three important Financial Statements that help in understanding the financial health of a firm.	02	1	1
	1.4	Distinguish between the terms Par Value and Coupon Rate in the context of Bond/Debenture.	02	2	2
	1.5	What is meant by the term "Yield to Maturity"?	02	2	2
	1.6	List the three distinct features of Capital Budgeting.	02	1	3
	1.7	Distinguish between the terms, Authorized Capital and Issued Capital.	02	2	4
	1.8	Why is Preference Capital considered as Hybrid form of Financing.	02	2	3
	1.9	List any two types of Intercorporate Deposits.	02	1	3
	1.10	What is meant by Factoring in the context of Working Capital Financing?	02	2	3

PART-B

2	a	Critically assess the contribution of Financial Intermediaries in the financial inclusion and economic development of a nation.	08	3	1
	b	Examine the recent growth trends in the Indian Financial System. In your answer, focus on to what extent digitization contributed to this growth.	08	3	1
3	a	Explain the key components of a firm's annual report.	05	2	2
	b	A company decides to deposit $Rs.20{,}000$ at the end of each year for 5 years in an account that earns 7% annual interest, compounded annually. Calculate the future value of this annuity.	05	3	2
	c	A firm expects to receive the following cash flows over the next 4 years: $Year$ | $Cash\ Flow\ (Rs)$ 1 | 5,000 2 | 7,000 3 | 6,000 4 | 8,000 If the discount rate is 10%, calculate the present value of these cash flows.	06	3	2

OR

4 a A bond has the following features:

- Par Value = $Rs.1{,}000$
- Coupon Rate = 8% (annual interest)
- Time to Maturity = 5 $years$
- Required Rate of Return (Discount Rate) = 10%

Calculate the value of the bond. 05 3 2

b A bond with a par value of $Rs.1{,}000$, coupon rate of 10% (annual coupon), and 5 years to maturity is currently priced at $Rs.920$. Calculate the approximate Yield To Maturity (YTM) using the formula approach. 05 3 2

c A firm is expected to grow at the rate of 6% per annum. The dividend expected on the firm's equity share a year hence is $Rs.2/-$. What price will you put on it if your Required Rate of Return for this share is 14%? 06 3 2

5 a Distinguish between Market Risk and Unique Risk. Support your explanation with suitable example for each. 06 2 2

b An investor is analyzing a stock that performs differently under various economic conditions. The following data summarizes the possible states of the economy, the probability of each state, and the expected return in each state:

State of Economy	*Probability*	*Rate of Return* (%)
Boom	0.3	20
Normal	0.5	12
Recession	0.2	−5

Calculate:

(i) The expected Rate of Return

(ii) The standard Deviation of Return. 10 3 2

OR

6 a Enumerate the key phases involved in Capital Budgeting Process. 06 2 3

b A project requires an initial investment of $Rs.50{,}000$ and is expected to generate the following cash flows over 4 years: Calculate the Internal Rate of Return (IRR) for the project using interpolation method.

Year	*Cash Flow* (*Rs.*)
1	15,000
2	18,000
3	20,000
4	25,000

10 3 3

7 a The choice of long-term finance depends heavily on the type, size, and stage of a business. Critically evaluate this statement by comparing the suitability of different long-term financing sources for: "A newly established tech startup". 08 4 4

b Highlight the major phases involved in raising long term finance through Term loans. 08 2 4

OR

8	a	What does a Stock Market Index represent? Explain the three important types of indices used.	08	2	4
	b	Explain the structure and features of the Government Securities (G-Sec) Market in India. Why are G-Secs considered low-risk investments, and what role do they play in the financial system?	08	3	4
9	a	"Working capital requirements vary significantly across industries and business models." Critically analyze this statement by identifying and evaluating four key factors that influence working capital needs, with relevant real-world examples to support your analysis.	08	4	3
	b	Explain the three strategies followed by the organizations as part of Current assets financing policy.	08	2	3
		OR			
10	a	"The length of the operating cycle and cash cycle directly affects a firm's liquidity and financing decisions." Evaluate how variations in these cycles impact working capital requirements and business sustainability.	08	3	3
	b	"Efficient working capital financing is critical for maintaining business liquidity without overburdening the firm with cost or risk." In the above context explain any four methods of working capital financing.	08	2	3

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B.E. Regular Examinations Sept/Oct – 2024

ELEMENTS OF FINANCIAL MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Usage of interest factor tables is permitted.

PART-A

			M	BT	CO
1	1.1	List the three broad areas of financial decision making.	02	01	01
	1.2	What is the primary function of a financial intermediary in Indian financial system?	02	02	01
	1.3	What is the meaning of "perpetuity" in the context of Time value of money?	02	02	01
	1.4	Distinguish between the terms, "Market Risk" and "Unique Risk".	02	02	01
	1.5	List the 3 distinct features of capital expenditure.	02	01	02
	1.6	What is meant by the term, "Cost of retained earnings"?	02	02	02
	1.7	Distinguish between 'primary Market' and 'Secondary market'.	02	02	03
	1.8	Distinguish between 'Initial Public Offer' and 'Follow on public offer'.	02	02	03
	1.9	Distinguish between the terms – Absorption and Consolidation in the context of Mergers.	02	02	04
	1.10	Distinguish between the terms – call option and put option.	02	02	04

PART-B

2	a	What is meant by a financial market? Summarize the categorization of financial markets based on several criterion.	08	03	01
	b	List any four types of financial intermediaries prevalent in India. Evaluate the benefits of having financial intermediaries in the financial system.	08	03	01
3	a	An amount of $Rs\,1000/-$ is annually deposited for 5 years and the deposit earns a compound interest rate of 10%. What will be the value at the end of 5 years? Assuming that each deposit occurs at the end of the year, determine the future of annuity.	05	03	01
	b	Your friend promises to give you $Rs.\,1000/-$ 6 years later. What is the present value of this amount if the interest rate is 10%?	05	03	01
	c	An investor expects to receive the following cash flows from an asset. *Year* | 1 | 2 | 3 *Cash Flow (Lakhs in rupees)* | 20 | 30 | 220 The discount rate for cash flows is 16%. Determine the value of the Asset.	06	03	01
		OR			
4	a	A $Rs\,1000/-$ par value bond, bearing a coupon rate of 14% will mature after 5 years. The required rate of return on this bond is 13%. What is the value of Bond?	05	03	01

	b	An equity share is expected to provide a dividend of $Rs\,2/-$ and fetch a price of $Rs\,18/-$ a year hence. What price would it sell for now if investors required rate of return is 12%?	05	03	01
	c	The expected dividend per share is Rs.5/-. The dividend is expected to grow at the rate of 6% per year. If the price share now is Rs.50/-, determine the expected rate of returns of the equity share.	06	03	01
5	a	Enumerate the steps involved in the process of capital budgeting.	10	02	02
	b	The cashflow of a project being considered by a company are as follows: *Year* | 0 | 1 | 2 | 3 | 4 *Cashflow (lakhs)* | 1 | 0.3 | 0.3 | 0.4 | 0.45 Calculate the Internal Rate of Returns of the project.	06	03	02
		OR			
6	a	Consider a debenture with a face value of Rs. 1000/-. The coupon rate is 12%. The remaining period to maturity is 4 years. The current market price of the debenture is Rs. 1040/-. Determine the cost of this debenture.	06	03	02
	b	The current market price of an equity is Rs. 80/-. The dividend expected a year hence from now is Rs. 4/-. The growth is expected to be at a constant rate of 10%. The floatation cost for new issue is 8%. Determine the cost of new equity.	05	03	02
	c	A project which is being evaluated by firm has a cost of capital of 12%. The initial investment is Rs. 1 lakh. The benefits over the years are as shown below: *Year* | 1 | 2 | 3 | 4 *Amount (Rs)* | 25,000 | 40,000 | 40,000 | 50,000 Determine the Benefit Cost Ratio of this investment and interpret whether this project is acceptable or not acceptable based on *BCR*?	05	04	02
7	a	Enumerate the five major 'sources' of long term finance available for organizations.	10	02	03
	b	Distinguish between 'Operating Cycle' and 'Cash Cycle' with the help of an illustration.	06	02	03
		OR			
8	a	These are three broad types of 'Indices' used to measure the movement of stocks in a market. Summarize the salient aspects of the three categories/types of indices.	06	02	03
	b	Enumerate the factors that have a crucial bearing on the working capital requirements of an organization.	10	02	03
9	a	Express the plausible reasons for mergers. Analyze how the mergers could contribute to enable organizations have a competitive edge over other organizations.	10	04	04
	b	"The warrant represents a potential source of capital in the future when the organization needs to raise additional capital without offering other bonds or stock". In the light of this statement, determine the features of Warrants.	06	02	04
		OR			
10	a	Examine any four ways of raising the foreign currency finance.	10	04	04
	b	Bring out the rationale of organizations going for 'Demerger' Strategy.	06	03	04

Facilities Planning, Design and Ergonomics

VI Semester SEE Question Papers

Course codes: IM365TDA, 18IM6C2, 16IM63, 16IM63 / 12IM71

- Aug 2025 (IM365TDA)
- Sep 2023 (18IM6C2)
- Aug 2022 (18IM6C2)
- Fast Track Oct 2020 (16IM63)
- Nov/Dec 2019 (16IM63 / 12IM71)
- Apr/May 2019 (16IM63)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations August-2025

Industrial Engineering and Management

FACILITIES PLANNING AND DESIGN (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	List any two steps in making a correct location choice.	02	2	2
	1.2	Process layout is also known as _____ layout and _____ layout.	02	1	1
	1.3	List any two factors that the transportation cost is dependent on.	02	2	2
	1.4	If the centers of four departments are (10,2), (2,6), (3,2), (7,6) and has a unit cost matrix as well as flow matrix, what is the total cost of the layout.	02	2	2
	1.5	List the tools which are applicable to just in time manufacturing.	02	3	3
	1.6	Flexible manufacturing systems allows for _____ and _____ product changes.	02	1	1
	1.7	The layout where the equipment, machinery, plant and people move as necessary is known as _____.	01	2	2
	1.8	_____ technique emphasizes transportation cost matrix in the determination of facility location.	01	2	2
	1.9	A self-service cafeteria is usually positioned as _____ layout.	01	3	3
	1.10	Visibility measures used in just-in-time (JIT) would include Samples of competitor's product, including _____ and _____.	02	1	2
	1.11	_____ advocated the Systematic Layout Planning framework.	01	1	1
	1.12	List the automation systems with examples.	02	1	1

PART-B

2	a	Enumerate and analyze the characteristics of all the facilities of a supply chain to achieve excellence.	08	2	1
	b	Apply the Winning facilities planning process methodology for facilities planning of a manufacturing organization, and provide your analysis summary.	08	3	3
3	a	Explain the concept of unit load. State at least five advantages of unit load.	08	2	1
	b	Given the following dimensions of a particular type of plastic reusable containers, a trailer with inside dimensions of $240'' \times 120'' \times 120''$ is used to transport these containers. The containers are not palletized. Assume that no clearance is needed between container and the walls of the trailers. Insider dim = $20'' \times 12'' \times 12''$; Outside dim = $22'' \times 14'' \times 14''$ Each nested container = $20'' \times 12'' \times 12''$ Determine the container space utilization and storage space efficiency if the dimension of the storage opening is $24'' \times 16'' \times 14''$.	08	2	2

OR

4	a	Explain the different material handling principles that are adopted by College-Industry Council on material handling education.	08	1	1
	b	Discuss the following categories of material handling equipment with examples: i) Fixed path equipment ii) Variable path equipment	08	1	1
5		Consider the following initial layout with unit cost matrix. Use the CRAFT pair wise interchange to obtain the desirable layout. 4 2 4 4 | 1 | 5 | 4 | 4 | 2 | 3 | Initial Layout | | 1 | 2 | 3 | 4 | | A | − | 1 | 1 | 3 | | B | 2 | − | 2 | 1 | | C | 0 | 2 | − | 2 | | D | 1 | 4 | 3 | − | Flow Matrix	16	3	4

OR

6	a	Differentiate between ALDEP AND CORELAP on the basis of selection, construction and placement.	08	2	2
	b	Explain batch picking and zone picking concepts of order picking in an warehouse operations.	08	1	1
7	a	Discuss with examples and sketches, the different types of flow in a layout.	08	2	2
	b	Explain the Material handling equipment categories.	08	1	1

OR

8	a	Explain the factors for production to be considered in space planning.	08	3	4
	b	Discuss the various function in the warehouse.	08	2	3
9	a	List and explain the different efficiency indices used in evaluating the layout.	08	3	3
	b	Discuss the alternative techniques in evaluating alternative plans.	08	2	2

OR

10	a	Explain deterministic model for single facility location models.	08	1	1
	b	Emulate the impact of just in time manufacturing on facilities design.	08	3	3

18IM6C2

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Examinations September-2023

Industrial Engineering and Management

FACILITIES PLANNING DESIGN AND ERGONOMICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	List any four objectives of Facilities Planning.	02
	1.2	List any four factors influencing plant location.	02
	1.3	List the phases of approach in Developing Alternative Material Handling System Designs.	02
	1.4	List any four elements in the definition of material handling systems.	02
	1.5	Differentiate between continuous location and discrete location problem.	02
	1.6	List the two principal approaches to order picking.	02
	1.7	List and define two main areas of Ergonomics.	02
	1.8	Differentiate between Cognitive and Physical Ergonomics.	02
	1.9	What is the difference between illuminance and luminance?	02
	1.10	List and define the two physiological responses to cold stress of the human body.	02

PART B

2	a	Explain the traditional engineering design process as applied to facilities planning.	08
	b	Elaborate the principles to be considered in designing a good plant layout.	08
3	a	Explain the classifications of various Material Handling Equipment	08
	b	List and explain the different material handling principles.	08
		OR	
4		Consider the following initial layout with unit cost matrix. Use the $CRAFT$ pairwise interchange technique to obtain the desirable layout. 6 4 4 A B 4 C D Initial Layout	

		<table><tr><th>To / From</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>A</td><td>--</td><td>1</td><td>1</td><td>3</td></tr><tr><td>B</td><td>2</td><td>– –</td><td>2</td><td>1</td></tr><tr><td>C</td><td>0</td><td>2</td><td>– –</td><td>2</td></tr><tr><td>D</td><td>1</td><td>4</td><td>3</td><td>– –</td></tr></table> Flow Matrix	16
5	a	Explain briefly the different type's location problems categorized on the basis of the distance between facilities.	08
	b	There are Five machines currently located in the department. They have the following coordinate locations: $P1-(1,1), P2-(6,2), P3-(2,8), P4-(3,6)$, and $P5-(8,4)$. The cost per unit distance traveled between the new machine and each existing machine is the same. The number of trips per day between the new machine and existing machines $1, 2, 3, 4$ and 5 are 10,20,25,20, and 25 respectively. Determine the optimum location for a single new facility when cost is proportional to rectilinear distance. Construct a contour line passing through the point having coordinates (3,4).	08
		OR	
6	a	Describe briefly the different types of Conventional Storage Models.	08
	b	Explain in detail the logic underlying in the simulation models for facility design.	08
7	a	Explain the three basic categories of Human-machine systems. With a Block diagram model, explain the interactions in a human-machine system.	08
	b	What are the three main components of the physical work environment? What is the difference between occupational safety and occupational health?	08
8	a	Explain briefly the factors affecting Visibility and Visual Performance.	08
	b	Provide an overview of the two general approaches that can be pursued to address the existence of a noise problem in a given facility.	08

18IM6C2

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Examinations August-2022

Industrial Engineering and Management

FACILITIES PLANNING DESIGN AND ERGONOMICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Distinguish between the terms 'facilities location' and 'facilities design'.	02
	1.2	Facilities planning process can be improved in 3 dimensions. What are they?	02
	1.3	Write a note on life cycle cost principle of material handling.	02
	1.4	List the two elements considered in determining the size of the unit load.	02
	1.5	When a facility location problem is categorized on the basis of the distance between facilities, 4 distance measures are used. What are they?	02
	1.6	Distinguish between total cover problem and partial cover problem in the context of facilities location models.	02
	1.7	List the 3 categories of Human-Machine systems from evolutionary point of view.	02
	1.8	Mention the functional activity of the cardiac muscle and smooth muscle in human body.	02
	1.9	What is meant by color rendering? Also mention the most widely used measure for color rendering.	02
	1.10	What is meant by "startle response" in the context of auditory environment and noise?	02

PART B

2	a	Proper facilities planning along the supply chain ensures that the product will follow to the satisfaction of the end customer. In the above context, elaborate on the characteristics that the facilities in the supply chain should possess.	08
	b	A new medical facility is to be located in Bangalore. The location factors, factor rating and scores for 2 potential sites are shown in the following table. Which is the best location based on factor rating method?	

Sl.No	Location factor	Factor rating	Rating	
			L_1	L_2
1	Facility utilization	8	3	5
2	Total patient/month	5	4	3
3	Average time/ emergency trip	6	4	5
4	Land and construction costs	3	1	2
5	Employee preferences	5	5	3

08

3	a	Illustrate the container/pallet flow in conventional system, container/pallet pooling system and integrated logistics and container/pallet pooling system. Also bring out the characteristics of container/pallet pooling system.	08
	b	To simulate the development of alternatives, the ideal systems approach was proposed by Nadler. This approach consists of 4 phases. Elaborate the phases.	08
		OR	
4	a	Elucidate the procedure adopted for improvement of an existing facility using CRAFT algorithm.	08
	b	What is ALDEP in the context of layout design? Elaborate on the following in the context of the same. i. Relationship chart ii. Sweep width.	08
5	a	Identify the condition in which the location-allocation problem can be solved using quadratic assignment problem. Represent the mathematical formulation of the problem.	08
	b	How is order picking system different from unit load storage/retrieval system? Review the 2 principal approaches to order picking system.	08
		OR	
6		Write short notes on the following. i. Block stacking ii. Deep lane storage iii. Single deep storage rack iv. Double deep storage rack.	16
7	a	The capacity of human body to use energy and apply forces depends on 3 factors. What are they? Review the same.	12
	b	Distinguish between rational decision models and descriptive decision models in the context of cognitive ergonomics.	04
8	a	Evaluate the possible physiological and psychological effects of noise on human beings in industrial work environment.	08
	b	Four primary variables in “climate” strongly affect the physical well-being and comfort levels of workers. What are they? Review the possible influences of the same on human well-being in work environment.	08

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

FACILITIES PLANNING AND DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	List the constituents of 'design' of a facility.	02
	1.2	What is referred to as facilities location?	02
	1.3	Write the equation for recommended material handling system.	02
	1.4	Identify the two elements in determining the size of unit load.	02
	1.5	Distinguish between 'construction type' and 'improvement type' layout procedures.	02
	1.6	There are two strategies to follow in developing a maximally weighted planar adjacency graph. Mention them.	02
	1.7	Expand the acronym $CRAFT$.	02
	1.8	What is the first objective of warehousing?	02
	1.9	What is referred to as 'Buffer area' in the context of warehouse operations?	02
	1.10	Groover identifies four capabilities to consider the manufacturing system as flexible. List them.	02

PART-B

2	a	Describe with an appropriate illustration, the three potential dimensions for improvement of facility planning process.	08
	b	What do you understand by 'plant location'? Examine the factors to be considered in plant location decision.	08
3	a	List the ten principles adopted by College-Industry council on material handling education. Describe in detail any four of the material handling principles.	08
	b	Categorize the material handling equipment broadly.	08
		OR	
4	a	Write a brief note on the following: i) Stackability ii) Nestability.	08
	b	Illustrate container/pallet flows in following systems: i) Conventional system ii) Pooling system iii) Integrated contract logistics and container/pallet pooling.	08

5	a	Write a note on the following: i) Procedural differences between $CORELAP$ and $ALDEP$. ii) Sweep width in the context of $ALDEP$. iii) Listing warehousing opportunities in a logistics network.	09
	b	Illustrate through a figure, different typical warehouse functions and flows.	07
		OR	
6	a	Analyze the steps required to determine the total space requirements for receiving and shipping areas.	08
	b	Write note on the following in the context of order picking: i) Discrete picking ii) Batch picking iii) Zone picking iv) Wave picking.	08
7	a	Choose ten main components of a kitchen (ex: oven, sink). Develop a relationship chart of the ten components. Sketch a kitchen containing the ten components and arrange them based upon relationship chart findings.	12
	b	List at least six factors for consideration in planning material flow.	04
8	a	Illustrate the following alternative layouts for a flexible manufacturing system: i) FMS with centralized layout ii) FMS based on robot handling system.	06
	b	Explain the five rules of thumb used in designing inprocess handling and storage systems.	10

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RV COLLEGE OF ENGINEERING®

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VI / VII Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

FACILITIES PLANNING AND DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Facilities planning are subdivided into the subjects of facilities _____ and facilities _____.	02
	1.2	Manufacturing of a standard product in large quantity is preferred in _____ layout.	02
	1.3	Principle of 'Unit load' states that _____ should be moved in _____.	02
	1.4	List the objectives of material handling.	02
	1.5	The appropriate goal of material handling is to design the most efficient _____ handling system at the _____ cost.	02
	1.6	List the inputs required for $CRAFT$ (Computerized Relative Allocation of Facilities Technique).	02
	1.7	Just in time was successfully implemented by _____ and invented by _____.	02
	1.8	Transfer lines are often used for _____ volume and highly _____ manufacturing environment.	02
	1.9	The effectiveness of the _____ layout and _____ handling influences the product mix and design.	02
	1.10	List the components of flexible manufacturing systems.	02

PART B

2	a	Analyze the following characteristics of the facilities planning in the context of a new manufacturing firm: i) Flexibility ii) Modularity iii) Upgradability iv) Adaptability.	08
	b	A manufacturer of a farm equipment is considering 3 locations A, B and C for a new plant. Cost studies show that the fixed costs per year at the sites are $Rs.2{,}40{,}000/-$, $Rs.2{,}70{,}000/-$ and $Rs.2{,}52{,}000/-$ respectively. The variable costs per year Rs. 100 per unit, $Rs.90$ per unit and $Rs.95$ per unit respectively. If the plant is designed to have an effective system capacity of 2500 units, find the most economic location.	08

3	a	Define material handling? List the objectives of material handling.	08
	b	State the guidelines in selecting material handling equipments?	08
		OR	
4	a	State and explain different principles of plant layout.	10
	b	Discuss the concept of unit load with an example.	06
5		An Auto Parts warehouse has requested that a new layout be designed for their main warehouse. The ware house has 8 major activity centers. The current building area can accommodate 10×24 unit squares. Develop the new layout using *ALDEP* using the data given below.	

Code	1	2	3	4	5	6	7	8
Area (m^2)	12800	8800	9200	12000	8800	13200	13200	8000

Storage	---						
Counter	*X*	---					
Paint Room	*O*	*O*	---				
Receiving	*A*	*X*	*U*	---			
Office	*U*	*E*	*U*	*U*	---		
Parts Bins	*O*	*A*	*O*	*U*	*U*	---	
Tailpipe Racks	*O*	*I*	*I*	*U*	*U*	*I*	---
Restroom	*O*	*O*	*O*	*U*	*O*	*O*	*O*	---

Closeness	*A*	*E*	*I*	*O*	*U*	*X*
Value	64	16	4	1	0	−1024

Minimum Department Preference (MDP) = I = 4; Sweep Width = 2; Number of iterations to be performed, $N = 1$; Assume Scale: 1 square = $400m^2$; Length × Width for the layout: 10×24. 16

OR

6	a	Explain functions of warehouse operations in the following areas: i) Receiving ii) Inspection and quality control.	08
	b	Discuss with example any two principles of order picking in warehouse operations.	08
7		A certain shop manufactures a product using different machines. The details of the machines are given below.	

Machine No.	*Machine Area* (m^2)	*No. of Machines*	*Machine Code*	*Production rate/hour*
1	4×3	1	*O*1	60
2	4.25×3	2	*O*5, *O*6	70
3	8.95×3	1	*O*8	80
4	7×3.5	1	*O*9	65
5	3.5×2.4	1	*O*11	50
6	4.6×3	2	*O*13, *O*14	60
7	11×4	1	*O*15	55
8	2.36×2	1	*O*17	45
9	8.43×3.57	1	*O*21	50
10	3.3×3	1	*O*23	45
11	4.48×2.9	1	*O*26	50
12	3.3×2	3	*O*28, *O*29, *O*30	60
13	6×4.15	1	*O*31	70

		The order of flow is $O8-O23-O15-O11-O14-O29-O17-O1-O6-O13-O28-O26-O31-O5-O21-O9-O30$ Maximum dimensions of a product = $30mm \times 500mm$; No. of hours per shift = 8; No. of shifts per day = 3; Category of material handling = Light; WIP details = 3 high stacking at each WIP area; Services to be included = Maintenance, Tool crib, Inspection, Supervision and Amenities. Draw an appropriate shop layout to a suitable scale and position machines for straight line flow within it, providing administrative area involving 5 executives and 12 supporting staff, a conference hall and a generator room.	16
8	a	Explain with an example fixed automation systems for a dial indexing machine.	08
	b	Analyze the applications for which flexible manufacturing systems is applied.	08

16IM63

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

FACILITIES PLANNING AND DESIGN

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	List the three potential dimensions for improvement of facility planning process.	02
	1.2	Distinguish between the terms "facilities location" and "facilities design".	02
	1.3	Mention the material handling system equation that helps in guiding the development of alternative material handling system designs.	02
	1.4	Identify the two elements in determining the size of the unit load.	02
	1.5	Distinguish between the features of layout for a product family department and layout for a process family department.	02
	1.6	What is meant by the term "sweep width" in solving a layout issue using *ALDEP* algorithm?	02
	1.7	Identify the equipment used as an interface between a dock at a given height and variable height carriers.	01
	1.8	What is meant by 'Honeycombing' in the context of warehouse operations?	02
	1.9	List the external factors that affect the facility planning process.	02
	1.10	List the basic components of a flexible manufacturing system.	02
	1.11	Expand *CRAFT*.	01

PART B

2	a	Evaluate the impact of facilities planning on the competitiveness of manufacturing facilities.	08
	b	Analyze the major factors that influence a typical plant layout.	08
3	a	Summarize the steps involved in the process of designing material handling systems.	08
	b	Illustrate and explain different stacking patterns for different pallet sizes.	08
		OR	
4	a	Categorize the material handling equipment based on the functions performed.	08

	b	For each of the following material handling principles, explain what it means and give a specific example of how it can be applied: i) Ergonomic principle. ii) Unit load principle. iii) Work principle. iv) Space utilization principle.	08
5	a	Bring out the basic differences between construction type and improvement type algorithms.	08
	b	Provide a review of the following: i) $COFAD$ ii) $PLANET$.	08
		OR	
6	a	Analyze the principles of 'order picking' operation.	08
	b	Evaluate the principles to be considered in accomplishing storage area objectives.	08
7		Consider a food restaurant you visit often. Describe in detail the arrangement of aisles for customers (the service line, the booth area, etc) and the arrangement of aisles for the workers (area behind the counter) by sketching a layout. Draw the flow of the customers and employees on the layout. Explain how the flow of the workers and the customers would change if the layout was different. Is there adequate space for the customers and employees to operate efficiently? Can any improvement in the layout create a more efficient flow? Describe any visual management approaches being used.	16
8	a	Analyze the impact of Just-in-time philosophy on the design of facility layout and material handling systems.	08
	b	Describe the single stage multimachine systems in terms of the resources involved.	08

Optimization Techniques

VI Semester SEE Question Papers

Course codes: IM266TER

- Aug 2025 (IM266TER)

IM266TER

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations August-2025

OPTIMIZATION TECHNIQUES

(Institutional Elective)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of Statistical table is permitted.

PART-A

			M	BT	CO
1	1.1	Operations research is a _____ approach to problem solving for executives.	1	1	1
	1.2	A linear programming technique improves the quality of _____.	1	1	2
	1.3	There are many real world problems that don't possess any _____ solution.	1	1	1
	1.4	If an optimal solution to a linear programming problem exists, it will lie at _____of the feasible solution.	1	2	2
	1.5	Define sensitivity analysis in linear programming.	2	1	2
	1.6	What does the shadow price in duality represent?	2	2	2
	1.7	Explain degeneracy in transportation problems.	2	2	2
	1.8	State the objective of the Traveling Salesman Problem.	2	2	2
	1.9	List the three time estimates used in PERT.	2	1	3
	1.10	What are the advantages of crashing a network?	2	1	3
	1.11	Differentiate between pure and mixed strategies.	2	1	3
	1.12	Explain the characteristics of a two-person zero-sum game.	2	1	3

PART-B

2	a	A company produces two products A and B. Each unit of A takes 2hrs of machine time and 3 units of labor. B takes 4hrs of machine time and 2 units of labor. The machine and labor availability are 100hrs and 90 units. The profit from A is ₹30 and from B is ₹50. Formulate the LPP.	06	3	2
	b	Explain in detail the seven-step model-building process with a real-life example for each step.	10	2	1
3	a	Solve the following LP using the Simplex Method: Maximize $Z = 4X_1 + 3X_2$ $subject\ to$: $2X_1 + X_2 \leq 10$, $X_1 + 2X_2 \leq 6$ $X_1, X_2 \geq 0$	10	3	2
	b	With suitable example, explain slack and surplus variables.	06	2	2
		OR			

4	a	Formulate the dual of the following problem and interpret the dual variables economically: Maximize $Z = 6X_1 + 8X_2$ $Subjected\ to$: $2X_1 + 3X_2 \leq 30$, $X_1 + 2X_2 \leq 20$, $X_1, X_2 \geq 0$	06	4	2
	b	Use the Simplex Method to solve the LP: Maximize $Z = 3x_1 + 2x_2$, subject to $x_1 + x_2 \leq 4$, $x_1 + 2x_2 \leq 6$, $x_1, x_2 \geq 0$	10	3	2
5	a	A manufacturer has 2 plants (P1 and P2) with production capacities of 100 and 150 units. These must be shipped to 3 warehouses (W1, W2, W3) requiring 80, 120, and 50 units, respectively. The cost of transportation per unit is as follows: $W1$ $W2$ $W3$ $P1$ 4 6 8 $P2$ 5 3 7 Formulate the transportation problem mathematically.	08	3	2
	b	In a textile sales emporium, four salesmen A, B, C and D are available to four counters W, X, Y and Z. Each salesman can handle any counter. The service (in hour) of each counter when manned by each salesman is given below: $Salesman$: A B C D $Counter$: W 41 72 39 52; X 22 29 49 65; Y 27 39 60 51; Z 45 50 48 52 How should the salesmen be allocated appropriate counters so as to minimize the service time? Each salesman must handle only one counter.	08	3	2
		OR			
6	a	Use the following data to find IBFS using: (i) North-West Corner Rule (NWCR) (ii) Least Cost Method (LCM) $D1$ $D2$ $D3$ $Supply$ $S1$ 2 3 1 30 $S2$ 5 4 8 50 $S3$ 5 6 8 20 $Demand$ 20 40 40	08	3	2
	b	A car rental company has one car at each of five depots I, II, III, IV and V. A customer requires a car in each town namely A, B, C, D and E. The distance (in km) between depots and towns are given in the following distance matrix. $Depots$: I II III IV V $Town$: A 160 130 175 190 200; B 135 120 130 160 175; C 140 110 155 170 185; D 50 50 80 80 110; E 55 35 70 80 105 How should the cars be assigned to the customers so as to minimize the distance travelled?	08	3	2

7 An established company has decided to add a new product to its line. It will buy the product from a manufacturing concern, package it, and sell it to a number of distributors that have been selected on a geographical basis. Market research has already indicated the volume expected and the size of sales force required. The steps shown in the following table are to be planned.

Activity	A	B	C	D	E	F	G	H	I	J	K	L	M
Duration	6	4	7	2	4	10	2	10	6	13	9	3	5
Predecessors	-	A	B	A	D	E	-	G	J,H	-	A	C,K	I,L

(i) Draw an arrow diagram for this project.
(ii) Indicate the critical path.
(iii) For each non-critical activity, find the total and free float — 16 | 4 | 3

OR

8 A small project involves 7 activities, and their time estimates are listed in the following table. Activities are identified by their beginning (i) and ending (j) node numbers.

Activity ($i-j$)	*Estimated Duration (weeks)*		
	Optimistic	*Most Likely*	*Pessimistic*
1 – 2	1	1	7
1 – 3	1	4	7
1 – 4	2	2	8
2 – 5	1	1	1
3 – 5	2	5	14
4 – 6	2	5	8
5 – 6	3	6	15

(i) Draw the network diagram of the activities in the project.
(ii) Find the expected duration and variance for each activity. What is the expected project length?
(iii) Calculate the variance and standard deviation of the project length. What is probability that the project will be completed:
I. at least 4 weeks earlier than expected time.
II. no more than 4 weeks later than expected time. — 16 | 4 | 3

9 a Use the graphical method to find the optimal strategy for both the players and also value of the game for the following pay – off matrix:

		Player B		
		B1	*B2*	*B3*
Player A	*A1*	1	3	12
	A2	8	6	2

10 | 3 | 4

b Explain: Minimax and Maximin principle used in the theory of games. 06 | 2 | 4

OR

10 a Use graphical method to find the optimal strategy for both the players and also value of the game for the following pay – off matrix:

		Player B				
		I	*II*	*III*	*IV*	*V*
Player A	*I*	3	0	6	–1	7
	II	–1	5	–2	2	1

10 | 3 | 4

b State the major limitations of the game theory. 06 | 2 | 4

Mathematical Modelling

VI Semester SEE Question Papers

Course codes: MA266TEU

- Jul/Aug 2026 (MA266TEU)
- Model Question Paper (2022 Scheme) (MA266TEU)

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RV COLLEGE OF ENGINEERING®

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VI Semester B.E. Regular / Supplementary Examination July / August 2026

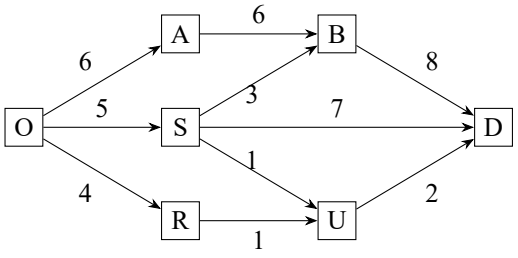
(Institutional Elective)

MATHEMATICAL MODELLING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

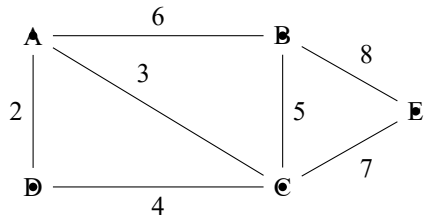
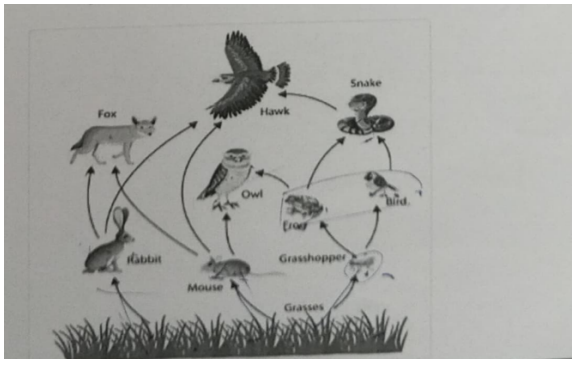
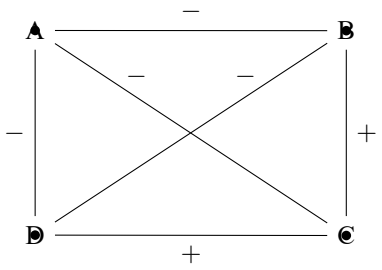
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	**M**	**BT**	**CO**
1	1.1	Write any two characteristics of mathematical modelling.	02	1	1
	1.2	The horizontal distance (range) covered by a soccer ball of mass 2 kg is kicked from the ground with an initial velocity of $30\,m/s$ at an angle of 30° to the horizontal is _____. (Take acceleration due to gravity as $g \approx 9.8\,m/s^2$)	02	2	2
	1.3	With respect to the characteristic of brown eyes, suppose the proportions of dominant, hybrid, and recessive individuals in the 9th generation are $p_9 = 0.36$, $q_9 = 0.48$, $r_9 = 0.16$. The proportion of recessives in the 10th generation (r_{10}) is _____ and proportion of hybrids in the 11th generation (q_{11}) is _____.	02	1	1
	1.4	Suppose the population of one-horned rhinoceros in foothills of Himalayas are decreasing at the rate of 2% per year and their population was 400 in 2015, then how many of them will be left approximately in 2024?	02	2	2
	1.5	Compute the fixed probability vector of the stochastic matrix $A = \begin{bmatrix} 0.5 & 0.5 \\ 0.1 & 0.9 \end{bmatrix}$.	02	1	1
	1.6	A travelling salesman **S** sells in 3 cities **A, B, C**. He never sells in the same city on successive days. If he sells in city A on a day, then the next day he sells in city B. However if he sells in either B or C, then the next day he is twice as likely to sell in city A as in the other city. Find the transition matrix of the chain of selling.	02	2	2
	1.7	Draw the bipartite graph $K_{3,5}$ and the complete graph K_4.	02	1	1
	1.8	Express the following transition probability matrix as a weighted digraph. $\begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0 & 0.5 \\ 0.25 & 0.35 & 0.4 \end{bmatrix}$	02	2	2
	1.9	In the following figure, with respect to dynamic programming, the value of $f_1^*(O)$ is _____. 	02	2	2

	1.10	The extremal of the functional $\int_0^3 x^3(y')^2\,dx$ is _____.	02	1	1

PART-B

2	a	Model and describe the second-order differential equations representing a mass-spring-dashpot mechanical system. Also highlight the analogies between the elements of the mechanical and electrical systems.	08	3	3
	b	Drug is induced in a patient's bloodstream at regular intervals of duration T each. Simultaneously the drug disappears at a rate proportional to the concentration $c(t)$ of the drug present at any time t. Determine the differential equation governing $c(t)$, solve it and interpret the results.	08	4	4
3	a	Discuss the Cobweb Model describing the variation of price and quantity of an agricultural commodity produced by farmers. Using a suitable graph, interpret the behaviour of the model.	08	3	3
	b	(i) A machine's value changes every year according to the relation $V_{n+1} = 0.80V_n - 2$, where V_n is the value in lakhs after n years. If the initial value $V_0 = 15$ lakh, find the value of the machine after n years. (ii) Solve the difference equation $2x_{t+2} - 5x_{t+1} + 2x_t = 0$.	08	2	2
		OR			
4	a	Model the population dynamics using difference equations and discuss the one period fixed points.	08	3	3
	b	Solve the difference equation $x_{t+2} = 4x_{t+1} - 5x_t$.	08	2	2
5	a	An e-commerce website tracks user navigation behaviour between four main sections: The Homepage (H), Product Catalog (C), Shopping Cart (W) and the Payment Gateway (P). From the Homepage (H): Users always navigate directly to the Product Catalog (C). From the Product Catalog (C): There is a 50% chance they stay browsing the Catalog (C) and a 50% chance they add an item to the Shopping Cart (W). From the Shopping Cart (W): There is a 60% chance they return to the Homepage (H), a 20% chance they proceed to the Payment Gateway (P) and a 20% chance they remain in the Cart adjusting items. From the Payment Gateway (P): After completing or cancelling a transaction, they are always redirected back to the Homepage (H). In the long run, what percentage of a user's active session time is spent on the Payment Gateway (P)?	08	4	4
	b	A company executive changes his bike every year. If he has Pulsar he changes to Splendor, if he has Splendor he changes over to Royal Enfield. If he has Royal Enfield, he is just as likely to change over to a bike Royal Enfield, Splendor or Pulsar. If he had Royal Enfield in 2020, find the probability that he will have a bike (i) Pulsar in 2022, (ii) Royal Enfield in 2022.	08	2	2
		OR			
6	a	Verify the transition probability matrix $P = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$ is irreducible or not. Also find its stationary probability vector.	08	2	2

	b	An automated robotic arm on a factory floor moves items between four inspection stations: Receiving (R), Sorting (S), Processing (P) and Re-inspection (M). From Receiving (R): Items always move directly to Sorting (S). From Sorting (S): There is a 40% chance an item stays in Sorting for deeper scanning and a 60% chance it moves to Processing (P). From Processing (P): There is a 70% chance it moves back to Receiving (R), a 10% chance it requires Re-inspection (M) and a 20% chance it stays in Processing (P). From Re-inspection (M): Once checked, it always moves back to Receiving (R). In the long run, what percentage of time does the robotic arm spend in Re-inspection (M)?	08	4	4
7	a	State the max-flow min-cut theorem. For the network shown below, determine the maximum flow between A and E by enumerating all the A-E cuts and the capacities therein. 	08	4	4
	b	Define the Harary measure of status. In the given food web, an edge from X to Y means X is food for Y. Find the status of each species according to the Harary measure. 	08	3	3
		OR			
8	a	Alvin (A) has invited three married couples to his summer house for a week: Bob (B) and Carrie (C) Hanson, David (D) and Edith (E) Irwin, and Frank (F) and Gena (G) Jackson. Since all six guests enjoy playing tennis match against every other guest except his/her spouse. In addition, Alvin is to play a match against each of David, Edith, Frank, and Gena. If no one is to play two matches on the same day, what is a schedule of matches over the smallest number of days.	08	4	4
	b	i. Is it possible to have a graph with the degree sequence 1,2,2,3,4,5? If yes, draw an example. If not, give reason. ii. Define the degree of balance of a signed graph. The following graph models the social relation between 4 people with '+' indicating friendship and '−' indicating enmity. Find the degree of balance of the graph. 	08	3	3

9	a	Find the extremals of the functional $\int_{1/10}^{1} y'(1+x^2y')\,dx$, if $y\left(\frac{1}{10}\right) = 19$, $y(1) = 1$.	08	3	3
	b	A company needs to transport goods from its main warehouse (location A) to its retail outlet (location J). Along the way, the goods must pass through a series of distribution centres (B, C, D, E, F, G, H, I). At each segment of the journey, transport costs are incurred. The company wants to minimize the total transportation cost from A to J. Using dynamic programming, determine the minimum transportation cost from A to J and identify the optimal route. The transportation costs between centres are given in the table below: A → B: 5, C: 3, D: 6 B → E: 4, F: 2, G: 5 C → E: 3, F: 4, G: 6 D → E: 5, F: 3, G: 2 E → H: 3, I: 4 F → H: 2, I: 5 G → H: 6, I: 3 H → J: 4 I → J: 3	08	2	2
		OR			
10	a	Model the problem of a uniform chain (cable or rope) of length $2l$, suspended under gravity between two points at equal height. Find the curve when the chain is in equilibrium.	08	4	4
	b	Discuss any five optimization techniques and any three optimization principles.	08	2	2

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VI Semester B.E. Examinations – Model Question Paper

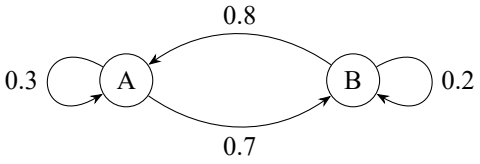
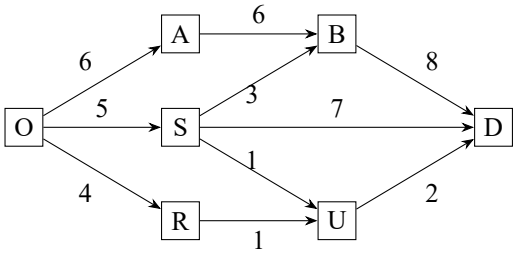
(Institutional Elective)

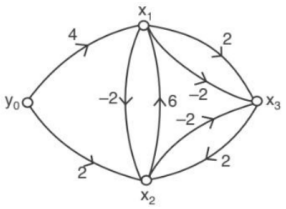
MATHEMATICAL MODELLING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

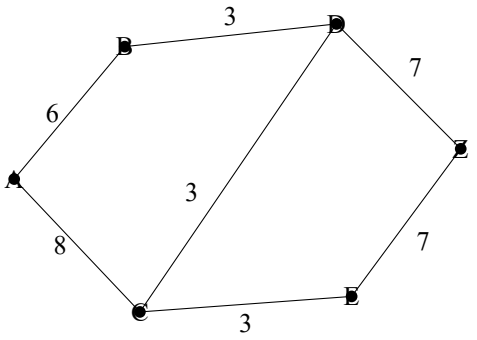
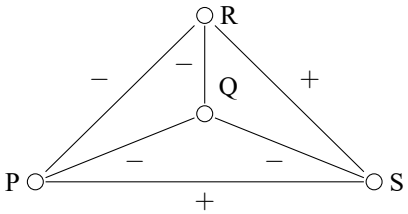
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

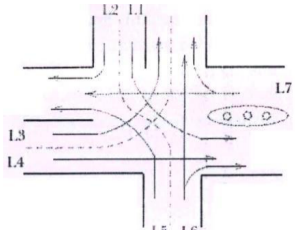
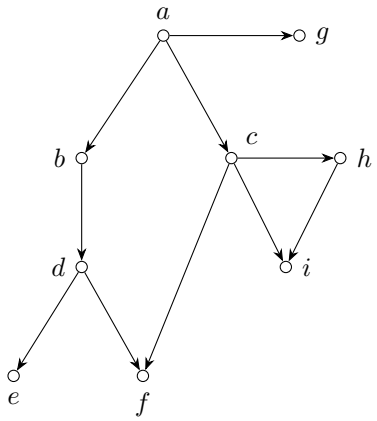
		PART-A	M	BT	CO
1	1.1	The population of California fox are decreasing at the rate of 5% per year. If their population was 400 in 2013, then how many of them will be left approximately in 2023?	02	2	1
	1.2	The growth of an organism follows the law $\frac{dx}{dt} = -\frac{1}{13}x$. If initially there were 200 of them, then after 13 years, their approximate number will be _____ and their half-life period is approximately _____ years.	02	2	3
	1.3	In terms of y_{t+1} and y_{t+2}, the arbitrary constant M in the equation $y_t = M2^t + N(-3)^t$ is equal to _____.	02	3	3
	1.4	The curve which extremizes the functional $\int_{10}^{11} \left[y^2 + x^2 y'\right] dx$ is _____.	02	2	2
	1.5	In the long run, the proportion of time that the given Markov chain will be in state A is _____ and state B is _____. 	02	2	2
	1.6	The angle of projection for a projectile to attain a) the maximum horizontal distance should be _____. b) the maximum height should be _____.	02	1	3
	1.7	In the following figure, with respect to dynamic programming, the value of $f_1^*(O)$ is _____. 	02	2	4

	1.8	Write the linear equations associated with the following signal-flow graph:  	02	2	4
	1.9	Given the initial state vector $(1, 0)$ and the transition matrix shown below, find the state vector corresponding to two steps later ($n = 2$). $\begin{bmatrix} 0.13 & 0.87 \\ 0.91 & 0.09 \end{bmatrix}$	02	1	1
	1.10	'Edge coloring' is the assigning of colors to the edges so that no two adjacent edges have same color. Consider a simple connected graph G in which the number of vertices is equal to the number of edges, and every vertex has degree 2. What is the minimum number of colors required to 'edge-color' G?	02	1	2

PART-B

2	a	Give a note on the classification of mathematical models giving at least four examples.	08	1	1
	b	Drug is induced in a patient's bloodstream at regular intervals of duration T each. Simultaneously the drug disappears at a rate proportional to the concentration $c(t)$ of the drug present at any time t. Determine the differential equation governing $c(t)$, solve it and interpret the results.	08	3	3
3	a	Mathematically establish the Hardy-Weinberg law in genetics.	08	3	2
	b	A person takes a loan of ₹S_0 with i as the compound interest rate per month. If the loan is to be paid back in n equal instalments by paying ₹R per month, then show that $R = S_0 \dfrac{i}{1-(1+i)^{-n}}$.	08	2	2
		OR			
4	a	Bring out the dynamics of price and quantity of a commodity produced by a farm according to the Cobweb model.	08	3	2
	b	i. Solve the difference equation $x_{t+2} = 4(x_{t+1} - x_t)$, $t \geq 0$, $x_0 = 1$, $x_1 = 3$. ii. The number of virus affected files in a system is 5000 (to start with) and this increases 200% every two hours. Model this using a difference equation and find the number of virus affected files in the system after one day.	08	2	2
5	a	Check if the transition probability matrix $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$ is irreducible. Also find its stationary probability vector.	08	2	1
	b	A city is served by two cable TV companies, Best TV and Cable Cast. Due to their aggressive sales tactics, each year 40% of Best TV customers switch to Cable Cast. On the other hand, 30% of the Cable Cast customers switch to Best TV. If the initial market share for Best TV is 20% and for Cable Cast is 80%, find the market share of each company after 5 years.	08	2	3
		OR			

6	a	Consider a network containing five individuals viz. a, b, c, d and e. The system a can communicate to only b and c and its chances of communicating with each are in the ratio 3:2 respectively. Similarly, b communicates with a and c in ratio 1:1; c can communicate with a, b and d in ratio 4:3:3; d communicates with c and e in ratio 3:7 and e can communicate to b only. Model the situation as a Markov chain and rank the five individuals in accordance with their importance in the network in the long run.	08	2	1
	b	A gambler has ₹2. He bets ₹1 at a time and wins ₹1 with probability $\frac{1}{2}$. He stops playing if he ends up with no money or wins ₹4. a. What is the TPM of the related Markov chain? b. What is the probability that he has lost his money at the end of 5 plays? What is the probability that the game lasts more than 7 plays?	08	3	3
7	a	At the end of the school day, all the students plan on driving from school (vertex A) to the concert (at vertex Z). The edges in the graph below represent one-way roads, and the weight of each edge represents the number of vehicles (in hundreds) that particular road can handle in one hour. What is the greatest number of vehicles that can get from school to the concert in one hour? How many vehicles should take each road?  	08	2	4
	b	i. Basavaraj has picked 8 friends for a project team such that they are all friends among themselves. Considering A and B being friends as a single friendship, is it possible that there are totally 38 friendships? Substantiate your answer. ii. Define the degree of balance of a signed graph. The following graph models the social relation between 4 people with '+' indicating friendship and '−' indicating enmity. Find the degree of balance of the graph.  	08	3	4
OR					

8	a	The Fig 8b shows an intersection of two streets where there is often heavy traffic. There are seven traffic lanes $L1, L2, \ldots L7$ where vehicles can enter the intersection of these two streets. A traffic light is located at this intersection. During a certain phase of this traffic light, those cars in lanes for which the light is green may proceed safely through the intersection. Model this situation by a graph giving out the details as to what the vertices and the edges represent. Use that graph to determine whether it is possible with four phases, for cars in all lanes to proceed safely through the intersection.  Fig 8b	08	2	4
	b	Define the Harary measure of status. Find the Harary measure of each individual in the organizational graph shown in Fig 9c.  Fig 9c	08	3	4
9	a	A person decides to travel from a place A to a place J. The journey would require traveling by a stagecoach through different cities on the paths of which there is danger of attack by thieves. Life insurance policies were offered to the passengers, the safest route would be the one with the cheapest total life insurance policy. The details of the insurance from one place to another are given below. Model it through a directed graph and find the route which minimizes the cost using dynamic programming. A: B 1, C 3, D 6 B: E 5, F 4, G 6 C: E 4, F 3, G 2 D: E 1, F 4, G 5 E: H 5, I 2 F: H 3, I 6 G: H 3, I 3 H: J 5 I: J 2	08	1	2
	b	Model the problem of light passing from a point A to a point B as a variational problem by applying Fermat's principle of least time. Show that a consequence of this modelling is the Snell's law of refraction.	08	3	4
		OR			
10	a	Discuss any five optimization techniques and any three optimization principles.	08	1	2
	b	A particle descends from one point to another point in the absence of friction under the action of gravity. Model the situation as a variational problem so as to find the path of the quickest descent.	08	3	4

Service Operations Management

VI Semester SEE Question Papers

Course codes: IM365TDB, 21IM64D2, 16IM6C2

- Aug 2025 (IM365TDB)
- Sep/Oct 2024 (21IM64D2)
- Grade Improvement Nov/Dec 2020 (16IM6C2)
- Apr/May 2019 (16IM6C2)

IM365TDB

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Regular Examinations August-2025

Industrial Engineering and Management

SERVICE OPERATIONS MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Define service blueprinting.	02	1	1
	1.2	List any two examples of services with high labor intensity and high interaction.	02	1	2
	1.3	What do you mean by a “moment of truth” in service delivery?	02	2	3
	1.4	List any four traits of excellent service personnel.	02	1	1
	1.5	What role does empathy play in customer service experience?	02	2	1
	1.6	What are the implications of poor capacity planning in service systems?	02	2	2
	1.7	Distinguish between front-stage and back-stage service activities.	02	2	3
	1.8	List two major benefits of technology in service design.	02	1	2
	1.9	Name two Industries where SERVQUAL can be effectively used.	02	2	1
	1.10	Mention any two types of customer expectations.	02	2	4

PART-B

2	a	Use hotel industry as an example, how do you differentiate Capability as opposed to Commodity. Give insights on Process styles, Service offer, Major challenges faced.	10	3	1
	b	Identify any 3 types of services and enumerate the challenges faced by these services?	06	1	1
3	a	You are hired as a consultant for improving the operations of a public transport system. Identify key tactical and strategic challenges you may encounter. Recommend a roadmap for improvement.	12	5	3
	b	Describe how “Service as a concept” can be used as a strategic tool.	04	2	1
		OR			
4	a	Explain the following customer types using a quadrant model: The Hostage, The Ally, The Anarchist, The Champion. Suggest one strategy to handle each.	10	3	3
	b	Stakeholder management is an important aspect of any service operations. Having many customers as allies can be the best thing that can happen to any service/business. Recommend how allies can be created among various customer types.	06	3	3
5	a	Illustrate with an example the downsides of expectation – perception approach to customer perceived quality.	04	3	2

	b	Statement 1: Satisfied customers are a sign of success and an impetus for productivity. Statement 2: Satisfied customer is someone who has availed the services provided. Considering the previous statements, explain the concept of "Service Confidence and how it can be enhanced" taking example of services offered by police.	06	4	2
	c	Identified any two internal causes for customer dissatisfaction and also represent following using appropriate block diagrams. i) The satisfaction continuum expressed in _5 to _5 scale. ii) Perception-expectation simplified gap model.	06	4	1
		OR			
6	a	Using a block diagram, explain the aspects of service quality using Airline Industry as an example with specific emphasis on any two of the below mentioned topics. i) Inputs ii) The service process iii) The service experience iv) Outcomes	08	3	3
	b	Assume a customer touch point example of your choice and briefly explain it. Defend your understanding of how customer satisfaction can be enhanced/maintained (Customer touch point considered) for the below mentioned use cases using _5 to _5 scale template. i) Adequate performance satisfies the customer ii) Using enhancers to delight the customer iii) A dissatisfied outcome iv) Enhancers compensate for failures	08	4	2
7	a	With respect to service product variety, discuss the following terms i) Runners ii) Repeaters iii) Strangers Also, comment on how organizations handle these runners, repeaters and strangers.	08	3	3
	b	Discuss how service processes can be repositioned.	08	3	3
		OR			
8	a	Service capacity is defined as the maximum level of value-added activity over a period of time that a service process can consistently achieve under normal operating conditions. Organization would keep experimenting with terms "Consistently" and "Under normal operating conditions". Comment from your life experience how these affect the Quality.	08	3	2
	b	Ensuring customer satisfaction enhances the performance of the organization. Customer satisfaction can be measured using several methods. Explain four such methods as per the below mentioned format.			

Sl. No.	Method Name	Explanation	Useful for organizations such as
1			
2			
3			
4			

			08	3	3

9	a	Depict a service and its challenges. Explain how you wish to use SERVQUAL and MoSCoW prioritization matrix to assess the requirements and measure it. Justify its usage to a Naïve Service Operations Manager running a typical service that you have chosen.	10	5	4
	b	How might a hotel chain's service operations manager use various performance measure to enhance overall performance?	06	3	3
		OR			
10	a	Difference between level and chase strategy with appropriate examples.	08	3	2
	b	What are the types of targets used in performance management, and how do they differ from each other?	08	4	3

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations Sept/Oct – 2024

Industrial Engineering and Management

SERVICE OPERATIONS MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	List two metrics used to judge the success of a service operation.	02	1	1
	1.2	Why are standardizing service processes often difficult in service operations management?	02	2	2
	1.3	How can customers be classified in service operations management?	02	2	3
	1.4	Why is customer retention important in service operations?	02	2	1
	1.5	What factors influence our confidence in a service before we receive it?	02	2	2
	1.6	Highlight the two key benefits of e-procurement over traditional procurement.	02	2	4
	1.7	What is Process mapping?	02	2	5
	1.8	Distinguish between commodity and capability processes.	02	2	1
	1.9	Compare a restaurant chain which has a centralized kitchen and many outlets across Bengaluru as opposed to a restaurant which has a decentralized kitchen in terms of advantages.	02	4	2
	1.10	List different types of Benchmarking.	02	2	3

PART-B

2	a	Explain the 'service concept' using examples from popular Indian food brands such as Haldiram's and MTR. How do these brands' service approaches illustrate the key elements of an effective service concept in the Indian market?	10	3	2
	b	Considering yourself as a Service Operations Manager for an online retail store, highlight on the challenges that you might encounter.	06	2	2
3	a	Illustrate the impact of a well-defined service concept on customer experience using examples from different industries. How does a clear service concept contribute to achieving competitive advantage?	08	3	1
	b	Evaluate the role of managing customer relationships in service operations in India. What practices can organizations adopt to effectively manage and improve these relationships?	08	4	1
		OR			
4	a	Identify and describe the different categories of customers based on their behavior or attitude. How do these classifications help businesses tailor their strategies?	10	2	1

	b	Explain how managers can develop good customer relationships in a $B2C$ service environment.	06	2	2
5	a	Discuss the factors contributing to gaps between customer expectations and perceptions in service delivery. What are the implications of these gaps on perceived service quality?	08	2	1
	b	What are service quality factors? Briefly describe any six service quality factor with example	08	2	4
		OR			
6	a	What strategies can be employed to effectively manage service supply chains?	08	2	3
	b	Outline the primary challenges that service operations managers face in managing complex and often messy supply chains, particularly when dealing with intermediaries. How can these challenges be effectively addressed?	08	2	2
7	a	Imagine a potential customer navigating the website of a fashion retail outlet to select and order a dress. How can understanding the nature of service processes help in designing a better customer experience? Provide specific examples of how this understanding can enhance the online shopping journey.	08	3	3
	b	With respect to service product variety, discuss the following terms i) Runners ii) Repeaters iii) Strangers Also, Comment on how organizations handle these runners, repeaters and strangers.	08	2	1
		OR			
8	a	For a service with significant front-office customer interaction, illustrate strategies to effectively manage bottlenecks and queues. How can these strategies improve the customer experience?	08	3	4
	b	Discuss how service processes can be repositioned.	08	2	1
9	a	Choose a specific service and describe its associated challenges. Explain how you would use $SERVQUAL$ prioritization matrix to assess and measure its requirements. Justify use of the tool to a novice service operations manager in the context of the chosen service.	06	3	5
		How can a hotel chain's service operations manager use various performance measures to improve overall operational performance? Provide examples of performance metrics and their impact.	10	2	4
		OR			
10	a	Differentiate between the level strategy and the chase strategy in capacity management, providing relevant examples for each approach.	06	4	1
	b	What are the different types of targets used in performance management, and how do they differ from one another? Explain the significance of each type in managing performance effectively.	10	2	1

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Grade Improvement Examinations Nov/Dec-2020

Industrial Engineering and Management

SERVICES OPERATIONS MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Identify two significant responsibilities of service operations manager.	02
	1.2	Comment on the consequences if a budget hotel follows capability process as per capability commodity model.	02
	1.3	Imagine you are going to watch an *IPL* cricket match at Namma Bengaluru's very own Chinnaswamy Cricket stadium, list 4 aspects that you associate regarding service experience.	02
	1.4	When distinguishing airlines based on the type of operations, Low cost airlines are typically associated with _____, _____, _____ & _____.	02
	1.5	Comment on challenges of running an *NGO*.	02
	1.6	Assume you are being interviewed for your short but successful career in the field of "Service operations management". One of the questions from a young enthusiast is "How to develop a culture of service quality in an organization?" Comment on at least 4 aspects.	02
	1.7	From your experience, give two examples where technology is used to enhance customer experience.	02
	1.8	List 2 differences between service operations and manufacturing operations organizations.	02
	1.9	Comment on the tools that you use to enhance customer satisfaction.	02
	1.10	Online shopping is the new trend among most of youngsters. Identify the 4 moments of truth for a product or service bought via online shopping as listed below. Zero Moment of truth: First Moment of truth: Second Moment of truth: Third Moment of truth:	02

PART B

2	a	Use hotel industry as an example, how you differentiate capability as opposed to commodity. Give insights on process styles, service offer, Major challenges faced.	10
	b	Identify any 3 types of services and enumerate the challenges faced by these services?	06

3	a	Assume yourself in a position of a service operations manager in a luxury resort in Maldives. Evaluate the challenges in managing the operations both tactically as well as strategically.	10
	b	Describe how "Service as a concept" can be used as a strategic tool.	06
		OR	
4	a	Explain how managers can develop Good customer relationship in a B2C service environment.	04
	b	Explain different types of customer types you might encounter in a service environment using appropriate block diagram.	06
	c	Stakeholder management is an important aspect of any service operations. Having many customers as allies can be the best thing that can happen to any service / business. Recommend how allies can be created among various customer types.	06
5	a	Illustrate with an example the downsides of expectation perception approach to customer perceived quality.	04
	b	Statement 1: Satisfied customers are a sign of success and an impetus for productivity Statement 2: Satisfied customer is someone who has availed the services provided. Considering the previous statements, Explain the concept of "Service Confidence and how it can be enhanced" taking example of services offered by police.	06
	c	Identify two internal causes for customer dissatisfaction and also represent following using appropriate block diagrams. i) The satisfaction continuum expressed in_5 to_5 scale. ii) Perception-Expectaion simplified gap model.	06
		OR	
6	a	Using a block diagram , Explain the aspects of service quality using Airline Industry as an example with specific emphasis on any two of the below mentioned topics i) Inputs ii) The Service process iii) The service Experience iv) Outcomes	08
	b	Assume a customer touch point example of your choice and briefly explain it. Defend your understanding of how customer satisfaction can be enhanced/maintained (Customer touch point considered) for the below mentioned use cases using _5 to_5 scale template. i) Adequate performance satisfies the customer ii) Using enhancers to delight the customer iii) A dissatisfied outcome iv) Enhancers compensate for failures v) Dissatisfaction shifts the zone of tolerance vi) Delight also shifts the zone of tolerance.	08

7	a	With respect to service product variety, discuss the following terms i) Runners ii) Repeaters iii) Strangers Also, Comment on how organizations handle these runners, repeaters and strangers.	08
	b	Discuss how service processes can be repositioned.	08
8	a	Service capacity is defined as the maximum level of value added activity over a period of time that a service process can consistently achieve under normal operating conditions. Organization would keep experimenting with terms "Consistently" and "Under normal Operating Conditions". Comment from your life experience how these affect the quality.	08
	b	Ensuring customer satisfaction enhances the performance of the organization. Customer satisfaction can be measured using several methods. Explain four such methods as per the below mentioned format.	08

Sl No	*Method Name*	*Explanation*	*Useful for Organizations Such as*
1			
2			
3			
4			

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

SERVICES OPERATIONS MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	With 2019 loksabha assembly elections around the corner, its our responsibility to vote for the right candidate who can serve better. Comment on any two challenges of providing public services ($G2C$).	02
	1.2	Distinguish between service operations with regard to a low-cost airline and a Full-service airline [any two differences] .	02
	1.3	Report the major challenges in providing service with regard to $B2B$ and $B2C$.	02
	1.4	Give any two implications of technological advances in educational sector.	02
	1.5	Expectations of a service are dynamic in nature and are influenced by _____ and _____.	02
	1.6	From your experience, give two examples where technology is used to enhance customer experience.	02
	1.7	Write any two points highlighting the importance of service process design.	02
	1.8	Assuming your family is running the food court at your college campus, where the challenges are different in comparison to a usual restaurant business in any business district. List the rewards and targets for your employees in such a scenario?	02
	1.9	Identify any two operational performance measures for smooth functioning of a typical bank.	02
	1.10	Identify any one reason why bench marking practice is criticized as industrial tourism.	01
	1.11	Identify any two types of targets any service firm should work towards.	01

PART B

2	a	Describe the six tactical challenges faced by service operations managers in a typical service environment.	08
	b	Your friend wants to be an entrepreneur and wants to provide a particular service in a tier *II* city like Mysore. He/she needs your suggestion to enhance the service experience. Explain the aspects of service experience that you would emphasize on in order to suggest your friend. Defend whether your suggestion would change to some extent if the same service were to be provided in a tier *I* city like Bengaluru.	08

3	a	A service concept is shared understanding of the nature of the service provided and received. Using an example of a theme park, explain the following: i) Organizing the idea. ii) The service experience iii) The service outcome iv) The service operation v) The value of the service.	10
	b	Describe how "Service as a concept" can be used as a strategic tool.	06
		OR	
4	a	Illustrate with an example how important is retaining good customer for the success of any business.	08
	b	Write short notes on(Use quadrants to showcase your understanding): i) Classification of customer types ii) Creating allies among the various customer types.	08
5	a	Sketch the touch points in the customer's journey of the following case. "A patient with an appointment to see a doctor for a routine medical examination".	05
	b	In consideration for the above case, illustrate how customer satisfaction can be enhanced at various touch points in the customer journey.	06
	c	Taking an example of a bank, recognize the importance of Hygiene factors, Critical factors, Neutral factors and Enhancing factors.	05
		OR	
6	a	Discuss the role of innovation and internet technology in developing the customer experience.	08
	b	One of the key decisions for procurement function is choosing/selecting a supply partner and particularly for those who handle critical elements of supply chain. Discuss the eight criterion for supplier selection excluding cost and quality.	08
7	a	Showcase your understanding of Runners, Repeaters and Strangers for providing following service in the given format below. Also, Comment on which among these (Runners, Repeaters and Strangers)is least efficient and difficult to handle and the reasons for it.	10
	b	Describe how can service processes be repositioned?	06
8	a	Write short notes on(Minimum four points): i) General rules for managing bottlenecks ii) Principles of queue management (perceived waiting time vs actual waiting time).	08
	b	Evaluate the need for "Metric Benchmarking" processes and the shortfalls while doing it.	08

Service	*Runner*	*Repeaters*	*Strangers*
Car service			
International Airline			
Hospital			

Entrepreneurship and Intellectual Property Rights

VI Semester SEE Question Papers

Course codes: 21HS61A, 18HSI61, 16HSI61, 12HSI61

- Sep/Oct 2024 (21HS61A)
- Sep 2023 (18HSI61)
- Aug 2022 (18HSI61)
- Apr/May 2019 (16HSI61)
- Apr/May 2018 (12HSI61)
- Apr/May 2017 (12HSI61)
- Apr/May 2016 (12HSI61)

21HS61A

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Regular Examinations Sep/Oct – 2024

Common to AS/BT/CH/CV/IM/ME

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Can any invention be patented after publication or display in the public exhibition?	01	2	1
	1.2	Is a substance obtained by a mere admixture resulting only in the aggregation of the properties of the components thereof or a process for producing such substance is patentable or not.	01	2	1
	1.3	Compulsory license is issued after the expiry of _____ years.	01	2	1
	1.4	BIS Hall mark is an example for _____ mark.	01	1	2
	1.5	In case of shorthand writer the copyright belongs to the _____.	01	2	2
	1.6	Entrepreneurial development is the key to achieve all-round _____ through acceleration of industrial and entrepreneurial activities.	01	2	3
	1.7	Define collective Trade Mark.	02	1	2
	1.8	Define cybercrime.	02	1	3
	1.9	A wildlife programme broadcast earlier by the Discovery channel is again broadcast by the Doordarshan without any license or consent from the discovery channel to do so. Is there an infringement of the Broadcasting reproduction right? Justify.	02	3	2
	1.10	Define eco labeling.	02	1	2
	1.11	List any two reasons which provide an individual with the motivation to start a new business venture?	02	2	4
	1.12	List any two characteristics of a successful entrepreneur?	02	2	3
	1.13	What is meant by Performers Right?	02	2	4

PART-B

2	a	How do you categorize the different types of Intellectual Property Rights, and explain each of them with an example.	06	2	1
	b	How can you draft a patent application to ensure it meets the requirements of different patent offices globally? How can partnerships and collaborations enhance the commercialization of a patented invention?	10	3	1
3	a	A technology company in India develops a unique algorithm that significantly enhances data processing speed. The company wants to protect this algorithm as a trade secret. Explain the steps the company should take to ensure the protection of its trade secret in accordance with Indian law. What legal tools and strategies should the company employ?	08	3	2
	b	Define “Trade Mark”. Enumerate the complete procedure for registration of a Trade Mark with the help of flowchart.	08	2	2

		OR			
4	a	An apparel company is considering the use of an Eco Label to enhance its brand image. What specific criteria should the company focus on to ensure its products meet the requirements for an Eco Label? Discuss how these criteria can be integrated into the company's production process.	08	3	2
	b	A local bakery is concerned that a new competitor is using a trade name and packaging design that closely resembles its own. What steps should the bakery take to protect its brand, and what legal remedies are available if the competitor's actions amount to passing off or trademark infringement?	08	4	2
5	a	Copyright protects expression and not the ideas. Discuss.	08	2	2
	b	Discuss the different types of cyber crimes under the IT Act.	08	2	2
		OR			
6	a	Define Industrial Design. Explain the key features of Industrial Design with an example.	08	2	2
	b	Explain the acts which do not constitute the infringement of Copy Right under the Copy Right Act.	08	3	2
7	a	What are some common challenges and obstacles that entrepreneurs often encounter in their business ventures? Explain.	08	3	3
	b	Sarah thinks that all entrepreneurs must have a groundbreaking idea to succeed. She observes her competitor, who started with a simple, existing idea and has achieved significant success. Discuss how Sarah's perception aligns or contrasts with the reality of entrepreneurship.	08	3	3
		OR			
8	a	Discuss how understanding the business environment can impact the strategic decisions of a startup.	08	2	3
	b	What are the characteristics of successful Entrepreneurs? Discuss and give examples of any two early Indian entrepreneurs and throw some light on their enterprises.	08	2	3
9	a	A tech startup failed after its initial product launch due to poor market fit and lack of customer feedback. Analyze the reasons for its failure and recommend steps the startup could have taken to avoid these pitfalls.	10	4	4
	b	What is Business model innovation? Give two examples of business model innovation in India.	06	3	4
		OR			
10	a	Describe the process of evaluating business opportunities.	08	2	4
	b	A startup is planning to establish a new business. Analyze the steps they should take to set up the enterprise, considering factors such as market research, legal requirements, financing, and operational planning.	08	4	4

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Examinations September-2023

COMMON TO AS/CH/CV/EI/IM/ME

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	An intellectual property right protects the use of information and ideas that are of _____ values.	01
	1.2	The first offence for infringement of copyright can be for a maximum of imprisonment for a term and fine of _____.	02
	1.3	Under which type of agreement royalty is paid based on the sale?	01
	1.4	On a school computer, Rama learned how to copy programs. A classmate asked her to copy a program for his home use. Her most ethical response would be _____.	02
	1.5	Patent can be granted for _____.	01
	1.6	International copyright is governed by the _____.	01
	1.7	The _____ is granted in the case of substances which are not food items or drugs as such, but are capable of being used as food and drugs.	01
	1.8	On what basis, the total royalty on the book sold is computed under the copyright agreement?	01
	1.9	If a person uses someone's song as a background in his movie, then such an act could be an example for _____.	01
	1.10	A business arrangement where one party allows another party to use a business name and sell its products or services is known as _____.	01
	1.11	List any two characteristics of an Entrepreneur.	02
	1.12	Why are small businesses important to country's economy?	02
	1.13	What are the knowledge requirements of entrepreneur?	02
	1.14	List the motivations to be an entrepreneur.	02

PART-B

2	a	Everyone has the right to the protection of the moral and material interests resulting from scientific, literary or artistic production of which he is the author. Justify the protection mechanism in India.	08
	b	List out the inventions that are not patentable in India.	08

3	a	Briefly explain the characteristics of a good trademark.	08
	b	Briefly explain the collective and certification trademark.	08
		OR	
4	a	Briefly explain the chances of refusal of a trademark application.	08
	b	Briefly explain the procedure for the registration of a good trademark.	08
5	a	Discuss briefly the need for the protection of an industrial design.	08
	b	Assume you have come up with a new programing language and have developed software. Briefly explain the procedure you would adopt for the protection of it.	08
		OR	
6	a	Discuss the rights that could be exercised by the broadcasting agency.	08
	b	Briefly explain the constraints of the design that cannot be protected by Industrial Design rights.	08
7	a	Entrepreneurial talents are the backbone of Indian economy. Justify.	08
	b	Briefly explain how an entrepreneur perceives opportunity, plan, organize resource and succeed the goal set.	08
8	a	Discuss briefly a case wherein an entrepreneur was successful in the beginning of the firm, but later lost in the race.	08
	b	Briefly explain how an entrepreneur decides on the path for success.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations August-2022

COMMON TO AS/CH/CV/EI/IM/ME

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	A patent application is filed on 1 January 1999. The application is published 18 months later on 1 July 2000 and granted on 30 May 2021. The patent is valid until _____.	01
	1.2	In most countries, how long does copyright last for?	01
	1.3	The broadcasting organizations like TV, Radio have a term of copyright for _____.	01
	1.4	An application for restoration of a lapsed patent shall be made within _____ months from the date of cessation of the patent.	01
	1.5	_____ entrepreneur's refusal to adopt opportunities to make changes in production formulae even at the cost of severely reduced returns.	02
	1.6	Who owns copyright in an article written by a staff member for a newspaper?	02
	1.7	Collective mark is registered by _____.	01
	1.8	Symbol of Maharaja of Air India is _____.	01
	1.9	Imagine a cricketer sets up his own company to sell his own range of clothes. What type of intellectual property can he use to show that the clothes are made by his company?	01
	1.10	Does trademark assure quality of the product?	02
	1.11	Information Technology Amendment Act in India came into force on _____.	01
	1.12	Design thinking typically helps in _____.	01
	1.13	Caitlin is creating a new product for university students. She takes a design-thinking approach. Her first step is addressing who she is creating the product for and conducts research on understanding this target market. What is this step in the design-thinking process?	01
	1.14	In order to register industrial design, design should be new and _____.	01
	1.15	What happens in the test stage of design-thinking?	02
	1.16	Individuals influencing an entrepreneur's career choice and style are known as _____.	01

PART-B

2	a	If you were to apply for an Indian Patent, how will you go about doing it stage by stage? Explain with indicative timelines for each stage?	08
	b	In order to be patentable, an invention must be worthy of a patent grant. The patent worthiness of an invention is assessed by the government through patentability requirements. Explain the essential requirements of patentability.	08
3	a	The plaintiff claimed that defendants are selling products under the trademark FEXIM that is deceptively similar to the plaintiff's mark PHEXIN which is used for pharmaceutical preparations. The defendants are selling anti-biotic tablets with the trademark 'FEXIM' with the packing material deceptively similar to that of the plaintiff whereby intending to not only to infringe the trademark but also to pass off the goods as that of the plaintiff, as the two marks are also phonetically similar. Decide referring to judicial decisions, the factors to be considered in deciding the question of deceptive similarity in passing off action?	08
	b	Enumerate the stages involved in the registration of trademarks in India.	08
		OR	
4	a	The licensing and assignment of Trademark is not subject to any control. After the grant of license or assignment, the license is free to use the trademark as he likes. Comment on this statement by explaining the difference between the licensing and assignment of Trademarks.	08
	b	Identify the grounds for refusal of registration of a trademark in India. What kind of trademarks can be registered under the (Indian) Trademarks Act, 1999?	08
5	a	The emergence of protection for industrial designs is intimately connected with the growth of industrialization and methods of mass production. In this context, explain the process of evolution of industrial design protection.	08
	b	Explain the rights conferred by copyrights to its owner. Elaborate with suitable examples.	08
		OR	
6	a	"Copyright is protection in form and not in idea". Elaborate the statement with suitable examples.	08
	b	List and explain the objectives and features of the Information Technology Act 2000.	08
7	a	Explain the risk-taking and resilience traits of Entrepreneurs with two relevant case examples.	08
	b	Discuss the importance of 'communication for a entrepreneurial process to the successful.	08
8	a	What are the major obstacles of Design Thinking these days? Discuss with relevant example.	08
	b	How Design Thinking works within a product development process? Explain with an example.	08

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Examinations April/May-19

Common to BT / CH / CV / EI / IM / ME

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	_____ is one of the requirements for patentability in India.	01
	1.2	Registered Industrial Design has protection for _____ years from the date of filing.	01
	1.3	_____ is transmitted orally and are deeply rooted in social and cultural systems.	01
	1.4	_____ and _____ are forms of intellectual property that can be used to protect innovations.	02
	1.5	The two characteristics of a good trade mark are _____ and _____.	02
	1.6	In India protection is given to literary works until _____	01
	1.7	Does style and specific manner in which a story is presented in a news article has any protection under intellectual property? If yes, name the same.	01
	1.8	*MNC* designs a shoe in Europe. But the shoe with similar design already sold in India. Does the manufacturer have to apply for registration in India too? Will the *MNC* get registration? Justify.	01
	1.9	List two categories of cyber – crime.	01
	1.10	Ram, a final year student, is from a well – off business family. He has spotted an opportunity in his home town related to production of electricity using wind. Can he become an entrepreneur and turn this into a venture? Which characteristics of entrepreneur help Ram to become entrepreneur?	02
	1.11	Which company did Danae Rengelmann, Slava Rubin, and Eric Schell start?	01
	1.12	Name any two myths about entrepreneurship.	01
	1.13	The first step in design thinking is _____.	01
	1.14	In the Maker style of entrepreneurship the main trait is _____.	01
	1.15	Which principle of 'show and tell' should be kept in mind, while meeting the customer for the first time?	01
	1.16	Empathy, observation and _____ are the main features of Design Thinking.	01
	1.17	The stage of the personal selling process when a startup gets a purchase commitment from a customer is _____.	01

PART B

2	a	Explain the procedure for the grant of a patent in India.	10
	b	Discuss the pros and cons of protecting an invention through patents and trade secrets.	06
3	a	Explain the procedural steps involved in registration of Trade Marks in India.	04
	b	A British manufacturer of biscuits has been selling cookies in Malaysia for more than 20 years, and has had a registered trade mark 'ChipsMore' for these goods during that time. Two years ago a Malaysian company started manufacturing and selling cookies under the brand 'ChipsPlus'. Analyze the case and comment on Infringement and Passing off.	06
	c	What are the reliefs available for infringement of registered trade mark to a proprietor?	06
		OR	
4	a	Enumerate the characteristics of a good trademark.	05
	b	A company *ABC* sells electronic goods and is using '*SPEEDELECTRO*' as trademark for his company for over 12 years. However *ABC* has not registered the same. A company *XYZ* decides to sell their electronic goods as '*SPEEDELECTRONICS*'. In the mind of the people who decide to buy the goods, they would associate '*SPEEDELECTRONIC*S' as *SPEEDELECTRO* and thus because of the acquired good will of *ABC's* company, they would purchase *XYZ's* products. Can *ABC* file a suit against *XYZ*? Analyze all possibilities of the case and arrive at an outcome.	05
	c	What are deceptively similar trademarks? List the factors to be taken into consideration when deciding the question of similarity.	06
5	a	Elucidate the classification of rights of a copyright owner.	06
	b	Explain the infringement of copyrights with a case example.	10
		OR	
6	a	Define cyber space. Explain the need for cyber laws with a relevant case.	06
	b	Discuss authorship issues in works used or made by computer under copyrights.	10
7	a	Discuss any five myths about entrepreneurship.	06
	b	How are entrepreneurship styles classified? List the main characteristic traits of Bill Gates, founder of Microsoft and Narayana Murthy, founder of Infosys and identify their style?	10
8	a	What is Design Thinking? Apply design thinking process to build a system to address the problem of pollution in your city.	10
	b	Identify the sales skills to become an effective entrepreneur.	06

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R. V. COLLEGE OF ENGINEERING

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VI Semester B. E. Examinations April/May-18

Common to EE / EC / IM / EI / TE

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART A

1	1.1	In copyright assignment, if duration is not stated then it is deemed as _____ years.	01
	1.2	Define performer's right.	02
	1.3	Right conferred by registration of design is called _____.	01
	1.4	Whether prime ministers speech can be copyrighted?	01
	1.5	Expand *MSME* and *GATT*.	02
	1.6	Compulsory license will be granted only after expiry of _____ years.	01
	1.7	Can the inventor of new process of bypass surgery claims a patent for new surgical method invented by him, can this be granted? Justify.	02
	1.8	*WIPO* was established in year _____, head quartered at _____.	02
	1.9	Mention three characteristic features of entrepreneurs.	02
	1.10	How social feasibility studies vary from technical feasibility studies?	01
	1.11	What are the two common reasons given by people who chose to become entrepreneur as the key drivers for their choice?	02
	1.12	*UPOV* stands for _____.	01
	1.13	Trademark should not be descriptive but _____.	01
	1.14	One man show generally is characteristic of _____ enterprise.	01
		PART B	
2	a	Traditional knowledge can be protected by the Indian Culture Intellects. Justify.	08
	b	Patents are not universal. Validate the statement with reference to inventions that can be patented.	08
		OR	
3	a	Elucidate scenario of *IPR* in India. With emphasis on various types of intellectual property.	08
	b	Elucidate on the process of patenting in India.	08
4	a	Explain the concept of different kind of trademarks with examples.	08
	b	Discuss the rights available to the owners of registered trademarks.	08
		OR	

5		Write short notes with proper illustrations: a) Infringement and acting trademark b) Passing off c) Eco-label d) Certification trademark.	 04 04 04 04
6	a	Explain the concept of design with reference to transfer of rights of interest.	08
	b	Elucidate various rights conferred by copyrights with emphasis on copyrights protection provisions in Indian context.	08
		OR	
7	a	Explain concept, scope and nature of copyright and allied rights.	08
	b	What is the object of registration of industrial design?	08
8	a	Entrepreneur and manager are two phases of same coin. Validate with the differentiating features among them.	08
	b	In the course of economic development, explain various types of entrepreneurs who help the development process.	08
		OR	
9	a	Elucidate the concept of entrepreneurship in India with an added note on intrapreneur.	08
	b	Explain the characteristics of entrepreneurs with emphasis in its function.	08
10		Give the critical note on: a) $MSME$ b) Feasibility studies.	 08 08
		OR	
11	a	Enumerate effect of WTO with focus on Indian micro, small and medium enterprise.	08
	b	What are various strategies for identification of business opportunities?	08

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R. V. COLLEGE OF ENGINEERING

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VI Semester B. E. Examinations April/May-17

Common to EE / EC / IM / EI / TE

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART A

1	1.1	*WIPO* stands for _____.	01
	1.2	Duration of patent is _____ years.	01
	1.3	The term of copyright is _____ years.	01
	1.4	Trade mark identify the _____ of the product.	01
	1.5	Passing off is a _____ remedy and it can be used to enforce _____ trade mark.	02
	1.6	'Agmark' is certification of mark used for _____.	01
	1.7	Collective mark means _____ with _____.	01
	1.8	In trade mark distinctiveness can be classified into _____ and _____.	02
	1.9	Duration of initial protection for designs under industrial Design Act 2000 is _____ years and registration can be renewed up to _____ years.	02
	1.10	_____ and _____ are the allied rights of copyright.	02
	1.11	Fictitious character can be protected under _____.	01
	1.12	To have copyright under copyright act 1957 work must be _____ and author must be _____.	02
	1.13	Intrapreneurs are usually _____ within the company and they are instructed to develop the company.	01
	1.14	Define an innovative entrepreneur.	01
	1.15	Pursuing a new venture is a _____ process.	01

PART B

2	a	Discuss the impact of *TRIPs* on Indian Patent Act 1970.	08
	b	Explain the basic requirements to grant patent under Indian Patent Act.	08
		OR	
3	a	Explain the tools for protection of Trade Secrets in India.	08
	b	Define traditional knowledge. Why traditional knowledge needs to be protected?	08
4	a	Define trade mark. Explain the features and functions of trade marks.	08
	b	Explain the enforcement of unregistered trade mark.	08
		OR	

5	a	What factors are to be considered for deciding deceptive similarity?	08
	b	Discuss the grounds on which registrar can be refused to register the trade mark.	08
6	a	Define industrial design. Why a design need to be registered under the Act.	08
	b	Discuss the rights conferred to the creator of the work under Copyright Act.	08
		OR	
7	a	Explain the rights available to the proprietor of a design by registration.	08
	b	Explain the works in which copyright existed under Copyright Act in India.	08
8	a	Define an entrepreneur. Distinguish the entrepreneur and a manager.	08
	b	Discuss the role of entrepreneur in economic development of a country.	08
		OR	
9	a	Explain the reasons for slow growth of entrepreneurship in India during British period.	08
	b	Discuss the evoluation of entrepreneur in India.	08
10	a	Discuss the impact of LPG on Micro small and medium enterprises.	08
	b	Explain the significance of governmental policies towards small scale industry in India.	08
		OR	
11	a	Discuss the reasons for entrepreneur to study the market, technical and social feasibility.	08
	b	Discuss the need and rationale of $MSME$ and its role in economic development.	08

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VI Semester B. E. Examinations April/May-16

Common to EE / EC / IM / IT / TE

INTELLECTUAL PROPERTY RIGHTS AND ENTREPRENEURSHIP

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	Wool Mask is an example of _____.	01
	1.2	In the absence of any agreement, for photograph taken, or a painting, engraving, or a cinematography film mode, _____ shall be the owner.	01
	1.3	The term period of protection of geographical Indication is _____ years.	01
	1.4	Compulsory license will be granted only after the expiry of _____ years.	01
	1.5	Trademark should not be descriptive but it should be _____.	01
	1.6	In case of posthumous work, the copyright subsists for _____ years from the beginning of next calendar year, after its first publication.	01
	1.7	The question papers in *UPSE* examination are the copyright of the paper setter. Justify.	02
	1.8	List the two essential documents to be generated and submitted by an inventor.	02
	1.9	Can a color by itself be subject matter of Registration of Industrial design? Justify.	02
	1.10	If the patent is expired, can that work be carried out by other company? Justify.	02
	1.11	_____ & _____ are regarded as the two basic elements involved in Entrepreneurship.	02
	1.12	The traits of Steve Jobs are an example for _____ type of entrepreneur.	01
	1.13	List the two rationale arguments in relationship with small scale industry development in India.	01
	1.14	One man show generally is a characteristic of _____ enterprise.	01
	1.15	Copyright Act came into force on _____.	01

PART-B

2	a	Explain in detail, difference between patentable and non-patentable inventions.	08
	b	Explain the significance of traditional knowledge with two instances of controversy.	08
		OR	

3	a	Explain the different methods of transfer of patent.	08
	b	In case you run a successful business venture and have a few crucial trade secrets that gives you an edge over the competition, what precautions would you take in order to protect your trade secrets from falling onto their hands?	08
4	a	What is a trademark? What are the various steps for registration of trade mark?	08
	b	Explain the essential features of infringement of trade mark with decided case studies.	08
		OR	
5	a	Explain the grounds for absolute refusal of registration of trademark.	08
	b	Explain the features and functions of Trade mark.	08
6	a	Explain the infringement of copyright with 2 case studies.	08
	b	Explain the essential features of Industrial design.	08
		OR	
7	a	Copyright is a bundle of rights. Describe various rights enjoyed by an owner of copyright.	08
	b	Explain different types of cybercrime.	08
8	a	Elaborate on the stages of the entrepreneurship process.	08
	b	Explain the characteristics and functions of an entrepreneur.	08
		OR	
9	a	In Indian context, explain the specific role that Entrepreneurship has fulfilled in the economic development of country.	08
	b	Distinguish between an entrepreneur and a manager.	08
10	a	Explain the impact of WTO and $GATT$ on trade system.	08
	b	List the characteristics of $MSME$.	08
		OR	
11	a	What are the steps to institute $MSME$ in the Indian context?	08
	b	Write short notes on: i) Financial feasibility studies ii) Technical feasibility studies.	08

Global Supply Chain Management

VI Semester SEE Question Papers

Course codes: 21IM62

- Sep/Oct 2024 (21IM62)

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VI Semester B. E. Examinations Sept/Oct– 2024

Industrial Engineering and Management

GLOBAL SUPPLY CHAIN MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical tables permitted.

PART-A

			M	BT	CO
1	1.1	The most limited scope over which the strategic fit is considered is one operation within a functional area within a company. This is referred to as _____ scope	01	2	1
	1.2	Draw the cost-responsiveness efficient frontier curve	02	2	1
	1.3	_____ cycle occurs at the manufacturer / supplier interface and includes all processes to ensure that materials are available for manufacturing to occur according to schedule	01	2	1
	1.4	The coordinate location of a manufacturer is $(40, 60)$. The coordinate location of its customer is $(20, 10)$. The manufacturer is required to supply 200 compressors to this customer. The cost of transporting one compressor is *Rs* 100. Using the gravity location model, determine the distance between the manufacturer and supplier, and the total transportation cost.	02	3	2
	1.5	Graphically depict variation of transportation cost and response time with number of facilities	02	2	2
	1.6	Enumerate the risks that are to be considered during network design.	02	2	2
	1.7	If the manager at B&M manages inventory such that the store does not run out of inventory in 8 out of 10 replenishment cycles, then stores achieve a CSL of _____%.	01	3	3
	1.8	What are the two factors which determine appropriate level of safety inventory?	02	2	3
	1.9	_____ contract allows the buyer to modify the order (within limits)as demand visibility increases closer to the point of sale	01	2	3
	1.10	Mention any two benefits of effective sourcing decisions	02	2	3
	1.11	What is continuous replenishment in supply chain management?	02	2	3
	1.12	Identify one example of a collaborative planning activity in CPFR	02	2	4

PART-B

2	a	Write the block diagram to depict the framework for structuring the drivers of supply chain performance. Choose an Indian organization and discuss the structuring of its drivers.	08	3	1
	b	Describe the cycle and push/pull views of a supply chain. Which of the two views is very useful when considering strategic and operational decisions related to Supply chain design	08	2	1

3	a	Discuss in detail about the role, factors influencing with respect to facility location for a pharmaceutical Supply chain.	08	3	2
	b	Illustrate and discuss three performance characteristics related to service for the following distribution network design options, using diagrams: (i) Manufacturer storage with direct shipping (ii) Distributor storage with last-mile delivery	08	2	2
		OR			
4	a	Identify and describe the factors influencing Network Design decisions	08	2	2
	b	Discuss how you evaluate the performance of a distribution network along the two dimensions namely: customer needs and the cost.	08	2	2
5	a	Weekly demand for Legos at Toys store is normally distributed with a mean of 2500 boxes and a standard deviation of 500. The replenishment lead time is two weeks. i. Assuming a continuous review replenishment policy, evaluate the safety inventory that the store should carry to achieve a CSL of 90% ii. If the store manager decides to follow a periodic review policy to manage inventory of Legos and plans to review inventory after every 4 weeks, for a CSL of 90%, how much of safety inventory should the store carry? What should its Order upto level (OUL) be? iii. Compare the above 2 replenishment policies and comment on requirement of safety inventory	10	3	3
	b	Weekly demand for 12" frames at the Frame Shop is normally distributed with a mean of 250 and a standard deviation of 150. The store manager continuously monitors inventory and currently orders 1,000 frames each time the inventory drops to 600 frames. The manufacturer currently takes two weeks to fill an order. How much safety inventory does the store carry?	06	3	3
		OR			
6	a	Explain the role of cycle inventory and safety inventory on supply chain with suitable examples.	10	3	3
	b	Weekly demand for microwave ovens at Best Buy store is normally distributed with mean of 2500 and standard deviation of 500. It takes two weeks to supply a Best Buy order. Best Buy targeting a CSL of 90 percent and monitors its inventory continuously. How much safety inventory of microwave ovens should Best Buy carry?	06	3	3
7	a	Analyze the factors affecting transportation decisions with the aid of a suitable example	10	2	3
	b	What are the reasons for outsourcing? Illustrate with an example	06	2	3
		OR			
8	a	Identify and discuss the factors considered when scoring and assessing suppliers in a supply chain context.	10	2	3
	b	Using examples, describe revenue-sharing and quantity flexibility contracts.	06	3	3

9	a	Describe supply chain co-ordination and the bull whip effect and their impact on supply chain performance	10	2	4
	b	How does Customer Relationship Management (CRM) contribute to supply chain efficiency?	06	2	4
		OR			
10	a	What is vendor managed inventory (VMI)? How it is different from continuous replenishment strategy? How is the manufacturer benefitted out of VMI?	10	2	4
	b	What does CPFR stand for in supply chain management? How does CPFR improve supply chain efficiency?	06	2	4

Lean Manufacturing Systems

VI Semester SEE Question Papers

Course codes: 21IM65E1, 12IM6D2

- Sep/Oct 2024 (21IM65E1)
- Apr/May 2018 (12IM6D2)
- Apr/May 2017 (12IM6D2)
- Apr/May 2016 (12IM6D2)

21IM65E1

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Examinations Sept/Oct – 2024

Industrial Engineering and Management

LEAN MANUFACTURING SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Define Lean Manufacturing.	02	2	1
	1.2	Mention the two pillars of the Toyota Production System (*TPS*)?	02	2	1
	1.3	Why is inventory needed in a manufacturing system?	02	2	1
	1.4	What is a Kanban system?	02	1	1
	1.5	List two benefits of lead-time reductions.	02	2	2
	1.6	What are the three fundamental issues of cultural change in lean implementation?	02	2	2
	1.7	What is the purpose of conducting a system-wide evaluation of the present state in lean implementation?	02	2	2
	1.8	What is Kaizen and its role in lean manufacturing?	02	2	2
	1.9	What is Hoshin-Kanri Planning?	02	1	1
	1.10	Why is it important to sustain the gains achieved through lean implementation?	02	2	2

PART-B

2	a	Explain the concept of Lean Manufacturing and discuss its primary goals.	08	2	1
	b	Discuss Taiichi Ohno's thoughts on the Toyota Production System and its significance in modern manufacturing.	08	2	1
3	a	Discuss the need for inventory in manufacturing and the disadvantages associated with high inventory levels.	08	3	4
	b	Explain the concept of Kanban and describe the process of Kanban calculations in a lean manufacturing system.	08	3	2
		OR			
4	a	Explain the need for inventory in a manufacturing system and discuss the various types of inventory that are typically maintained.	08	3	4
	b	How do finished goods inventory calculations help in lean manufacturing, and what factors are considered in these calculations?	08	3	2
5	a	Analyze the benefits of lead-time reduction in manufacturing and describe techniques used to achieve these reductions.	08	3	2
	b	Discuss the cultural change fundamentals necessary for implementing lean manufacturing and the challenges associated with these changes.	08	2	3

		OR			
6	a	Trace the historical evolution of lead time in manufacturing and discuss its changing significance over time.	08	3	3
	b	Examine some cultural aspects that are essential for a successful lean implementation.	08	3	3
7	a	Outline the steps involved in implementing a lean project and discuss the importance of each step in the process.	08	2	3
	b	"After implementing lean practices, an organization has seen significant improvements. However, there is concern about sustaining these gains over time. Discuss the importance of sustaining the gains in a Lean environment. How would you address the existence of process gains and losses to ensure long-term success?"	08	4	4
		OR			
8	a	"A manufacturing company is struggling with inefficiencies in its production line. As a Lean consultant, you are asked to guide the company through the Lean implementation process. Discuss how you would assess the three fundamental issues related to cultural change in the company, and outline the steps you would take to complete a system-wide evaluation of the present state."	08	4	4
	b	Explain the concept of Kaizen and its role in the continuous improvement process within lean manufacturing.	08	2	3
9	a	"A manufacturing company is transitioning to Lean 4.0. Explain how the traditional dimensions of lean manufacturing, such as waste reduction, just-in-time production, and continuous improvement, are influenced by the technologies and principles of Industry 4.0. Provide specific examples of how these dimensions can be enhanced by digital transformation."	08	4	3
	b	Describe Hoshin-Kanri Planning and its importance in aligning organizational goals with lean initiatives.	08	2	3
		OR			
10	a	"During the implementation of *Lean* 4.0, a manufacturing company faces several challenges, including resistance to change, data management issues, and high implementation costs. Discuss these challenges in detail and propose practical solutions to overcome them."	08	3	3
	b	Discus the importance of sustaining the gains achieved through lean implementation and it strategies to prevent process loss.	08	2	3

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VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

LEAN MANUFACTURING SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART A

1	1.1	At the heart of *JIT* is a deep understanding and control of _____.	01
	1.2	The makeup of the typical production cell line and how _____ is handled is a key manufacturing aspect of *TPS*.	01
	1.3	Define buffer stock.	01
	1.4	State Little's law.	01
	1.5	As per *KP* methodology, what is a problem?	01
	1.6	Name one of the three requisite skills that are required for the leader of lean initiative.	01
	1.7	Cite a reason to implement lean using one value stream at a time approach.	01
	1.8	Future state value stream map will _____ production internally.	01
	1.9	The key uniqueness in the Hoshinkanri planning is the concept of _____.	01
	1.10	The two strongest techniques to sustain the gains once they are found are through _____ and _____ simplification.	01
	1.11	Write two differences between *KANBAN* and *CONWIP*.	02
	1.12	In a manufacturing organization the production time is 23.3 hours, planning time in 24 hours delivery time is 2 hours and the typical queue of 16 hours. Assuming $\alpha = 0.05$ calculate cycle stock. takt time is 1 minute.	02
	1.13	What are the approaches to reduce first piece lead time and total lead time?	02
	1.14	_____ and _____ are the two extremely powerful detractors that will rear their ugly hands time and again as an organization pursues the lean journey.	02
	1.15	Name any two characteristics of 'Goals' that are created during lean implementation.	02

PART B

2	a	Write a detailed note on pull production technology.	08
	b	Discuss the following revolutionary concept used in the *TPS*: The supply of value to the customer.	08
		OR	
3	a	The *TPS* is not a complete manufacturing system. Critically analyze the statement.	08
	b	Prepare a comparative analysis between lean and the Toyota production system.	08

4	a	"To reduce inventory levels reduce variation". Justify the statement.	06
	b	Enumerate the six rules of *KANBAN* and the several unique functions that are provided.	10
		OR	
5	a	Summarize the people aspects of the basic needs to execute the *TPS* (quality control).	09
	b	Elaborate the concept of Jidhoka as a pillar of quantity control. List the techniques under Jidhoka pillar.	07
6	a	List and explain briefly the benefits of lead time reductions. Illustrate with examples.	08
	b	Bring out the importance of presence of talented problem solvers as a fundamental issue of cultural change to be addressed during lean implementation.	08
		OR	
7	a	The interdependence of activities is a unique aspect of lean implementation initiative. As a team member of lean implementation group how do you handle this issue?	04
	b	Select a process of your choice with minimum three stages. Develop the present state value stream map with usual notations and symbols. Make suitable assumptions about the data.	12
8		For a small scale manufacturing organization of about 100 employees how do you perform the assessment of the three fundamental issues to cultural change? Prepare a report to be submitted to the management for lean implementation.	16
		OR	
9		Assume that you are chosen as the sensei. How do you evaluate a manufacturing company based on the five precursors to implementing a lean initiative? Prepare a report of evaluation which becomes a significant part of the implementation plan.	16
10	a	Why are goals and goal deployment so important? Discuss.	08
	b	Illustrate with a sketch the Hoshinkanri planning model Enumerate unique strengths of Hoshinkanri planning.	08
		OR	
11	a	"Transparency is the concept that lets you see in real time right now what is happening in your process". Elaborate with examples.	08
	b	Explain the various means of achieving process gains by either reducing the variation in the product, the process or both.	08

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VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

LEAN MANUFACTURING SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

6 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

7 Answer FIVE full questions from Part B.

PART A

1	1.1	The two pillars of TPS are _____ and _____.	02
	1.2	Lean is also about _____ and _____.	02
	1.3	As per Little's law lead time is the ratio of _____ and _____.	02
	1.4	Total number of finished goods Kanban is the sum of (_____) + (_____) + (_____), divided by _____.	02
	1.5	Expand $SMED$ and OEE.	02
	1.6	Takt is the ratio of _____ and _____.	02
	1.7	List any four out of seven wastes.	02
	1.8	Define a push system.	02
	1.9	The three fundamental issues to be assessed to cultural changes are i) Do we have the _____ to make this a success? ii) Do we have the _____ to make this a success? iii) Do we have the _____ to make this a success?	02
	1.10	Muda,, Mura and Muri means _____, _____ and _____ respectively.	02

PART B

2	a	Define Lean.	02
	b	Discuss the ohno's thought about the Toyota production system.	06
	c	Explain in detail where lean will not work?	08
		OR	
3	a	Explain clearly the two pillars of the TPS.	08
	b	Write a note on: i) TOC ii) Crosby's approach to zero defects.	08
4	a	An insurance company processes 9600 claims per year. The company will not work for 4 weeks/year. Calculate the WIP if the average processing time of a claim is 21 days.	06
	b	What is so bad about Inventory? In this context, discuss the two basic questions which are frequently overlooked by the managers.	10
		OR	

5	a	Discuss the following: i) $5S$ techniques ii) MSA.	08
	b	What is TPM? Explain the five pillars of TPM.	08
6	a	Illustrate the role played by $SMED$.	04
	b	Describe the seven techniques to reduce lead time.	12
		OR	
7	a	With a sketch explain Jung's scales for personality types.	07
	b	Discuss the three basic issues of cultural change.	09
8		Write a note on the following i) Seven wastes ii) VSM iii) Spaghetti diagram.	06 05 05
		OR	
9	a	List all the steps used as a prescription for the lean implementation and explain any one of them.	10
	b	With a sketch explain Kaizen spiral.	06
10		Write a note on the following: a) Process gain b) Poka – yoke c) Sustaining the gains d) Goal characteristics.	16
		OR	
11	a	Explain Hoshin – Kawi planning model.	08
	b	Analyze the concept "how do we know there is a loss?"	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

LEAN MANUFACTURING SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART-A

1	1.1	What is the popular definition of lean?	02
	1.2	What did Ohno say about the Toyota Production system?	02
	1.3	To reduce inventory levels, reduce_____.	01
	1.4	List the three forms of buffers.	02
	1.5	The foundation of high quality has two strategies. First is the _____ of the workforce.	01
	1.6	Two key characteristics that few businesses measure in any form but that all businesses want are flexibility and_____.	01
	1.7	Lead time as a metric was not discussed much at all until the book _____ (Lean Enterprise Institute, 1998) heightened peoples awareness of this powerful concept.	01
	1.8	The _____ is a simple yet powerful tool to visualize movement and transportation in a shop floor.	01
	1.9	On the balancing chart, one can quickly see three things: how much time is wasted _____ and the bottleneck.	01
	1.10	The leader and the_____ are far and away the two most important people in the lean implementation effort.	01
	1.11	As part of the complete a system wide evaluation of the present state, the leadership makes_____ evaluation of the entire manufacturing system.	01
	1.12	Ohno says that_____ is the basic condition. It is the foundation of the Toyota Production System (TPS).	01
	1.13	What is the alternate name for future state value stream map?	01
	1.14	Sketch the H-K planning model.	02
	1.15	What is Poka-Yoke?	02
		PART-B	
2	a	Explain the two pillars of Toyota Production system.	08
	b	Provide comparative analysis of mass production system and Toyota production system considering any four aspects.	08
		OR	
3	a	"The supply of value to the customer is a revolutionary concept as used in the TPS". Justify the statement.	08
	b	Outline the four basic concepts, lacking any of these, an enterprise is poorly suited to use Lean as a primary business philosophy.	08

4	a	Enumerate the six rules of kanban management and discus the unique functions of these rules.	08
	b	Critically examine the factors that cause the need for the three types of inventory in an organization.	08
		OR	
5	a	Explain in detail the method of calculating KANBANs.	08
	b	List and discuss the tactics, and skills required for achieving quantity control under the Jidoka pillar.	08
6	a	"Reducing the lead time is not only a result of reducing variation; it is a variation reduction technique of extreme power". Comment.	08
	b	What are the three fundamental issues must be addressed, any time any major culture-changing initiative is undertaken, such as the implementation of a lean initiative? Evaluate.	08
		OR	
7	a	As an enlightened manager highlight the cultural issues that you may have to address while implementing lean in an organization.	08
	b	An organization plans manufacturing spread over two ten-hour shifts and each shift includes a 30-minute launch and two ten-minute breaks. The normal work schedule is five days per week and there are nine holidays. The customer has a contractual agreement to purchase 500,000 units per year. Compute the takt time.	04
	c	Discuss the importance of one of the four key strategies part of the overall approach to make a value stream lean.	04
8	a	In what situations you recommend the following lean implementation approaches? Illustrate with reasons. i. Clean sweep ii. One-value-stream-at-a-time	08
	b	Evaluate the five procedures to implementing a lean initiative in an organization.	08
		OR	
9		Write notes on the following lean techniques: i. Single minute exchange of dies. ii. Present state value stream map. iii. Future state value steam map. iv. Spaghetti diagram.	16
10	a	Why are goals and goal development so important? Discuss.	08
	b	Explain the purposes and characteristics of goal that are developed as part of policy deployment.	08
		OR	
11	a	The ultimate purpose of transparency, starting with situation assessment is rapid response PDCA. Elaborate.	06
	b	Discuss the five step prescription on how to sustain the gains from lean implementation.	10

Quality Assurance

VI Semester SEE Question Papers

Course codes: 21IM63

- Sep/Oct 2024 (21IM63)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Examinations Sept/Oct – 2024

Industrial Engineering and Management

QUALITY ASSURANCE

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical tables and formula handbook permitted.

PART-A

			M	BT	CO
1	1.1	What is the primary purpose of using the Pareto chart in quality improvement?	01	1	1
	1.2	What is the main objective of quality audit?	01	1	1
	1.3	A company incurs the following quality costs: Prevention cost= $10000, Appraisal costs = $15000, Internal failure costs = $20000 and external failure costs = $25000. Calculate the total quality cost and percentage of each type of cost relative to the total.	02	2	2
	1.4	What is an assignable cause of variation?	01	1	1
	1.5	What is the purpose of using control limits in control charts?	01	1	1
	1.6	A control chart for a process mean is set with upper and lower control limits of 12 and 8, respectively. A sample of size 4 has the following measurements: 9, 10, 11 and 12. Determine if this sample is in control. Justify your answer by calculating the sample mean and comparing it with the control limits.	02	3	3
	1.7	List the purpose of cumulative sum (CUSUM) control chart.	01	1	1
	1.8	What does the Average Outgoing Quality (AOQ) indicate in acceptance sampling?	01	1	1
	1.9	An acceptance sampling plan has a lot size of 1000 items and uses a single sampling plan with a sample size of 50 and an acceptance number of 2. If the defect rate is 3%, calculate the expected number of defects in the sample and determine if the lot would be accepted if the actual number of defects found is 2.	02	3	2
	1.10	What is a completely randomized design (CRD) in experimentation?	01	1	1
	1.11	What is the primary advantage of using factorial experiments?	01	1	1
	1.12	In a 2^2 factorial experiment, the effect of two factors A and B on the output was studied. The following averages were observed: y_{11}= 30, y_{10}= 20, y_{01}= 25 and y_{00}= 15. Calculate the main effects of A and B.	02	2	3
	1.13	Expand MTBF . Why is it important?	01	1	1
	1.14	What is the typical shape of the failure rate curve in reliability studies?	01	1	1
	1.15	A system consists of two components in series with reliabilities of 0.95 and 0.90 and a parallel component with a reliability of 0.85. Calculate the overall system reliability.	02	3	3

PART-B

2	a	In a production environment, what are the seven essential quality control tools? And how they can be used to optimize processes and ensure consistent product quality?	08	1	1
	b	Explain the importance of analyzing patterns on control charts in quality management. Identify three patterns that may appear, their potential causes and their impact on production. How can these patterns be addressed to maintain process control?	08	1	1
3		Construct and interpret $\bar{x}$ *and* s charts using piston ring inside diameter measurements in the Table.			

Sample Number	Observations				
1	74.030	74.002	74.019	73.992	74.008
2	73.995	73.992	74.001	74.011	74.004
3	73.988	74.024	74.021	74.005	74.002
4	74.002	73.996	73.993	74.015	74.009
5	73.992	74.007	74.015	73.989	74.014
6	74.009	73.994	73.997	73.985	73.993
7	73.995	74.006	73.994	74.000	74.005
8	73.985	74.003	73.993	74.015	73.988
9	74.008	73.995	74.009	74.005	74.004
10	73.998	74.000	73.990	74.007	73.995
11	73.994	73.998	73.994	73.995	73.990
12	74.004	74.000	74.007	74.000	73.996
13	73.983	74.002	73.998	73.997	74.012
14	74.006	73.967	73.994	74.000	73.984
15	74.012	74.014	73.998	73.999	74.007
16	74.000	73.984	74.005	73.998	73.996
17	73.994	74.012	73.986	74.005	74.007
18	74.006	74.010	74.018	74.003	74.000
19	73.984	74.002	74.003	74.005	73.997
20	74.000	74.010	74.013	74.020	74.003
21	73.982	74.001	74.015	74.005	73.996
22	74.004	73.999	73.990	74.006	74.009
23	74.010	73.989	73.990	74.009	74.014
24	74.015	74.008	73.993	74.000	74.010
25	73.982	73.984	73.995	74.017	74.013

										16	3	3
		OR										
4	a	Frozen orange juice concentrate is packed in 6-oz cardboard cans. These cans are formed on a machine by spinning them from cardboard stock and attaching a metal bottom panel. By inspection of a can, we may determine whether, when filled, it could possibly leak either on the side seam or around the bottom joint. Such a non-conforming can has an improper seal on either the side seam or bottom panel. Set up a control chart to improve the fraction of non-conforming cans produced by this machine.										
		Sample Number	1	2	3	4	5	6	7	8		
		Number of nonconforming cans, D	12	15	8	10	4	7	16	9		
		Sample Number	9	10	11	12	13	14	15	16		
		Number of nonconforming cans, D	14	10	5	6	17	12	22	8	10	3	3
	b	Explain the difference between C9p and Cpk in process capability analysis.									06	1	1
5	a	Explain various features of the Group Control Chart									06	1	1
	b	A single sampling plain is as follows: $N = 5000$; $n = 100$; $c = 3$. Plot the operating characteristic curve for the plan.									10	3	2
		OR											

6	a	Compare single, double and multiple sampling plans.	06	1	1
	b	A manufacturer uses a single sampling plan to inspect lots of 1000 items each. The sampling plan involves selecting a sample of 100 items from each lot and accepting the lot if there are 2 or fewer defective items in the sample. The incoming quality (proportion defective) is 0.02 (or 2%). Calculate the probability of accepting a lot using the binomial distribution. (i) Determine the Average Outgoing Quality (AOQ) (ii) Find the Average Outgoing Quality Level (AOQL) (iii) Calculate the Average Total Inspection (ATI) (iv) Calculate the Producer's Risk (Type I Error) (v) Calculate the Consumer's Risk (Type II Error)	10	3	2
7	a	List the principles of experimentation and provide guidelines for designing experiments.	4	2	1
	b	A chemist wishes to test the effect of four chemical agents on the strength of a particulate type of cloth. Because there might be variability from one bolt to another, the chemist decides to use a randomized block design, with the bolts of cloth considered as blocks. She selects five bolts and applies all four chemicals in random order to each bolt. The resulting tensile strengths follows.			

	Bolt				
Chemical	1	2	3	4	5
1	73	68	74	71	67
2	73	67	75	72	70
3	75	68	78	73	68
4	73	71	75	75	69

Analyze the data from this experiment (use a= 0.05) and draw appropriate conclusions. 12 3 4

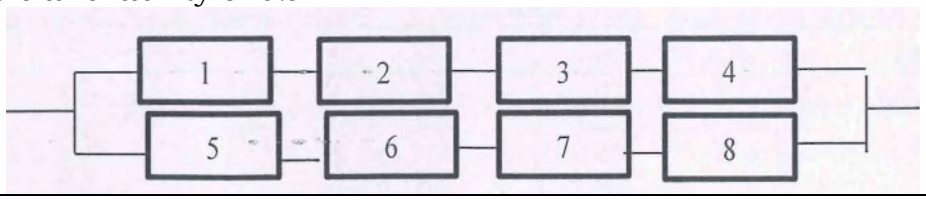
OR

8 An experiment was conducted to study the life (in hours) of two different brands of batteries in three different devices (radio, camera and portable DVD player). A completely randomized two factorial experiment was conducted and the following data resulted.

Brand of Battery	*Device*		
	Radio	*Camera*	*DVD player*
A	8.6	7.9	5.4
	8.2	8.4	5.7
B	9.4	8.5	5.8
	8.8	8.9	5.9

(i) Analyze the data and draw conclusions using | 0.05.
(ii) Which brand of batteries would you recommend? 16 3 4

9	a	Sketch the General failure rate curve (Bath tub curve) of an electronic product and identify the causes of failures in the various zones.	08	2	2
	b	In a manufacturing process, two compressors are connected to supply air to a furnace using a gas fired system. One of the compressors is standby equipment. The compressor system includes a piston, motor, coupling and compressor with the following reliabilities: $0.88, 0.90, 0.87$ *and* 0.94 respectively. The gas fired system has a reliability of 0.98 and the furnace has a reliability of 0.93 for an operating period of 150 hours. Draw the system configuration and determine the reliability of the total system for 150 hours of operation.	08	3	4

		OR			
10	a	The reliability of each component for a 20 hour operating period is as follows: • $Component\ A: RA = 0.8187\ R_A = 0.8187\ RA = 0.8187$ • $Component\ B: RB = 0.7788\ R_B = 0.7788\ RB = 0.7788$ • $Component\ C: RC = 0.6703\ R_C = 0.6703\ RC = 0.6703$ • $Component\ D: RD = 0.6703\ R_D = 0.6703\ RD = 0.6703$ • $Component\ E: RE = 0.6065\ R_E = 0.6065\ RC = 0.6065$	08	3	3
	b	Derive a general expression for the reliability of the model shown below and hence evaluate the system reliability if all components have a reliability of 0.9 	08	3	3

Total Quality Management

VI Semester SEE Question Papers

Course codes: 21IM65E2, 12IM6C2

- Sep/Oct 2024 (21IM65E2)
- Fast Track Jul 2019 (12IM6C2)
- Fast Track Jul 2018 (12IM6C2)
- Apr/May 2018 (12IM6C2)
- Fast Track Jul 2017 (12IM6C2)
- Apr/May 2017 (12IM6C2)
- Apr/May 2016 (12IM6C2)

21IM65E2

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Examinations Sept/Oct – 2024

Industrial Engineering and Management

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Expand the acronym PDCA.	02	1	1
	1.2	List the three elements of Joseph Juran's Quality Trilogy.	02	1	1
	1.3	Distinguish between Market-in and Product- out concepts.	02	2	2
	1.4	List the seven management tools that help companies identify latent needs and translate those needs into plans for products.	02	1	2
	1.5	Mention the two major ideas on which the concept of continuous improvement is based.	02	1	3
	1.6	WV model illustrates three types of problem solving. What are they?	02	1	3
	1.7	List the three phases in Hoshin Management.	02	1	1
	1.8	Expand the acronym DMAIC in six sigma context.	02	1	1
	1.9	What is meant by Lean Six sigma?	02	2	4
	1.10	Write a note on 'Rolled Throughput Yield' as a metric for six sigma.	02	2	4

PART-B

2	a	Illustrate and explain the evolution and development of Four Fitnesse's of Quality.	10	3	1
	b	Enumerate Philip Crosby's Four Absolutes.	06	2	1
3	a	"The market-in concept includes the idea of a process for improvement aimed at continuing customer satisfaction"- Analyze the statement.	10	4	2
	b	Illustrate and explain the elements of "management by process".	06	3	2
		OR			
4	a	"WV model illustrates the interplay between thought and experience"- Justify the statement.	10	4	3
	b	Distinguish between 'Reactive Improvement' and 'Proactive Improvement' methodologies in Total Quality Management.	06	2	3
5	a	"Joining of daily work and improvement work everywhere in the organization is the idea of dual function of work". Illustrate and explain.	06	3	2
	b	Summarize the seven principles for activating teamwork in organizations.	10	2	2

		OR			
6	a	Enumerate the seven elements of organizational infrastructure for implementing Total Quality Management.	10	2	2
	b	"Total Quality Management as a learning system aims at developing individual, team, company and national skill". Elaborate.	06	3	2
7	a	Examine different levels of expertise in terms of belts and their contribution to the implementation of six sigma.	10	4	1
	b	Write a note on the critical to Quality concept in six sigma.	06	2	2
		OR			
8	a	Enumerate any five benefits six sigma can offer to organizations.	06	2	2
	b	Performance metrics are critical for measuring and improving process performance in six sigma. In the above context discuss the relevance of the following: (i) Process capability indices. (ii) First pass yield. (iii) Cost of poor quality. (iv) Mean time between failures.	10	3	2
9	a	A leading automotive parts manufacturer is facing problem of high defect rates in the assembly line, leading to increased costs and customer dissatisfaction. Which six sigma methodology the organization should use to reduce the defects and improve the customer satisfaction? Discuss the methodology in detail in the context of above case.	16	5	4
		OR			
10	a	There are two types of problem solving methodology in six sigma. What are they? Also, discuss when to use each of those methodologies.	10	4	4
	b	Summarize the benefits of Lean six sigma to organizations.	06	2	4

12IM6C2

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Fast Track Examinations July-19

Electrical and Electronics Engineering

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from PartA. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	When should Six Sigma be used?	02
	1.2	Provide any four obstacles associated with TQM implementation.	02
	1.3	How are cause and effect diagrams useful?	02
	1.4	The main contribution of Masaki Imai is the concept of _____.	01
	1.5	The fundamental focus of the Quality of design as a concept is to define different _____ of quality for achieving the same functional requirement of the customers.	01
	1.6	As per the 6th principle for activating team work(Create an environment for team learning). Draw the sketch of physical arrangement for setting up meetings and justify.	02
	1.7	The major philosophical change between the 'market in' concept and the 'customer in' concept is to put the emphasis on _____ aspect of quality.	01
	1.8	Among the four fitness of quality; which fitnesses aim towards meeting quality of conformance and which ones for quality of design.	02
	1.9	A project that is $SMART$ is _____, _____, _____, _____, _____.	02
	1.10	Rewrite the following problem statement by applying guidelines for "Clear statement of theme". Understand customer deliver, quality and pricing requirement.	02
	1.12	The concept of _____ emphasizes the participation of all the employees in the organization is the quality improvement initiatives taken up by the top management.	01
	1.13	What are the three main kinds of teams as per TQM?	02
		PART-B	
2	a	Compare and contrast quality of conformance and quality of design. Illustrate with examples.	08
	b	Summarize Deming's approach to quality management in manufacturing industries.	08
		OR	
3	a	'Juraninsists that Quality does not happen by accident'. Identify the Quality planning roadmap as proposed by Juran and the other two aspects of Juran'strilogy namely Quality Control and Quality improvement. Give an example to clarify the three aspects of Juran trilogy.	08
	b	Explain in detail the quality concepts that characterizes the eras of fitness to standard and fitness to use.	08

4	a	What is semantics? How semantics helps in bridging the dual function of language? Elaborate.	08
	b	Using WV model, illustrate the three types of problem solving.	08
		OR	
5	a	Critically examine the steps in constructing the Affinity diagram.	08
	b	Shewart, Deming, Ishikawa and other practitioners of Statistical Process Control have proposed a model for Process Control. Describe this model for process control and mention the tools that help in execution of this model in the shop floor Quality context.	08
6	a	As a team lead of 10 members, how would you conduct meetings so that they are very effective?	04
	b	Analyze the importance of CEO involvement in TQM mobilization. Elaborate Domino theory of implementation.	08
	c	What is Hoshin Management? What are the phases in Hoshin Management?	04
		OR	
7	a	Bring out the comparison between Management by objectives and Hoshin Management. How does these two approaches distinguish the American and Japanese philosophers to Quality Management?	08
	b	Identify the elements of infrastructure for networking. Elaborate.	08
8	a	Take the example of Piston-Cylinder assemblies in Engines and identify the critical to Quality parameters. What are the consequences of process capabilities of the piston and cylinder manufacturing processes in this context.	08
	b	What are the common myths about six sigma? Why do you call them myths?	08
		OR	
9	a	List and explain the guidelines that helps to set up, measure and get results from the metrics.	08
	b	Explain what do you understand by the term "Process in Statistical Control". How does this reflect the stability of the manufacturing process in the mass production of components and sub-assemblies? Give a quantitative feel in the answer.	08
10	a	Evaluate $DMAIC, DFSS, DMADV$ methodologies with illustrations.	08
	b	Critically analyze the role of the Measure phase of the $DMAIC$ project methodology of the Six Sigma quality improvements. What are the measurement tools commonly used in $DMAIC$?	08
		OR	
11	a	Develop the framework for evaluating and prioritizing six sigma projects. Highlight the project selection criteria and significance ratings.	08
	b	On what basis a good lean six sigma project is identified?	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VISemester B. E. ExaminationsFast Track July-18

Industrial Engineering and Management

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	Fitness of cost means high _____ and _____ cost.	01
	1.2	Fitness to use and fitness to latent requirements have to do with _____ integration of the organization.	01
	1.3	Continuous improvement is based on _____ improvement and _____ improvement.	01
	1.4	Give a reason for presenting and diagnosing the quality improvement stories.	01
	1.5	Name the type of team that has a larger inherent risk of conflict than other types of teams.	01
	1.6	Hoshin management is also known as _____.	01
	1.7	Answer in one line: Why six sigma is known as a problem solving methodology?	01
	1.8	For a process of six sigma quality level the value of C_p =_____.	01
	1.9	A project selection matrix is a _____ and _____ decision matrix.	01
	1.10	What are the perspectives used to derive the basic criteria for selecting a six sigma project.	01
	1.11	Define quality of conformance.	02
	1.12	Illustrate with a sketch the optimum balance between too little and too much process.	02
	1.13	Meetings are important part of team learning. List substitutes for meetings.	02
	1.14	Provide a list of topics for training champion under six sigma.	02
	1.15	What is growth management system?	02

PART B

2	a	Explicate Dr. Armand V Feigenbaum's concept of total quality control.	06
	b	Trace the evolution of quality concepts. Elaborate the concepts and cite examples.	10
		OR	
3	a	Write a detailed note on Ishikawan's concept of company wide quality control.	06
	b	What are future fitnesses? Elucidate the importance in the Indian context.	06
	c	Compare and contrast quality conformance and quality of design.	04

4	a	Enumerate the key features of market in concept.	06
	b	The team leader of a quality improvement team is currently leading the team on a reactive improvement project. Suggest suitable tools and techniques for data collection, data analysis and cause analysis. You are also required to educate the team on the tools.	10
		OR	
5	a	List the general guidelines for managers diagnosing a quality improvement story. How do these guidelines help the QI team in developing problem solving skill?	06
	b	An entrepreneur in collecting data for a proactive improvement project. Evaluate and suggest the semantics methods for the purpose.	10
6	a	Evaluate the types of work groups in Japanese companies and three kinds of teams as per TQM.	06
	b	Develop strategy for TQM introduction for a small scale manufacturing company. The strategy may be based on the learning's from the CQM companies: Teradyne, BBN and digital.	10
		OR	
7	a	Write anHoshin indicating your goal for next year. Illustrate the alignment achieved by Hoshin management.	08
	b	Identify six elements of the infrastructure for networking. Develop a block diagram illustrating the interaction of these elements.	08
8	a	Analyze the costs of six sigma implementation.	04
	b	Elaborate the importance of assessing readiness of a company for implementing six sigma.	04
	c	Six sigma is about techniques and tools and statistics – but the results of six sigma depend on the people applying the tools etc. Write notes on the key roles played by executive leaders and the champions.	08
		OR	
9	a	Develop a plan for the six sigma initiatives in an organization. Detail the communication initiative, deployment plan, training for members and projects.	08
	b	Explain the guidelines to set up the proper metrics for six sigma implementation.	08
10	a	List the four aspects that make a project a poor choice with regards to its overall impact on business. Provide critical analysis of the above aspects.	08
	b	Develop a project selection matrix for evaluating and prioritizing projects.	08
		OR	
11	a	Provide a summary of DMA DV method of design for six sigma.	08
	b	Illustrate with a sketch the basic concept of growth management system.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	Juran has made the greatest contribution to the management literature of any quality professional. He is altogether the author of _____ books and his _____ book has been translated into 13 languages.	01
	1.2	The fitness _____ requirement has made the concept of quality multidimensional.	01
	1.3	Who modified Kawakita's 'W' model for application to TQM and developed 'WV' model?	01
	1.4	A quality circle team is solving a quality problem. The team has to establish relationship between two parameters influencing the process. Suggest a suitable tool for the purpose.	01
	1.5	What is dual function of work?	01
	1.6	_____ are catalysts and one of the three elements necessary for successful societal diffusion of TQM.	01
	1.7	Six sigma is a program that focuses on the control of a process to the point of _____ from a center line.	01
	1.8	Define first time yield.	01
	1.9	The fundamental criterion for selection of six sigma projects is the opportunity to turn the _____ into the bottom line savings.	01
	1.10	_____ barrier is the point at which the efforts to improve products and services are limited by the capability of designs for products, services and processes.	01
	1.11	Among the four fitness of quality; which fitnesses aim towards meeting quality of conformance and which ones for quality of design.	02
	1.12	Rewrite the following statement with weakness orientation "Understand customer quality requirement".	02
	1.13	What are the three main kinds of teams as per TQM?	02
	1.14	Mention two important roles in which the master black belts are involved in an organization.	02
	1.15	List two reasons that companies give for the failure of new products.	02

PART B

2	a	Summarize Deming's approach to quality management in manufacturing industries.	08
	b	During the evolution of quality concepts, how did various organizations achieve fitness of cost? Elaborate with a detailed discussion.	08
		OR	

3	a	Develop the Juran's trilogy diagram. Explain the three phases with examples.	08
	b	Offer your comments on the following. i) Quality of conformance versus quality of design. ii) From devaitions to weakness to opportunities.	08
4	a	Illustrate the three types of problem solving with a sketch of 'WV' model. Write a note on process control.	08
	b	What is semantics? How semantics helps in bridging the dual function of language? Elaborate.	08
		OR	
5	a	Explain the market-in concept. How is it different from product-out concept?	08
	b	Select a problem of your choice. Illustrate the application of $7QC$ tools and $7QC$ steps.	08
6	a	Explain the seven principles for activating teamwork.	12
	b	What is Hoshin Management? What are the phases in Hoshin Management?	04
		OR	
7	a	Explain in detail, the Teradyne strategy for TQM introduction.	08
	b	What are the elements that are necessary for successful diffusion of TQM? Develop the sketch illustrating infrastructure for networking and explain.	08
8	a	"There are many misunderstandings about six sigma." List the myths and elaborate the truth.	08
	b	Explain the following in the context of using statistics in six sigma approach to management: i) Causes of variation. ii) Process capability.	08
		OR	
9	a	Develop a deployment plan for implementing six sigma approach in a medium scale manufacturing organization. List the elements of the plan and discuss.	08
	b	Elucidate the basic steps in setting metrics for an organization.	08
10	a	What are the perspectives that guide an organization in developing the list of possible projects for six sigma? Critically analyze.	08
	b	Provide an overview of $DMADV$ method of design for six sigma. Mention two differences between $DMADV$ and $DMAIC$.	08
		OR	
11	a	Develop the framework for evaluating and prioritizing six sigma projects. Highlight the project selection criteria and significance ratings.	08
	b	Identify the fundamental building blocks of growth management system. Elaborate the importance of knowledge management and lean six sigma.	08

12IM6C2

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	Japanese companies worry about _____, not _____.	01
	1.2	_____ indicates the degrees to which a product or service confirms to standard.	01
	1.3	Poincare said "it is by logic we prove, but it is by _____ we discover.	01
	1.4	Two essential components to a TQM organization: _____ and _____.	01
	1.5	What is Hoshin management?	02
	1.6	Innovation is our competitor advantage, and process will stifle innovation. Comment.	02
	1.7	List the future fitness of quality.	02
	1.8	What are the four levels of TQM practice implicit in the four revolutions of management thinking?	02
	1.9	Differentiate between $SDCA$ and $PDCA$ cycle.	02
	1.10	What is product out concept?	02
	1.11	What is the purpose of using affinity diagram?	02
	1.12	List the myths about six sigma.	02

PART-B

2	a	Discuss how Juran's quality triology provide a successful frame work for achieving company's quality objectives.	10
	b	Bring out the contributions of Imai for Kaizen management practices.	06
		OR	
3	a	Illustrate the key features and weaknesses of the four fitnesses of quality.	06
	b	Each company must find its own way to implement TQM, but four areas of concepts and practices are common to most successful implementations, they represent the four revolutions in management thinking. Elaborate the four revolutions.	10

4	a	Explain how market-in concept helps the companies to focus on customer satisfaction as the purpose of work.	06
	b	WV model of continuous improvement guides in doing things right. Discuss the three types of improvements through the WV model.	10
		OR	
5	a	Discuss with case examples the seven steps in reactive improvement.	11
	b	Explain JiroKawakita's five principles for collecting data for proactive improvement.	05
6	a	Teams and team work are fundamental components of the way TQM organizes work. Comment on this statement.	09
	b	Explain briefly with example the 7 principles for activating team work.	07
		OR	
7	a	'Goals must be set for the company's TQM implementation and for the company's business'. Comment.	10
	b	Elaborate on the six elements of infrastructure for networking.	06
8	a	Discuss Six Sigma concept.	10
	b	Give an example to explain CTQ.	06
		OR	
9	a	Explain briefly Dos and Don'ts while implementing six sigma.	08
	b	State the benefits and draw backs of following six sigma methodology.	08
10	a	Explain $DMAIC$ approach to six sigma. List various tools used in each phase of $DMAIC$ cycle.	10
	b	How to sustain six sigma in an organization? Discuss.	06
		OR	
11	a	With a case example, explain the $DFSS$ methodology.	10
	b	Implementing lean six sigma approach has become a necessity in today's business environment. Justify.	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	List any two deadly sins and diseases advocated by W Edwards Deming.	02
	1.2	"Japanese companies worry about cost, not process". Justify.	02
	1.3	The TQM slogan is, "Next process is your _____".	01
	1.4	Customer needs determine the _____ output whereas the process used determine the _____ output.	01
	1.5	Rewrite the following problem statement by applying guidelines for "Clear statement of theme". Understand customer deliver, quality and pricing requirement.	02
	1.6	As per the 6th principle for activating team work(Create an environment for team learning). Draw the sketch of physical arrangement for setting up meetings and justify.	02
	1.7	Write full form of the acronym $MITI$.	01
	1.8	At each lower level the Hoshins have the same format but are more _____.	01
	1.9	Define six sigma.	02
	1.10	Critical to quality concept in six sigma allows you to focus on improving quality from the perspective of the _____.	01
	1.11	List a benefit of six sigma that do not translate into Rupees, at least not directly.	01
	1.12	A project that is $SMART$ is _____, _____, _____, _____, _____.	02
	1.13	$DFSS$ initially requires a driving impulse from the _____ level in an company, well beyond verbal support.	01
	1.14	Six sigma enhances _____ and lean improves _____ of production processes in an organization.	01

PART B

2	a	Elucidate the salient features of Ishikawa's company-wide quality control.	08
	b	Summarize Philip Crosby's diagnosis repot of a troubled company. List the Crosby's absolutes for Quality Management.	08
		OR	

3	a	Explain in detail the quality concepts that characterizes the eras of fitness to standard and fitness to use.	08
	b	"Although each company must find its own way to implement TQM, four areas of concepts and practices are common to most successful implementations and these differ from practices in many non-TQM companies". Examine the statement, identify the four revolutions and discuss.	08
4	a	"TQM can be thought of as management in the face of rapid change". Illustrate with the help of a block diagram and highlight the importance of market-in concept.	08
	b	Compare the following: i) Process control and process improvement ii) Process versus creativity.	08
		OR	
5	a	Sketch the model of the steps in reactive problem solving. Illustrate the steps in identifying the problem with examples.	08
	b	Critically examine the steps in constructing the Affinity diagram.	08
6	a	Analyze the importance of CEO involvement in TQM mobilization. Elaborate Domino theory of implementation.	08
	b	Write brief notes on the three typical phases of TQM phase-in.	08
		OR	
7	a	"Hoshin management has three alignment purposes". Evaluate the statement and illustrate the three alignments.	08
	b	Construct the TQM model for skill development. Compile information about the complementary elements of indirect and direct learning.	08
8	a	Examine the six sigma methodology that helps to focus on engaging people and changing processes.	08
	b	Investigate why six sigma is not another quality program. Identify the myths and misunderstandings about six sigma.	08
		OR	
9	a	List and explain the guidelines that helps to set up, measure and get results from the metrics.	08
	b	Enumerate and discuss the basic steps in setting metrics.	08
10	a	Outline the $DMAIC$ methodology of six sigma approach to achieve process excellence.	08
	b	Evaluate $DMAIC, DFSS, DMADV$ methodologies with illustrations.	08
		OR	
11	a	Design a strategy for implementing six sigma in an organization.Summarize the steps involved.	08
	b	Present and defend your opinion on application of lean and six sigma methodologies for effectively managing the supply chains.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

TOTAL QUALITY MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART-A

1	1.1	Deming's contribution to the Quality Management is recognized through the use of _____ cycle for continuous Quality Improvement process.	01
	1.2	Juran's trilogy has three main aspects of _____ Quality Control and Quality Improvement.	01
	1.3	The main contribution of Masaxi Imai is the concept of _____.	01
	1.4	The fundamental focus of the Quality of design as a concept is to define different _____ of quality for achieving the same functional requirement of the customers.	01
	1.5	The term focus on customers as one of the four major revolution in management thinking refers to identifying the customer's _____ in a scientific process of capturing the customer's expectation from product or service offerings.	01
	1.6	The two types of improvement methodologies commonly adapted in Quality Management processes are Reactive Improvement and _____ Improvement.	01
	1.7	While the seven basic tools for Quality Control are called the seven QC tools, the seven tools used in the planning and Management of Quality are called the _____.	01
	1.8	Shoji Shiba –A Quality Guru has proposed a model for continuous improvement known as the _____.	01
	1.9	The identification of the Quality gaps in the product or process designs, provides an opportunity for Quality teams to initiate the process of _____, to improve the quality goals attained.	01
	1.10	The major philosophical change between the 'market in' concept and the 'customer in' concept is to put the emphasis on _____ aspect of quality.	01
	1.11	The concept of _____ emphasizes the participation of all the employees in the organization is the quality improvement initiatives taken up by the top management.	01
	1.12	Hoshinmanagement is a technique which emphasizes the _____.	01
	1.13	Define Societal Networking.	01

	1.14	One of the cultural aspects of Quality Improvement projects, when the Quality teams are engaged in problem solving in the contribution to _____ learning among the team members.	01
	1.15	The concept of $DPMO$ in the context of Six Sigma project stands for _____.	01
	1.16	The fundamental difference between the process capability approach to Quality of rejections and rolled throughput yield approach to Quality is in the _____ of the first pass yield, without the need for segregation according to the specification targets.	01
	1.17	$DFSS$ is an acronym which stands for _____.	01
	1.18	The phase in which the improvements, are instituted in the system as part of $DMAIC$ cycle and sustained is the _____ phase.	01
	1.19	The $DMADV$ method is the design approach to Six – Sigma is characterized by the D and v phase which stands for _____ and verify part of Quality Improvement projects.	01
	1.20	The lean six sigma approaches are driven by the simultaneous attack on the process inefficiencies and focus on _____ improvements in processes generating the product features.	01

PART-B

2	a	Trace the evolution of the Quality concept using the four fitnesses. Briefly characterize the concept of Quality as enshrined in the four quality concepts.	08
	b	Distinguish between Quality of Design and Quality of Conformance. Take the example of a Mobile handset and the features of a smartphone to bring out the distinction between these two aspects of Quality.	08
		OR	
3	a	Examine the Demings 14 points for Management. List the 14 points with a very brief explanation of each of these points with a link to Quality initiatives and the responsibility of the Top Management.	08
	b	'Juran insists that Quality does not happen by accident'. Identify the Quality planning roadmap as proposed by Juran and the other two aspects of Juran's trilogy namely Quality Control and Quality improvement. Give an example to clarify the three aspects of Juran trilogy.	08
4	a	Distinguish between the product – out concept and the Market – in concept. Why is the market – in necessary in the face of rapid change? Give an example.	08
	b	Justify the use of the $W-V$ Model for problem solving. How does this model help organizations to be guided through the three types of improvement?	08
		OR	

5	a	Shewart, Deming, Ishikawa and other practitioners of Statistical Process Control have proposed a model for Process Control. Describe this model for process control and mention the tools that help in execution of this model in the shop floor Quality context.	08
	b	Design the scenarios which promotes the interaction between the $PDCA$ cycle and the $SDCA$ cycle. Graphically depict the interaction between the $SDCA$ cycle and the Reactive Improvement Cycle.	08
6	a	Teams and Team work are a fundamental component of the way TQM organizes work. Explain as to why these components are important for Quality Improvement projects in the face of complexity in business.	08
	b	Cross functional teams involved in Quality Improvement projects using the TQM approaches can benefit by treating TQM as a learning system and TQM Model for skill development. Take an example of improvements on an existing product or service solution and explain the benefits for the teams.	08
		OR	
7	a	What is Hoshin Management? List the three alignment purposes Hoshin Management. Illustratively depict the alignments in Hoshin Principles.	08
	b	Bring out the comparison between Management by objectives and Hoshin Management. How does these two approaches distinguish the American and Japanese philosophers to Quality Management?	08
8	a	How can any organization assess the readiness for the Six Sigma programs? What are the steps in planning for Six Sigma?	08
	b	Take the example of Piston-Cylinder assemblies in Engines and identify the critical to Quality parameters. What are the consequences of process capabilities of the piston and cylinder manufacturing processes in this context.	08
		OR	
9	a	Explain what do you understand by the term “Process in Statistical Control”. How does this reflect the stability of the manufacturing process in the mass production of components and sub-assemblies? Give a quantitative feel in the answer.	08
	b	Distinguish between baselines and Benchmarking as Quality Concepts. How do these concepts drive Quality Improvement Projects?	08
10	a	Critically analyze the role of the Measure phase of the $DMAIC$ project methodology of the Six Sigma quality improvements. What are the measurement tools commonly used in $DMAIC$?	08
	b	How can one assess the business impact of a Six Sigma project? Give an example to illustrate the need to attach priority to selecting projects with high business impacts.	08
		OR	

11	a	"The Goal of Lean is elimination of waste and the goal of Six Sigma is to minimize defects". How can these two goals be combined using Lean Six Sigma project framework? Explain with an example.	08
	b	How can large organizations help its suppliers in Quality Improvement Processes? Why is it necessary to focus on supplier's Quality program in the Supply Chain context?	08

Design of Experiments

VI Semester SEE Question Papers

Course codes: 18IM6D4, 16IM6D3

- Sep 2023 (18IM6D4)
- Apr/May 2019 (16IM6D3)

18IM6D4

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Examinations September-2023

Industrial Engineering and Management

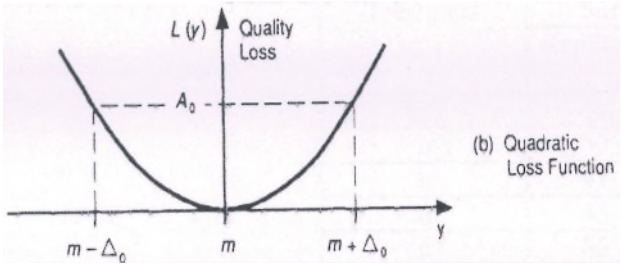
DESIGN OF EXPERIMENTS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	State the major disadvantage of the $OFAT$ strategy.	01
	1.2	_____ developed the foundations of robust design and validated its basic philosophies by applying them in the development of many products.	01
	1.3	_____ table is used to modify the Standard linear graph.	01
	1.4	What is the meaning of the notation:$L162^{15}$.	01
	1.5	What is principal fraction with reference to fractional factorial designs?	01
	1.6	Which is the design that has highest resolution among the Resolution III, IV and V designs?	01
	1.7	Select a manufacturing process of your choice; identify controllable factors, noise factors and response.	02
	1.8	List the three steps strategy that has as intuitive appeal for the engineering problem of optimizing a product or process design.	02
	1.9	Sketch the general model of a process.	02
	1.10	Identify the quadratic loss function plotted in the Fig 1.10. Write the expression for the same.  Fig 1.10	02
	1.11	Illustrate with a sketch the treatment combinations in the 2^2 design.	02
	1.12	Suppose the functional limits for the colour density of television sets are m$\pm$7. Let the repair of a television set in the field cost \$98. What is the average quality loss incurred by the customers receiving sets with colour density m+4.	02
	1.13	Develop the required linear graph for a situation in which there are four factors $(A, B, C \,\&\, D)$ at two levels each. All the main effects and interaction effect AB and CD are to be estimated.	02

PART B

2	a	Explain with examples the three basic principles of experimental design.	08
	b	Classify the noise factors and explain with a product of your choice.	08
3	a	How do you carry out data analysis from a single replicate of 2^k design? Discuss.	04
	b	A chemical engineer has conducted an experiment to study the effect of temperature and pressure on the reaction time of a chemical process. Let the two factors temperature and pressure are denoted by A and B respectively. Suppose the following results are obtained from this study. The number of replications is two.	

$Pressure\ (B)$	$Temperature\ (A)$	
	Low (60°C)	$High$ (70°C)
Low (100 MPa)	40	43
	37	50
$High$ (150 MPa)	59	37
	54	43

i) Estimate the factor effects. Which effects are large?

ii) Perform an analysis of variance, what are your conclusions? 12

OR

4 An experiment was conducted to study the effect of tool type (A), speed (B) and feed (C) on the surface finish of a machined part. Each factor was studied at two levels and obtained two replications. The order of experiment was random. The results obtained ate given in Table.

$Treatment$	$Surface\ Finish$	
	$Replicate\ 1$	$Replicate\ 2$
(1)	54	73
a	60	53
b	41	51
ab	43	49
c	86	66
ac	82	73
bc	63	65
abc	66	65

i) Estimate the factor effects. Which effects are large?

ii) Carryout analysis of variance, what are your conclusions? 16

5	a	Illustrate with an example confounding 2^3 factorial design in to 2 blocks.	06
	b	Explain with an example partial confounding	05
	c	Write a note on confounding 2^k factorial design in four blocks.	05
		OR	
6	a	The successful use of fractional factorial designs is based on three key ideas. Elaborate.	06
	b	Discuss the importance resolution III, IV and V experimental designs.	06
	c	Write a note on one quarter fraction of the 2^k design.	04

7	a	Explain with illustration the Dummy level technique.	06
	b	Consider an experiment with four factors $(A, B, C \text{ and } D)$ each at two levels. Also, the interactions AB, AD and BD are of interest to the experimenter. Design the Orthogonal Array experiment layout.	10
8	a	List the steps in robust design.	06
	b	Explain any four signal-to-noise ratios for static problems. Illustrate with sketches.	10

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RV COLLEGE OF ENGINEERING®

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VISemester B. E. Examinations April/May-19

Industrial Engineering and Management

DESIGN OF EXPERIMENTS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of statistical tables and orthogonal array tables permitted.

PART A

1	1.1	In the early years of statistically designed experimentation the ideas of efficient design for conducting research was applied to the field of agriculture. What was the performance measure in this case?	01
	1.2	Sketch the Ideal Curve for the quality loss function.	01
	1.3	The total number of treatment combination when 4 factors are investigated each at 2 levels will be?	01
	1.4	Where in the P diagram do you indicate the Noise factors?	01
	1.5	Mention any one merits of the factorial experimentation.	
	1.6	What is the significance of listing treatments as per Yate's order? (standard order).	01
	1.7	If there is a factor A at 3 levels and factor B at 3 levels, what will be the degree of freedom for the interaction term AB?	01
	1.8	Mention any one benefit of representing the results of a technical experimentation using $ANOVA$ table.	01
	1.9	What is the advantage of the blocking as a strategy of experimentation?	01
	1.10	If effects are confounded, then what would be the impact on the analysis of main effects and interaction effects?	01
	1.11	Define a Resolution III design precisely.	01
	1.12	What is the advantage of a fractional factorial design over a full factorial design for experimentation in R & D situations?	01
	1.13	What is the meaning of L_9^{34} notation?	01
	1.14	Give the typical structure of a Linear graph?	01
	1.15	Do you agree that the interaction table and Linear graph serve the same purpose while constructing orthogonal array for a experimental situation.	01
	1.16	Which technique is most suited for accommodating a factor with 4 levels in a 2 level orthogonal array layout design?	01
	1.17	Write the S/N ratio for Nominal the best quality characteristic type of problem.	01
	1.18	What is an outer array?	01
	1.19	Graphically represent the concept of robustness through the principle of exploiting the non-linearity of the responsive curve (input parameters versus response characteristic curve).	01
	1.20	What does Grey denote in the Grey Taguchi technique?	01

PART B

2 a "Offline methods of quality improvement and online methods of quality control need to be initiated to take care of the product and process quality requirements in organization". Elaborate on the principles, approaches and techniques related to online and offline quality control and improvement with examples. 08

b A TV manufacturer has observed that the average cost to repair the deviation in a particular parameter X is $Rs.1500$. If the parameter X has a tolerance Zone as $50 <= x <= 90$, with the target as 70. Derive the corresponding Quadratic loss function.
For the above situation, estimate the average loss for $5\,TV$ sets with measurements for parameter X yielding 58,62,70,52,75. Sketch your results graphically. 08

3 a One of the critical measures of some raw paper is the ability to minimize the absorption of water for a specific time and then reweighing the sample. A percentage weight gain is the measure of the water absorption characteristic. In an effort to minimize the absorption, two factors have been selected for experimentation

Factor A- Type of sizing chemical used
Levels being normal and enhanced
Factor B- Amount of the sizing chemical Levels being 5% and 8%

A 2^2 factorial design was run. Four replicates were taken. The result is in the Table 3(a) below:

Table 3(a)- Data for example

		$\boldsymbol{B}(Sizing\ type)$			
$(B)Amount\ of$		$Normal(-1)$		$Enhanced(+)$	
	5%	1.34	5.24	10.38	6.66
	$(-)$	6.56	4.20	6.68	9.56
	8%	4.98	3.49	6.22	11.0
	$(+)$	3.92	5.29	4.72	8.36

i) Calculate the effects.
ii) Create pareto chart of the effects.
iii) Carry $ANOVA$ and identify the significant effects. 10

b Using the data from part (a) above, derive an appropriate regression model and compute the residuals. Comment on the residual values. 06

OR

4 a A mechanical engineer is studying the thrust force developed by a drill press. He suspects that the drilling speed and feed rate of the material are the most important factors. He selects four feed rates and uses a high and low drill speed chosen to represent the extreme operating conditions. He obtains the following results. Analyze the data and draw conclusions. Use $\alpha = 0.05$

	$Feed\ Rate$			
$Drill\ Speed$	0.015	0.030	0.045	0.060
125	2.70	2.45	2.60	2.75
	2.78	2.49	2.72	2.86
200	2.83	2.85	2.86	2.94
	2.86	2.80	2.87	2.88

10

	b	Write down the structure for the Analysis of Variance of a general 2^K design. What will be the total degrees of freedom?	06
5	a	What is the meaning of Confounding? Can there be partial Confounding as well? Justify.	06
	b	Construct a 2^{5-1} design. Show the design may be run in two blocks of eight observations each. Are any main effects or two factor interactions confounded with blocks?	10
		OR	
6	a	A 2^{5-2} design was run to investigate the effect of A-Condensation temperature B- Amount of material-1 C-Solvent Volume D-Condensation time E- Amount of material 2 The yield results obtained are as follows $e = 23.2 \;\; ad = 16.9 \;\; cd = 23.8 \;\; bde = 16.8$ $ab = 15.5 \; bc = 16.2 \; ace = 23.4 \;\; abcde = 18.1$ i) Verify the design generates used where $I = ACE$ and $I = BDE$ ii) Write down the complete defining relation and alias for this design iii) Estimate the main effects	10
	b	Explain the process of arriving at the resolution of a design. What is the meaning of a Resolution III, IV, V designs?	06
7	a	Outline the steps for the construction of Orthogonal arrays for experimental investigation with the help of an example of your own. How do you accommodate a three level factor in a 2 level design?	06
	b	In an Investigative study on the stability of power factor of capacitor, it was decided to examine the following factors: $Sl.No.$ | Factors | $Levels$ 1 | $Leakage\ Current\ of\ foil(A)$ | $<= 2mA \quad >= 2mA$ 2 | $Power\ Factor(B)$ | $<= 3\% \quad >= 3\%$ 3 | $Capacitance\ value\ of\ coil(C)$ | $8-10\mu f \quad 11-13\mu f$ 4 | $Soaking\ time\ during\ \ Impregnation(D)$ | $2hrs \quad 4hrs$ 5 | $Loading\ Pattern\ (E)$ | $70\% \quad 100\%$ Suspected interactions effects are AB, BC. Develop the Orthogonal Layout.	10
8	a	Write down the Signal to Noise ratios for any two cases of Static Quality characteristics and for any two cases of Dynamic Quality characteristics problems. Give examples for each of the above type of problems.	08
	b	Compute the Signal to Noise ratio corresponding to a Quality characteristic with 5 replication values as $1.0, 0.80, 0.92, 1.02$ and 0.83. Compute the S/N ratio under the following assumptions for the state problem with Quality characteristics being treated as: i) Smaller the better type ii) Nominal the best type iii) Larger the better type.	08

Financial Accounting and Costing

VI Semester SEE Question Papers

Course codes: 18IM62

- Sep 2023 (18IM62)
- Aug 2022 (18IM62)

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Examinations September-2023

Industrial Engineering and Management

FINANCIAL ACCOUNTING AND COSTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Differentiate between Book-keeping and Accounting.	02
	1.2	_____ is the basis of accounting which records expenses when resources are consumed, regardless of when payment is made.	01
	1.3	_____ is a financial statement which is driven by the need to ascertain the gross profit from the business transactions.	01
	1.4	Based on the three elements of cost, the cost variances are classified into _____, _____ and _____ variance.	02
	1.5	Variance represents the difference between _____ and _____ costs.	02
	1.6	What is process costing?	02
	1.7	During the month of March, the standard cost of actual hours worked amounted to *Rs.*42000, the standard hours allowed for actual production were 2000 and the direct labor efficiency variance amounted to *Rs.*2000 unfavorable. The standard direct labor rate per hour was _____.	02
	1.8	Differentiate between sales budget and production budget.	02
	1.9	What do you mean by Double entry systems of book keeping?	02
	1.10	If total cost of 100 units is *Rs.*5000 and those of 101 units is *Rs.*5030, then increase of *Rs.*30 in total cost is termed as _____ cost.	01
	1.11	Mention any four advantages of preparation of cost sheet.	02
	1.12	Aggregate of all direct cost is known as _____ cost.	01

PART B

2	a	Explain the different kinds of accounts and state the rules of debit and credit for them with examples.	07
	b	Journalize the following transactions in the books of Mr. Ram.	09

			Rs.
2019	*Jan* 1	*Mr. Ram started his business*	*Rs* 80,000
	Jan 4	*Brought goods for cash*	*Rs* 20,000
	Jan 6	*Deposited in Bank*	*Rs* 10,000
	Jan 8	*Sold goods to Krishna*	*Rs* 10,000
	Jan 10	*Purchased goods from Kumar*	*Rs.* 15,000
	Jan 15	*Brought furniture for cash*	*Rs.* 5000
	Jan 20	*Received commission*	*Rs.* 500
	Jan 28	*Withdrew cash*	*Rs.* 1000
	Jan 31	*Paid Rent*	*Rs.* 2000

3 a There are several mistakes in *ABC* company (Pvt) Ltd Trial Balance. You are requested to identify errors and make corrected Trial Balance.

Sl.No	*Heads of accounts*	*Debit*	*Credit*
1	*Owner equity*		1551
2	*Drawings*	560	
3	*Equipment's*	2850	
4	*Sales*		2850
5	*Due from customers*		530
6	*Purchases*	1260	
7	*Purchase return*	364	
8	*Bank loan*		996
9	*Creditors*	528	
10	*Taxes*	720	
11	*Cash in hand*	226	
12	*Note payable*	680	
13	*Inventory*		264
14	*Repair*	461	
15	*Return Inward*		98
		Rs. 7649	*Rs.* 6289

12

b Comment on the provisions for bad debts, cash discounts in the balance sheet preparation. Explain these items. 04

OR

4 Following is the Trial Balance of *M/S BMW* & Associates for the year ended 31st *July* 2021. Prepare trading, Profit & Loss *A/C* and Balance Sheet.

Particulars	*Rs.*	*Particulars*	*Rs.*
Capital	813400	*Creditors*	39000
Sales	1090000	*Purchase returns*	24500
Plant & machinery	356000	*Stock*	150500
Purchases	437000	*Discount allowed*	40000
Freight Inward	35000	*Insurance*	15000
Salaries	110000	*Advertisements*	56000
Other administration expenses	60000	*Building*	650000
Loan from IDBI	350000	*Bad debts*	2000
Furniture	75000	*Investment in NSC's*	90000
Wages	153900	*Cash*	6500
Depreciation on machinery	25000	*Bank*	13000
Debtors	42000		
Closing stock as on 31/07/2021			*Rs.* 57000

16

5 Prepare a cost sheet for the year 31/3/20216 from the following figures extracted from book of Best Engineering company.
Opening stock:
i) Raw material – 40,350
ii) Work in progress -15,000 and
iii) Finished stock -35590

Cost incurred during the period:

Materials purchased	2,50,000
Wages paid	2,00,000
Carriage inward	2,000
Consumable stores	10,000
Wages of storekeeper	7,000
Depreciation of plant & machinery	10,000
Materials destroyed by fire	5,000
Repairs and renewals	5,010
Office Managers salary	10,000
Salary to office staff	20,500
Printing and stationary	10,000
Power	10,500
Lighting for office building	2,000
Carriage outwards	3,000
Freight	5,000
Entertainment	2,500
Warehousing charges	1,500
Legal charges	2,000
Expenses for participating in industrial exhibition	6,000

Closing stock:

i) Raw material: 35,000
ii) *WIP* : 14,500 and
iii) Finished stock : 40,030
iv) Profit 25% on cost — 16

OR

6 The following are the details of Process X, process Y and process Z.

	Process X	*Process Y*	*Process Z*
Input units	2000	1840	1740
Normal loss unit	10%	5%	10%
Direct material	40,000	60,400	69,240
Wages	70,000	84,520	1,00,000
Overheads	30,000	40,000	60,000
Scrap value per unit	25	50	60
Output of Process Z is 1600*units. Prepare the necessary accounts.*			

16

7 a From the following data, calculate the labour cost variance.

Particular	*SH/unit*	*SR/hr*	*Total actual hours*	*Actual rate/hr.*
Skilled	5000	1.5	4500	2
Unskilled	8000	0.5	10,000	0.45
Semiskilled	4000	0.75	4200	0.75

Number of units produced = 1000 — 10

b Standard cost is defined as "the predetermined cost that is calculated at the management's standards of efficient operations and the relevant necessary expenditure". Elaborate on the statement with examples to illustrate your points. — 06

8	a	Expense for budgeted production of 10,000 units in a factory are given below:	

$Material$	50
$Labour$	20
$Variable\ OH$	10
$Fixed\ OH\ (Rs.50{,}000)$	5
$Variable\ expense\ (Direct)$	2
$Selling\ expense\ (10\%\ fixed)$	10
$Distribution\ expense\ (20\%\ fixed)$	5
$Administration\ expense\ (Rs.40000)$	4
$Total\ cost/unit$	106

Prepare a flexible budget for production of 8000 units indicating each items cost/unit. 10

b In what respect does the production budget contribute to managerial:

i) Planning
ii) Coordination
iii) Control

Explain. 06

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VI Semester B. E. Examinations August-2022

Industrial Engineering and Management

FINANCIAL ACCOUNTING AND COSTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Journal is commonly known as the book of _____ entry.	01
	1.2	The liabilities of a firm are $Rs.8{,}000$ and the capital of the proprietor is $Rs.10750/-$. The total assets are $Rs.$ _____.	02
	1.3	Use Debit or credit in the blanks below: a) Increases in expenses are _____; decreases are _____. b) Increases in assets are _____; decreases are _____.	02
	1.4	Pushpa and Company's Trial Balance contains the following information: 12% Bank loan, $Rs\,40{,}000$, Interest paid $Rs.3{,}800$. Interest debited to the profit & loss account is _____.	02
	1.5	While finalizing the current year accounts, the company realized that an error was made in the calculation of closing stock of the previous year. In the previous year, closing stock was valued more by $50{,}000/-$. As a result previous year profit is _____ and the current year profit is _____ stated.	02
	1.6	Income tax paid by Mr. A amount to $Rs.3000/-$. The accounting treatment is to be _____ from the capital.	02
	1.7	State any two examples for indirect costs	02
	1.8	A flexible budget recognizes the difference in behavior between _____ and _____ in relation to fluctuations in output, turnover or other variable factors etc.	02
	1.9	Material cost variance is the sum of _____ and _____.	02
	1.10	_____ budgets are concerned with expected cash receipts/ disbursement.	01
	1.11	List any four categories of budgets.	02

PART B

2	a	Differentiate between single entry and double entry systems of book keeping.	04

	b	Journalize the following transactions in the books of Somu 2020 March 1. Commence the business with cash *Rs* 80,000 4. Brought goods for cash for *Rs* 20,000/− 6. Deposit *Rs* 10,000 in Bank 8. Purchased goods from Kumar for *Rs* 15,000 10. Sold goods to Krishna worth *Rs.* 10,000 15. Brought furniture of *Rs.* 5000 for cash 20. Received commissions of *Rs* 500 28. Withdraw cash of *Rs* 1000 for person value 31. Paid Rent of *Rs* 2000	12
3	a	Prepare a Trial balance from the following. Name of the account — Balances in Rs Capital 10,000 Debtors 2,700 Purchases 9,500 Wages 5,000 Goods in trade 2,000 Sales 14,500 Carriage 150 Machinery 3,500 Creditors 1,400 Bank 1,500 Rent 450 Drawings 900 Sundry-expenses 200	12
	b	How is profit and loss account different from trading account? Explain	04
		OR	
4		From the trial balance of a company prepare a trading, P&L account for the year ending 30*th June* 2019 and balance sheet as on that date. Debit Balances. Rs. — Credit Balances Rs. Opening stock 750 — Capital 2625 Purchases 1490 — Returns outwards 45 Returns inwards 40 — Sales 3810 Duty on import goods 260 — Commission earned 200 Carriage on □urchases 140 — B/P 1500 Carriage on sales 200 — Sundry creditors 770 Office salaries 240 Drawing 400 — Adjustments: Rent paid 180 — Stock in hand/ Closing stock 985 General expenses 150 — Rent due but not paid 30 Bank balance 300 Cash in Hand 100 Sundry Debtors 1000 Buildings 2000 M/C 1000 B/R 250 Depreciation 200 Horses & Carts 150 Internet 90 Discount 10	16

5 a The product of a manufacturing concern passes through two processes A & B and then to finished stock. It is ascertained that in each process 5% of the total weight is lost. 10% scrap which from process 'A' & 'B' realizes $Rs.80/-$ per tonne & $Rs.200/-$ per tonne respectively. The following are the figures relating to both the processes:

	Process A	Process B
Materials (tonnes)	1000	70
Cost of Materials (Rs/tonne)	125	200
Wages (Rs)	28,000	10,000
Manfr. Expenses (Rs)	8000	5250
Output (tonnes)	830	780

Preapre the process cost account showing cost/tonne of each process. There was no stock or WIP in any process. 12

b Highlight the differences between Job costing or Process costing. 04

OR

6 The annual demand of a product is 2400 units. It is produced in batches and largest size of a single batch is 6000 units. After each batch is complete the setup cost is $Rs.750$ and annual carrying cost is $Rs.\ 2.25/unit$. Assume average inventory as $q/2$ of n. of units made in each batch. Selecting $4, 6, 8, 12$ & 24 batches/annum, determine the annual costs of each batch and state the optimum number of batches to minimize total cost. 16

7 a From the following information. Calculate total:

i) Material cost variance
ii) Material price variance
iii) Material usage variance

Material	Standard		Actual	
	Qty (kg)	Cost, Rs/kg	Qty (kg)	Cost, Rs/kg
X	10	4	12	3.75
Y	15	2	18	4.50

10

b Discuss the classification of variances. 06

8 a Expense for budgeted production of 10,000 units in a factory are given below:

Material	50	Distribution expense (20% fixed)	5
Labour	20	Administration expense (Rs. 40000)	4
Variable OH	10		
Fixed OH (Rs. 50,000)	5		
Variable expense (Direct)	2	Total cost/unit	106
Selling expense (10% fixed)	10		

Prepare a flexible budget for production of 8000 units indicating each items cost/unit. 10

b Differentiate between:

i) Sales budget and Production budget
ii) Flexible budget and Master budget 06

Multicriteria Decision Modelling

VI Semester SEE Question Papers

Course codes: 18IM6C3

- Sep 2023 (18IM6C3)
- Aug 2022 (18IM6C3)

18IM6C3

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VI Semester B. E. Examinations September-2023

Industrial Engineering and Management

MULTI CRITERIA DECISION MODELLING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	State the description for the terminologies- alternative, decision maker in decision making environment.	02
	1.2	What two requirements should a problem satisfy in order to be suitable for solving it by a Genetic Algorithm?	02
	1.3	What are the four types of $MCDM$ methods in Discrete Multicriterion Decision Making?	02
	1.4	What is the meaning of a deviational variable in Goal programming?	02
	1.5	What is the significance of 1,5,7,9 in Saaty's Scale?	02
	1.6	Mention any one of the techniques under the category of: a) Non-traditional optimization methods b) Traditional optimization methods.	02
	1.7	Expand the acronym $MAUT$.	02
	1.8	State the principle of $TOPSIS$.	02
	1.9	What range of values vary in degree of similarity and which is considered the best?	02
	1.10	What is Signal to Noise ratio? And state three possible types of computations in it.	02

PART B

2	a	What is structured decision making? Illustrate the structured decision making situations in the field of engineering and management with an example.	06
	b	Outline the importance of simulated annealing and simulated quenching with a help of a flow chart.	10
3	a	Explain Multi-Objective Differential Evolution ($MODE$) with necessary steps.	08
	b	State some problem areas in management where goal programming might be applicable.	08
		OR	

4	a	List the methods falling in Plan Generation and Selection. Discuss any one.	08
	b	An office equipment manufacturer produces two types of products: chairs and lamps. The production of either a chair or a lamp requires one hour of production capacity in the plant. The plant has a maximum production capacity of 50 hours per week. Because of the limited sales capacity, the maximum numbers of chairs and lamps that can be sold are 6 and 8 per week respectively. The gross margin from the sale of a chair is $Rs.90$ and from the sale of a lamp is $Rs.60$. The plant manager desires to determine the number of units of each product that should be produced per week in consideration of the following set of goals. Goal 1: Available production capacity should be utilized as much as possible but should not exceed 50 hours per week. Goal 2: Sales of two products should be as much as possible. Goal 3: Overtime should not exceed 20% of available production time. Formulate this problem as a GP model so that plant manager may achieve his goals as closely as possible.	08
5	a	List the steps involved in Multicriteria Decision making ($MCDM$) methodology.	06
	b	Discuss the Analytical Hierarchy Process for a packaging problem as an example with a neat block diagram.	10
		OR	
6	a	Explain Composite Programming of distance based Multicriterion Decision Making Methods.	06
	b	Compute priority of alternatives for the payoff matrix presented in the following table using weighted average method. Assume weights for each criterion as $0.1, 0.12, 0.15, 0.20, 0.33$ and 0.1. Use normalization method 4 for analysis. $Alternate$ | $Criteria$: C_1 | C_2 | C_3 | C_4 | C_5 | C_6 A_1 | 8 | 9 | 6 | 5 | 2 | 4 A_2 | 6.5 | 5.8 | 2 | 9 | 6 | 3 A_3 | 8.3 | 6 | 4 | 7 | 4 | 4 A_4 | 4 | 8 | 8 | 7 | 3 | 8 A_5 | 5.2 | 7.2 | 4 | 2 | 1 | 6 A_6 | 9.4 | 9 | 6 | 3.2 | 2 | 4 Hint : Normalized method 4 $= \frac{f_j(a)}{\sqrt{[\sum_{a=1}^{N} f_j(a)^2]}}$	10
7	a	In your opinion which fuzzy number based approach is advantageous in decision making scenario? Triangular or trapezoidal? Why?	06

	b	Three infrastructure projects A_1 to A_3 are evaluated on four criteria namely C_1, C_2 C_3 and C_4. Evaluations are presented in table below as trapezoidal fuzzy numbers in the normalized payoff matrix. All the criteria are maximization in nature. Assume crisp weights of criteria as $0.10, 0.20, 0.25$ and 0.45. Determine which infrastructure project is the best. Explore three defuzzification methods.	10

	Criteria															
	$C_1(0.10)$				$C_2(0.20)$				$C_3(0.25)$				$C_4(0.45)$			
Alt	a_1	b_1	c_1	d_1	a_2	b_2	c_2	d_2	a_3	b_3	c_3	d_3	a_4	b_4	c_4	d_4
A_1	0.40	0.50	0.60	0.70	0.20	0.40	0.60	0.80	0.70	0.85	0.90	1.00	0.10	0.30	0.40	0.45
A_2	0.15	0.25	0.35	0.45	0.20	0.30	0.40	0.50	0.25	0.35	0.40	0.45	0.30	0.50	0.70	0.90
A_3	0.10	0.20	0.30	0.40	0.52	0.30	0.50	0.70	0.40	0.50	0.70	0.75	0.35	0.50	0.55	0.60

8	a	Describe Taguchi methodology and its role in Multicriteria Decision making.	06
	b	Five irrigation farms were analyzed for the cost per unit of groundwater pumping. Three variations of costs of groundwater pumping were found in each farm and presented in Table below.	

Irrigation farm	*Cost of groundwater for scenario*		
	1	2	3
1	270	290	285
2	300	320	240
3	280	260	270
4	210	230	320
5	280	295	240

Select which farm you prefer as pilot project for improving other farms. 10

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VI Semester B. E. Examinations August-2022

Industrial Engineering and Management

MULTICRITERIA DECISION MODELLING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Define the term pay off matrix	01
	1.2	Mention any one difference between Multi criteria decision making and multi attribute decision making.	01
	1.3	Mention any of the techniques under the category of Non Traditional Optimization methods.	01
	1.4	The tem used for the initial population of a string of variables goes by which name (As applied to Genetic algorithm).	01
	1.5	Write down the mathematical expression for the weighted sums of various objectives for the weightage method.	01
	1.6	The Acronym MOGA has its full form as what?	01
	1.7	What is the meaning of a deviational variable in the context of Goal programming?	01
	1.8	Mention any two groups under the methods falling in the plan selection category as part of multi objective optimization.	01
	1.9	Define compromise programming.	01
	1.10	What kind of approach is adapted in the case of co-operative Game theory?	01
	1.11	Expand the acronym TOPSIS.	01
	1.12	Write down the expression for the separation measure Da^+ in f dimensional Euclidean distance for each alternative a	01
	1.13	Sketch a typical fuzzy number $\bar{A}$ with triangular membership function.	01
	1.14	Write the expression for relative closeness C_i of each alternate i with reference to negative ideal measure $D_{\bar{\imath}}$	01
	1.15	Fuzzy compromise programming is an extension of which technique in fuzzy environment.	01
	1.16	Expand the acronym PROMOTHEE.	01
	1.17	Can the Data envelopment analysis be considered as a special form of Linear programming? How?	01
	1.18	What is the central idea in the Taguchi methodology?	01
	1.19	Define Swarm Intelligence.	01
	1.20	Who is the propounder of particle swarm optimization?	01

PART B

2 a Explain the necessity of structured decision making? Outline the steps and methodical aspects of structured decision making. 08

b Solve the single objective LPP problem

$\text{Max}\, Z = 2x_1 + 3.5x_2$

$s/t \quad 4x_1 + 3x_2 \geq 6$

$5x_1 + x_2 \leq 1.5$

$x_1 \geq 1.5$

$x_2 \geq 1.0$ 08

3 a Two crops x_1 and x_2 are planned in a command area where water and fertilizer are major constraints whereas sufficient land is available for planning (ie land is not a constraint). Three objectives are the proposed for the command area planning scenario viz Net benefits, agricultural production and labour employment which are all of maximization in nature. Details of the crops such as net benefits (Rs/ha) , agricultural production (T/ha)., Labour employment (man-days/ha) , Fertilizer requirement (kg/ha), water requirement (ha-m/ha) and minimum bands on area under crops are presented in table below.

Various parameter of the planning problem

Characteristics	Crop x_1	Crop x_2	Resources
Net benefits	400	500	---
Agricultural production (T/ha).	2	5	--
Labour employment (man-days/ha)	10	5	--
Fertilizer requirement (kg/ha)	0.41	0.32	28kg
Water Requirement (ha-m/ha)	0.22	0.48	30 ha-m
Minimum bands on area under crops (ha)	25	30	

Solve the problem in multi objective framework using weightage method with weight sets (0.2,0.3,0.5) (0.4,0.2,0.4)(0.1,0.2,0.1) assigned to net benefits (B) , agricultural production (P) and Labour employment (L). Use normalization method for the analysis. 12

b Briefly highlight the ideas behind the multi objective genetic algorithms (Any 3 ideas central to its working would suffice). 04

OR

4 a Alpha company produces two kinds of fancy products, pen holder and paper trays. Production of either of them requires 1 hr. production capacity in the plant. The plant has a maximum production capacity of 12 hours per week. The maximum number of pen holders and paper trays that can be sold are 7 and 10 respectively. The gross margin from the sale of pen holder is Rs.90 and Rs.45 for paper tray. The plant manager has set the following goals in order of importance.

		P1: He wants to avoid underutilization of production capacity P2: He wants to limit overtime hours to 3 hrs P3: He wants to sell as many as pen holders and pen twice as possible. Since the gross margin from the sale of a pen holder is set as twice the amount of the profit from paper tray the manager has twice much desire to achieve the sales goals for pen holders as for paper trays P4: The manager wishes to minimize the overtime operation of the plant as much as possible Formulate the goal programming problem based on the above information	10
	b	Differentiate between (i) Goal programming ad Compromise programming (ii) Goal programming and fuzzy goal programming.	06
5	a	Tom schmid is thinking about buying a Nordic Sci machine. The three important features to him are price, Ease of use and the ability to store the exercise equipment in a closet when he is done using it. Given the following data, help Tom determine the best machine for him. <u>Factor Weights</u> Factor — Importance Weight Price — 0.9 Ease of Use — 0.75 Storage — 0.6 <u>Factor Evaluation</u> Factor — Professional Nordic Skier — Econo Nordic skier Price — 0.5 — 0.8 Ease of Use — 0.95 — 0.6 Storage — 0.9 — 0.7	08
	b	George is about to buy a compact stereo casette player. He is currently considering three brands Sun, Histen and Surgo. The three important factors to George are the price, color , warranty, size of the unit and the brand name. George has determined factor of weights of 0.4,0.1,0.1 0.1 and 0.3 respectively. Furthermore George has determined factor evaluations for all of the three different manufacturers of that unit he is considering. The sun unit has factor evaluation of 0.7,0.9,0.8,0.8 and 0.9 for the price , color warranty, size and brand name factors. The Hitek unit has factor evaluation of 0.6,0.9,0.9,0.8 and 0.9 of these factors. Finally Surgo has factor evaluation of 0.8,0.4,0.4,0.2 and 0.6. For the same features of price, color, warranty , size and brand name. Determine the total weighted evaluation for the three manufacturers. Which one should George select?	08
		OR	

6 a Compute the ranking pattern of alternatives given the following pair of matrix using the TOPSIS method

Criteria / Alternate	C_1	C_2	C_3	C_4	C_5	C_6
A_1	8	9	6	5	2	4
A_2	6.5	5.8	2	9	6	3
A_3	8.3	6	4	7	4	4
A_4	4	8	8	7	3	8
A_5	5.2	7.2	4	2	1	6
A_6	9.4	9	6	3.2	2	4

Assume the weights of the criteria as 0.1,0.12,0.15.0.20.0.33,0.10. Use normalization method 4 for analysis. 10

b Compute the ranking portion for the pay off matrix presented in the table below using compromise programming method for p=1,2. Assume weights for criteria as 0.35, 0.40, 0.25. All the criterion are of maximization in nature.

Criteria / Alternate	C_1	C_2	C_3
A_1	35.250	0.77718	1152.00
A_2	36.300	0.77250	1301.80
A_3	38.492	0.76386	1439.50
A_4	40.331	0.74119	1586.50
A_5	43.3239	0.66374	1599.40
A_6	45.500	0.58989	1478.70

06

7 Four infrastructure projects A1 to A4 are evaluated in fuzzy environment on three criteria namely net monetary benefits (C_1), social benefits (C_2) and environmental construction (C_3). All criteria are of maximization in nature. Evaluations are rated as triangular fuzzy number in the payoff matrix based on Chen's methodology. Assume weights of criteria as (0.25,0.30,0.35), (0.40,0.45,0.50) (0.50,0.55,0.60). Identify substitute infrastructure projects using fuzzy TOPSIS.

Criteria / Alternate	C_1			C_2			C_3		
	a_1	b_1	c_1	a_2	b_2	c_2	a_3	b_3	c_3
A_1	3	4	5	3	4	6	3	5	7
A_2	1	3	5	4	5	7	4	5	6
A_3	2	4	6	2	4	6	3	4	6
A_4	5	7	8	4	8	10	5	7	9

16

8 a Describe advantages and limitations of Data Envelopment Analysis (DEA). Compare Data Envelopment Analysis with multicriteria Decision making (MCDM). 08

b Differentiate Ant colony optimization , particle swarm optimization and Genetic algorithms on the bases of at least three characteristics. 08

Advanced Manufacturing Process

VI Semester SEE Question Papers

Course codes: 18IM6C5

- Aug 2022 (18IM6C5)

18IM6C5

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations August-2022

Industrial Engineering and Management

ADVANCED MANUFACTURING PROCESS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	In shell moulding sand preparation, the resin is combined with a catalyst (hexa) to develop the _____ characteristics.	01
	1.2	Dental castings and turbine blades are produced by _____ casting.	01
	1.3	In plaster mould castings, talc is added to the plaster while mixing to prevent _____ of mould.	01
	1.4	The thickness of shells in shell moulding is ranging from _____ to _____.	01
	1.5	Petrol engine cylinders are made by _____ moulding.	01
	1.6	Use of runner, risers and cores are eliminated in _____ casting.	01
	1.7	Strip drawing test is a method which is mainly used for evaluating the characteristics of _____.	01
	1.8	The flow stress _____ with strain rates during High Energy Rate Forming (HERF) Processes.	01
	1.9	Explosive forming is a metalworking technique in which an _____ charge is used instead of a punch.	01
	1.10	Metal forming is a process in which the metal is deformed _____ to get into the desired shape.	01
	1.11	A proper care should be taken while cooling or solidification of castings to avoid _____ formation.	01
	1.12	Rapid heat extraction from the casting causes narrowing of liquid-solid zone which sweeps itself across the _____ metal.	01
	1.13	An alloy, unlike a pure metal, does not have a sharply defined _____ temperature.	01
	1.14	Composite materials are classified based on the type of _____.	01
	1.15	Fibers or particles embedded in matrix of another material are the examples of _____ material.	01
	1.16	Fiber Reinforces Composites are composed of fibers embedded in _____ material.	01
	1.17	Cellular manufacturing is an approach whereby production can be done in _____ batches.	01
	1.18	Cellular manufacturing is also known as _____ Technology.	01
	1.19	The main objective(s) of Industrial robot is to _____ the labour requirement.	01

	1.20	The Robot designed with cylindrical coordinate systems has _____ linear and one rotational movement.	01

PART B

2	a	Differentiate between true centrifugal casting and semi centrifugal casting?	06
	b	Explain the process along with the principle involved in vacuum mould casting.	10
3	a	Discuss the process variables and advantages associated with explosive forming.	08
	b	Enumerate the importance of sticking friction in metal working and why is friction generally undesirable in metal forming operations?	08
		OR	
4	a	Discuss the different operations involved in bulk deformation process.	06
	b	Explain the principle and Process Characteristics of Electro hydraulic Forming.	10
5	a	Analyze the use of Computer Aided Engineering (CAE) concepts in a product manufacturing.	08
	b	List the application of CAE techniques in Thermal analysis of Heat-Affected Zone.	08
		OR	
6	a	Elaborate on the application of CAE technique in the analysis of forging process.	08
	b	Explain the role CAE techniques in the simulation of molten metal flow.	08
7	a	Explain the importance of injection moulding with respect to the following: i. Three-plate mould ii. Two-plate mould	08
	b	Explain the four major steps typically involved in the making of composite products.	08
8	a	What is a flexible manufacturing system and state the advantages of flexible manufacturing system technology, compared to conventional batch operations.	08
	b	Define group technology and explain the “composite part concept” in group technology.	08

Digital Manufacturing

VI Semester SEE Question Papers

Course codes: 18IM6D5, 16IM6C1

- Aug 2022 (18IM6D5)
- Apr/May 2019 (16IM6C1)

18IM6D5

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)
VI Semester B. E. Examinations August-2022
Industrial Engineering and Management

DIGITAL MANUFACTURING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	____ is a modeling tool applied in discrete asynchronous concurrent systems and ____ is a set of objects having same attributes and services.	02
	1.2	Reverse engineering is also known as _____.	01
	1.3	___ has become the indispensable drive factor in the product life cycle of the manufacturing industry.	01
	1.4	Basic modeling method is also called _____.	01
	1.5	What is initial stage of manufacturing where manufacturing existed as a skill?	02
	1.6	What is the relevance of LAN in Digital Manufacturing?	01
	1.7	Mention any two key technologies of DM.	02
	1.8	The middle layer in the hierarchical structure of intelligent sensing system is also called ____.	01
	1.9	Human factors engineering is termed as ____.	01
	1.10	Name any one hybrid diagram-Analysis model.	01
	1.11	_____ System is in a state of nonlinear system or nonlinear region.	01
	1.12	Simulating the manufacturing activity of human experts through computers is called _____.	01
	1.13	Cleaning the data to eliminate noise and inconsistencies is called ____.	01
	1.14	What is Metadata?	02
	1.15	Mention any two sensors deployed in machining process monitoring.	02

PART B

2	a	Elaborate on the key features of digital manufacturing system.	06
	b	Illustrate and Explain with a flow chart the Architecture in digital manufacturing system.	10
3	a	"Materialization of manufacturing information is a special transmission and transformation of original manufacturing information". Elaborate.	10
	b	Brief about the operation process of virtual manufacturing alliance organization with Process flow diagram.	06

		OR	
4	a	What is Virtual prototyping? Illustrate its frameworks.	10
	b	Explain the various applications of reverse engineering in digital manufacturing.	06
5	a	Explain the framework of functions to realize the autonomy of manufacturing system.	10
	b	Explain the process of data mining.	06
		OR	
6	a	Discuss the concepts and features of Intelligent manufacturing.	06
	b	Explain the general process to design the model of management of technology(MOT)	10
7	a	State the differences between CAPP/CAM.	06
	b	Explain how bottleneck problem of information resource sharing is done in digital manufacturing.	10
8	a	Explain the role of Digital manufacturing technology in Micro-Nano manufacturing.	06
	b	With the help of a sketch explain the relationship between the manufacturing digitalization and manufacturing green revolution.	10

16IM6C1

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

DIGITAL MANUFACTURING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	What are the "3*S*" of an enterprise to be analyzed before digitizing?	02
	1.2	What are the three changes of Digital Marketing?	02
	1.3	What are the components that perform as a package in Digital Marketing?	02
	1.4	What is metadata?	02
	1.5	List the different types of network environments needed in implementation of Digital Marketing.	02
	1.6	Wire frame modeling, surface modeling and Solid Modeling are _____ modeling forms.	01
	1.7	*VLSI* stands for _____.	01
	1.8	Manufacturing systems can be classified as _____ and _____ manufacturing systems.	02
	1.9	*WAN* stands for _____.	01
	1.10	Expand *MEMS*.	02
	1.11	List the technologies involved in Digital Recognition and Integration.	02
	1.12	What is extreme manufacturing?	01

PART B

2	a	What are the various stages of development of traditional, nontraditional and digital manufacturing systems?	08
	b	Illustrate the Iterate-Adjust-Optimize loop. Support your answer with suitable live example.	08
3		Highlight the foundation of digital manufacturing and list the modeling methods of digital manufacturing system and explain any one in detail.	16
		OR	
4		Considering the Chip manufacturer for a *SIM* card as an example build an *IDEF* modeling method.	16

5	What are the various parameters to be considered while establishing computational method? Explain the theoretical units in manufacturing computing.	16
	OR	
6	Elaborate the connotation of *MOT* with respect to the statement quoted below: "Technology economics and business management reflect a relation between the theory and commercial application which is similar to the relation between management and economics".	16
7	Illustrate the relationship among the entities involving sales activities and explain the fundamental characteristics of manufacturing activities.	16
8	Explain in detail the Micro-Nano-Electro Mechanical system. Support your answer with suitable examples.	16

Employability Skills and Professional Development

VI Semester SEE Question Papers

Course codes: 18HSI68

- Aug 2022 (18HSI68)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations August-2022

COMMON TO AS/CH/CV/EI/IM/ME

COMMON TO 2018 AND 2016 SCHEME

EMPLOYABILITY SKILLS AND PROFESSIONAL DEVELOPMENT OF ENGINEERS

Time: 02 Hours *Maximum Marks: 50*

Instructions to candidates:

Answer any 25 questions. Write the question number and the correct answer against the question number

1	1	Priya must survey all of the people in her department to get their ideas on how to launch a new product. The technique that would be the best to use is _____.	01
	2	The problem solving model includes _____ steps .	01
	3	The _____ is a suitable method for priotizing ideas.	01
	4	NAF technique is _____.	01
	5	The five step design thinking process popularized by Standford University's d.school includes _____.	01
	6	_____ is a method used to compare two designs (web site, app, etc.)?	01
	7	Shoe swapping is a tool for _____.	01
	8	The coin landed on heads 5 times ; its definitely going to be tails next time. The fallacy is _____.	01
	9	After a car accident , both drivers blame the other for causing the crash. Both , or at least one of them has the _____ bias.	01
	10	The scientist who provided crucial clues to the structure of DNA with the help of X-ray diffraction is _____.	01
	11	What are some ways that you can increase your creativity ? a. Practice b. Organize your life c. Educate Yourself d. All of the above	01
	12	Creativity and innovation are basically the same concepts a. True b. False c. Both a and B d. None of the above	01
	13	Sheila and her colleague are generating ideas by acting as another person. This is an example of a. Role storming b. Brainstorming c. Stepladder technique d. Brain writing	01

14.	Donald Trump was the business man behind the success of the Four seasons chain of hotels. a. True b. False c. Both a and B d. None of the above	01
15	The fishbone is a tool that helps to : a. Highlight links between ideas b. Increase creativity c. Analyze cause and effect d. None of the above	01
16	Which of the following is not one of the five phases of the creative problem-solving model? a. Preparation b. Analysis c. Incubation d. Evaluation	01
17	Creativity can be defined as a. Generation of ideas b. Identification of business ideas c. A problem- solving tool d. None of the above	01
18	Creativity is for only designers a. True b. False c. Both a and b d. None of the above	01
19	Which of the following standards must be adopted before making a decision? a. Encourage and explore different interpretations of user problems b. Work in isolation c. User research d. Make decision based on the data available	01
20	Which of the following is not a technique to generate ideas? a. Fishbone b. Brainstorming c. Mind mapping d. Shoe swap	01
21	How many steps are involved in defining a problem? a. Five b. Seven c. Eight d. Nine	01
22	Problem solving techniques can be used in a. Business problems b. Social problems c. Economical problems d. All of the above	01

23	Which of the following can help you increase your creativity? a. Practice b. Give yourself time c. Get some distance d. All of the above	01
24	Which step should you take before starting a brainstorming session? a. Define your purpose b. Define your user's feedback c. Define your solutions d. Define your profit margin	01
25	Empathy is the last stage of design thinking a. True b. False c. Both a and b d. None of the above	01
26	POV in design thinking is a. Point of View b. Position of View c. Both a and b d. None of the above	01
27	Brainstorming can be done with a. Only one person b. A group c. Random users d. Specific users	01
28	Black Hat refers to a. Devil's advocate b. Cool and in control c. Both and and b d. None of the above	01
29	Design thinking is a problem-solving technique that is used to design products and services based on the knowledge you have of what the customer needs a. True b. False c. Both a and b d. None of the above	01
30	The technique that was developed to allow everyone to be involved , to address the limitations of brain storming a. Brain reading b. Brain writing c. Brain dumping d. Reverse brain storming	01
31	Which prototypes are more basic early-stage representations of the final product or service? a. No-fidelity b. Low-fidelity c. Medium-fidelity d. High-fidelity	01

32	The greatest value of De Bono's hats technique is that it : a. Allows you to know the perspective of each person by the color of their hat b. It is fun to see the different hats c. It ensures that all perspectives are expressed d. It is an interesting team building activity	01
33	Which is not the tool for Ideation? a. Mind-map b. Brainstorming c. Brain-Writing d. Idea Probe	01
34	Design Thinking is similar to Creative Thinking a. True b. False c. Both a and b d. None of the above	01
35	What happens at testing stage? a. User Research b. User Feedback c. User Prototyping d. User Empathy	01
36	High Fidelity prototypes are a. Easy to make b. Less expensive c. Functional representation of solution d. Final products	01
37	What is counter example? a. Premises are TRUE and Conclusion is also TRUE b. Premises are FALSE and Conclusion is also TRUE c. Premises are FALSE and Conclusion is also FALSE d. Premises are TRUE and Conclusion is also FALSE	01
38	Is the following statement TRUE or FALSE? Arguments can be valid even when there is no connection between the premises and the conclusion a. False b. True c. Both a and b d. None of the above	01
39	An argument is SOUND when : a. Premises are TRUE and Conclusion is FALSE b. Premises are FALSE and Conclusion is TRUE c. All Premises are TRUE and argument is VALID d. All the above	01
40	Which of the following arguments is INVALID? a. The smartest students all go to engineering. Hence, I know that engineering students are so smart. b. Zombies do nothing but eat brains. So, Zombies are always hungry. c. Tall peaks are hard to climb. Mount Everest is a tall peak. It is hard to climb Mt Everest. d. You can trust dogs more that cats. Mr Woof is a dog and Miss Meow is a cat. You can trust Mr Woof more than Miss Meow.	01

41	The following argument is a circular argument. True or False? You can't understand something the first instant you hear it because it takes time to understand things. a. False b. True c. Both a and b d. None of the above	01
42	Is the following argument a good argument or a bad argument? Cricket keeps you fascinated with the tension. That's why the tension in cricket is so interesting. **Premise** : Cricket's tension is interesting **Conclusion** : Cricket has interesting tension a. Good argument b. Bad argument c. Both a and b d. None of the above	01
43	Is the following argument VALID, INVALID, SOUND, CIRCULAR or FINE? I just know that India will lose the match. They always do. They always lose their football matches and almost always lose cricket. I'll be more heartbroken if they lose their cricket match. I care about it more. **Premise 1** : If India loses their football match, I'll be less upset if they lose their cricket match **Premise 2:** India always loses football matches and always ALWAYS loses cricket matches. **Conclusion:** I will be upset. a. Valid b. Invalid c. Circular d. Fine	01
44	Is the following argument Valid or Invalid **Premise:** Quantum mechanics is true **Conclusion:** Therefore the predictions of quantum mechanics are all correct. a. Valid b. Invalid c. Both a and b d. None of the above	01
45	Is the following argument a case of inference to the best explanation? People who get the vaccine don't contract the disease. Therefore the vaccine prepares the immune system to fight off the disease. **Premise:** People who get the vaccine don't contract the disease **Conclusion :** The vaccine prepares the immune system to fight off the disease. a. No, it is not an inference to the best explanation b. Yes, it is an inference to the best explanation c. Both a and b d. None of the above	01

46	Is the following argument a case of inference to the best explanation? **Premise:** His parents are both doctors **Conclusion :** He will become a doctor a. Yes, it is an inference to the best explanation b. No, it is not an inference to the best explanation c. Both a and b d. None of the above	01
47	Is the following argument a good argument from evidence or a bad argument form evidence, according to Bayesianism? **Premise:** This owlet is brown. That owlet is brown. That other owlet is brown. **Conclusion :** So they all are brown. a. Good argument from evidence b. Bad argument from evidence c. Both a and b d. None of the above	01
48	Look at the argument and its four possible breakdowns. Select the best interpretation. Do not quit half-way through! College work only makes your life better if you actually graduate and finish the degree. It is just pointless to give up before you have your certificate. **a. Interpretation [1]** **Premise 1:** College will not make life better if you quit it halfway through. **Premise 2:** You should not quit something if it will not make your life better. **Conclusion :** Do not quit college half way through! **b. Interpretation [2]** **Premise 1:** Quitting college means you will not get your certificate **Premise 2:** If you will not get certificate then it is pointless **Conclusion :** College is pointless. **c. Interpretation [3]** **Premise 1:** If you quit college, you will not get your certificate **Premise 2:** If you quit college, you will not have understood what you have studied **Conclusion :** You should not quit college. **d. Interpretation [4]** **Premise 1:** If you quit college, then you will not get your certificate **Premise 2:** If you do not get your certificate then college will have been pointless **Premise 3 :** You shouldn't do something if it will be pointless. **Conclusion :** You should not quit college	01

49	Is the argument valid or invalid? The number of people in the room is x. x>3 Therefore, x>2 a. Valid argument b. Invalid argument c. Both a and b d. None of the above	01
50	Identify the fallacies in the following: Ajay: Yes, I believe there could be extraterrestrial life in the universe. Mehul : Ha! So you believe that alien UFOs have been visiting Earth for thousands of years. That's clearly studpid, there is no real evidence to support that. Ajay : I din't say anything about UFO's a. Slippery slope b. Strawman Fallacy c. Post Hoc Fallacy d. Hasty Generalization	01

Supply Chain and Logistics Management

VI Semester SEE Question Papers

Course codes: 18IM63, 16IM64, 16IM64 / 12IM72

- Grade Improvement Nov 2021 (18IM63)
- Grade Improvement Nov 2020 (16IM64)
- Fast Track Oct 2020 (16IM64 / 12IM72)
- Fast Track Jul 2019 (16IM64)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Grade ImprovementExaminationsNov-2021

Industrial Engineering and Management

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

Answer any FIVE full questions out of TEN. Each carries 20 marks

Use of statistical tables permitted

1	1.1	Graphically show intracompany inter functional scope of strategic fit.	02
	1.2	As the number of facilities in a supply chain increases, total transportation cost increases upto a point and then decreases. State the reason for this.	02
	1.3	Identify any two models used in analysis of network design decisions..	02
	1.4	What are the four drivers of supply chain performance?	02
	1.5	As safety inventory increases both ____ and ____ increases.	02
	1.6	If demand increases by a factor of K, the optimal lot size __ by a factor of _____.	02
	1.7	Define tailored transportation.	02
	1.8	Identify any two benefits of effective sourcing decisions.	02
	1.9	List the supply chain macroprocesses.	02
	1.10	With one example classify which are the parties involved in B2C e-business.	02
2	a	Describe the cycle and push/pull views of a supply chain. How do you relate them to supply chain of L.L Bean?	10
	b	Define supply chain management. Explain the decision phases in a supply chain.	10
3	a	Explain with suitable example, why achieving strategic fit is critical to a company's overall success.	12
	b	What are the abilities that supply chain should have if it is to be more responsive? Explain	08
4	a	List and explain the factors influencing network design decisions.	10
	b	Draw the network of the following distribution design options. Also bring out the advantages and limitations. i. Manufacturer storage with diret shipping. ii. Distributor storage with last mile delivery.	10
5	a	Being the head of TVS motors, discuss the role of network design in their supply chain.	10
	b	In the context of designing distribution network, elaborate on impact of e-business on customer service.	10
6	a	Explain the role of cycle inventory and safety inventory in supply chain practices.	10
	b	The mean demand/period for a product is 2500units with standard deviation/ period being 500units. The lead line taken is 2 weeks and review interval is 4 weeks. Assume periodic review policy and cycle service level of 90%. Find the safety stock and order upto level.	10

<table>
<tr><td>7</td><td>a</td><td>In the context of multiechelon supply chains,
Explain:

i. Nested inventory policy
ii. Integer replenishment policy.</td><td>10</td></tr>
<tr><td></td><td>b</td><td>Consider the following all unit discount pricing schedule. Annual demand is 7000 units ordering cost is Rs 60. Holiday charges are 15%. Evaluate the number of bottles that should be ordered in each lot.
<table>
<tr><td>Order quantity</td><td>Unit price in Rs</td></tr>
<tr><td>0-3000</td><td>4.0</td></tr>
<tr><td>3000-6000</td><td>3.35</td></tr>
<tr><td>6000 or above</td><td>3.05</td></tr>
</table></td><td>10</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>8</td><td>a</td><td>What are some ways that a firm like Walmart benefits from good sourcing decisions?</td><td>10</td></tr>
<tr><td></td><td>b</td><td>Identify pros and cons of different transportation networks.</td><td>10</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>9</td><td>a</td><td>Explain in detail tailored transportation network by customer density and distance.</td><td>10</td></tr>
<tr><td></td><td>b</td><td>What are the reasons for outsourcing? Illustrate with suitable examples?</td><td>10</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>10</td><td>a</td><td>What is bullwhip effect? What are the managerial levers that help achieve coordination in the supply chain? Explain.</td><td>10</td></tr>
<tr><td></td><td>b</td><td>What is vendor managed inventory? How it is different from continuous replenishment strategy? How manufacturer benefited out of VMI?</td><td>10</td></tr>
</table>

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Grade Improvement Examinations Nov-2020

Industrial Engineering and Management

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of statistical tables permitted.

PART A

1	1.1	What are the four drives of supply chain performance?	02
	1.2	Draw cost-responsiveness efficient frontier curve.	02
	1.3	Graphically depict variation of transportation cost and response time with number of facilities.	02
	1.4	What are the two factors which determine appropriate level of safety inventory?	02
	1.5	List the supply chain macro processes.	02
	1.6	Write a note on inventory turnover ratio.	02
	1.7	If demand increases by a factor of K, the optimal lot size _____ by a factor of _____.	02
	1.8	What is Milkrun in the context of transportation?	02
	1.9	Identify any two benefits of effective sourcing of decisions.	02
	1.10	What is tailored transportation?	02

PART B

2	a	List and explain the broad steps involved in achieving a strategic fit between the company's competitive strategy and SC strategy.	10
	b	Describe the framework for structuring the drivers with the help of a block diagram.	06
3	a	Draw the network of the following distribution design options. Also bring out the advantages and limitations of the same. i) Manufacturer storage with direct slipping. ii) Distributor storage with last mile delivery.	10
	b	Identify and describe the factors influencing network design decisions.	06
		OR	

4	a	Demand allocation decisions have to be made for plants in a supply chain in which many manufacturing plants serve different markets. Multiple plants may be used to satisfy demand at a market. Perform the following to formulate a mathematical model that minimizes total cost of production, inventory and transportation. i) Identify the inputs required for mathematical model. ii) Identify the design decision variables. iii) Formulate objective function. iv) Formulate the constraint equations for the model.	10
	b	What are the six key factors to be considered when designing the distribution network? Explain briefly.	06
5	a	What is the role of cycle and safety inventory in supply chain?	06
	b	Best buy sells 3 models of computers, the Litepro, Medpro and Heavypro. Annual demands for the 3 products are 12000 *units*, 1200 *units* and 120 *units* respectively. Each model costs Best buy 500. A fixed transportation cost of 4000 is incurred each time an order is delivered. For each model ordered and delivered on the same truck, an additional fixed cost of 1000 is incurred for receiving and storage. Best buy incurs a holding cost of 20%. Evaluate the lot size that the Best buy manager should order if lots for each product are ordered and delivered jointly for all 3 models. Also evaluate the annual cost of such a policy.	10
		OR	
6	a	W.W Grainger sources from hundreds of suppliers and is considering the aggregation of inbound shipments to lower costs. Truckload shipping costs 500/*truck* along with 100 per pickup. Average annual demand from each supplier is 10,000 *units*. Each unit costs 50 and Grainger incurs a holding cost of 20%. What is the optimal order frequency and order size of Grainger decides to aggregate four suppliers/truck? What is the optimal order size and frequency if each truck has a capacity of 2500 *units*?	10
	b	Identify managerial levers that reduce lot size and cycle inventory in a supply chain without increasing cost.	06
7	a	What are the different modes of transportation? Explain briefly the important characteristics of any three of them.	10
	b	List and explain the risks of using a third party.	06
8	a	Describe supply chain coordination, the bullwhip effect and their impact on SC performance.	10
	b	What is vendor managed inventory? How it is different from continuous replenishment strategy?	06

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Fast track Examinations Oct-2020

Industrial Engineering and Management

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	Mention the process views of a supply chain.	02
	1.2	List the steps involved in achieving strategic fit between the supply chain and competitive strategies?	02
	1.3	Outbound transportation costs per unit tend to be higher than inbound costs. Do you agree with the above statement? Give suitable example to justify your answer.	02
	1.4	Draw a graph showcasing variation in logistics cost and response time with number of facilities.	02
	1.5	Mention the components of ordering cost.	02
	1.6	List the factors to be considered to determine appropriate level of safety inventory.	02
	1.7	Differentiate between the terms offshoring and outsourcing.	02
	1.8	What is the purpose of supplier scoring and assessment?	02
	1.9	Highlight the various funds that will impact IT in the supply chain.	02
	1.10	Mention the important characteristics of information that will support effective supply chain decisions.	02

PART-B

2	a	Consider the supply chain involved when a customer purchases a book at a book store. Identify the cycles in this supply chain and the location of the push/pull boundary.	08
	b	Give arguments to support the statement that Walmart has achieved good strategic fit between its competitive and supply chain strategies. What challenges does it face as it works to open smaller format stores in urban environments.	08
3	a	Explain the basic steps involved in decision tree analysis to evaluate supply chain decisions under uncertainty.	08
	b	Trips logistics, a third party logistics firm that provides warehousing and other services is facing a decision regarding the amount of space to lease for the upcoming three year period. The general manager has forecast that Trips logistics will need to handle a demand of 100,000 units for each of the next three years. Historically, Trips logistics has required 1,000 sq.ft of warehouse space for every 1000 units of demand. For the purposes of the discussion, the only cost Trips logistics faces is the cost for the warehouse.	

		Trips logistics receives revenue of \$1.22 for each unit of demand. The general manager must decide whether to sign a three year lease or obtain warehousing space on the spot market each year. The three year lease will cost \$1 per square foot per year, and the spot market rate expected to be \$1.20 per square foot year for each of the three years. Trip logistics has a discount rate of $K = 0.1$.	08
		OR	
4	a	How do the location and size of the warehouses affect the performance of firm such as Amazon? What factors should Amazon take into account when deciding where and how big its warehouses should be?	08
	b	Conduct *SWOT* analysis on supply chain management of Mumbai Dabbawalla.	08
5	a	Demand for Red Four mobile at best buy is 1,000 units per month. Best buy incurs a fixed order placement, transportation and receiving cost of \$4000 each time an order is placed. Each mobile costs best buy \$500 and the retailer has a holding cost of 20 percent. Evaluate the number of mobiles that store manager should order in each replenishment lot.	08
	b	What is the role of cycle and safety inventory in supply chain?	08
		OR	
6	a	Assume that weekly demand for *TVs* at SamSon office supplies is normally distributed, with a mean of 2500 and standard deviation of 500. The manufacturer takes two week to fill an order placed by the Samson manager. The store manager currently orders 10,000 *TVs* when the inventory on hand drops to 6000. Evaluate the safety inventory and the average inventory carried by Samson. Also evaluate the average time a *TV* spends at Samson.	08
	b	Explain the major components of inventory holding cost with suitable examples.	08
7	a	Highlight the Pros and Cons of different transportation networks.	08
	b	What is Tailored transportation? Mention some transportation options based on customer density and distance.	08
8	a	With neat sketch elucidate the Bull Whip effect in supply chain.	08
	b	Identify the major processes involved in effectively managing supplier Relationship.	08

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Fast Track Examinations July-19

Industrial Engineering and Management

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Consider an example of a customer purchasing a wireless router for Best Buy and paying $60. The difference between the $60 that the customer paid and the sum of all costs incurred by the supply chain to produce and distribute the router represents the supply chain is known as _____.	01
	1.2	_____ are the actual physical locations in the supply chain network where product is stored, assembled or fabricated.	01
	1.3	What is the fundamental trade-off that managers achieve when making facilities' decisions?	02
	1.4	Amazon stocks the medium to fast moving items at their intermediate warehouses and uses package carriers to transport products from the warehouses to the final customer. This is an example for _____ type of distribution network design option.	01
	1.5	Write only the objective function to minimize total supply chain cost for capacitated plant location model.	01
	1.6	List any four factors that influence network design decisions.	02
	1.7	Trade promotions generally increase cycle inventory in a supply chain and hurt performance. This realization has led many firms, including the world's largest retailer, to adopt _____, Here the price is fixed over time and no short-term discounts are offered. This eliminates any incentive for forward buying and all stages of the supply chain purchase in quantities that match demand.	01
	1.8	If demand increases by factor of k, the optimal lot size also increases by a factor of _____.	01
	1.9	The store manager at Jean-Mart currently purchases in lots of 1,000 pairs. Then Jean-Mart carries a cycle inventory of _____ pairs of jeans.	02
	1.10	Define 'Milk Run' in the context of transportation in SCM.	01
	1.11	In _____ contracts, the manufacturer charges the retailer a low wholesale price c, and shares a fraction of retailer's revenue.	01
	1.12	Define the postponement strategy.	02
	1.13	A supply chain lacks _____ if each stage optimizes only its local objective, without considering the impact on the complete chain.	01
	1.14	Fluctuations in orders increase as they move up the supply chain from retailers to wholesalers to manufacturers to suppliers is called _____.	01
	1.15	What are the five key processes involved in Internal Supply Chain Management?	02

PART B

2	a	There is a close connection between the design and management of supply chain flows (product information, and funds) and the success of a supply chain. In this context, justify "Wal-Mart' has been a leader at using supply chain design, planning and operation to achieve success.	06
	b	Outline the basic steps for achieving strategic fit between the company's competitive strategy and supply chain strategy. Further elaborate on these steps citing suitable example.	10
3	a	Draw the diagram of 'Manufacturer storage with direct shipping' type design option for transportation. Identify it's any three performance characteristics connected to service.	08
	b	During network design, managers need a methodology that allows them to estimate the uncertainty in their forecast of demand and price and then incorporate this uncertainty in the decision-making process. In this context, explain such a methodology, for evaluating network design decisions under uncertainty.	08
		OR	
4	a	With the help of block diagram, explain a framework for supply chain network design decisions.	08
	b	Justify the statement: "Network decisions have a significant impact on performance of the supply chain".	08
5	a	Demand for vitamins is 10000 bottles per month in Drugs Online (DO). DO incurs a fixed order placement, transportation and receiving cost of $Rs.100$ each time an order for vitamins is placed with the manufacturer. DO incurs a holding cost 20%. The price charged by the manufacturer varies according to the all unit discount pricing schedule shown. Evaluate the no. of bottles that DO manager should order in each lot. $Order\ quantity$ | $Unit\ price$ $0-5000$ | $Rs.3.0$ $5000-10000$ | $Rs.2.96$ $10000\ or\ above$ | $Rs.2.92$	12
	b	Describe different measures of product availability.	04
		OR	
6	a	A manufacturing company purchases 9000 parts of a machine for annual requirements ordering for month usage at a time, each part costs $Rs.20$. The ordering cost per order is $Rs.15$ and carrying charges are 15% of the average inventory per year. You have been assigned to suggest a more economical purchase policy for the company. What advice do you offer and how much would it save to the company per year?	12
	b	Give reasons for the following: i) Amazon.com has lower levels of book and music inventories than bookstore chains such as Borders and Barnes & Noble, which must keep inventory in every retail store. ii) Consider the sale of Palm personal Digital Assistants ($PDAs$) at $B\&M$ office supplies. When a new palm model is introduced, $B\&M$ carries a much higher level of safety inventory relative to demand. After few days of sales, it reduces level of safety inventory carried relative to demand.	04

7	a	Discuss by giving example. "Firms planning inventory aggregation must consider the trade-offs among transportation, inventory and facility costs".	06
	b	Identify and explain the assessment factors that are to be considered while scoring and assessing the supplies (Any ten of the factors to be identified and explained).	10
8	a	What is vendor managed inventory? How it is different from continuous replenishment strategy? How manufacturer is benefited out of VMI?	10
	b	What does $CPFR$ stand for? Elaborate on key features of $CPFR$ model.	06

E-Commerce

VI Semester SEE Question Papers

Course codes: 16IM6D5

- Grade Improvement Nov 2020 (16IM6D5)
- Apr/May 2019 (16IM6D5)

16IM6D5

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Grade Improvement Examinations Nov-2020

Industrial Engineering and Management

E-COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Enumerate the merits and demerits of E-Commerce.	02
	1.2	Identify the components of E-Commerce.	02
	1.3	Explain the architecture of E-Commerce.	02
	1.4	Enumerate different applications of E-Commerce.	02
	1.5	List the different models of E-Commerce.	02
	1.6	Summarize the $B2C$ model.	02
	1.7	Find the features of E-Commerce website.	02
	1.8	List the problems faced by online sellers.	02
	1.9	Identify the security issues of E-Commerce.	02
	1.10	What are emerging trends in E-Commerce?	02

PART B

2	a	Analyze transparency issues in $B2B$ models and substantiate your answers.	08
	b	Distinguish between $B2B$ and $B2C$ in terms of serviceability, control and order completion cycle.	08
3	a	Discuss the TCP layer process with a neat block diagram.	08
	b	Differentiate between $IPv4$ and $IPv6$.	08
		OR	
4	a	Segregate with examples between Frames, Forms and Style sheets.	08
	b	Define $HTML$ and why do we need it in E-Commerce applications?	08
5	a	Classify the critical three primary processes which are enhanced in E-business transactions. Give examples.	08
	b	Elucidate with examples, the major different types of E-commerce applications.	08
		OR	

6	a	Assess the impact of $B2B$ markets on the economy of developing countries.	08
	b	How is E-Commerce useful to developing new Entrepreneurs? Substantiate your answers.	08
7	a	"For E-Commerce to be successful, the distribution channel needs to be redesigned"- Comment on this statement with illustrations.	08
	b	How can you use E-Commerce for effective integration of multichannel business? Discuss with suitable examples.	08
8	a	Estimate with reasoning, the trends and the challenges of social commerce.	08
	b	Discern the external drivers from internal drivers for E-Commerce.	08

16IM6D5

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

E - COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Give one example each for horizontal portals and vertical portals.	02
	1.2	What is the purpose of affinity portals?	02
	1.3	Draw basic network architecture for the internet connections.	02
	1.4	Provide general representation of Uniform Resource Locator.	02
	1.5	List different types of e-commerce.	02
	1.6	Give any two examples of more commonly used $B2B$ examples and best practice models.	02
	1.7	What is meant by dynamic pricing?	02
	1.8	"You cannot push a square peg through a round hole". What does this statement means?	02
	1.9	How does the social commerce help the retailers? List any 4 points.	02
	1.10	List the factors affecting environmental and organizational drivers of online selling diffusion (any four factors).	02

PART B

2	a	Web 2.0 characterized especially by the change from static web pages to dynamic or user generated content. How this technology advancement impacted e – commerce? Discuss with a suitable case study.	08
	b	Briefly explain the history of e-commerce by highlighting the chronological events occurred.	08
3	a	With the help of a neat block diagram, explain the basic architecture of internet networking.	08
	b	Explain the web components between the front end and back end activities of a web portal.	08
		OR	
4	a	How does transmission control protocol layer processing work? Use a neat block diagram to explain the mechanism of TCP layer processing.	10
	b	Explain different steps to be followed before a web page is designed to get a good end result.	06

5	a	"The existing legal systems in most developing countries are not sufficient to protect those engaged in e-commerce". Do you agree with above statement? Explain the common legal issues faced by many developing countries.	08
	b	With the suitable examples explain any four models of e-commerce.	08
		OR	
6	a	Briefly outline major players and corresponding requisites involved in a typical e – commerce transaction loop.	10
	b	Elaborate the typical components of e – business model.	06
7	a	With the help of a suitable case study explain the benefits of web enabled channels.	08
	b	What is the importance of dynamic pricing for e – commerce? Explain with the help of an example.	08
8	a	"Social commerce sometimes known to be effective only for impulse purchases". In evident to this statement, explain future of social commerce in a country like India, which has diversified cultural and economic background.	08
	b	Analyze the environmental and organizational drivers of online selling diffusion.	08

Enterprise Information Systems

VI Semester SEE Question Papers

Course codes: 16IM62

- Grade Improvement Nov 2020 (16IM62)
- Fast Track Oct 2020 (16IM62)
- Apr/May 2019 (16IM62)

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Grade Improvement Examinations Nov-2020

Industrial Engineering and Management

ENTERPRISE INFORMATION SYSTEMS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of statistical tables permitted.

PART A

1	1.1	List the four key board rationales for implementing *ERP* systems.	02
	1.2	What is meant by Electronic Data Interchange?	02
	1.3	Identify the three driving forces behind reengineering.	02
	1.4	What is Business Process Reengineering?	02
	1.5	Generally datamarts exist in three forms. List them	02
	1.6	What constitutes closed loop *MRP*?	02
	1.7	What is Business intelligence as defined by Gartner Group?	02
	1.8	Summarize the two broad ways in which *EIS* applications help in effectively delivering *SCM*.	02
	1.9	What is the meaning of 'Best of Breed' in the context of Enterprise system Development and Deployment?	02
	1.10	Data warehouse projects can fail for a number of reasons. But according to Wu.J, what is one of the most common reason?	02

PART B

2	a	What is Materials Requirement Planning? Tabulate the information needed and obtained in Material Requirement planning.	08
	b	EIS is not merely integration of diverse enterprise processes, but can also possess key characteristics to meet the requirements. In the light of above statement describe any four common features found in *EIS*.	08
3	a	Reengineering is concerned with questioning the fundamentals of what is done. Illustrate how risk increases with the scope of work involved. Also outline atleast five conditions under which Reengineering works best.	08
	b	With the help of an illustration, explain different levels of *BPR* attainment with respect to process quality design.	08

OR

4	a	"*BPR* has unfortunately been associated with "downsizing" or "rightsizing", both of which means laying off workers" Comment on the above statement.	08
	b	Furnish specific examples (two each) of how *IT* has changed business practices and consumer expectations.	08
5	a	Explain the evolution of business intelligence systems with the aid of an appropriate sketch.	08
	b	Describe the four stages in business intelligence development along with an illustration.	08
		OR	
6	a	'Datawarehouses are centre points for storage for many systems' In the light of above statement, elaborate on datawarehouse linkages with information and business intelligence, with an appropriate illustration.	08
	b	Business Intelligence includes two major components-the business intelligence platform and end-user tools. Justify your answer.	08
7	a	Tabulate the main differences between *EIS* and *SCM* against the following points of comparison: Comprehensiveness, sourcing tables, processing speed and managing of constraints.	08
	b	Web enabled` technologies have revolutionized the way businesses are conducted. They have a huge impact on *SCM*. Elaborate on the above statement.	08
8	a	Review the *USP* of nest generation of Enterprise software.	08
	b	Describe atleast four features of the next generation of *EIS* software.	08

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RV COLLEGE OF ENGINEERING®

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VI Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

ENTERPRISE INFORMATION SYSTEMS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	List any four common features of *EIS*.	02
	1.2	What re the key inputs to *MRP* system?	02
	1.3	Mention any two driving forces behind reengineering.	02
	1.4	What is business intelligence?	02
	1.5	List any four application areas in data mining.	02
	1.6	What is data integrity?	02
	1.7	What is product lifecycle management?	02
	1.8	How does *EIS* applications help in effective delivering of supply chain management.	02
	1.9	List any two new technologies which enhances *EIS* implementation.	02
	1.10	Mention any two features of next generation *EIS* software.	02

PART-B

2	a	Identify and explain four key rationales for implementing Enterprise Resource planning systems.	08
	b	Highlight the Pros and Cons of a typical Enterprise Information systems.	08
3	a	Draw a typical flow of inputs converted to outputs through a process, with the focus on measures of process effectiveness.	08
	b	Provide specific examples (two each) of how *IT* has changed business practices and consumer expectations.	08
		OR	
4	a	Analyze the various conditions under which reengineering works better.	08
	b	With the help of suitable examples highlight different levels of *BPR* attainment with respect to process quality design.	08
5	a	Establish a roadmap to embrace the evolution of business intelligence from simple data to support business decision making.	08
	b	With the help of suitable examples bring out the difference between data warehousing and data mining.	08

		OR	
6	a	Establish relationship of data warehouses and business intelligence.	08
	b	Mention the various applications of data mining.	08
7	a	Compare Enterprise Information systems and supply chain management based on several dimensions.	08
	b	Analyze the impact of web enabled enterprise information systems/ Enterprise resource planning on supply chain management.	08
8	a	Analyze the various features of Next generation of enterprise software.	08
	b	Explain few of the futuristic trends in the development of enterprise information systems.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

ENTERPRISE INFORMATION SYSTEMS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	List any four common features of EIS.	02
	1.2	Mention three $C's$ behind Reengineering process.	02
	1.3	What are the three key steps in Business Intelligence?	02
	1.4	List any two of the benefits of PLM.	02
	1.5	List the various components of a successful BI architecture.	02
	1.6	What are the different forms of data marts?	02
	1.7	Give four examples of data mining applications.	02
	1.8	Differentiate between $OLAP$ and $OLTP$.	02
	1.9	Draw a typical block diagram of MRP system.	02
	1.10	Expand $MRP-III$.	02

PART B

2	a	Highlight the Pros and cons of a typical EIS system.	08
	b	Explain the various reasons for implementing ERP.	08
3	a	Analyze the various conditions under which Reengineering works better.	08
	b	With the help of suitable examples display different levels of BPR attainment with respect to process quality design.	08
		OR	
4	a	Draw a typical flow of inputs converted to outputs through a process, with the focus on measures of process effectiveness.	08
	b	Highlight some of many processes involved in accomplishing operational functions within an organization.	08
5	a	Differentiate between Data ware house and Business intelligence.	08
	b	Explain the various stages of BI and Road map of BI.	08
		OR	

6	a	With the help of a block diagram explain the business intelligence system architecture.	10
	b	Mention the various applications of data mining.	06
7	a	Through an illustration, explain how the EIS/ERP and SCM can be synchronized?	08
	b	Compare EIS and SCM based on several dimensions.	08
8	a	Analyze the various features of next generation of EIS software.	08
	b	Explain few of the futuristic trends in the development of EIS.	08

Data Mining Technologies

VI Semester SEE Question Papers

Course codes: 16IM6C5 / 12IM7E5, 16IM6C5

- Fast Track Oct 2020 (16IM6C5 / 12IM7E5)
- Apr/May 2019 (16IM6C5)

16IM6C5 / 12IM7E5

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

DATA MINING TECHNOLOGIES (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

			02
1	1.1	Distinguish between 'descriptive' and 'predictive' data mining tasks.	02
	1.2	What is meant by 'topic model' in the context of data mining?	02
	1.3	Mention the three attributes considered for determining the data security.	02
	1.4	Identify the algorithm that uses prior knowledge for mining frequent item sets.	02
	1.5	Mention the two steps involved in data classification.	02
	1.6	Identify a heuristic for selecting the splitting criterion that separates a given data partition.	02
	1.7	What is meant by network planning?	02
	1.8	Mention the appropriate method used for classification to discover structural relationships within noisy data.	02
	1.9	Distinguish between Agglomerative and Divisive hierarchical clustering.	02
	1.10	Categorize the types of attributes that represent the features of a data object.	02

PART B

2	a	Review with suitable examples, several disciplines that strongly influence the development of data mining methods.	08
	b	Examine different approaches to data visualization.	08
3	a	Analyze the major steps involved in data preprocessing.	08
	b	Provide an overview of data reduction strategies.	08
		OR	
4	a	Explain different data smoothing techniques with suitable illustrations.	08
	b	In the real world data, tuples with missing values for some attributes are a common occurrence. Review the methods generally followed to fill in the missing values for the attribute in tuples.	08
5	a	The attribute selection measures provide a ranking for each attribute describing the given training tuples. In the above context, provide an overview of different attribute selection measures that are commonly used for building decision trees.	08

	b	Categorize the metrics for evaluating the performance of 'classifiers'. Also tabulate the measures and the corresponding formulae.	08
		OR	
6	a	Different ensemble methods are employed to improve classification accuracy. Provide a brief note on the same.	08
	b	The class imbalance problem occurs when the main class of interest is represented by only a few tuples. Review the strategies to address this problem.	08
7	a	Illustrate a simple Bayesian belief network for the variable 'Lung Cancer'.	08
	b	Distinguish between Eager Learners and Lazy Learners. Also explain the features of Case Based Reasoning.	08
8	a	Characterize different methods of clustering.	08
	b	Write short notes on the following: i) Visual data mining versus Audio data mining ii) Ubiquitous data mining versus Invisible data mining.	08

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

DATA MINING TECHNOLOGIES (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Write a note on Data Mining.	02
	1.2	What is data visualization?	02
	1.3	Identify the seminal algorithm for mining frequent item sets in data mining.	02
	1.4	What is meant by Association rule mining?	02
	1.5	What is dimension table?	02
	1.6	Classify different Bayes methods.	02
	1.7	What are Lazy Learners?	02
	1.8	What is the purpose of cluster analysis?	02
	1.9	List different partitioning methods	02
	1.10	Provide an overview on Density based methods.	02

PART B

2	a	Explain any two types of data used for mining.	05
	b	List the applications of data mining: Explain any two in detail	05
	c	What are Data objects & Attribute types? Provide examples.	06
3	a	Explain the four major tasks in Data Preprocessing.	08
	b	Explain the necessity of data integration in data mining.	08
		OR	
4	a	Explain the strategies, “Data transformation” & “Data discretization” in data mining.	08
	b	Briefly explain different methods of mining frequent patterns.	08
5	a	What are basic concepts of classification? Explain the two steps involved in classification.	08
	b	Explain the methodology of decision tree induction in data mining.	08
		OR	

6	a	Explain in detail the two methods of Bayes classification.	08
	b	Explain different methods of Rule based classification.	08
7	a	Explain Bayesian belief networks. Also write a note on training bayesian belief networks.	08
	b	Explain the two methods of classification using frequent patterns.	08
8	a	Explain any two partitioning methods in cluster analysis.	08
	b	Explain any four data mining applications with respect to research frontiers.	08

Financial Management

VI Semester SEE Question Papers

Course codes: 16IM6C4

- Apr/May 2019 (16IM6C4)

16IM6C4

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Examinations April/May-19

Industrial Engineering and Management

FINANCIAL MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.
3. Use of interest factor tables permitted. Tables for factors leading to discounting and compounding conversions permitted.

PART A

1	1.1	Mention any two main areas of decision making as relevant to the area of financial management.	01
	1.2	Write the formula for sinking fund factor with usual notations.	01
	1.3	Distinguish between regular annuity and deferred annuity.	01
	1.4	Mention any one reason as to why the concept of time value of money be considered for all calculations in financial decision making.	01
	1.5	Bring out any significant difference between an ordinary share and a preference share.	01
	1.6	What is the most appropriate measure of risk with respect to a single asset?	01
	1.7	Mention any one of the reasons that make capital budgeting decisions important.	01
	1.8	Write down the governing equation of the Capital Asset Pricing Model($CAPM$).	01
	1.9	Mention any one of the uses of liquidity ratios.	01
	1.10	Write down the equation for any one of the ratios related to capital structure (for leverage).	01
	1.11	Mention any one of the application of profitability ratio for financial analysis.	01
	1.12	Mention any one of the problems related to ratio analysis?	01
	1.13	What is the meaning of the rights issue of equity shares?	01
	1.14	Mention any one advantage of lease financing over hire purchase financing.	01
	1.15	Mention any two methods of venture capital financing.	01
	1.16	Highlight any one significant trend indicating development of venture capital in India.	01
	1.17	Graphically illustrate the cash to cash cycle as applied to working capital assessment.	01
	1.18	Why is inventory levels in the organization included in the working capital calculations?	01
	1.19	What is spontaneous financing in the context of working capital management.	01
	1.20	Distinguish between permanent working capital and temporary working capital.	01

PART B

2 a Discuss the three broad areas of financial decision making. 08

b Explain and illustrate the concept of yield to maturity. 04

c A $Rs.100/-$ per value bond bearing a coupon rate of 12 percent will mature after 7 years. What is the value of the bond if the discount rate is 14 percent? 12 percent? 04

3 a You are considering purchasing the equity stock of MVM company. The current price per share is $.10/-$. You expect the dividend a year hence to be $Rs.100/-$. You expect the price per share of MVM stock a year to have the following probability distribution.

Price a year hence	$Rs.10/-$	$Rs.11/-$	$Rs.12/-$
Probability	0.4	0.4	0.2

i) What is the expected price per share a year hence?

ii) What is the probability distribution of the rate of return on $MVM's$ equity stocks? 08

b A Machine purchased six years back for $Rs.1,50,000$ has been depreciated to a book value of $Rs.90,000/-$. It originally had a projected life of fifteen years and zero salvage value. A new machine will cost $Rs.2,50,000/-$ and result in a reduced operating cost of $Rs.30,000/-$ per year for the next nine years. The older machine can be sold for $Rs.50,000/-$. The machine also will be depreciated on a straight line method on nine year life with salvage value of $Rs.25,000/-$. The company's tax rate is 50% and cost of capital is 10%. Determine whether the old machine should be replaced. 08

OR

4 a Vardhman Limited's earnings and dividends have been growing at a rate of 18 percent per annum. This growth is expected to continue for 4 years . After that the growth rate will fall to 12 percent for the next 4 years. Thereafter the growth rate is expected to be 6 percent forever. If the last dividend per share was $Rs.2.00$ and the investors required rate of return on Vardhaman's equity is 15 percent, what is the intrinsic value per share? 08

b The cash flows associated with three projects P, Q and R are given below.

Year	*Net cash flow*		
	P	Q	R
0	(2000)	(2000)	(2000)
1	1,400	500	500
2	600	1,100	500
3	400	900	1600

Calculate the net present value of each project at discount rate of 0 percent, 5 percent, 10 percent, 15 percent, 25 percent and 30 percent. Plot the results on a graph paper and interpret. 08

5 a Explain the important profit margin ratios. 04

b Determine the sales (i.e revenues from operations) of a firm with the following financial data.

Current ratio	1:5
Acid test ratio	1:2
Current liabilities	8,00,000
Inventory turnover ratio	5 *times*

08

	c	What are percentage format financial statements? How are they useful?	04
		OR	
6	a	What is the importance of ratio analysis? Briefly discuss the importance of the following accounting ratios. i) Liquidity ratio ii) Debt equity ratio iii) Stock turnover rate iv) Ratio of debtors to turnover.	08
	b	Complete the balance sheet and sales data (fill in the blanks) of a firm with the following financial data. Debt/equity ratio= 0.6 Acid test ratio= 1.2 Total assets to turnover ratio = 1.5 Day's sales outstanding recievables in trade= $40 days$ cost of goods sold as a percentage of total revenue 80 percent inventory turnover ratio = 5 Balance sheet	

Equity capital	50,000
Retained earnings	60,000
Plant and equipment	– – – –
Inventories	– – – –
Trade recievables	– – – –
Cash and cash equivalents	– – – –
Revenue from operations	– – – –
other income	*NIL*

08

7	a	Define the following terms in relation to an equity share: Par value, issue price, book value and market value.	04
	b	What are the features of venture capital?	04
	c	The Hypothetical Manufacturers Ltd (HML) has taken a plant on lease, valued at $Rs.20$ crores. The lease arrangement is in the form of leveraged lease. HML is the equity participant and the Hypothetical Bank Limited (HBL) is the loan participant. They find the investment in the ratio of 2:8. The loan from HBL carries a fixed rate of interest of 19 percent, payable in 6 equated installments. The lease term is 6 years with lease rental payable annually in arrear. i) Compute the equated annual installment from the point of view of HBL. ii) If the lease rate is unknown and $HBL's$ pre-tax yield is 25 percent, what is the minimum lease rate that must be quoted?	08
8	a	The following annual figures relate to XYZ Co.	

Sales (at two month's credit)	*Rs* 3600,000/–
Materials consumed (suppliers extend two months credit)	*Rs* 9,00,000/–
Wages paid (Monthly in arrears)	*Rs* 7,20,000/–
Manufacturing expenses outstanding at the end of the year (cash expenses are paid month in arrear)	*Rs* 80,000/–
Total administrative expenses paid as above	2,40,000/–
Sales promotion expenses, paid quarterly in advance	1,20,000/–

	The company sells its products on a gross profit of 25% consisting depreciation as part of the cost of production. It keeps one month's stock each of raw materials and finished goods and a cash balance of $Rs.1{,}00{,}000/-$. Assuming 20% safety margin, work out the working capital requirements of the company on 'cash cost basis'. Ignore work in process.	10
b	Discuss the important forms of working capital advance given by banks. What is the meaning of an letter of credit arrangement?	06

Operations Management

VI Semester SEE Question Papers

Course codes: 12IM62, 12IM62 / 10IM53

- Fast Track Jul 2018 (12IM62)
- Apr/May 2018 (12IM62)
- Apr/May 2017 (12IM62)
- Fast Track Jul 2016 (12IM62 / 10IM53)
- Apr/May 2016 (12IM62)

12IM62

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VISemester B. E. Fast Track Examinations July-18

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

4 Answer FIVE full questions from Part B.

PART-A

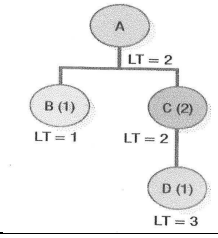
		Give one word for the following.	
1	1.1	A group of resources performing all or part of one or more processes.	01
	1.2	The means by which operations implements corporate strategy and helps to build a customer-driven firm.	01
	1.3	The strategy that specifies the pattern of decisions made in managing processes so that the processes will achieve their competitive priorities.	01
	1.4	Determination of the process type relative to the kinds of resources needed, how resources are partitioned between them, and their key characteristics.	01
	1.5	The degree to which a resource such as equipment, space, or the workforce is currently being used, and is measured as the ratio of average output rate to maximum capacity.	01
	1.6	It is what its capacity should be for some future time period to meet the demand of the firm's customers (external or internal), given the firm's desired capacity cushion.	01
	1.7	A factor that limits the performance of a system and restricts its output (capacity).	01
	1.8	The maximum time allowed for a job / work on a unit at each station in a line assembly.	01
	1.9	The repeated observations of demand for a service or product in their order of occurrence to form a pattern.	01
	1.10	An individual item or product that is held in inventory somewhere along the supply chain, such as in a distribution center.	01
	1.11	A common forecasting practice, whereby the forecast for the nest period $[F_{t+1}]$ equals the demand for the current period $[D_t]$	01
	1.12	It is a measure that indicates whether a method of forecasting is accurately predicting actual changes in demand.	01
	1.13	The process of making sure that demand and supply plans are in balance, from the aggregate level down to the short-term scheduling level.	01
	1.14	An additional cost involved in order to expedite past due orders.	01
	1.15	The cost of lost sales and the potential cost of loosing customer to a competitor.	01
	1.16	The summation of finish time and time since job arrived at workstation.	01
	1.17	A system that translates the master production schedule and other sources of demand into the requirements for all subassemblies and components needed to produce the required parent items.	01

	1.18	A record of all the components of an item, the parent-component relationships, and the usage quantities derived from engineering and process designs.	01
	1.19	A process that details how many end items will be produced within specified periods of time and breaks the sales and operations plan into specific product schedules.	01
	1.20	The difference between the customer orders already booked and the quantity that operations is planning to produce.	01

PART-B

2	a	"Hyundai Motors Private Limited could act as both manufacturing and service provider". Justify the statement with reasons.	06
	b	Consider Amazon, whose website enjoys millions of hits each day and puts customers in touch with millions of services and products. What are Amazon's competitive priorities and what should its operations strategy focus on?	10
		OR	
3	A	How do the process strategy decisions of Flipkart and Burger King differ, and how do their process structures affect customer satisfaction? Demonstrate using customer-contact matrix.	10
	b	Dr. Chad is an orthodontist. He estimates that adding two new chairs will increase fixed costs by $150,000, including the annual equivalent cost of the capital investment and the salary of one more technician. Each new patient is expected to bring in $ 3,000 per year in additional revenue, with variable costs estimated at $ 1,000 per patient. The two new chairs will allow Dr. Chad to expand his practice by as many as 200 patients annually. How many patients would have to be added for the new process to break even?	06
4	a	A young boy sets up a lemonade stand at the end of a College Street. Temperatures in the area climb at 40℃ during the summer. The intersection is near a major university and a large construction site. Briefly discuss to this young entrepreneur how his / her business might benefit from economies of scale.	06
	b	Identify an industry in which expansionist strategy has generally been followed by most firms in the past. Under which conditions will it be better for a firm to follow the wait-and-see strategy rather than expansionist strategy? Then identify a firm or an industry that has done so successfully.	10
		OR	
5	a	Take a process that you encounter on a daily basis, such as the lunch cafeteria, and identify the bottlenecks that limit the throughput of this process. Also examine the ways in which the bottlenecks can be relieved, keeping in mind – the theory of constraints.	08
	b	A company is setting up an assembly line to produce 192 units per 8 hour shift. The following table identifies the work elements, times, and immediate predecessors: i) Draw the precedence diagram. ii) What is the desired cycle time (in sec)? iii) What is the minimum number of stations required? iv) What is the balance delay of the system?	

		$Work\ Element$ | $Time\ (Sec)$ | $Immediate\ Predecessor(s)$ A | 40 | $None$ B | 80 | A C | 30 | D, E, F D | 25 | B E | 20 | B F | 15 | B G | 120 | A H | 145 | G I | 130 | H J | 115 | C, I $Total$ 720	08
6	a	Before using forecasting techniques, a manager must make two decisions about what to forecast and what type of forecasting technique to select for different items. Elucidate.	06
	b	Illustrate, with examples, all the steps involved in the judgment methods of forecasting.	10
		OR	
7	a	Chicken Palace periodically offers carryout five-piece chicken dinners at special prices. Let Y be the number of dinners sold and X be the price. Based on the historical observations and calculations in the following table, determine the regression equation, correlation coefficient, and coefficient of determination. How many dinners can Chicken Palace expert to sell at \$3.00 each? $Observation$ | $Price\ (X)$ | $Dinner\ Sold\ (Y)$ 1 | \$ 2.70 | 760 2 | \$ 3.50 | 510 3 | \$ 2.00 | 980 4 | \$ 4.20 | 250 5 | \$ 3.10 | 320 6 | \$ 4.05 | 480	06
	b	The Northville Post Office experiences a seasonal pattern of daily mail volume every week. The following data for two representative weeks are expressed in thousands of pieces of mail: Day | $Week\ 1$ | $Week\ 2$ $Sunday$ | 5 | 8 $Monday$ | 20 | 15 $Tuesday$ | 30 | 32 $Wednesday$ | 35 | 30 $Thursday$ | 49 | 45 $Friday$ | 70 | 70 $Saturday$ | 15 | 10 $Total$ | 224 | 210 i) Calculate a seasonal factor for each day of the week. ii) If the postmaster estimates, 230,000 pieces of mail to be sorted next week, forecast the volume for each day of the week.	10
8	a	Examine the stages of operations planning and scheduling for a manufacturing company.	10

	b	When does an organization opt for either chase or level strategy? Elaborate.	06
		OR	
9	a	Currently a consulting company has five jobs in its backlog. The time since the order was placed, processing time, and promised due dates are given in the following table: $Customer$ | $Time\ Since\ Order\ Arrived\ (days\ ago)$ | $Processing\ Time\ (days)$ | $Due\ Date\ (days\ from\ now)$ A | 15 | 25 | 29 B | 12 | 16 | 27 C | 5 | 14 | 68 D | 10 | 10 | 48 E | 0 | 12 | 80 Create a schedule using the EDD rule, calculating the average days past due and flow time. In this case, does EDD outperform the $FCFS$ rule?	08
	b	Consider the following two machines and six jobs flow shop-scheduling problem. Using Johnson's algorithm, obtain the optimal sequence and the makespan. $Job\ (1)$ | $Machine\ Number$: $Machine\ 1$ | $Machine\ 2$ 1 | 5 | 4 2 | 2 | 3 3 | 13 | 14 4 | 10 | 1 5 | 8 | 9 6 | 12 | 11	08
10	a	Consider a service provider that is in the delivery business, such as UPS or FedEx. How can the principles of MRP be useful to such a company?	06
	b	Consider a master production schedule for a major automobile company, such a Hyundai. Discuss the ways in which MPS is formulated.	10
		OR	
11	a	A seat assembly consists of a seat frame and a seat cushion. The seat frame consists of 4 seat frame boards. Draw the BOM (in the form of a tree structure) for 20 such seat assemblies.	04
	b	The MPS start quantities for product A calls for the assembly department to begin final assembly according to the following schedule: 100 units in week 2; 200 units in week 4; 120 units in week 6; 180 units in week 7; and 60 units in week 8. Develop a material requirements plan for the next 8 weeks for items B, C, and D. The BOM for A is shown in the following figure, and data from the inventory records are shown in the following table. **$Data\ Category$** | **B** | **C** | **D** $Lot-sizing\ rule$ | $POQ\ (P=1)$ | $L4L$ | $FOQ = 500\ units$ $Lead\ time\ (LT)$ | $1\ week$ | $2\ weeks$ | $3\ weeks$ $Scheduled\ receipts$ | $None$ | $200\ (week\ 1)$ | $None$ $Beginning\ (on-hand)\ inventory$ | 20 | 0 | 425 	12

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

8 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

9 Answer FIVE full questions from Part B.

PART A

1	1.1	Distinguish order winner from an order qualifier.	02
	1.2	List the three elements of product process matrix.	02
	1.3	Define utilization and write the formula to find the utilization in capacity planning.	02
	1.4	State atleast two key practical applications of the theory of constraints.	02
	1.5	Define tracking signal and write the formula to find tracking signal.	02
	1.6	With an example, distinguish between dependent variable and independent variable.	02
	1.7	Distinguish between MRP and $MRPII$.	02
	1.8	Give 2 level BOM for a ladder Back Chair.	02
	1.9	Distinguish between Backlogs and Backorders.	02
	1.10	State the two priority sequencing rules that are commonly used in practice.	02

PART B

2	a	What are the key capabilities that operations must develop to complete successfully in a market segment?	10
	b	How do the process strategies of e Bay and McDonald's differ, and how do their choices relate to customer introduced variability?	06
		OR	
3	a	With suitable examples, explain the four process choices of manufacturing process structuring.	10
	b	Define Break even analysis. A company manufactures diesel filters. Fixed cost of the company is Rs 5 lakhs and total variable costs are 40% of the sales. i) To gain a profit of Rs 50,000 what should be the magnitude of sales? ii) What is the profit gained for a sale of Rs 4,00,000?	06

4 a A copy center in an office building prepares bound reports for 2 clients. The center makes multiple copies of each report. The processing time to run, collate and bind each copy depends on among other factors, the number of pages. The center operates 250 days/yr, with one 8 hour shift. Management believes that a capacity cushion of 15% is best. It currently has 3 copy machines. Based on the following table of information, determine how many machines are needed at the copy center.

$Item$	$Client\ X$	$Client\ Y$
$Annual\ demand\ forecast(copies)$	2000	6000
$Standard\ processing\ time(hr/copy)$	0.5	0.7
$Average\ lot\ size(copies/report)$	20	30
$Standard\ setup\ time(hrs)$	0.25	0.4

06

b Explain the 7 principles of the Theory of constraints. 10

OR

5 a A company is setting up an assembly line to produce 192 units per 8 hr shift.

$Work\ element$	A	B	C	D	E	F	G	H	I	J
$Time(sec)$	40	80	30	25	20	15	120	145	130	115
$Predecessor(s)$	-	A	D,E,F	B	B	B	A	G	H	C,I

i) What is the desired cycle time?
ii) What is the minimum number of stations? 10
iii) What are the efficiency and balance delay?

b List and explain the steps involved in a systematic approach to long term capacity decisions. 06

6 a Explain the four judgment methods used to forecast. 08

b A lathe manufacturer had noted the sales invoices from January to August and noticed the sales during each month as $10, 11, 10, 11, 14, 15, 14$ and 15 units respectively. What would be his estimate of sales for September?

i) If he had followed 4 month weighted moving average(with weights $0.4, 0.3, 0.2$ & 0.1)
ii) If he has followed simple exponential smoothing with $\alpha = 0.2$ & his estimated sales for January is 11. 08

OR

7 a The forecast and the actual demand for the 6 months are given below. Compute the various forecast errors:

i) MAD
ii) $RSFE$
iii) Tracking signal
iv) MSE
v) MFE
vi) $MAPE$.

$Month$	$Actual\ Demand$	$Forecast\ Demand$
1	71	78
2	80	75
3	101	83

$Month$	$Actual\ Demand$	$Forecast\ Demand$
4	84	84
5	60	88
6	73	85

12

b Explain the patterns of demand. 04

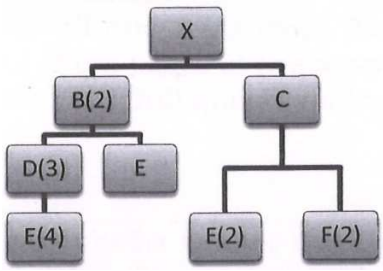
8 a Explain the time phased information contained in the inventory record. 10

b Item A is made from components B, C and D. Item B is a subassembly that requires 2 units of C and 1 unit of E. Item D is an intermediate item made from F. All other usage quantities are 2. Draw the BOM for item A. 06

OR

9 a What are the three major inputs for an MRP system? Explain with the help of a block diagram. 08

b Following is the problem of MRP. The product tree structure for end item 'X' is as follows:



i) For the product tree structure shown above find the quantities of B, C, D, E & F to assemble one 'X'.

ii) Determine the quantities of these components that will be required to assemble 200 '$X's$'. 08

10 a Explain scheduling strategies(any four). 08

b Find the best makespan using the CDS heuristic method for the following flow shop problem.

Job	M/C 1	M/C 2	M/C 3	M/C 4
1	4	3	7	8
2	3	7	2	5
3	1	2	4	7
4	3	4	3	2

08

OR

11 a Identify the performance measures that are important in selecting a schedule. 04

b Find the sequence that minimizes the total elapsed time required to complete the following tasks on two machines.

$Task$	A	B	C	D	E	F	G	H	I
$M/C\ I$	2	5	4	9	6	8	7	5	4
$M/C\ II$	6	8	7	4	3	9	3	8	11

12

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

10 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

11 Answer FIVE full questions from Part B.

PART A

1	1.1	Differentiate between process and an operation.	02
	1.2	State three principles concerning the implementation of process strategy.	02
	1.3	What is the relationship between utilization and capacity cushion?	02
	1.4	List all the steps that help managers make sound capacity decisions.	02
	1.5	What is line balancing?	02
	1.6	Draw a neat sketch to exhibit trend demand pattern with respect to quantity and time.	02
	1.7	State the two key decisions that are considered for making effective forecasts.	02
	1.8	Distinguish dependent demand from independent demand with an example.	02
	1.9	Differentiate between a backorder and a stockout.	02
	1.10	Differentiate between forward scheduling and backward scheduling.	02

PART B

2	a	Explain the differences between manufacturing and service organizations with suitable examples.	08
	b	Calculate the productivity for the following operations. i) Three employees process 600 insurance policies in a week. They work 8 hrs/day, 5 days/week. ii) A team of workers makes 400 units if a product which is sold in the market for $Rs.10$ each. The accounting department reports that for this job the actual costs are $Rs.400$ for labour, $Rs.1000$ for material and $Rs.300$ for overhead.	08
		OR	
3	a	What are the key capabilities that operations must develop to complete successfully in a market segment?	10
	b	Explain the three process strategy decisions.	06
4	a	List and explain the steps involved in a systematic approach to longterm capacity decisions.	10
	b	Define theory of constraints. How the firms operational measures relate to its financial measures.	06
		OR	

5 a A company is setting up an assembly line to produce 192 units/8hrs shift. The following table identifies the work elements, times and immediate predecessors.

Work element	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*	*I*	*J*
Time(sec)	40	80	30	25	20	15	120	145	130	115
Predecessors	-	*A*	*D, E, F*	*B*	*B*	*B*	*A*	*G*	*H*	*C, I*

i) What is the desired cycle time?
ii) What is the theoretical minimum number of stations?
iii) Use trial and error to work out a solution and show your solution on a precedence diagram.
iv) What are the efficiency and balance delay of the solution found? 10

b List the tools used for capacity planning and explain any one in detail. 06

6 a Explain the patterns of demand. 06

b A lathe manufacture had noted the sales invoices from Jan to Aug and noticed the sales during each month as 10,11,10,11,14,15,14,15 units respectively. What would be his estimate of sales for sep?
i) If he had followed 4 month weighted moving average (with wts. $0.4, 0.3, 0.2, 0.1$)
ii) If he had followed *SES* with $\propto = 0.2$ and his estimated sales for Jan is 11. 10

OR

7 a Explain any two judgement methods used to forecast. 06

b Alpha company has the following sales pattern. Compute the sales forecast for the year 10.

Year	1	2	3	4	5	6	7	8	9
Sales (Lakhs)	6	8	11	23	29	34	40	45	56

10

8 a Explain the time phased information contained in the inventory record. 08

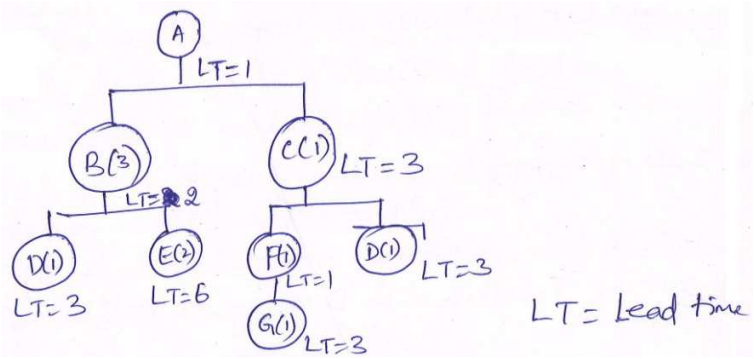
b A company has forecast the requirements for a component as shown in the table. The company maintains a safety stock of 50. Complete the *MRP*. Determine the inventory on hand at the end of period 12.

Order quantity $= 80$ *Lead time* $= 4$ *S.stock* $= 50$	1	2	3	4	5	6	7	8	9	10	11	12
Projected requirement	25	25	30	25	25	30	25	25	35	30	30	30
Receipts		80										
On hand at the end of period $= 75$												
Planned order volume												

08

OR

9 a What are the three major inputs for a *MRP* system? Explain with a block diagram. 10

	b	If there is no existing inventory and no scheduled receipts, how many units of items G, E and D must be purchased to produce five units if end item A?  Fig 9(b)	06
10	a	The following data is available with respect to 5 jobs: Job | A | B | C | D | E $Processing\ time$ | 4 | 17 | 14 | 9 | 11 $Due\ date$ | 6 | 20 | 18 | 12 | 12 Sequence the jobs using following Priority rules: i) First come first served ii) Earliest due date iii) Shortest processing time iv) Largest processing time v) Least slack. Compare $FCFS$ and SPT based on: x) Flow time y) Average number of jobs each day.	10
	b	Explain the types of schedulers. (any two)	06
		OR	
11		Consider the following problem in single machine scheduling with independent jobs. Job | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 t_j | 5 | 12 | 8 | 10 | 3 | 15 | 8 | 6 d_j | 10 | 16 | 11 | 16 | 6 | 25 | 12 | 14 w_j | 2 | 1 | 1 | 2 | 3 | 4 | 2 | 3 Obtain the optimal schedule for each of the following performance measure a) To minimize mean flow time. b) To minimize the maximum lateness. c) To minimize weighted mean flow time. Also compute $\bar{F}$, L_{max} and $\overline{F_w}$	16

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V / VI Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

4 Answer FIVE full questions from Part B.

PART-A

1	1.1	Define operations management.	01
	1.2	Differentiate between market segmentation and market needs assessment.	02
	1.3	Student tuition at a university is $Rs.100$ per semester credit hour. The state supplements revenue by matching student tuition, rupee for rupee. Average class size for a typical three credit course is 50 students. Labor costs are $Rs.4000/-$ class, material costs are $Rs.20$ per student per class and overhead costs are $Rs.25{,}000$ per class. Multifactor productivity ratio is _____.	02
	1.4	The unique resources and strengths that an organization's management considers when formulating strategy is known as _____.	01
	1.5	A strategy used by manufacturers that make products to customer specifications in low volume is known as _____.	01
	1.6	A plant manager just received marketing's latest forecasts of product A sales for the next year. The manager wants its production line to be designed to make 2400 products per week for at least the next 3 months. The plant will operate 40 hours per week. If total production time is 244 seconds. a) What should be the line's cycle time? b) What is the smallest number of workstations that the manager could hope for in designing the line for this cycle time?	02
	1.7	_____ method is a process of gaining consensus from a group of experts while maintaining their anonymity.	01
	1.8	a) Compute three week moving average forecast for the arrival of medical clinic patients in week 4.the numbers of arrival for the past three weeks were $Week$ | 1 | 2 | 3 $Patient\ arrivals$ | 400 | 380 | 411 b) If the actual number of patient/arrivals in week 4 is 415. What is the forecast error for week 4?	02
	1.9	Explain and define $CPFR$.	02
	1.10	Distinguish between Forward Scheduling and backward Scheduling.	02

	1.11	What is dependent demand and independent demand in the context of MRP?	02
	1.12	List any two examples for $FIFO$ and $LIFO$ rules of scheduling in manufacturing/service sectors.	02

PART-B

2	a	List and explain the challenges faced by operations manager in an organization to prepare for competition with global players.	08
	b	A haircut organization operates with eight haircut shops on a $250\ days/year$, $8\ hours/day$ basis. They charge $Rs.50$ for a haircut. One shop has annual fixed costs of $Rs.3{,}00{,}000/-$ and variable costs estimated at $Rs.20$ per customer. i) What is the contribution per customer? ii) How many customers per hour must the shop average in order to break even?	08
		OR	
3	a	What are the major process decisions, operations managers must consider while dealing with processes? Explain.	10
	b	Distinguish between Jobs, Batch and line processes.	06
4	a	Explain four step procedure that help managers to make capacity related decisions?	09
	b	State the seven key principles of the theory of constraints.	07
		OR	
5	a	Illustrate the two strategies for expanding capacity.	08
	b	XYZ fast foods have enjoyed considerable success by using call box ordering stations and multilane channels. It can serve 32 cars per hour in one lane of its drive thrus. Now they are planning for a new drive thru facility in a city. Market research data suggests it may have the peak hourly demands estimated below.	

	Number of cars per hour				
	$0 < 40$	$40 < 80$	$80 < 120$	$120 < 160$	$160 < 200$
Probability (% chance)	0.10	0.35	0.4	0.13	0.02
cumulative probability	0.10	0.45	0.85	0.98	1.00

		the operations manager is considering two alternatives for deciding how much capacity to install. What capacity is required to meet 85% of the estimated peak hourly demands. accommodate 110% of the estimated average demand plus a 25% allowance of growth?	08
6	a	State the conditions under which Delphi method is used and at least three disadvantages resulting from its use.	04

	b	A lathe manufacturer had noted the sales invoices from January to August and noticed the sales during each month as $10, 11, 10, 11, 14, 15, 14$ and 15 units respectively. What would be his estimate of sales for September? i) If he has followed 4 month weighted moving average (with weights $0.4, 0.3, 0.2$ and 0.1). ii) If he has followed simple exponential smoothing with $\propto = 0.2$ and his estimated sales for January is 11.	12
		OR	
7	a	What is the difference between qualitative and quantitative techniques of forecasting? When is a quantitative model more appropriate?	06
	b	The following table shows the actual sales of upholstered chairs for a furniture manufacturer and the forecasts made for each of the last eight months. Calculate CFE, MSE, MAD and $MAPE$ for this product.	10

Month	***Demand***	***Forecast***
1	200	225
2	240	220
3	300	285
4	270	290
5	230	250
6	260	240
7	210	250
8	275	240

8	a	Identify the nature and importance of the information of that MRP can provide to each of the functional areas. Explain the processing logic of MRP with a suitable example. What are the limitations of MRP?	10
	b	Explain the key inputs of an MRP system with the help of block diagram.	06
		OR	
9	a	How can the concept of dependent demand be useful to service providers?	06
	b	Enumerate the key functions of Master Production Schedule.	04
	c	List and explain the strategies used to absorb fluctuations in demand.	06
10	a	Explain the term scheduling. What are its objectives?	05
	b	How is the sales and operations planning process similar to the forecasting process?	05
	c	Explain the key differences between the FOQ, POQ and $L4L$ lot sizing rules.	06
		OR	

11	a	The following data pertains to manufacturing activity: Job, j | $Processing\ time, t_j$ | $Weight\ W_j$ 1 | 7 | 1 2 | 18 | 2 3 | 6 | 1 4 | 8 | 2 5 | 12 | 3 Determine the sequence which will minimize the weighted mean flow time of the problem. Also find the weighted mean flow time.	10
	b	What does an advanced planning system do and what are the functions of its major components?	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

OPERATIONS MANAGEMENT

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

5 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

6 Answer FIVE full questions from Part B.

PART-A

1	1.1	A conversion process that includes manufacturing yields _____.	01
	1.2	Decision tree analysis is helpful in solving the _____ problem of facility expansion.	01
	1.3	The capability is the rate of _____ of a facility.	01
	1.4	The _____ package is used for computerized 8inventory and scheduling activities.	01
	1.5	Master scheduling follows _____ in production activities.	01
	1.6	_____ scheduling is an approach where the customer orders are processed as late as possible so that they are finished and delivered exactly on their due date.	01
	1.7	If the sequencing of jobs is done based on increasing order of due dates, then it is known as _____ rule.	01
	1.8	Given the following data: $Week$ | 1 | 2 | 3 $Demand$ | 40 | 36 | 41 a) The actual demand in week 4 is 45. Forecast for week 4 by three week moving average is _____. b) The forecast error for week 4 is _____.	02
	1.9	The amount of time job spends in the system is _____.	01
	1.10	List the process views of a supply chain.	01
	1.11	A typical supply chain should consist of _____.	02
	1.12	The four different types of errors that are generally computed in forecasting are:	02
	1.13	Write the formula for system capacity and system efficiency.	02
	1.14	List the basic inputs for MRP.	02
	1.15	Describe scheduling.	01

PART-B

2	a	Explain briefly the process views of a supply chain.	08
	b	Explain briefly the three steps top achieve the strategic fit.	08
		OR	
3	a	Differentiate between manufacturing and service operation.	06
	b	List the different strategies in operation management for strategic management.	10
4	a	Explain design capacity and system capacity.	06
	b	Explain briefly the capacity planning strategies.	10
		OR	
5	a	Briefly explain the theory of constraints.	06
	b	List and explain the various models used in decision making related to capacity planning.	10
6	a	List the types of forecasting in decision making.	08
	b	Explain the different demand patterns.	08
		OR	
7	a	Alpha company has the following sales pattern. Compute the sales forecast for the year 10 using linear regression.	

$Year$	1	2	3	4	5	6	7	8	9
$Sales\ (in\ lakhs)$	6	8	11	23	29	34	40	45	56

10

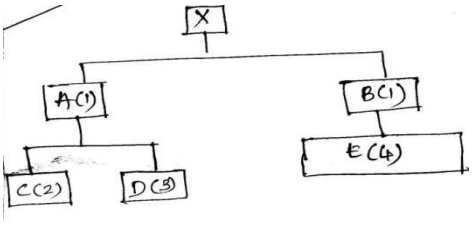
	b	Explain the Delphi method of forecasting.	06
8	a	Explain the Master production scheduler with a flow chart showing relationship of MPS to other manufacturing planning and control activities.	08
	b	Explain materials requirements planning support your answer with an examples.	08
		OR	
9	a	Given the forecast requirements for end item y, complete MRP. Note that a scheduled receipt of 60 units is due in period 2 and a safety stock of 25 is to be maintained. Lead time $= 2$ weeks and order quantity is 60 units.	

$Week$	1	2	3	4	5	6	7	8	9	10
$Gross\ requirement$	20	20	20	30	20	20	20	25	20	35
$Scheduled/planned\ receipts$		60								
$On\ hand/at\ end\ period\ (50)$										
$Planned\ order\ release$										

10

	b	With the given product structure tree and inventory compute the net requirements for A, B, C, D and E to produce 50 units of x.	

Components	*A*	*B*	*C*	*D*	*E*
Inventory or hands on order	20	10	15	30	100



06

10	a	Given the following information set the aggregate planning problem as a transportation problem.	

Demand / Capacity	*Period* 1 500	2 700	3 750
Regular production	500	500	500
Overtime	50	50	50
Subcontract	120	120	120

		Beginning inventory = 100 units (In pounds), Regular production = 60/unit(In pounds), overtime = 80/unit(In pounds), Subcontracting = 90 per unit(In pounds), inventory carrying cost = 1 per unit period(In pounds). Back ordering is not permitted. Final inventory = 100 units.	12
	b	What is scheduling? Explain with respect to single machine concept.	04

OR

11	a	A machine operator has to perform 3 operations turning, threading and knurling on number of jobs in that order. Determine the optimal schedule and the idle times for the 3 machines.	

Job	*M/C* 1 *Turning (min)*	*M/C* 2 *Threading (min)*	*M/C* 3 *Knurling (min)*
1	3	8	13
2	12	6	14
3	5	4	9
4	2	6	12
5	9	3	8
6	11	1	13

10

	b	Consider the following single machine scheduling problem.	

Job (j)	1	2	3	4	5
Processing time (ti) hrs	4	5	8	4	10
Due date (di)	5	8	12	6	11

		Determine maximum lateness using *EDD* rule.	06

Simulation Modeling and Analysis

VI Semester SEE Question Papers

Course codes: 12IM64

- Fast Track Jul 2018 (12IM64)
- Apr/May 2018 (12IM64)
- Fast Track Jul 2017 (12IM64)
- Apr/May 2017 (12IM64)
- Apr/May 2016 (12IM64)

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VISemester B. E. ExaminationsFast Track July-18

Industrial Engineering and Management

SIMULATION MODELING AND ANALYSIS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2 Answer FIVE full questions from Part B.
3 Use of statistical tables permitted.

PART A

1	1.1	The collection of variables to describe the system at any time, relative to the objective of study is knows as _____.	01
	1.2	The activities and events occurring within a system are called as _____.	01
	1.3	_____ is the determination that a model is an accurate representation of the real system.	01
	1.4	Define system.	02
	1.5	When $C = 0$ in a linear congruential recursive relationship, it is known as _____.	01
	1.6	A sequence of random number generated must have two important statistical properties _____ and _____.	02
	1.7	The random number when generated by some deterministic process but they qualify the pre-determined, Statistics test for randomness are _____.	01
	1.8	Direct transformation method is used to generate random variate for _____ and _____ distribution.	01
	1.9	The cumulative distribution function$F(x)$ of an exponentially distributed random variable X is given by _____ for $X \geq 0$.	01
	1.10	The uniform $(20, 38)$ random variate corresponding to a random number 0.2451 is _____.	02
	1.11	Quantile – Quantile plots are plotted with _____ on x axis and _____ on y axis.	02
	1.12	_____ test provides guidance for evaluating the suitability of input model.	01
	1.13	_____ software of deneb robotics is used for robotic simulation and programming and _____ for ergonomic analysis.	02
	1.14	Arena is built on the _____ simulation language.	01
	1.15	Steady state simulation is sued to study the steady state behavior of a _____ system.	01

PART B

2	a	Explain with help of flow chart the various step involved in a simulation study.	08
	b	List the area of application of simulation. Explain any three of them.	08

OR

3 a A transport cooperation operates 15 long distance buses each with 1 driver. The cooperation has a policy of keeping 3 reserve drivers on call to replace those who report sick. If a driver is not available, the bus trip is cancelled. The probability distribution for daily number of sick driver is as follows.

Number of sick	0	1	2	3	4	5
Probability	0.17	0.25	0.23	0.15	0.10	0.10

Use montecarlo to determine the utilization of reserve drier and also the probability that at least one trip is cancelled. Use following random numbers: $91, 14, 23, 51, 84, 43, 92, 19, 49, 37, 65, 95, 89, 54, 45$. 08

b Explain the different types of simulation models. 08

4 a Explain the steps in time advance event scheduling algorithm with help of example. 08

b Determine whether there is an excessive number of runs above and below the mean for the sequence of numbers given in the following table. Use $\alpha = 0.05$

0.41	0.68	0.89	0.94	0.74	0.91	0.55	0.62	0.36	0.27
0.19	0.72	0.75	0.08	0.54	0.02	0.01	0.36	0.16	0.28
0.18	0.01	0.95	0.69	0.18	0.47	0.23	0.32	0.82	0.53
0.31	0.42	0.73	0.04	0.83	0.45	0.13	0.57	0.63	0.29

08

OR

5 a Given the following sequence of numbers, test whether 2nd, 9th and 16th number in sequence are auto-correlated where $\alpha = 0.05$

0.30	0.48	0.36	0.01	0.54	0.34	0.96	0.06	0.61	0.85
0.48	0.86	0.14	0.86	0.89	0.37	0.49	0.60	0.04	0.83
0.42	0.83	0.37	0.21	0.90	0.89	0.91	0.79	0.57	0.99
0.95	0.27	0.41	0.81	0.96	0.31	0.09	0.06	0.23	0.77
0.73	0.47	0.13	0.55	0.11	0.75	0.36	0.25	0.23	0.72
0.60	0.84	0.70	0.30	0.26	0.38	0.05	0.19	0.73	0.44

08

b The sequence of number $0.54, 0.73, 0.98, 0.11$ & 0.68 has been generated. Use the Kolmogorov – Smirnov test with $\alpha = 0.05$. To determine if the hypothesis that the number are uniformly distributed on the interval $(0, 1)$ can be rejected. 08

6 a Develop a generator for a triangular distribution with range $(2, 6)$ and mode at $x = 3$. Use $R = 0.36, 0.38, 0.89, 0.01, 0.24, 0.18$. 10

b Explain the inverse transformation technique for generation of a random variable, sampling from exponential distribution. 06

OR

7 a Lead times have been found to be exponentially distributed with mean 3.7 day generate 5 random mean times with the distribution $0.931, 0.417, 0.612, 0.079, 0.826$. 08

b Explain the factor that needs to be considered in optimization via simulation. 08

8	a	Explain the three step approach formulated by Nayler and tinger to aid in validation process.	08
	b	Records pertaining to the monthly number of job related injuries at an underground coal mine were being studied by a government agency. The value for the part 100 months were as follows:	

Injuries/month	0	1	2	3	4	5	6
Frequency of occurrence	35	40	13	6	4	1	1

		Apply the chi-square test to these data to test the hypothesis that the underlying distribution is Poisson. Use a level of significance of $\alpha = 0.05$.	08
		OR	
9	a	List the steps in the development of useful model of input data.	08
	b	Consider the following frequency table of observation on the random variable X	

Value	0	1	2	3	4
Observed frequency	24	30	31	4	4

		Based on these 100 observations, is a Poisson distribution with a mean of 1.2 an appropriate model? Perform a goodness of fit procedure with $\alpha = 0.05$.	08
10	a	Explain the factors to be considered while selecting a simulation software package.	10
	b	Differentiate between terminating and non-terminating simulation with help of an example.	06
		OR	
11	a	Explain the way by which the initialization bias in the output estimate canbe reduced in a steady state simulation.	08
	b	Explain the trends in simulation software.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

SIMULATION MODELING AND ANALYSIS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

12 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

13 Answer FIVE full questions from Part B.

14 Use of statistical tables permitted.

PART A

1	1.1	The collection of entities that interact together over time to accomplish one or more goals is known as _____.	01
	1.2	A system is often affected by changes occurring outside the system. Such changes are said to occur in the _____.	01
	1.3	Simulation models that contain no random variable are classified as _____.	01
	1.4	A queuing system is described by its _____ population.	01
	1.5	_____ a variable represents simulation time.	01
	1.6	Changing simulation settings like advancing the clock and building new snapshot is called _____.	01
	1.7	_____ method produces a sequence of integers between zero and m-1.	01
	1.8	_____ tests are concerned with the testing of dependence between numbers in a sequence.	01
	1.9	The method of generating external arrival stream in simulation is called _____.	
	1.10	The important part of model assumption used in the validation of simulation is called _____.	01
	1.11	The objective of _____ simulation is to study long run behavior of the system.	01
	1.12	Sample mean and sample variance are used to estimate the parameters of a _____.	01
	1.13	The properties of random numbers are _____ and _____.	01
	1.14	In multiplicative congruential method the value of C is _____.	01
	1.15	Stochastic models in simulation depend upon _____.	01
	1.16	For evaluating the stability of a potential input model _____ test provides helpful guidance.	01
	1.17	_____ Test is used to check the frequency with which certain digits are repeated in a series of numbers.	01
	1.18	_____ simulation software is offered by the Lanner group.	01
	1.19	*ARENA* comes with the _____ analyzer and _____ analyzer.	01
	1.20	_____ Simulation software is offered by Deneb Robotics.	01

PART B

2 a Define simulation. What are the advantages and disadvantages of simulation? 05

b Explain the steps involved in a simulation study. 05

c A small grocery store has only one checkout counter. Customers arrive at the checkout counter at random from 1 to 8 minutes apart. Each possible value of inter arrival time has the same probability of occurrence. The service times vary from 1 to 6 months with the probability as given below:

Service –time distribution

Service time (minutes)	*Probability*
1	$0-10$
2	$0-20$
3	$0-30$
4	$0-25$
5	$0-10$
6	0.05

The problem is to analyze the system by simulating the arrival and service of 5 customers.

Random number for IAT: $913, 727, 15, 948, 309$

Random number for service time: $84, 10, 74, 53, 17$

Compute:

i) Average service time

ii) Probability of server being idle

iii) Average time customer spends in the system. 06

OR

3 a Explain simulation of inventory system. 08

b Given A, B and C, three independent random variables, variable A is normally distributed with $\mu = 100$ and $\sigma^2 = 400$. Variable B is a discrete uniformly distributed with probability distribution given by $p(b) = 1/5$ with $b = 0,1,2,3$ and 4. Variable c is distributed in accordance with the following table:

Value of C	*Probability*
10	0.10
20	0.25
30	0.50
40	0.15

Use simulation to estimate the mean of a new variable D, defined as $D = (A - 25B)/2C$

Use a sample size 10

Random normal number for A : $-0.84. 1.03, 0.16, 2.04, -1.34$

Random number for B : $0.24, 0.83, 0.64, 0.95, 0.06$

Random number for A: $0.18, 0.89, 0.38, 0.64, 0.04$. 08

4 a Explain the basic concepts in discrete event simulation. 08

b List the properties of random numbers. 04

c Use the linear congruential method to generate a sequence of random numbers with $X_o = 27, a = 17, c = 43$ and $m = 100$ 04

OR

5	a	Define Time-advance algorithm. Briefly explain the steps in the algorithm.	08
	b	Perform a test of uniformity using the Kolmogorov Smirnov test with a level of significance α of 0.05 for the random numbers generated. $0.44, 0.81, 0.14, 0.05, 0.93$.	08
6	a	Generate random variate for the exponential distribution that has pdf given by: $f(x) = \begin{cases} \lambda e^{-\lambda x} & , x > 0 \\ 0 & , x < 0 \end{cases}$ Use inverse transform technique.	08
	b	Generate random variate for triangular distribution which has pdf $f(x) = \begin{cases} x & , \quad 0 \le x \le 1 \\ 2 - x & , \quad 1 < x \le 2 \\ 0 & , \quad otherwise \end{cases}$ Use inverse transform technique.	08
		OR	
7	a	Apply A_R method for gamma distribution with mean 2.2 minutes and variance $2 - 10$ minutes. Generate random variates using the random numbers: $R_1 = 0.832, 0.434$ $R_2 = 0.021, 0.716$	08
	b	Briefly explain the meaning of optimization via simulation.	04
	c	Discuss in brief the steps in random search algorithm.	04
8	a	Explain the steps involved in the development of useful model of input data.	08
	b	The values for the past 1`00 months of records pertaining to the monthly number of job related injuries are as follows: $Injuries/month$ | $Frequency\ of\ occurrence$ 0 | 35 1 | 40 2 | 13 3 | 6 4 | 4 5 | 1 6 | 1 Apply chi-square test to the data to test the hypothesis that the underlying distribution is Poisson. Use a level of significance of $\alpha = 0.05$.	08
		OR	
9	a	Explain model building, verification and validation of simulation models.	08
	b	Explain the 3-step approach formulated by Naylor and Finger to aid in the validation process.	08
10	a	Explain the measures of performance and their estimation for output analysis of simulation models.	08
	b	Briefly explain output analysis of steady state simulation.	08
		OR	
11	a	List and explain the types of simulation with respect to output analysis.	08
	b	Explain briefly trends in simulation software.	04
	c	Explain the features if $ARENA$ simulation software.	04

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

SIMULATION MODELING AND ANALYSIS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

5 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
6 Answer FIVE full questions from Part B.
7 Use of statistical tables permitted.

PART-A

1	1.1	In _____ type of model one or more input variables and output variables are random in nature.	01
	1.2	Random numbers generated using spreadsheet are called _____.	01
	1.3	Differentiate between verification and validation?	02
	1.4	Consider the following sequence generated by tossing a coin 10 times: *HTTHHTTTHT*. In this example, length of the run is _____.	01
	1.5	_____ test is used to determine the significance of interval between recurrences of same digit.	01
	1.6	Random numbers have two statistical properties namely _____ and _____.	02
	1.7	Random variate for uniform distribution is given by _____.	01
	1.8	The _____ method generates random variates by adding together two or more random variables to obtain a new random variable with the desired distribution.	01
	1.9	List the standard compromises optimization via simulation makes due to existence of sampling variability.	02
	1.10	What is calibration?	01
	1.11	What is the meaning of standard statistic 'current contents' as applied to a system having items of some kind flowing through it?	01
	1.12	List the broad steps in model building.	02
	1.13	Non-Terminating simulation is alos known as _____ simulation.	01
	1.14	Optimization tool 'SimRunner' is developed by simulation software named _____.	01
	1.15	What are the Common characteristics of simulation packages?	02

PART-B

2 a What are disadvantages of simulation? How these disadvantages can be offset? 08

b XYZ spare parts company wishes to determine the level of stock it should carry for items in its range. Demand is not certain and there is a lead time for stock replenishment. For one item X, the following information is obtained.

$Demand\ (Units/day)$	3	4	5	6	7
$Probability$	0.10	0.20	0.30	0.30	0.10
$Carrying\ Cost\ (Per\ unit\ Per\ day)$	$Rs.2$				
$Ordering\ Cost\ (Per\ order)$	$Rs.50$				
$Lead\ Time\ for\ Replenishment$	$3\ days$				

Stock on hand at the beginning of the simulation exercise was 20 units. Carry out a simulation run over a period of 10 days with the objective of evaluating the inventory rule: Order 15 units when present inventory plus outstanding falls below 15 units. The sequence of random numbers to be used is: 0,9,1,1,5,1,8,6,3,5,7,1,1,2,9 using the first number for day one. Calculate total cost of operating this inventory rule for 10 days. 08

OR

3 a Name entities, attributes, activities, events and state variables for the following systems:

i) A small appliance repair shop
ii) A cafeteria
iii) A grocery store
iv) A fast-food restaurant. 08

b A company manufactures 30 units per day. The sale of these items depends upon demand which has the following distribution:

$Sale\ (Units)$	$Probability$
27	0.10
28	0.15
29	0.20
30	0.35
31	0.15
32	0.05

The production cost and sale price of each unit are $Rs.40$ and $Rs.50$ respectively. Any unsold product is to be disposed off at a loss of $Rs.15$ per unit. There is a penalty of $Rs.5$ per unit if the demand is not met. Using the following random numbers, estimate the total profit/loss for the company for the next 10 days.

Use the Random Numbers: $10, 99, 65, 99, 95, 01, 79, 11, 16$ and 20. 08

4 a What is world view? Explain three phases of activity scanning approach. 08

b Given the following sequence of numbers, test whether $4^{th}, 9^{th}, 14^{th}$ number in the sequence are auto correlated, where $\alpha = 0.05$.

0.41 0.68 0.89 0.94 0.74 0.91 0.55 0.62 0.36 0.27
0.19 0.72 0.75 0.08 0.54 0.02 0.01 0.36 0.16 0.28
0.18 0.01 0.95 0.69 0.18 0.47 0.23 0.32 0.82 0.53
0.31 0.42 0.73 0.04 0.83 0.45 0.13 0.57 0.63 0.29 08

		OR	
5	a	Provide the detailed flow chart of a typical arrival event and a departure event in a single channel queuing system.	08
	b	Based on Runs up and Runs down, determine whether the following sequence of 50 numbers is such that the hypothesis of independence can be rejected where $\alpha = 0.05$. $0.99\ 0.17\ 0.99\ 0.46\ 0.05\ 0.66\ 0.10\ 0.42\ 0.18\ 0.49$ $0.37\ 0.51\ 0.54\ 0.01\ 0.81\ 0.28\ 0.69\ 0.34\ 0.75\ 0.49$ $0.72\ 0.43\ 0.56\ 0.97\ 0.30\ 0.94\ 0.96\ 0.58\ 0.73\ 0.05$ $0.06\ 0.39\ 0.84\ 0.24\ 0.40\ 0.64\ 0.40\ 0.19\ 0.79\ 0.62$ $0.18\ 0.26\ 0.97\ 0.88\ 0.64\ 0.47\ 0.60\ 0.11\ 0.29\ 0.78$	08
6	a	What is random variate? How would you generate random variates for the exponential distribution?	06
	b	A machine is taken out of production if it fails or after 5 hours, which ever comes first. By running similar machines until failure, it has been found that time to failure, X has the Weibull distribution with $\alpha = 8, \beta = 0.75$ and $\gamma = 0$. Thus, the machine is taken out of production can be represented as $Y = Min\{x, 5\}$. Develop a step-by=step procedure for generating Y. use $R_i = 0.13;\ 0.04;\ 0.65;\ 0.79;\ 0.76$.	10
		OR	
7	a	Write the algorithm for random search.	06
	b	Develop a generator for a triangular distribution with range $(1, 10)$ and mode at $x = 4$. $R = 0.66, 0.38, 0.64, 0.87, 0.52, 0.12, 0.18, 0.22, 0.14, 0.27, 0.08$	10
8	a	Daily demands for transmission overhaul kits for the $D-3$ dragline were maintained by Earth moving Tractor Company with the following results. $0\ 2\ 0\ 0\ 0$ $1\ 0\ 1\ 1\ 1$ $0\ 1\ 0\ 0\ 0$ $2\ 0\ 1\ 0\ 1$ $0\ 1\ 0\ 0\ 2$ $1\ 0\ 1\ 0\ 0$ $0\ 0\ 0\ 0\ 0$ $1\ 0\ 1\ 0\ 1$ $0\ 0\ 3\ 0\ 1$ $1\ 0\ 0\ 0\ 0$ How are the daily demands distributed? Develop and test an appropriate model.	10
	b	Explain the methods available for obtaining information when data is not available.	06
		OR	

9	a	Consider the following frequency table of observations on the random variables X: $Values:$ 0 1 2 3 4 $Frequency$: 4 21 10 13 2 Based on these 50 observations, is a binomial distribution with $n = 6$ and $p = 0.25$ an appropriate model? Perform a goodness-of-fit procedure with $\alpha = 0.05$.	09
	b	Site an example where lead time and demand are not independent random variables. Develop a suitable model for such a case.	07
10	a	Differentiate between terminating and steady state simulation with respect to output analysis with an example.	08
	b	Mention some important points which you would consider in selecting simulation software.	05
	c	Explain the features of Arena's output analyser.	03
		OR	
11	a	Explain briefly point estimate and interval estimate as measures of performance.	08
	b	List the trends in simulation software.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

SIMULATION MODELING AND ANALYSIS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

12 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

13 Answer FIVE full questions from Part B.

14 Use of statistical tables permitted.

PART A

1	1.1	Generation of artificial history of the system is called _____.	01
	1.2	Stochastic models in simulation depend upon _____.	01
	1.3	In a simulation study two types of documentation are _____ and _____.	01
	1.4	If there is already one customer in the queue and one customer in the system who is leaving, then there will be _____ number of customers in the queue at the time of the next imminent event of arrival.	01
	1.5	Random numbers are distributed _____ and independently on the interval _____.	01
	1.6	The probability distribution of a sum of two or more independent random variables is called a _____ of the distributions of the original variables.	01
	1.7	State the suggested estimators for the following distribution which are frequently used in simulation a) Exponential b) Poisson c) Gamma.	02
	1.8	_____ event is the immediate next event occurring in the system simulation.	01
	1.9	State two examples of dependence between random numbers.	01
	1.10	In a triangular distribution, if $a = 0, b = 2$ and $c = 4$, mean, mode and median are _____, _____ and _____.	02
	1.11	What do you mean by replication of runs?	01
	1.12	List the goals of validation process.	02
	1.13	_____ and _____ are two prominent examples of robust heuristics which can be applied to problems with mixed types of decision variables.	02
	1.14	Whether a simulation is considered to be terminating or not depends on _____ and the _____.	02
	1.15	_____simulation occurs when simulation models are placed in other software.	01

PART B

2 a Assuming suitable example describe the steps involved in conduction of a simulation study. 08

b Consider a continuously operating job shop. Inter arrival times of jobs are distributed as follows:

$Time\ between\ arrivals(hrs)$	0	1	2	3
$Probability$	0.23	0.37	0.28	0.12

Processing times for jobs are normally distributed with mean 50 minutes and standard deviation 4 minutes. Construct a simulation table, and perform a simulation for 8 new customers. Assume that when the simulation begins there is one job being processed (scheduled to be completed in 20 minutes) and there is one job with a 15 minutes processing time in the queue. Use the following random numbers for inter arrival times and service time respectively- $79, 39, 7, 17, 91, 99, 80, 57$ and $448, 326, 51, 73, 81, 648, 924, 267$

i) What is the average time in the queue for the 8 new jobs?
ii) What is the average processing time of the 8 new jobs?
iii) What is the maximum time in the system for 8 new jobs? 08

OR

3 a Distinguish between the following , with examples:
i) Exogenous Activities and Endogenous activities.
ii) Deterministic and stochastic models.
iii) Discrete system and continuous system. 06

b Given A, B and C as three independent random variables. Variable A is normally distributed with mean $= 100$ and variance $= 400$. Variable B is discrete uniformly distributed with probability distribution given by $P(b) = \frac{1}{5}$, with $b = 0,1,2,3$ and 4. Variable C is distributed in accordance with the following table:

$Value\ of\ C$	10	20	30	40
$Probability$	0.10	0.25	0.50	0.15

Use simulation to estimate the mean of a new variable D, defined as $D = (A - 25B)/2C$.
Use a sample size of 5.
Random normal numbers for A: $-0.84, 1.03, 0.16, 2.04, -1.34$
Random normal numbers for B: $0.24, 0.83, 0.54, 0.95, 0.06$
Random normal numbers for C: $0.18, 0.89, 0.38, 0.64, 0.04$ 10

4 a List the random number tests used in practice. The sequence of numbers $0.43, 0.70, 0.60, 0.38$ and 0.89 has been generated. Use the Kolmogorov – Smirnov test with significance level of 0.05 to determine if the hypothesis that the numbers are uniformly distributed can be rejected. 06

b Based on the length of runs above and below the mean test, determine whether the following sequence of 60 two digit random numbers is such that the hypothesis of independence can be rejected with a level of significance , $\propto = 0.05$

		0.30	0.48	0.36	0.01	0.54	0.34	0.96	0.06	0.61	0.85
		0.48	0.86	0.14	0.86	0.89	0.37	0.49	0.60	0.04	0.83
		0.42	0.83	0.37	0.21	0.90	0.89	0.91	0.79	0.57	0.99
		0.95	0.27	0.41	0.81	0.96	0.31	0.09	0.06	0.23	0.77
		0.73	0.47	0.13	0.55	0.11	0.75	0.36	0.25	0.23	0.72
		0.60	0.84	0.70	0.30	0.26	0.38	0.05	0.19	0.73	0.44

10

OR

5	a	Discuss event-to-event simulation in comparison with fixed time step simulation using a flow chart.	08
	b	What is event scheduling/ time advance algorithm? Illustrate with an example.	08
6	a	Enumerate on convolution method and direct transformation method of generating random variates.	10
	b	What is "optimization through simulation" and what is the difficulty involved in achieving it?	06
		OR	
7	a	Explain the inverse transform technique used for generating random variates of the following distributions: i) Uniform distribution. ii) Triangular distribution.	10
	b	Down time of machine is found to be gamma distributed with mean 2.2 minutes and variance 2.1 minutes. Generate a gamma distribution random variate using acceptance rejection technique. Use the following random numbers $0.832, 0.021, 0.434, 0.716$.	06
8		Records pertaining to the monthly number of job related injuries at an underground coal mine were studied by a federal agency. The values for the past 100 months were as follows:	

Injuries/month	0	1	2	3	4	5	6
Frequency of occurrence	35	40	13	6	4	1	1

a) Apply the chi-square test to these data to test the hypothesis that the underlying distribution is Poisson. Use a level of significance of 0.05. 10

b) Explain the steps in the development of useful model of input data. 06

OR

9	a	Explain the three step approach formulated by Naylor and Finger to aid in the validation process.	09
	b	Define x as the number of underfilled bottles from a filling operation in a carton of 24 bottles. 60 cartons are inspected and the following observations on x are recorded.	

Values	0	1	2	3
Frequency	39	23	12	1

Based on these 75 observations, is a binomial distribution an appropriate model? Perform a goodness of fit procedure with $\propto = 0.05$. 07

10	a	Explain “point estimation” and “interval estimation” with respect to output analysis of simulation model.	10
	b	Write a note on initialization bias in steady state simulations.	06
		OR	
11	a	Describe the trends in simulation software.	08
	b	List the factors considered when evaluating and selecting simulation software.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

SIMULATION MODELING AND ANALYSIS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

9 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

10 Answer FIVE full questions from Part B.

PART-A

1	1.1	_____ is an abstraction of reality.	01
	1.2	Instantaneous occurrence that may change the state of the system is called _____.	01
	1.3	The sequence of actions in which a simulation language performs to advance the clock and build a new snapshot is called _____.	01
	1.4	In general, chi-square test can be applied for testing uniformity of random numbers if the sample size is _____.	01
	1.5	Arena is written in _____ simulation language.	01
	1.6	Acceptance rejection technique is used to generate _____.	01
	1.7	A simulation system that runs continuously for long period of time is called _____.	01
	1.8	_____ is the first step in input modeling.	01
	1.9	_____ is the determination that a model is an accurate representation of the real system.	01
	1.10	Sim Runner works with _____.	01
	1.11	Define a system.	02
	1.12	The properties of random numbers are _____ and _____.	02
	1.13	The two parameters that are necessary to describe a normal distribution are _____ and _____.	02
	1.14	Define a run with reference to runs test.	02
	1.15	What is a convolution of a distribution?	02

PART-B

2	a	List the areas of application of simulation. Explain any three of them.	08
	b	Demand for widgets follows the probability distribution as shown:	

Daily demand	0	1	2	3	4
Probability	0.33	0.25	0.20	0.12	0.10

Stock is examined every day and if the inventory level has reached 6 units or less, then an order of 10 widgets are placed. The lead time is probabilistic and follows the following distribution.

Lead Time (days)	1	2	3
Probability	0.3	0.5	0.2

when the simulation begins, 12 widgets are on hand and no orders have been back ordered. Simulate 2 cycles of operations (Assume 5 days in a cycle).
Analyse the system and determine:
i) Average ending inventory.
ii) The number of days shortage condition occurs.
RD for lead time: $5, 1$
RD for demand : $24, 35, 65, 81, 54, 93, 81, 00, 95, 40.$ 08

OR

3 a With a flowchart explain the steps involved in a simulation study. 08

b Lead time for a stock item is normally distributed with a mean of 7 days and a standard deviation of 2 days. Daily demand is distributed as follows:

Daily demand	0	1	2	3	4
Probability	0.367	0.368	0.184	0.062	0.019

Determine the lead time demand for 2 order cycles. Use $258, 071$ as random digits to generate lead time for the two cycles and $551, 675, 530, 200, 602, 865, 916, 635, 619, 246, 671, 450$ to generate demand. 08

4 a Explain the concepts in discrete event simulation. 08

b A sequence of random numbers $0.44, 0.81, 0.14, 0.05, 0.93$ has been generated. Use Kolmogorov Smirnov test with a level of significance 0.10 to determine if the hypothesis that the numbers are uniformly distributed on the interval (0,1) can be rejected. 08

OR

5 a List the steps in event-scheduling algorithm. 05

b Use the mixed congruential method to generate a sequence of three two-digit random integers between 0 and 24 with $X_0 = 13, a = 9$ and $c = 35$. 03

c Based on the runs up and down test, determine whether the following sequence of 30 two-digit random numbers is such that the hypothesis of independence can be rejected with the level of significance $\propto = 0.05$

0.97	0.87	0.63	0.66	0.30	0.69	0.37	0.01	0.66	0.51
0.92	0.36	0.47	0.80	0.94	0.31	0.07	0.09	0.19	0.29
0.94	0.87	0.85	0.66	0.78	0.94	0.56	0.67	0.07	0.43

08

6 a What is a random variate? Explain the inverse transformation technique for generation of a random variable, sampling from exponential distribution. 08

b Consider the 30 numbers given in the table for question 5(c). Test whether the $2^{nd}, 6^{th}, 10^{th}$ and so on, numbers in the sequence are auto correlated, where $\propto = 0.05$. 08

OR

7	a	Explain briefly meaning of optimization via simulation.	08
	b	Develop a random-variate generator for triangular distribution with range $(2, 6)$ and mode at 3. Generate four values of random variates. Use $R = 0.36, 0.18, 0.89, 0.60$	08
8	a	Explain the three step approach formulated by Nayler and Finger to aid in the validation process of a simulation model.	08
	b	Records pertaining to monthly number of job-related injuries at an underground coal mine were being studied by a federal agency. The values for the past 100 months were as follows:	

Injuries/month	0	1	2	3	4	5	6
Frequency of occurrence	35	40	13	6	4	1	1

Apply the chi-square test to these data to test the hypothesis that the underlying distribution is poisson. Use a level of significance of $\propto = 0.05$. 08

OR

9	a	Explain the different steps in model building.	06
	b	The following data are available on the processing time at a machine (in minutes): $0.64, 0.59, 1.1, 3.3, 0.54, 0.04, 0.45, 0.25, 4.4, 2.7, 2.4, 1.1, 3.6, 0.61, 0.20, 1.0, 0.27, 1.7, 0.04, 0.34$. Develop an input model for the processing time.	10
10	a	List and explain the types of simulation with reference to output analysis.	08
	b	Explain the trends in simulation software.	08

OR

11	a	List the various simulation packages. Explain any three of them.	08
	b	Explain briefly point estimate and interval estimate as measures of performance.	08

Advanced Operations Research

VI Semester SEE Question Papers

Course codes: 12IM6D3

- Apr/May 2018 (12IM6D3)
- Apr/May 2017 (12IM6D3)
- Apr/May 2016 (12IM6D3)

12IM6D3

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

ADVANCED OPERATIONS RESEARCH (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

6 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

7 Answer FIVE full questions from Part B.

PART A

1	1.1	In the two phase method, an _____ variable is never considered for re-entry into the basis.	01
	1.2	If in the integer *LP* problem, few of the basic variables do not possess non-negative integer value, then an additional constraint known as _____.	01
	1.3	Goal programming is used for ballistic _____ management.	01
	1.4	In Goal Programming, a goal constraint having over achievement variable is expressed as a _____ constraint.	01
	1.5	Resource leveling is the process of _____ the utilization of resources in a project.	01
	1.6	Crashing is the process of reducing the total time that it takes to complete a project by expending _____.	01
	1.7	The mean of exponential distribution is the _____ time between arrivals.	01
	1.8	the payment of telephone bills by cheque or cash is the example of _____ discipline.	01
	1.9	In Markov analysis, state probabilities must _____ to one.	01
	1.10	A zero element in the transition matrix indicates that the transition is _____.	01
	1.11	Optimally is indicated for a maximization problem when all elements in the $c_j - z_j$ rows are _____, while for minimization problem all elements must be _____.	02
	1.12	In integer programming, each fractional cut that is made, requires the addition of a _____ or _____ constraint.	02
	1.13	Network is the graphical display of a project that contains both _____ and _____.	02
	1.14	The dynamic arrival process is controlled by _____ facility and the _____.	02
	1.15	A problem is classified as Markov chain provided states are collectively _____ and _____ exclusive.	02

PART B

2	a	Define slack and surplus variables in a linear programming problem.	04
	b	A company sells two types of fertilizers, one is liquid and the other is dry. The liquid fertilizer contains 2 units of chemical A and 4 units of chemical B per jar the dry fertilizer contains 3 units of each of the chemicals A and B per carton. The liquid fertilizer sells $Rs.3$ per jar and dry fertilizer sells for $Rs.4$ per carton. A farmer requires at least 90 units of chemical A and 120 units of chemical B for his farm. How many of each fertilizer should the farmer purchase to minimize the cost while meeting the requirements. Solve by using Two Phase Simplex Method.	12
		OR	
3		Solve the following integer linear programming problem by Branch and Bound method. Max $Z = 6x_1 + 8x_2$ subject to $x_1 + 4x_2 \leq 8$ $7x_1 + 2x_2 \leq 14$ where x_1, x_2 are non-negative integers.	16
4	a	Why all goal linear programming problems are minimization problem? Give reasons.	06
	b	The elastic firm has two products coming on the market: Zigs and Zags. To make a Zig, the firm needs 10 units of product A and 15 units of product B. To make a Zag, they need 20 units of product A and 15 units of product B. there are only 2,000 units of product A and 3200 units of product B available to the firm. The profit on a Zig is $Rs.4$ and on a Zig it is $Rs.6$. Management objectives in the order of their priority are: i) Produce exactly 50 Zigs ii) Achieve a target profit of at least $Rs.750$ iii) Use all of the product B available. Formulate this as a goal programming problem.	10
		OR	
5	a	Explain Goal Programming applications with respect to the following areas: i) Marketing ii) Finance iii) Production iv) Human Resource Development.	08
	b	Use the Wolfe's method to solve the quadratic programming problem: Max $Z = 4x_1 + 6x_2 - 2x_1^2 - 2x_1x_2 - 2x_2^2$ subject to $x_1 + 2x_2 \leq 2$ and $x_1, x_2 \geq 0$	08
6	a	Discuss the general algorithm used in solving problem of dynamic programming approaches.	10
	b	Regardless of the type or size, following terms are common in dynamic programming problem: discuss them: i) Stage ii) State iii) Return Function.	06

OR

7 The following table gives data on normal time, cost and crash time and cost for a project:

Activity	Normal		Crash	
	Time (weeks)	Cost (Rs)	Time (weeks)	Cost (Rs)
1 – 2	3	300	2	400
2 – 3	3	20	3	30
2 – 4	7	420	5	580
2 – 5	9	720	7	810
3 – 5	5	250	4	300
4 – 5	0	0	0	0
5 – 6	6	320	4	410
6 – 7	4	400	3	470
6 – 8	13	780	10	900
7 – 8	10	1000	9	1200

Indirect cost is $Rs.50$ per week.

a) Draw the network and identify the critical path with double line
b) What are the normal project duration and associated cost?
c) Find out the total float associated with each activity
d) Crash the relevant activities symmetrically determine the optimal project completion time and cost. 16

8 a Explain the following features of generalized queuing system:
i) Length of the queue
ii) System
iii) Waiting time
iv) Total time. 08

b Arrivals at a telephone booth are considered to be Poisson with an average time of 10 minutes between one arrival and the next. The length of phone calls is assumed to be distributed exponentially with mean 3 minutes.
i) What is the probability that a person arriving at the booth will have to wait?
ii) What is the average length of the queues that forms from time to time?
iii) The telephone department will install a second booth when convinced that an arrival would expect to have to least three minutes for the phone. By how much the flow of arrivals is to be increased in order to justify a second booth? 08

OR

9 a Discuss the following with respect to a customer's behavior in a queue:
i) Balking
ii) Collusion
iii) Jockeying. 06

b A person repairing radios finds that the time spent on radio sets has exponential distribution with mean 20 minutes. If the radios are repaired in the order in which they come in and their arrival is approximately Poisson with an average rate of 15 for 8 hour day, what is the repairman's expected idle time each day? How many jobs are ahead of the average set just bought in? 10

10	a	State and discuss the characteristics of a Markov Chain.	10
	b	State the steps for constructing Matrix of Transition Probabilities of Markov Chain.	06
		OR	
11	a	State the application of Markov analysis.	04
	b	The school of international studies for population find out, though its survey, that the mobility of the population (in percent) of a state to a village, town and city is in the following percentages.	

		TO		
		$Village$	$Town$	$City$
$From$	$Village$	40	30	30
	$Town$	20	60	20
	$City$	20	40	40

What will he the proportion of population in village, town and city after two years, given that the percent population has proportions of $0.6, 0.3$ and 0.1 in village, town and city respectively? 12

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

ADVANCED OPERATIONS RESEARCH (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

8 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

9 Answer FIVE full questions from Part B.

PART A

In this part, there are statements made based on the concepts related to this course. You are required to provide one line of reasoning to substantiate theses statements (why? Part of understanding these concepts is required to be articulated) and its relevance to problem solving operation's research modeling frameworks. (you may provide case situations, diagrams, illustrations to reason out as necessary)

1	1.1	The revised simplex method substantially reduces the computational effort as compared to the Dantzig's simplex method.	01
	1.2	Shadow prices with respect to resource constraints have their utility in evaluating the resource priorities and economic significance in managing the resources.	01
	1.3	The two phase simplex method diagnoses the condition of infeasibility at the end of Phase I and hence is efficient compared to the regular simplex method.	01
	1.4	The branch and bound algorithm is the most widely used method for solving both pure and mixed integer programming problems in practice. Most commercially available computer codes for solving integer programs are based on this approach.	01
	1.5	In goal programming, all the objectives are assigned target levels for achievement and a relative priority on achieving these levels. Goal programming treats these targets as goals to aspire for and not as absolute constraints.	01
	1.6	Inmost engineering problems, -especially real world formulations- the objective is to optimize (maximize or minimize) a criterion (or objective) function subject to several constraints. These could be linear or non linear in nature.	01
	1.7	The mathematical technique of Lagrange multipliers has been developed to convert constrained optimization problems into unconstrained optimization problems.	01
	1.8	The optimal set of decisions in a multistage decision process problem has the property that whatever the initial stage, state and decisions are, the remaining decisions must constitute an optimal sequence of decisions for the remaining problem, with stage and state resulting from the first decisions (or occurring naturally) considered as initial conditions.	01

1.9	In dynamic programming, one solves multivariable optimization problems sequentially, or one stage at a time. Hence it will be necessary to keep track of all the returns accumulated in our decision process as we proceed from stage to stage.	01
1.10	If the search for the optimal solution of a given LPP(Linear Programming Problem) starts with a dual feasible basis (A canonical tableau where the optimality conditions are satisfied), and search for a primal feasible basis by moving from one dual feasible tableau to another dual feasible tableau . the approach is called the dual simplex method.	01
1.11	When the process is irreducible, recurrent, positive and a periodic the Markov chain modeling is applied the resultant Markov chain is ergodic.	01
1.12	If you are modeling a process that does not terminate after some length of time, the model will take the form of so called absorbing Markov chain.	01
1.13	Given a transition matrix, we can construct a unique transition diagram; Given a transition diagram we can construct a unique transition matrix.	01
1.14	In the context of queuing models, a customer is anything that arrives occupies a server for some period of time, and departs. It need not be a human, or for that matter a physical object. It could be a job, an order a transaction, a message or some similar abstraction.	01
1.15	The relationship linking L(Mean length of system), W(Mean waiting time in the system) and R (mean throughput rate) is expressed as simple direct relationship $L = RW$ (little's formula). This formula applies on a very broad scale to all kinds of problems. (say for conveyor assembly line for example).	01
1.16	The simplest arrival process to deal with is a Poisson process, in which the times between arrivals are independent negatively exponentially distributed random variables. The same process viewed another way gives the random variables for the number of customers arriving in any interval (hence the name for the process).	01
1.17	One of the most efficient algorithms for solving shortest route problems is known as the Dijkstra's algorithm.	01
1.18	The time required to complete the various activities in a research and development project is generally not known a priori. Thus $PERT$ incorporates uncertainties in activity times in the analysis.	01
1.19	For large projects the approach to solving CPM problems deployed is the mathematical modeling methods. The basic among them is the LPP models.	01
1.20	The basic assumption in CPM models is that the activity times are proportional to the level of resources allocated to them. By assigning the additional resources (Capital, People, Materials and Machines) to an activity, its duration can be reduced to a certain extent.	01

PART B

2	a	A factory manufactures three products. Three resources- technical services, labor and administration are required to produce these products. The following table gives the requirements on each of the resources for the three products.	

$Product$	$Resources(hours)$			$Unit\ Profit$ (\$)
	$Technical\ service$	$Labor$	$Administration$	
1	1	10	2	10
2	1	4	2	6
3	1	5	6	4

There are 100 hours of technical services available, 600 hours of labor, and 300 hours of administration , In order to determine the optimal produce mix which will maximize the total profit, the following linear program was solved:

Maximize $Z = 10x_1 + 6x_2 + 4x_3$

Standards to $x_1 + x_2 + x_3 \leq 100$ $(technical)$

$10x_1 + 4x_2 + 5x_3 \leq 600$ $(labor)$

$2x_1 + 2x_2 + 6x_3 \leq 300$ (administration)

$x_1, x_2, x_3 \geq 0.$

Where $x_1, x_2,$ and x_3 are the number of product 1, product 2 and product 3 produced. The optimal solution is given by the following tableau, where $x_4, x_5,$ and x_6 are the slack variables:

C_j	10	6	4	0	0	0	$Constants$
$Basis$	x_1	x_2	x_3	x_4	x_5	x_6	
x_2	0	1	5/6	10/6	−1/6	0	400/6
x_1	1	0	1/6	−4/6	1/6	0	200/6
x_6	0	0	4	−2	0	1	100
$C\ Row$	0	0	−16/6	−20/6	−4/6	0	$Z = 4400/6$

Using sensitivity analysis, answer the following with respect to the above optimal tableau:

i) What should be the profit of product 3 before it becomes worthwhile to manufacture? Find the most profitable product mix if the profit on product 3 were increased to \$50/6.

ii) What is the range on the profit of product 1 so that the current solution is still optimal?

iii) It is believed that the estimates of the available hours of technical services might be wrong. The correct estimate is $10 + 10\lambda$ where λ is some unknown parameter. Find the range of values λ within which the given product mix is still optimal.

iv) Determine the shadow prices of all the resources. 16

OR

3 a Use Two-Phase simplex method to

Maximize, $Z = 2x_1 + x_2 + x_3$

Subject to $4x_1 + 6x_2 + 3x_3 \leq 8$

$3x_1 - 6x_2 - 4x_3 \leq 1$

$2x_1 + 3x_2 - 5x_3 \geq 4$

$x_1, x_2, x_3 \geq 0.$ 08

b Express this LPP in the format of revised Simplex method. Carry out first two iterations using the revised simplex method.

Maximize, $Z = x_1 + 5x_2 + 3x_3$

Subject to $x_1 + 2x_2 + x_3 = 3$

$2x_1 - x_2 = 4$

$x_1, x_2, x_3 \geq 0$ 08

4 a Camyo Manufacturing produces four parts that require the use of a lathe and a drill press. The two machines operate 10 hours a day. The following table provides the time in minutes required by each part:

7 The following tables gives the activities in a construction project and other relevant information:

Activity	Description	Immediate Predecessor	Normal		Crash		Cost slope	Rank
			Time	cost	Time	cost		
A	Dismantle and inspect	-	2	100	1	140	40	II
B	Subcontract repairs	A	6	200	2	500	75	IV
C	Buy spares	A	3	100	1	140	20	I
D	Rebuilt /repair	A	4	500	2	800	150	V
E	Assemble	B, C, D	2	200	1	250	50	III
F	Trial run	E	1	50	Not possible	250	-	-

Determine the least cost schedule for the project. 16

8 a The queue discipline is a parameter that explains how the customers arrive at a service facility. The commonly referred structures of queue disciplines are

i) First in first out ($FIFO$)
ii) First in last out($FILO$)
iii) Served in random order ($SIRO$)
iv) Priority scheduling
v) Processor (or Time) sharing.

Give one practical example for each one of the disciplines mentioned above. 08

b With usual Notation and based on the notion of birth Death Processes. Bring out the relationships between L, W, Lq and Wq: for a typical $M/M/1$ queuing system. 08

OR

9 a A blood donation drive organized in a small college operates from a room with 2 beds, and on an average it takes each donor 20 minutes to enter the room, finish the screenings, donate blood and leave. There is enough space to seat any number of waiting students outside the room. The students arrive according to a Poisson process with a mean inter arrival time of 25 minutes. What is the probability of an arriving student having to wait to enter the room?

What is the utilization of the beds in the room? 10

b How would you use the Kendall & Lee's notation to describe the characteristics of various types of queuing system? Give an example. 06

10 a ABC hospital operated on a charity basis. All expenses are paid by the government. Recently the board of governors of the hospital has been complaining about the size of the budget and is in insisting that the hospital cut expenses. The major area of concern has been the cost of keeping the patients in intensive care. This cost has averaged $Rs.1000$ per week per person compared to only $Rs.500$ per week per person for keeping patients in the ICU at the beginning of the week, 50% will be there at the end of the week and 50% will be moved to a ward. Of the patients in the ward at the beginning of the week, 50% will be there at the end of the week, 10% will get worse and be transferred to ICU: and 40% will become outpatients.

Of those persons who are outpatients at he beginning of the week; 85% will remain outpatients at the end of the week; 10% will be admitted to a ward and 5% will be admitted to ICU. The board of the government believes the criteria for keeping patients in ICU are strict and has instructed the ICU staff to relax the criteria so that only 40% of the ICU patients remain in ICU each week and 60% are transferred to ward. The staff insists that if this is done, 20% of the ward patients will be going into ICU each week and only 30% will be transferred to out patients status. The percentage ward patient remaining will be still be 50%. There will be no change in outpatient status. Will the policy advocated by the board of governors actually save money? 12

b List out any four important areas of applications of the Markov Chain. Modeling approach. 04

OR

11 a A market survey has been conducted to determine the movements of people between types of residences. Two hundred apartment dwellers were asked if their previous residence was an apartment, a townhouse, their own house, or a rented home. Similarly, 200 townhouse dwellers were asked about their previous residences, and so on. The results of the survey are tabulated below:

Present residence	*Previous residence*				
	Apt.	*Townhouse*	*Own house*	*Rented House*	*Total*
Apartments	100	20	40	40	200
Townhouse	150	40	0	10	200
Own house	50	20	120	10	200
Rented house	100	20	20	60	200
Total	400	100	180	120	800

The data are believed to be representative of the behavior of the population at large. Develop a model and answer the following:

i) What is the probability that someone who is now is an apartment will own his own home after two moves?
ii) What is the probability that the same person will own his own home after two moves but not sooner?
iii) How many moves will occur on the average, before a homeowner will live in a townhouse?
iv) What, if anything can you say about the long term demand for townhouses, relative to the other residence types?
v) Consider the validity of your assumption. 12

b Discuss the property of ergodicity and its significance to modeling using Markov chain structures. 04

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

ADVANCED OPERATIONS RESEARCH (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

4 Answer FIVE full questions from Part B.

PART-A

1	1.1	The advantage of the two phase simplex method over the regular simplex method is that the _____ condition in the solution to a given LPP is identified quickly (earlier) during the process of solving the LPP.	01
	1.2	The revised simplex method requires the expression of the given LPP in the _____ structure to initiate the process of solving using the matrix inversion technique to change the basis at every iteration.	01
	1.3	The integer programming formulation which requires that all decision variables have necessarily to be integers is known as the _____ integer program.	01
	1.4	The shadow price is defined as the _____ worth of one additional unit of the resources expressed as constraints in a given LPP model.	01
	1.5	The branch and bound technique of solving a given integer programming problem requires the solution of many _____ to the original problem to generate the branches and bound the optimal solution.	01
	1.6	Goal programming formulation requires the specification of decision variables and _____ with respect to each of the goals to be met in a given problem situation.	01
	1.7	The goal programming formulation involves the objective function which involves the _____ of the sum of deviational variables.	01
	1.8	The Kuhn Tucker conditions for the non linear programming problem can be said to be the necessary and _____ conditions for the optimal solution of the non linear program.	01
	1.9	The general structure of a quadratic programming problem involves the non linear objective function and _____ constraints.	01
	1.10	If the constraint of a quadratic programming problem is $\sum_{j=1}^{n} a_{ij}x_j \leq b_j$ and the non negativity restraints are $x_j \geq 0$. The slack variables _____ are introduced into the constraints.	01
	1.11	The dynamic programming problem involves the division of the problem into _____ and define the state transition function to link the various stages of the solution methodology.	01
	1.12	The principles of optimality which form the basis for solving a given problem using dynamic problem formulation is known as the _____ principle of optimality.	01

	1.13	The schedule of a project network which drives the total cost of the project inclusive of the direct cost and indirect cost to the absolute minimum is known as the _____ schedule.	01
	1.14	One of the most efficient algorithms to determine the solution for the shortest route problem is known as the _____ algorithm.	01
	1.15	The _____ property which is one of the basis for realizing the continuous time process for modeling the queuing process, helps in assuming the exponential distribution for time between arrivals.	01
	1.16	Nearly all simple queuing models (as well as a number of non queuing models) are special cases of the _____ process which is itself a special case of the general continuous time Markov process.	01
	1.17	If for all n $p\{X_n = j_n/X_{n-1} = j_{n-1}, X_{n-2} = j_{n-2} - - - - - - X_0 = j_0\} = p\{X_n = j_n/X_{n-1} = j_{n-1}\}$ the process is said to be discrete state, discrete parameter Markov process or more generally known as _____.	01
	1.18	The property of _____ is exhibited by a discrete p-state, discrete parameter process, when the process is irreducible, recurrent, positive and aperiodic.	01
	1.19	If you are modeling a process that does terminate after some length of time the model will usually take the form of a so called _____ Markov chain.	01
	1.20	$p^{(2)}$ is called _____ transition matrix; its elements, two –step transition probabilities. They give conditional probabilities for states at time 2 under varying possible conditions for the state at time zero.	01

PART-B

2	a	Solve the following LPP use the two phase method Minimize $Z = 4x_1 + x_2$ Subject to the constraints: $3x_1 + x_2 = 3$ $4x_1 + 3x_2 \geq 6$ $x_1 + 2x_2 \leq 4$ $x_1 x_2 \geq 0$	08
	b	Use revised simplex method to solve the following LPP Minimize $Z = x_1 + 2x_2$ Subject to $2x_1 + 5x_2 \geq 6$ $x_1 + x_2 \geq 2$ $x_1, x_2 \geq 0$	08
		OR	
3	a	Consider a diet problem in which a college student is interested in finding a minimum cost diet that provides at least 21 units of vitamin A and 12 units of vitamin B from five foods of the following properties:	

$Food$	1	2	3	4	5
$Vitamin\ A\ content$	1	0	1	1	2
$Vitamin\ B\ content$	0	1	2	1	1
$Cost\ per\ unit$	20	20	31	11	12

The corresponding linear program is
Minimize $Z = 20x_1 + 20x_2 + 31x_3 + 11x_4 + 12x_5$
Subject to
$x_1 + x_3 + x_4 + 2x_5 \geq 21$ (Vitamin A constraint)
$x_2 + 2x_3 + x_4 + x_5 \geq 12$ (Vitamin B constraint)
Where x_i- amount of food 'j' bought for $j = 1, 2, - - -5$. Solution obtained for the LP using the computer program is given below:
Optimal solution $x_1 = x_2 = x_3 = 0, x_4 = 3, x_5 = 9$
Optimal value Min $Z = 141$
Shadow prices 1.0 and 10.0 for constraints 1 and 2 respectively
Opportunity costs $19.0, 10.0, 10.0$ and 0.0 for variables x_1, x_2, x_3, x_4 and x_5 respectively.

Ranges on objective function coefficients

Variable	*Lower Limit*	*Present Value*	*Upper Limit*
x_1	1	20	∞
x_2	10	20	∞
x_3	21	31	∞
x_4	6	11	12
x_5	11	12	22

Ranges on RHS constants

Row	*Lower Limit*	*Present Value*	*Upper Limit*
1	12	21	24
2	10.5	12	21

Using the above information, answer the following:

i) For what range of values of $C3$ (cost of food 3) is the current solution still optimal?

ii) Determine the optimal solution is the minimum diet cost of food 1 is reduced to 5 cents per unit.

iii) What is the optimal solution and the minimum diet cost if the unit cost of food 4 is reduced to 7 cents.

iv) A student's favorite is food 2 and she would like to include 2 units of it in the diet. Explain how this will affect the minimum cost diet.

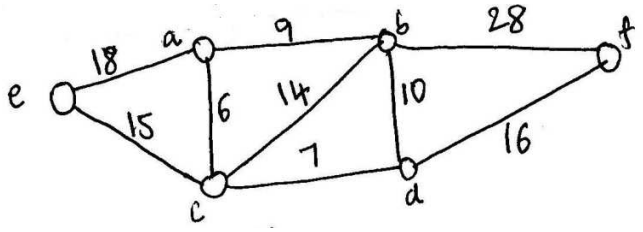
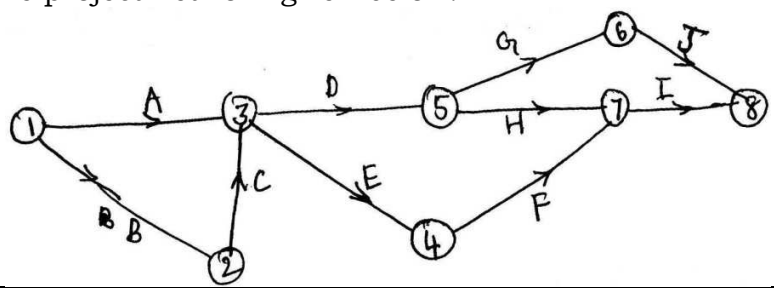
v) Suppose the vitamin A requirement is increased from 12 to 15 units. A local pharmacist offers the additional 3 units of vitamin B at a cost of 12 cents per unit. Would you accept the offer? Why or why not? 12

b List the steps of the Gomory's cutting plane method for solving an integer programming problem. 04

4 a Camyo manufacturing produces four parts that require the use of lathe and a drill press. The two machines operate 10 hours a day. The following table provides the time in minutes required by each part

Production time in minutes

Part	*Lathe*	*Drill Press*
1	5	3
2	6	2
3	4	6
4	7	4

		It is derives to balance the two machines by limiting the differences between their total operations to at most 30 minutes. The market demand for each part is atleast 10 units. Additionally the number of units of part 1 may not exceed that of part 2. Formulate the problem as a goal programming model.	08
	b	Use Wolfe's method to solve the quadratic programming problem $f(x) = x_1^2 + x_2^2 - 2x_1 - 4x_2$ Subject to $x_1 + 4x_2 \leq 5$ $2x_1 + 3x_2 \leq 6$ $x_1, x_2 \geq 0$	08
		OR	
5	a	Solve the following problem using only the Kuhn-Tucker conditions $f(x) = 100 - 1.2x_1 - 1.5x_2 + 0.3x_1^2 + 0.05x_2^2$ Subject to: $g_1(x) = x_1 + x_2 \geq 35$ $g_2(x) = x_1 \geq 0$ $g_3(x) = x_2 \geq 0$	08
	b	A company has two machines that can manufacture a product. Machine 1 makes 2 units per hour while machine 2 makes 3 per hour. The company has an order for 160 units. Each machine can be run for 40 hours on regular time; however overtime is available. The company's goal in order of importance are to: i) Avoid falling short of meeting the demand. ii) Avoid any overtime of machine 2 beyond 10 hours. iii) Minimize the sum of overtime (assign differential weights according to the relative costs of overtime hours, assuming that the operating cost of the two machines are the same). iv) Avoid under-utilisation of regular working hours (assign weights according to the productivity of the machines).	08
6	a	List the main features of a dynamic programming model. Briefly bring out its relevance.	04
	b	Apply Dijkastra's algorithm to find the shortest route from s to t in the graph given below:  	12
		OR	
7	a	Consider the project network given below:  	

The data for normal times, crash times and crashing costs are given as follows:

Job	$Normal\ Time\ (days)$	$Crash\ Time\ (days)$	$Cost\ of\ crashing\ per\ day$ (\$)
A	10	7	4
B	5	4	2
C	3	2	2
D	4	3	3
E	5	3	3
F	6	2	5
G	5	4	1
H	6	4	4
I	6	3	3
J	4	3	3

Let T represent the earliest completion time of the project

i) Determine the maximum and minimum value of T

ii) Given that the overhead costs as \$5 per day, determine the optimal duration of the project in terms of both the crashing and overhead costs by direct enumeration method. 14

b Mention any two applications of the minimal spanning tree problem with brief of the problem and suitability of the minimal spanning tree algorithm. 02

8 a What is a Markovian queuing model? 03

b Explain your understanding of the relationship between the arrival rate λ and the average inter arrival time. What are the units describing each variable? 03

c In a bank operation, the arrival rate is 2 customers per minute. Determine the following:

i) The average number of arrivals during 5 minutes.

ii) The probability that no arrivals will occur during the next 5 minutes.

iii) The probability that at least one arrival will occur the next 5 minutes.

iv) The probability that the time between successive arrivals is at least 3 minutes. 10

OR

9 a Describe Kendalls and lee's notation for identifying queuing models. Give two examples to clarify the notation 04

b What is Little's formula and what is its use? 04

c Consider an airport with two runways, one of which is used solely for take off's and other solely for landings. Assume that a plane, whether landing or take off will occupy a runway for an average of 2 minutes where "by occupy" we mean "prevent use by any other plane". Delay on the ground although unpleasant for passengers, does not pose a safety hazard; delay on the other hand is of serious concern. Suppose that the regulation authority specify that the mean delay in the air (i.e w_q) must not exceed 10 minutes. Assume that the planes arrive according to a poisson process and construct a model from the information given above to answer the following questions:

		i) What is the maximum tolerable load on the airport in terms of the mean number of planes that can arrive per hour? ii) Would it make much difference if you were told that the standard deviation of the time a plane will occupy a run way is 1 minute? iii) What practical suggestion can you make as to what might be done to increase the load capacity of the airport, short of building new runways?	08
10	a	Explain how to construct a n step transition probabilities of a typical Markov chain. Give an example to illustrate the construction of TPM.	04
	b	A manufacturing company has a certain piece of equipment that is inspected at the end of each day and classified as just overhauled, good, fair or inoperative. If the item is inoperative it is overhauled, a procedure that takes one day. Let us denote the classification as states $1, 2, 3$ and 4 respectively. Assume that the working conditions of the equipment follows a Markov chain with the following transition matrix Tomorrow $$P = Today \begin{matrix} \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{matrix} \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{bmatrix} 0 & 3/4 & 1/4 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$ If it costs Rs $125/-$ to overhaul a machine (including cost time) on the average and Rs $75/-$ as production lost if a machine is found inoperative. Using the steady – state probabilities. Compute the expected cost per day for maintenance of the piece of equipment.	12
		OR	
11	a	Define the following terms (Give illustrations wherever necessary) i) Absorbing states ii) Ergodic Markov chain iii) Irreducible Markov chain iv) Chapman- kolmogrov equation	06
	b	Suppose that new razor blades were introduced in the market by three companies at the same time. When they introduced each company had an equal share of the number, but during the first year the following changes took place: i) Company A retained 90 percent of its customers, lost 3 percent to B and 7 percent to C. ii) Company B retained 70 percent of its customers, lost 10 percent to A and 20 percent to C. iii) Company C retained 80 percent of its customers, lost 10 percent to A and 10 percent to B. Assuming that the buying habits of the customer remains same. What are the market shares of the three companies at the end of first year? The second year? What are the long-run market shares of the three companies?	10

Applied Statistics

VI Semester SEE Question Papers

Course codes: 12IM6C3

- Apr/May 2018 (12IM6C3)
- Apr/May 2017 (12IM6C3)
- Fast Track Jul 2016 (12IM6C3)
- Apr/May 2016 (12IM6C3)

12IM6C3

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

APPLIED STATISTICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

4 Answer FIVE full questions from Part B.

5 Use of statistical tables permitted.

PART A

1	1.1	if $P(x) = \begin{cases} \frac{x}{15} & x = 1,2,3,4,5 \\ \theta & elsewhere \end{cases}$ then, $P(x = 0 \; or \; x = 2) =$ _____.	01
	1.2	Thye moment generating function of the random variable whose moments are $\mu_r^{'} = (r+1)!\, 2^r$	02
	1.3	In a book of 520 pages, 390 typo-graphical errors occur. Assuming Poisson law for the number of errors per page, the probability that a random sample of 5 pages will contain no error _____.	02
	1.4	X is a Gamma distribution with parameter $\lambda = 2$. Mean and variance are related as _____.	01
	1.5	The joint probability density function of a two dimensional random variable (x, y) is given by $f(x,y) = \begin{cases} 2, & \phi < x < 1, \phi < y < x \\ \phi, & otherwise \end{cases}$. The marginal density of x is _____.	01
	1.6	The test statistic used in a test for proposition of defectives based on *'d'* defectives observed in parts is _____.	02
	1.7	Type – II error is defined as _____.	01
	1.8	What is the difference between contingency table test and goodness of fit test?	02
	1.9	In testing for equality of two means the pooled variance (i.e., unbiased estimator of the common population variance σ^2) is given by _____.	01
	1.10	The expression for determining sample size while estimating mean is _____.	01
	1.11	Which hypothesis is tested by Wilcoxon Rank sum test?	01
	1.12	Least squares estimate for β in matrix form is _____.	01
	1.13	What are the hypothesis statements in a test of regression with 3 independent variables?	02
	1.14	What is coefficient of multiple determination?	02

PART B

2	a	A perfect coin is tossed twice. Find the moment generating function of the number of heads. Hence find the mean and variance.	08
	b	Derive moment generating function of $f(x) = \theta e^{-\theta x}, x \geq \theta$. Find the density function, mean and variance.	08
		OR	
3	a	The diameter of an electric cable X is continuous random variable with density $$f(x) = \begin{cases} kx(-1x), & if\ \theta \leq x < 1 \\ \theta, & elsewhere \end{cases}$$ Find: i) k ii) $F(x)$ iii) $p\left(x < \frac{1}{2} \middle| \frac{1}{3} < x < \frac{2}{3}\right)$.	08
	b	The distribution function of a random variable X is given by $F(x) = 1 - (1 + x)e^{-x}, x \geq \theta$. Find the density function, mean and variance.	08
4	a	In a component manufacturing industry, there is a small probability of $\frac{1}{500}$ for any component to be defective. The components are supplied in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing: i) Atleast one defective and ii) At most one defective in a consignment of 1000 packets.	08
	b	Suppose that the life time of a certain kind of an emergency backup battery (in hours) is a random variable X having the Weibull distribution with $\alpha = 0.01$ and $\beta = 0.5$. Find the probability that such a battery will: i) Last more than 300 hours ii) Not last 100 hours Also find the mean life time of these batteries.	08
		OR	
5	a	Suppose that the reaction time in seconds of a person can be modeled by a log normal distribution with parameter values $\mu = -0.35$ and $\sigma = 0.2$. i) Find the probability that the reaction time is less than $0.6s$ ii) Find the reaction time that is exceeded by 95% of the population.	08
	b	A case of wine has 12 bottles, 3 of which contain spoiled wine. A sample of 4 bottles is randomly selected from the case. i) Find the probability distribution for x, the number of bottles spoiled wine in the sample ii) What is the probability that at least one bottle is spoiled? iii) Obtain the mean and variance of x.	08
6	a	If the joint density function of the two random variables X and Y be $$f(x,y) = \begin{cases} e^{-(x+y)} & x \geq 0, y \geq 0 \\ 0 & otherwise \end{cases}$$ Find: i) $P(x < 1)$ ii) $P(x + y < 1)$	08

	b	Let the random variable X denote the time until a computer server connects to your machine (in milli seconds) and let Y denote the time until the server authorizes you as a valid user (in milli seconds). Each of these random variables measures the wait from a common starting time and $x < y$. The joint density function of X and Y is $$f_{XY}(x,y) = 6 \times 10^{-6} e^{-0.001x-0.002y} \; for \; x < y$$ i) Obtain marginal distribution of X and Y ii) Find $P(x \leq 1000, Y \leq 2000), P(Y > 2000)$	08
		OR	
7	a	Let X be a Bernoulli random variable with probability mass function $$f(x,p) = p^x(1-p)^{1-x}, x = 0,1$$ where p is the parameter to be estimated. Obtain the maximum likelihood estimator of p.	08
	b	i) An electronic company manufactures resistors that have mean resistance of 100Ω and a standard deviation of 10Ω. The distribution of resistance is normal. Find the probability that a random sample of $n = 25$ resistors will have an eaverage resistance of fewer than 95Ω. ii) An article in the journal Materials Engineering describes the results of tensile adhesion tests on 22 $U-700$ alloy specimens. The load at specimen failure is as follows (in MPa)	

19.8	10.1	14.9	7.5	15.4	15.4	15.4	18.5	7.9	12.7	11.9
11.4	11.4	14.1	17.6	16.7	15.8	19.5	8.8	13.6	11.9	11.4

	Construct 95% confidence interval for μ	08
8 a	Consider the following frequency table of observations on the random variable X.	

$Values$:	0	1	2	3	4
$Observed\ Frequency$:	24	30	31	11	4

	i) Based on these 100 observations, is a Poisson distribution with a mean of 1.2 an appropriate model? Perform a goodness of fit procedure with $\alpha = 0.05$ ii) Calculate the P-value for this test	08
b	Ten samples are taken from a plating bath used in an electronics manufacturing process and the pH of the bath was determined. The sample pH values are 7.91, 7.85, 6.82, 8.01, 7.46, 6.95, 7.05, 7.35, 7.25 and 7.42. Manufacturing engineering believes pH has a median value of 7.0. i) Do the sample data indicate that this statement is correct? Use sign test with $\alpha = 0.05$ to investigate this hypothesis. Find the p-value for this test ii) Use the normal approximation for the sign test to test $H_o: \tilde{\mu} = 7.0 \; vs \; H_1: \tilde{\mu} \neq 7.0$	08
	OR	
9 a	An article is knee surgery considered arthoscopic meniscal repair with an absorbable screw. Results showed that for tears greater than $25mm$, 14 of 18 repairs were successful, but for shorter tears, 22 of 30 reapairs were successful. Is there evidence that the success rate is greater for longer tears? Use $\alpha = 0.05$. what is the p-value?	08

	b	Grades in a statistics course and an operations research course taken simultaneously were as follows for a group of students $Operation\ research\ grade$ $Statistics\ Grade$ | A | B | C | $Other$ A | 25 | 6 | 17 | 13 B | 17 | 16 | 15 | 6 C | 18 | 4 | 18 | 10 $Other$ | 10 | 8 | 11 | 20 Are the grades in statistics and operations research related? Use $\alpha = 0.01$ in reaching your conclusion.	08
10	a	In semiconductor manufacturing, wet chemical etching is often used to remove silicon from the backs of wafers prior to metallization. The etch rate is an important characteristic in this process and known to follow a normal distribution. Two different etching solutions have been compared using two random samples of 10 wafers for each solution. The observed etch rates are as follows $Solution\ I$ | 9.9 | 9.4 | 9.3 | 9.6 | 10.2 | 10.6 | 10.3 | 10.0 | 10.3 | 10.1 $Solution\ II$ | 10.2 | 10.6 | 10.7 | 10.4 | 10.5 | 10.0 | 10.2 | 10.7 | 10.4 | 10.3 i) Do the data support the claim that the mean etch rate is the same for both solutions? In reaching your conclusions, use $\alpha = 0.05$ and assume that both population variances are equal. ii) Find 95% confidence interval on the difference in mean etch rates.	08
	b	Derive normal equations for fitting regression equation $y = \beta_0 + \beta_1 x$ using method foe last squares.	08
		OR	
11	a	An article in the Journal of Environmental Engineering took a sample from the Holston River below king port, Tennessee during 1997. The biochemical oxygen demand (BOD) test is conducted over a period of time in days. The resulting data follow: $Time\ (days)$: | 1 | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 BOD: | 0.6 | 0.7 | 1.5 | 1.9 | 2.1 | 2.6 | 2.9 | 3.7 | 3.5 | 3.7 | 3.8 i) Assuming that a simple linear regression model is appropriate, fit the regression model relating to BOD to the time. What is the estimate of σ^2? ii) What is the estimate of expected BOD level when the time is 15 days?	08
	b	Describe matrix approach to fit a multiple linear regression with usual notations.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

APPLIED STATISTICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4 Answer FIVE full questions from Part B.
5 Use of statistical tables permitted.

PART A

1	1.1	Probability that at least 4 tosses of a fair coin are required for 3 heads is _____.	01
	1.2	There are 4 types of contaminants A, B, C and D found in a bottle of chemical. A is twice likely to be found as B, C and D are equally to be found, and each are twice as likely as A. Probability of C being found is _____.	02
	1.3	X is lognormal with parameters $\mu = 4$ and $\sigma^2 = 1.5$. Y is lognormal with parameters $\mu = 8$ and $\sigma^2 = 1.5$. $E(X)$ and $E(Y)$ are related as _____.	02
	1.4	Estimation techniques using sample data as well as information about parameters available beforehand are known as ___estimation.	01
	1.5	The likelihood function for a sample of 10 values drawn from a Poisson random variable with mean rate of 8 is _____.	01
	1.6	The test statistics used in a test for proportion of defectives based on 'd' defectives observed in parts is _____.	02
	1.7	The rejection region in a test for variance of a Normal population, $H_o: \sigma = \sigma_o$ vs $H_1: \sigma > \sigma_o$ is given by_____.	01
	1.8	What is the difference between contingency table test and goodness of fit tests?	02
	1.9	What is the model to be used to study the effect of interactions in a study with 2 independent variables?	01
	1.10	Which hypothesis is tested by the Wilcoxon Rank Sum test?	01
	1.11	What are the hypothesis statements in a test of significance of regression with 3 independent variables?	02
	1.12	How is multi collinearity among regressor variables estimated?	02
	1.13	The least square normal relations in matrix form is given as: _____ (mention content of each matrix)	02

PART B

2	a	Derive the moment generating function $M_X(t)$ of a random variable X representing the no. of successes in a set of n trials, with probability p of a trial being a success. Using $M_X(t)$, derive the expectation and variance of X.	08

	b	A random variable X has its probability density function as $f(x) = (1/2)x^2e^{-x},\ x > 0$. Find its moment-generating function $MX(t)$ and hence find its expression.	08
		OR	
3	a	X is equally likely between the interval 1 and 2. How is $Y = e^{-x}$ distributed? Compute the probability of Y being at most 0.25. find the expectation of Y.	08
	b	Suppose a random variable Y has the probability density function given by $f(y) = 6y(1-y)$ for $0 \leq y \leq 1$. Find $F(y)$, its mean and median. Find $P(0.5 \leq y \leq 0.8)$	08
4	a	Suppose $Mx(t) = 9/(3-t)^2$. Derive an expression for $E(x^n)$. What is the $var(x)$.	08
	b	Proportion of down-time of a machine, P is known to be Beta distributed with probability density $f(p) = 2(1-p)$. Find the expectation and variance of the cost of down-time W (due to lost production, maintenance and re-set up costs), given by $W = 10 + 20P$.	08
		OR	
5	a	The life time of a projector lamps follows a log-normal distribution with a co-efficient of variation (ratio of std dev to mean) of 2. The expected life time is 5000 hours of operation. i) Determine the parameters of the distribution and the probability that a lamp will last 6000 hours. ii) How much of is the lifetime 95% of the lamps exceed?	08
	b	The time between severe earthquakes at a given region follows a log-normal distribution with a coefficient of variation (ratio of std dev to mean) of 40%. The expected time between severe earthquakes is 80 years. i) Determine the parameters of this log-normally distributed interval time T. ii) Determine the probability that a severe earthquake will occur within 20 years from previous one. iii) Suppose the last severe earthquake in the region took place 100 years ago. What is the probability that a severe earthquake will occur the next year.	08
6	a	Determine the value of c that makes the function $f(x,y) = ce^{(-2x-3y)}$ a joint probability density function over the range $0 < x < y$. Determine the following: i) Marginal probability distribution of X. ii) Conditional probability distribution of Y given $X = 1$. iii) $E(Y\|X = 1)$ iv) Are X and Y independent? Why or why not?	08
	b	Two methods of measuring surface smoothness are used to evaluate a paper product. The measurements X and Y are recorded as deviations from the normal surface smoothness in coded units. The joint probability distribution of the two measurements is a uniform distribution over the region $0 < x < 4, 0 < y$ and $x - 1 < y < x + 1$.	

		Determine the following: i) $P(X < 0.5, Y < 0.5)$ ii) $P(X < 0.5)$ iii) Marginal probability distribution of X iv) Conditional probability distribution of Y given $X = 1$. v) $E(Y/X = 1)$	08
		OR	
7	a	Three electronic ovens that were dropped during shipment are inspected and classified as containing either a major, a minor, or no defect. In the past, 60% of dropped ovens had a major defect, 30% had a minor defect, and 10% had no defect. Assume that the defects on the three ovens occur independently. i) What is the probability distribution of the count of ovens in each category ii) Derive the joint probability mass function of the number of ovens with different types of defects. iii) Compute the expected number of ovens with a major defect.	08
	b	i) Let X be a geometric random variable with parameter p. find the maximum likelihood estimator of p, based on a random sample of size n. ii) $\overline{Y_1}$ and S_1^2 are the sample mean and sample variance from a population with mean μ_1 and variance σ_1^2. Similarly $\overline{Y_2}$ and S_2^2 are the sample mean and sample variance from a second independent population with mean μ_2 and variance σ_2^2. The sample sizes are n_1 and n_2, respectively. Show that $\overline{Y_1}$- $\overline{Y_2}$ is an unbiased estimators of $\mu_1 - \mu_2$. Find the standard error of $\overline{Y_1} - \overline{Y_2}$.	08
8	a	A company operates four machines three shifts each day. From production records, the following data on the number of breakdowns are collected: $Shift$ | $Machines$: A | B | C | D 1 | 41 | 20 | 12 | 16 2 | 31 | 11 | 9 | 14 3 | 15 | 17 | 16 | 10 Test the hypothesis at 5% significance that breakdowns are independent of the shift. How will you find the P-value for this test?	08
	b	The titanium intent in an aircraft-grade alloy is an important determinant of strength. A sample of 20 test coupons reveals the following titanium content(in percent): 8.32,8.05,8.93,8.65,8.25,8.46,8.52,8.35,8.36,8.41,8.42,8.30,8.71,8.75,8.60,8.83, 8.50,8.38, 8.29,8.46 i) The median titanium content should be 8.5%. Use the sign test with significance of 0.05 to investigate the hypothesis. ii) Suppose that the distribution of titanium content is symmetric and continuous. Use Wilcoxon signed-rank test to test if median is 8.5%.	08
		OR	

9 a Grades in a statistics course and an operation research course taken simultaneously were as follow for a group of students

Grade in statistics	*Operation research grade*			
	A	*B*	*C*	*other*
A	25	6	17	13
B	17	16	15	6
C	18	4	18	10
Other	10	8	11	20

Are the grades in statistics and operation research related? Use 1% significance in reaching your conclusion. 08

b Ten samples were taken from plating bath used in an electronics manufacturing process, and the bath pH was determined. The sample pH values are $7.91, 7.85, 6.82, 8.01, 7.46, 6.95, 7.05, 7.35, 7.25, 7.42$. Manufacturing engineering believes that pH has a median value of 7.0. Do the sample data indicate that this statement is correct? Use the sign test with 5% significance to investigate this hypothesis. How will you find the p-value for this test? Use the normal approximation for the sign test to see if the median is different from a pH value of 7.0. 08

10 a Two machines are used for filing bottles with a net volume of $16.0\,ml$ of a drug. The fill volume can be assumed normal, with standard deviation $\sigma_1 = 0.020$ and $\sigma_2 = 0.025ml$. A quality engineering staff suspects that both machines fill to the same mean net volume of $16.0\,ml$. a random sample of 10 bottles is taken from the output of each machine.

Machine 1		*Machine* 2	
16.03	16.01	16.02	16.03
16.04	15.96	15.97	16.04
16.05	15.98	15.96	16.02
16.05	16.02	16.01	16.01
16.02	15.99	15.99	16.00

i) Do you think the engineer is correct? Use $\alpha = 0.05$
ii) What is the p-value for this test?
iii) Find 95% confidence interval on the difference in means. Provide a practical interpretation of this interval. 08

b Enumerate the various techniques employed in variable selection in multiple linear regressions. How are the techniques to choose the best set of variables? 08

OR

11 a A procurement specialist has purchased 25 resistors from vendor1 and 35 resistors from vendor 2. Each resistror's resistance is measured with the following results:
Vendor1: $96.8, 100.0, 100.3, 98.5, 98.3, 98.2, 99.6, 99.4, 99.9, 101.1, 103.7, 97.7, 98.6$
Vendor2:$106.8, 106.8, 104.7, 104.7, 108.0, 102.2, 103.2, 103.7, 106.8, 105.1, 104.0, 106.2, 102.6, 100.3, 104.0, 107.0, 104.3, 105.8$
Perform an appropriate statistical hypothesis-testing procedure to determine whether the procurement specialist can claim that the variance of resistance of product from vendor 1 is significantly different from variance of resistance of product from vendor 2. 08

b How do you investigate whether an independent variable affects the output or not? Explain with the help of an example, using multiple regression techniques and indicator variables. 08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

APPLIED STATISTICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART-A

1	1.1	When shape and scale parameters of a Gamma distribution are 3 and 4.5 respectively, its expectation is _____.	01
	1.2	When the parameters of a Beta distribution are equal, its expectation is _____.	01
	1.3	The random variable representing the number of tosses of a coin required to get 3 Heads, is distributed as _____ distribution.	01
	1.4	For a random variable x, its variance can be expressed in terms of its moment generating function $M_x(t)$ as _____.	01
	1.5	X is lognormal with parameters $\mu = 4$ and $\sigma^2 = 1.5$ the mean and variance of X are _____ and _____ respectively.	02
	1.6	In a lot of 850 parts, 50 are defective. 10 parts are chosen without replacement for inspection. The expectation and variance of the number of defective parts out of 10 are _____ and _____ respectively.	02
	1.7	The test statistic used in the signed rank test is computed as _____. (mention what the terms mean)	01
	1.8	The likelihood function for a sample of 10 values drawn from a Poisson distributed population with a mean of 8 is _____. (The sample is $x_1, x_2, \ldots\ldots\ldots x_{10}$)	01
	1.9	An inspection process involves picking 20 parts and classifying them as Good/Fair/Poor. The number of ways of sequencing 9 Good, 8 Fair and 3 Poor parts for further analysis is _____.	01
	1.10	The covariance between two variables X and Y can take on values in the range _____.	01
	1.11	When can a paired t-test be used when comparing the means of two populations?	01
	1.12	The test statistic used to compare variances of two populations is _____.	01
	1.13	What do the diagonal entries of the covariance matrix contain?	01
	1.14	The error degree of freedom in a multiple linear regression with 50 observed values y_i and 7 regressor variables is _____.	01
	1.15	Type II error is committed when the conclusion is _____ the null hypothesis when the null hypothesis is _____.	01

	1.16	What are the hypothesis statements in a test of significance of regression, involving two independent variables?	01
	1.17	What is the pooled variance in a two-sample test drawn with sample sizes $n_1$and n_2, and when the population variances are unknown but equal?	02

PART-B

2	a	How is the moment generating function $M_x(t)$ defined for a discrete distribution and how is it used to compute the mean and variance?	06
	b	Given a moment generating function $M_x(t) = \frac{1}{8}e^{-5t} + \frac{1}{4}e^{t} + \frac{3}{8}e^{7t} + k$, compute the value of k and write the mass function. Derive $E[X^n]$, and hence find the mean and variance.	10
		OR	
3	a	The overtime T in hours, in a production shop floor varies continuously between 0 to 4 hours, with the pdf $(t) = \frac{t}{8}$. The total wages w, including the base pay is $200\,t + 400$. Find the pdf of W, $f(w)$.	08
	b	In a game, a circular hoop is thrown towards a few poles at a distance. The hoops land over a pole with a probability p. Players keep throwing hoops one by one till a hoop lands over a pole. The cost of the game $= X^2$, where X = number of throws of the hoop. Find the probability distribution of C, $f_c(c)$.	08
4		A state run lottery requires 3 numbers to be chosen from 20 numbers without replacement. A jackpot is won if all 3 chosen numbers match the state's sample.	
		a) What is the probability of a jackpot win?	03
		b) If a player plays this lottery every week, what is the expected number of weeks to play to hit a jackpot win?	03
		c) What is the probability that the player achieves at least one matching within 3 weeks of play?	10
		OR	
5		Lifetime of an electronic gadget used in a communication equipment is log normally distributed with parameters $\mu = 10$ hrs and $\sigma = 1.5$ hrs (not the mean and standard deviation)	
		a) What is the probability that the lifetime exceeds 10.000 hours?	04
		b) What is the lifetime exceeded by 99% of the gadgets?	05
		c) What is the probability that exactly 5 gadgets have to be tested for an inspector to encounter a gadget having a lifetime in excess of 20,000 hours?	07

6 A company uses two different inspection devices to test 4 randomly selected parts. The first device detects defects 99.3% of the time and the second, 99.7% of the time. The devices operate independently of each other. Let X and Y denote the number of defective parts identified by device 1 and 2 respectively. Determine:

a) Range of values of X and Y. Are X and Y independent? Why or why not? 02

b) The joint probability mass function $f_{X,Y}(x,y)$, as an expression 02

c) Marginal probability mass functions $f(x)$ and $f(y)$. 04

d) Expectation and variance of Y, given that device 1 has identified 2 parts as defective. 04

e) What is the probability that the 1st device identifies 2 defectives and 2nd identifies 3 defectives? 04

OR

7 Suppose $X_1, X_2, \ldots X_n$ is a random sample of size n drawn from a normal distribution with unknown mean and variance μ and σ^2 respectively. $f(x) = \frac{1}{\sigma\sqrt{2\pi}} l^{-\frac{(x-\mu)^2}{2\sigma^2}}$, $-\infty < x < \infty$. Derive and compare estimators for μ and σ^2 using

a) Method of moments, and

b) Method of maximum likelihood. 16

8 a Eleven identical units of an electronic hardware were tested and number of hours of continuous operation before overheating and shutting down was recorded as: $1.5, 2.2, 0.9, 1.3, 2.0, 1.6, 1.8, 1.5, 2.0, 1.2, 1.7$ Test at 5% level whether the mean length of continuous operation is 1.8 hours or not. Use Wilcoxon signed rank test. 08

b Strength measurements on 9 specific steel plates used in construction were made using two methods for comparison, as shown below:

Plate	1	2	3	4	5	6	7	8	9
*Method*1	1.19	1.15	1.32	1.34	1.2	1.4	1.37	1.54	1.56
Method 2	1.06	0.99	1.06	1.06	1.07	1.18	1.04	1.09	1.05

i) Construct a 95% confidence interval on the difference in strength from the two methods.

ii) Test if there is any difference between the two methods, at 5% significance. Is the result consistent with the findings of part (i). Explain why. 08

OR

9 a Failure data for electronic devices has the following observations categorized according to type of failure and mounting positions for 134 devices:

Failure Type		*A*	*B*	*C*	*D*
Mounting Portion	1	22	46	18	9
	2	4	17	6	12

Is failure type independent of mounting position? Test at 1% significance. 06

b	In order to evaluate on time performance of cab companies A and B, the number of minutes late is recorded for the last 6 journeys of each company as below in Table 9(b). Negative values indicate early arrival. Is there evidence that either company has better on-time performance than the other? Test using Wilcoxon rank-sum test at 1% significance.	06
c	In a random sample of 85 bearings, 10 have poor surface finish. Construct a 95% 2-sided confidence interval for the population proportion of bearings with poor surface finish.	04
10	An analyst has collected information about an output y for different settings of x, and the summary quantities are: $n = 20$, $\sum x_i = 23.92$, $\sum y_i = 1843.21$, $\sum x_i^2 = 29.29$, $\sum y_i^2 = 170044.21$, $\sum x_i y_i = 2214.66$ a) Estimate the slope and intercept assuming a linear relation b) Test for significance of regression, using $ANOVA$, at 1% significance c) The analyst suspects that the true linear relation involves a slope of $\beta_1 = 12$. Test the claim at 1% significance.	04 06 06
	OR	
11	A construction firm studied soil conditions depth in feet x_1, and moisture content in %, x_2, affecting the shear strength y of the soil. Summary quantities are computed from 10 observations as follows: $\sum x_{i1} = 223, \sum x_{i2} = 553, \sum x_{i1}^2 = 5200.9, \sum x_{i2}^2 = 31729, \sum x_{i1}x_{i2} = 12352, \sum y_i^2 = 371595.6, \sum x_{i1}y_i = 43550.8 \sum x_{i2}y_i = 104736.8$ Few entries of the matrix $(X'X)^{-1}$ are: $\begin{bmatrix} 4.76 & -.09 & -.05 \\ & .004 & 7.7 \times 10^{-5} \\ & & .009 \end{bmatrix}$ a) Write the normal equations for least square regression b) Estimate coefficients of regression $\beta_0, \beta_1, \beta_2$ c) Estimate the soil shear strength at 18 feet with 43% moisture.	08 06 02

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

APPLIED STATISTICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART-A

1	1.1	Determine the range (or possible values) of the random variable in the number of options selected while ordering for an automobile which is sold in the market with a base model and the options to add any number of 15 optional features.	01
	1.2	Suppose $f(x) = x/8$ for $3 < x < 5$, determine the mean and variance for X.	02
	1.3	If X and Y are independent, normal random variables with $E(X) = 0, V(Y) = 4, E(Y) = 10, V(Y) = 9$. Determine $E(2X + 3Y)$.	01
	1.4	The number of times a transistor in a computer memory changes state in one operation is a _____ random variable.	01
	1.5	If X denote the number of trials until the first success. Then X is a _____ random variable with parameters $0 < p < 1$ and $f(x) = (1-p)^{x-1} * p$ for $x = 1,2 \ldots$	01
	1.6	If X is a negative binomial random variable with parameters p and r, then the mean of the random variable $\mu = E(X) =$ _____.	01
	1.7	The net weight of a packaged chemical herbicide is uniform for $49.75 < X < 50.25$ pounds. The mean of the weight of the packages is _____.	01
	1.8	If X is a gamma random variable with parameters λ and γ then the mean of the random variable is given as $\mu = E(X) =$ _____.	01
	1.9	If the weibull distribution has parameters δ(scale parameters) $\delta > 0$ and shape parameters $\beta(\beta > 0)$ then the cumulative distribution is given by the expression $F(x) =$ _____.	01
	1.10	The random variables $X_1, X_2 \ldots X_K$ that denote the number of trials that result in class1,class2…class$k$ respectively have a _____ distribution with a defined joint probability mass function.	01
	1.11	A point estimate of some population parameter θ is a single numerical value θ of a statistic θ. The statistic θ is called the _____.	01
	1.12	If we consider all unbiased estimators of θ, then the one with smallest variance is called the _____ ($MVUE$ for short).	01
	1.13	The function $L(\theta) = f(x_1:\theta).f(x_2:\theta)\ldots..f(x_n:\theta)$ is known as the _____ function of the sample.	01
	1.14	The _____ value is the smallest level of significance that would lead to reflection of the null hypothesis H_0 with the given data.	01

	1.15	The test statistic for the Goodness of fit test on distribution is given as $X_O^2 =$ _____, where O_i is the observed frequency and E_i is the expected frequency in the ith class.	01
	1.16	If data is collected in pairs and we need to compare the differences between the pair of valves, then a special case of t test called the _____ is applied.	01
	1.17	Regression models are constructed based on data collected from the field or the process, this data driven modeling approach is known as _____ model.	01
	1.18	The residuals $e = y_i - \hat{y}_i$ are used to obtain an estimate of σ^2. The sum of squares of the residuals is often called the _____, given by the expression $\delta\delta_E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$.	01
	1.19	The two component source of variation, when the analysis of variance procedure is applied on the regression model are the _____ sum of squares and the Error sum of squares.	01

PART-B

2	a	In an automated filling operation, the probability of an in correct fill when the process is operated at a low speed is 0.001. when the process is operated at high speed the probability of an incorrect fill is 0.01. assume that 30% of the containers are filled when the process is operated at a high speed and the remainder are filled when the process is operated at a low speed. i) What is the probability of an incorrectly filled container? ii) If an incorrectly filled container is found, what is the probability that it was filled during the high speed operation?	06
	b	An assembly consists of three mechanical components. Suppose that the probabilities that the first, second and third c components meet specifications are 0.95 0.98 and 0.99. Assume that the components are independent. Determine the probability mass function of the number of components in the assembly that meet specifications.	06
	c	Define a Moment generating function. How do you determine the moments using the moment generating function? Give an example.	04
		OR	
3	a	A manufacturer of a consumer electronics product expects 2% of units to fail during the warranty period. A sample of 500 independent units is tracked for warranty performance i) What is the probability that none fails during the warranty period? ii) What is the expected number of failures during the warranty period? iii) What is the probability that more than two units fail during the warranty period?	06
	b	The lifetime of a semiconductor laser has a log-normal distribution and it is known that the mean and standard deviation of lifetime are 10,000 hours and 20,000 hoursrespectively. i) Calculate the parameters of the log normal distribution ii) Determine the probability that a life time exceeds 10,000 hours. iii) Determine the lifetime that is exceeded by 90% of lasers.	06

	c	Assume that the life of a roller bearing follows a Weibull distribution with parameters $\beta = 2$ and $\delta = 10{,}000$hours. i) Determine the probability that a bearing lasts at least 8000 hours. ii) Determine the mean time until failure of a bearing.	04
4	a	It is known that two defective copies of a commercial software program were erroneously sent to a shipping lot that has how a total of 75 copies of the program. A sample of copies will be selected from a lot without replacement. i) If three copies of the software are inspected determine the probability that exactly one of the defective copies will be found ii) If three copies of the software are inspected, determine the probability that both defective copies will be found.	08
	b	Let the random variables X_1 and X_2 denote the length and width respectively of a manufactured part. Assume that X_1 is normal with $E(X_1) = 2$ centimeters and standard deviation 0.1 centimeters and that X_2 is normal with $E(X_2) = 5$ centimeters and standard deviation 0.2 centimeters. Also assume that X_1 and X_2 are independent. Determine the probability that the perimeter exceeds 14.5 centimeters.	08
		OR	
5	a	Suppose that the log-ins to a computer network follow a poisson process with an average of 3 counts per minute i) What is the mean time between counts? ii) What is the standard deviation of the time between counts? iii) Determine the value of x such that probability that atleast one count occurs before time x minutes is 0.95.	08
	b	Assume that life of a packaged magnetic disk exposed to corrosive gases has a weibull distribution with $\beta = 0.5$ and the mean life of 600 hours. i) Determine the probability that a packaged disk last atleast 500 hours. ii) Determine the probability that a packaged disk fails before 400 hours.	06
	c	What are the uses of moment generating function?	02
6	a	Four electronic ovens that were dropped during shipment are inspected and classified as contacting either a major, minor or no defect. In the past 60% of dropped ovens had a major defect, 30% had a minor defect and 10% had no defect. Assume that the defects on the four ovens occur independently. i) Is the probability distribution of the count of ovens in each category multinomial? Why or why not? ii) What is the probability that of the four dropped ovens two have a major defect and two have a minor defect? iii) What is the probability that no oven has defect?	08

	b	Suppose we have a random sample of size $2n$ from a population denoted by X_1 and $E(X) = \mu$ and $V(X) = \sigma^2$. Let $\bar{X}_1 = \frac{1}{2n}\sum_{i=1}^{2n} X_i$ and $\bar{X}_2 = \frac{1}{n}\sum_{i=1}^{n} X_i$ be two estimates of μ. Which is the better estimator of μ. Explain your choice.	06
	c	Define the likelihood function.	02
		OR	
7	a	The weight of adobe bricks used for construction are normally distributed with a mean of 3 pounds and a standard deviation of 0.25 pound. Assume that the weights of the bricks are independent and that a random sample of 20 bricks is selected. i) What is the probability that all the bricks in the sample exceeds 2.75 pounds? ii) What is the probability that the heaviest brick in the sample exceeds 3.75 pounds?	08
	b	Suppose that $\widehat{\theta_1}$and $\widehat{\theta_2}$ are unbiased estimators of the parameter θ. We know that $V(\widehat{\theta_1}) = 10$ and $V(\widehat{\theta_2}) = 4$. Which estimator is best and in what sense it is best?	04
	c	Explain the method of moments as an estimating procedure?	04
8	a	A manufacturer of electronic calculator takes a random sample of 1200 calculators and finds that there are eight defective units. i) Construct a 95% confidence interval on the population proportion. ii) Is there a evidence to support a claim that the fraction of defective units produced is 1% or less?	06
	b	Two companies manufacture a rubber material intended for use in automotive application. The part will be subject to abrasive wear in the field application so we decide to compare the material produced by each company in a test. Twenty five samples of material from each company are tested in an abrasion test and the amount of wear after 1000 cycles is observed. For company 1 the sample mean and standard deviation of wear are $\overline{X_1} = 20$ milligram/1000cycles and $s_1 = 2$ milligram/1000 cycles, while for company 2 we obtain $\overline{X_2} = 15$ multigrams/1000 cycles and $s_2 = 8$ milligram /1000 cycles i) Do the data support the claim that the two companies produce material with different mean wear? Use $X = 0.05$ and assume each population is normally distributed but the variances are not equal? ii) What is the P-value of the test?	10
		OR	
9	a	The manager of a fleet of automobiles is testing two brands of radial tires and assigns one size of each brand at random to the two rear wheels of eight cars and runs the cars until the tyres wear out.	

Car	$Brand\ 1$	$Brand\ 2$
1	36,925	34,318
2	45,300	42,280
3	36,240	35,500
4	32,100	31,950
5	37,210	38,015
6	48,360	47,800
7	38,200	37,810
8	33,500	33,215

The data (in kilometers) follow. Find a 99% confidence interval on the difference in mean life. Which brand would you prefer based on this calculation. 08

b Oxide layers on semiconductor wafers are etched in a mixture of gases to achieve the proper thickness. The variability in the thickness of these oxide layers is a critical characteristic of the wafer and low variability is desirable for subsequent processing steps. Two different mixture of gases are being studied to determine whether one is superior in reducing the variability of oxide thickness. Twenty wafers are etched in each gas. The sample standard deviation of oxide thickness are $s_1 = 1.96$ angstoms and $s_2 = 2.13$ angstroms respectively. Is there any evidence to indicate that either gas is preferable? Use $\alpha = 0.05$. 08

10 The grams of solid removed from a material (y) is thought to be related to the drying time. Ten observations obtained from an experimental study follow.

y	4.3	1.5	1.8	4.9	4.2	4.8	5.8	6.2	7.0	7.9
x	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0	6.5	7.0

a) Construct a scatter diagram for these data
b) Fit a simple linear regression model
c) Test the significance of regression
d) Based on the data, what is your estimate of the mean grams of solid removed at 4.25 hours? Find a 95% confidence interval on the mean
e) Analyze the residuals and comment on the model adequacy. 16

OR

11 Simple regression model was used to express drain currently(in milli amperes) as a function of ground-to-source voltage x in volts in the data obtained are as follows:

y	x	y	x
0.734	1.1	1.50	1.6
0.886	1.2	1.66	1.7
1.04	1.3	1.81	1.8
1.19	1.4	1.97	1.9
1.35	1.5	2.12	2.0

a) Draw a scatter diagram of these data. Does a straight line relationship seem possible?
b) Fit a simple linear regression model to these data?
c) Test for significance of regression using $\alpha = 0.05$. What is the P-value for this test?
d) Find a 95% confidence interval estimate on slope.
e) Test the hypothesis $H_o\beta_o = 0$ versus $H_1\beta_o \neq 0$ using $\alpha = 0.05$. What conclusions can you draw? 16

Computer Integrated Manufacturing

VI Semester SEE Question Papers

Course codes: 12IM63

- Apr/May 2018 (12IM63)
- Fast Track Jul 2017 (12IM63)
- Apr/May 2017 (12IM63)
- Apr/May 2016 (12IM63)

12IM63

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

COMPUTER INTEGRATED MANUFACTURING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

10 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

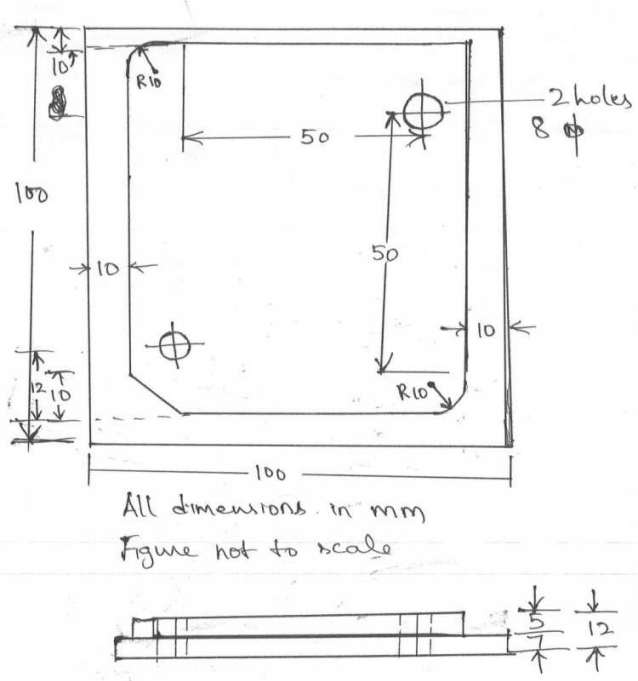
11 Answer FIVE full questions from Part B.

PART A

1	1.1	The use of computer software packages to aid in the analytical, development, costing and ergonomic problems associated with the product of design is known as _____.	01
	1.2	The different phases that a product undergoes from the conceptualization of the product until the end product reaches the customer is known as _____.	01
	1.3	The final stage of implementation of CAD system is _____.	01
	1.4	The use of computer system to plan, manage and control the operations of a manufacturing plant through direct computer interface with the production is known as _____.	01
	1.5	CNC drilling machine is considered to be a _____ controlled machine.	01
	1.6	Fixed automation is characterized by _____ rates.	01
	1.7	List the basic components of DNC system.	02
	1.8	What are two types of bar code technology?	02
	1.9	Mention any two limitations of NC machines.	02
	1.10	Name any two automation strategies that will result in reduction of scrap.	02
	1.11	Name any two types of $AGV's$.	02
	1.12	Define: i) MLT ii) Production rate.	02
	1.13	Write G/M codes for the following.	02

PART B

2	a	Define CAD and explain the role of computers in manufacturing.	08
	b	Discuss about advantages gained by the adoption of CAD and CAM.	08
		OR	
3	a	Explain the process of design characterized by Shigley.	06
	b	Discuss interaction between CAD and CAM database with the help of block diagram.	06
	c	List any four advantages of computer aided design.	04

4	a	Classify numerical control system and explain NC positioning system.	08
	b	What are the advantages and disadvantages of DNC?	08
		OR	
5	a	Explain: i) Canned cycles ii) Radius and length compensation.	06
	b	Write a manual part program for the following component. Assuming missing data suitably. 	10
6	a	Define automation. Why is automation necessary? Explain how automation strategies improve productivity.	08
	b	Draw a neat sketch of information processing cycle and explain its functions.	08
		OR	
7	a	A machine shop has 3 machines producing same part. The shop operates 14 shits per week at an average of 7 hrs/shift and average production rate of each machine is 20 units per hour. i) Determine the production capacity per week. ii) If the availability of the machine is 95% and the utilization of the machine is 85%, find the expected plant output.	05
	b	Write a note on buffer storage in production.	05
	c	A rotating work table is driven by a geneva mechanism with 5 slots. The driver rotates at 25 rev/min. Find i) Cycle time. ii) Available processing time. iii) Indexing time or repositioning time.	06
8	a	With sketches, explain any two types of automated assembly systems.	10
	b	Write a note on multi station assembly machines.	06
		OR	

9	a	Explain $AGV's$ applications.	06
	b	Explain with a neat diagram, the elements of parts delivery system at an assembly work station.	10
10	a	With a suitable block diagram, explain retrieval type of $CAPP$. How is it different from generative type of $CAPP$.	10
	b	Discuss three selection criteria for CAD/CAM implementation.	06
		OR	
11	a	What is FMS? Explain the components of FMS.	10
	b	Explain the application of bar code technology and use of bar code reader in industrial scenario.	06

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

COMPUTER INTEGRATED MANUFACTURING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

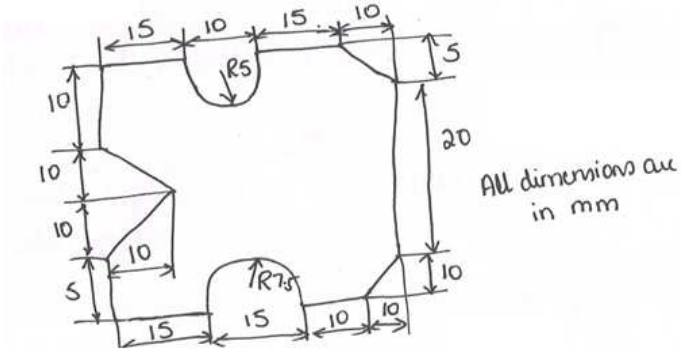
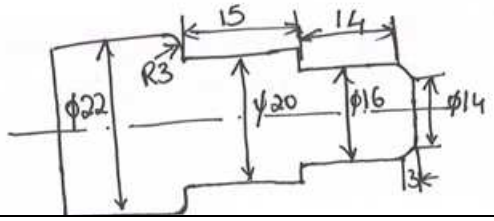
3 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

4 Answer FIVE full questions from Part B.

PART-A

1	1.1	General codes are also called as _____.	01
	1.2	In insert nomenclature $TNMG110408.T$ & 04 stands for _____ & _____.	02
	1.3	What are the different information flows that are followed in an industry?	02
	1.4	High initial investment for custom engineered equipment and high production rate are typical features of _____ type of automation.	01
	1.5	Asynchronous transfer system is also known as _____ & _____ system.	01
	1.6	Draw the part drawing which was turned on a lathe using following part program. $GO1\ X15$ $GO1\ Z-20$ $GO3\ X20\ Z-25\ R5$ $GO1\ Z-35$ $GO1\ X25\ Z-40$ The program is written in absolute system. The largest diameter of the part is $25\ mm$.	02
	1.7	List the two approaches used in computer aided process planning.	02
	1.8	Operation time for machining operation is composed of three elements, the actual machining time, _____ & tool handling time per work piece.	01
	1.9	Define product flexibility.	02
	1.10	State the difference between home zero and machine zero.	02
	1.11	What kind of attenuation would you recommend for manufacturing of: i) Cement ii) Light bulbs.	02
	1.12	A total of 800 shafts must be produced in the lathe section of the machine shop during a particular week. Each shaft is identical and requires a m/c cycle time $T_C = 11.5\ min$. all of the lathes in the department are equivalent in terms of their capability to produce the shaft in the specified cycle time. How many lathes must be devoted to shaft production during the given week; if there are 40 hs of available time on each lathe?	02

PART-B

2	a	The process of designing is characterized by shigley as an interactive procedure which consists of six identifiable steps. Elaborate. Support your explanation with a block diagram.	10
	b	Explain briefly the two categories of CAM.	06
		OR	
3	a	With the help of block diagram, explain the procedure for creating the database in a manufacturing system.	08
	b	List the benefits and advantages of CAD/CAM.	08
4	a	Explain the different types of NC motion control systems.	06
	b	Write the manual part program for the component shown below. 	10
		OR	
5	a	Derive the equations of manufacturing lead time for the following situations: i) All operation time, setup times and non operation times are equal. ii) For job shop in which batch size is 1.	06
	b	Write the manual part program for the component shown below. 	10
6	a	With the help of block diagram, explain the information processing in manufacturing environment.	10
	b	Discuss the two important reasons for the use of buffer storage zones.	06
		OR	
7	a	Define an automated flow line. Discuss any one type of automated flow line with a neat block diagram.	08
	b	Explain the automation strategies that are employed to improve productivity in manufacturing operations.	08

8	a	List and explain any two types of automated assembly systems.	08
	b	Elaborate on the four main components of an automated guided vehicle.	08
		OR	
9	a	With the help of neat sketch explain different components in part feeding devices in an automated assembly system.	10
	b	Write a note on the following: i) $AGVS$ pallet trucks ii) $AGVS$ fork lift trucks.	06
10	a	Explain the functions of material handling and storage system in an FMS.	06
	b	Illustrate the generative $CAPP$ system of automated process planning with the components needed in it.	10
		OR	
11	a	If CAD/CAM system is required for a company; what is the general procedure and guidelines that are to be followed for the implementation of the system.	08
	b	Explain the two types of Barcode technology used for shop floor control in manufacturing industries.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

COMPUTER INTEGRATED MANUFACTURING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	Categorize the applications of CAD/CAM into two broad areas.	02
	1.2	Identify the type of production activity generally carried out in the manufacturing of prototypes, aircraft and other such equipment.	02
	1.3	What is meant by Part program manuscript in the context of numerically controlled manufacturing?	02
	1.4	What is meant by Direct Numerical Control?	02
	1.5	Mention the two categories of automated elements of a typical production system.	02
	1.6	List the subsystems of a typical automated assembly system.	02
	1.7	Identify the two main reasons for the occurrence of partially automated production lines in industries.	02
	1.8	Distinguish between Dedicated Flexible Manufacturing System and Random-order Flexible Manufacturing System.	02
	1.9	Identify the four reasonable tests of flexibility in an automated manufacturing system.	02
	1.10	Distinguish between width modulated and height modulated bar code symbologies.	02

PART B

2	a	Illustrate the relationship among four types of production in terms of product variety and production quantities.	08
	b	"The process of designing something is characterized by Shigley as an iterative process". Justify the above statement.	08
		OR	
3	a	Illustrate the relationship between CAD/CAM database and design and manufacturing in a typical production oriented company.	08
	b	Analyze the application of computers for various design related tasks.	08

4	a	Explain the three basic components of an operational numerical control system.	08
	b	Discuss with relevant sketches, the basic types of numerically controlled motion control systems.	08
		OR	
5	a	Elaborate on the functions and advantages of Direct Numerical Control Systems.	08
	b	Explain different components of Direct Numerical Control Systems.	08
6	a	Explain different levels of automation and control in a typical manufacturing environment.	08
	b	Critically review the strategies for automation and production as advocated by M.P.Groover.	08
		OR	
7	a	Classify the system configurations in automated production lines. Also write a brief note about each.	08
	b	Evaluate the reasons for incorporating storage buffers in automated production lines.	08
8	a	Explain the hardware constituents of a typical parts delivery work station.	08
	b	Illustrate the principal physical configurations of Automated assembly systems.	08
		OR	
9	a	Explain the principles that can be applied in product design to facilitate automated assembly.	08
	b	Identify the three categories of automated guided vehicles. Explain the applications of each of the categories.	08
10	a	Discuss the types of work stations commonly found in an *FMS* environment.	08
	b	Explain the functions of material handling and storage systems in an *FMS* environment.	08
		OR	
11	a	Explain briefly different methods used to collect data from the shop floor.	08
	b	Distinguish between the features of retrieval and generative approaches to computer aided process planning.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

COMPUTER INTEGRATED MANUFACTURING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

7 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

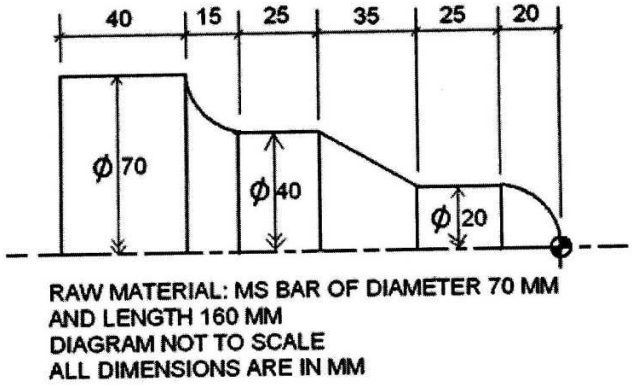
8 Answer FIVE full questions from Part B.

PART-A

1	1.1	Define CAD/CAM.	02
	1.2	State two reasons for using a computer aided design system to support the engineering design function.	02
	1.3	List three basic components of an NC system.	02
	1.4	In part programming, G and M codes represents _____ and _____ functions.	02
	1.5	List different types of automation.	02
	1.6	Classify automated assembly systems by their physical configurations.	02
	1.7	Define rotary indexing machine of a automated flow line system.	02
	1.8	List the concepts and technologies that can be included in an FMS.	02
	1.9	Computer Aided Process Planning consists of _____ and _____ systems.	02
	1.10	What is bar code reader?	02

PART-B

2	a	Explain with a neat block diagram of a product cycle with the emphasis on CAD/CAM.	08
	b	Explain with a neat block diagram the general design process of CAD/CAM.	08
		OR	
3	a	Explain the fundamental reasons for implementing a computer aided design systems.	08
	b	Explain the following applications of computer for design. i) Geometric modeling ii) Engineering analysis iii) Design review and evaluation iv) Automated drafting	08
4	a	Explain at least four advantages and four disadvantages of Numerical Control machines.	08
	b	Explain the following block format of the CNC programming: $N135\ G01\ X1.0\ Y1.0\ Z0.125\ F5.0$	08

		OR	
5		Write the part program for turning operations using G and M codes for the following:  RAW MATERIAL: MS BAR OF DIAMETER 70 MM AND LENGTH 160 MM DIAGRAM NOT TO SCALE ALL DIMENSIONS ARE IN MM Fig 5a	16
6	a	Define automation and list the reasons for automation.	08
	b	A production machine is operated $65 hr/week$ at full capacity. Its production rate si$20 units/hr$. During a certain week, the machine produces 1000 good parts and was idle the remaining time. i) Determine the production capacity of the machine ii) What was the utilization of the machine during the week under consideration?	08
		OR	
7	a	Provide a brief note on automated flow lines to meet the objectives of the use of flow line automation.	08
	b	Enumerate different rotary transfer mechanism and explain anyone with a neat sketch.	08
8	a	Explain with a neat diagram the elements of parts delivery system at an assembly workstation.	12
	b	Classify the automated assembly systems.	04
		OR	
9	a	Explain different types of material handling equipment and illustrate them with example.	12
	b	List the applications of Automated guided vehicle systems.	04
10	a	Provide a brief note on the retrieval and generative type computer aided process planning.	10
	b	List the benefits of computer aided process planning.	06
		OR	
11	a	What is an Flexible Manufacturing System and explain the components of Flexible Manufacturing System.	10
	b	Discuss a typical shop floor control system.	06

Enterprise Resource Planning and E-Commerce

VI Semester SEE Question Papers

Course codes: 12IM6D5

- Apr/May 2018 (12IM6D5)
- Apr/May 2017 (12IM6D5)
- Fast Track Jul 2016 (12IM6D5)
- Apr/May 2016 (12IM6D5)

12IM6D5

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

ENTERPRISE RESOURCE PLANNING AND E-COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	Differentiate between traditional commerce and electronics commerce.	02
	1.2	What are the characteristics of supply chain management in an e-commerce environment?	02
	1.3	What are the types of security features used in client server types of network?	02
	1.4	Why firewalls are used?	02
	1.5	Expand the acronyms EFT and EDI.	02
	1.6	Differentiate between $MRP-I$ and $MRP-II$.	02
	1.7	List the categories of e-commerce.	02
	1.8	Explain the application of e-commerce in the home banking.	02
	1.9	Define e-commerce. What are the benefits of using e-commerce?	02
	1.10	Describe the process of data mining.	02

PART B

2	a	Define ERP and explain the importance of ERP.	08
	b	Explain the methodology used in ERP implementation.	08
		OR	
3	a	Do you justify ERP investments in startup companies? Give examples.	08
	b	"Supply Chain is vastly benefited by enabling ERP". Explain by providing illustrations.	08
4	a	Discuss online analytical processing with suitable examples. Sketch a neat block diagram.	08
	b	Identify and analyze various internet business models with examples.	08
		OR	

5	a	How relevant is Business Process Reengineering in today's context? Explain different phases with their advantages and examples.	08
	b	Identify and explain any four modules of *ERP* system with suitable examples.	08
6	a	What are the problems faced during *ERP* implementation? How to overcome them? Give examples.	08
	b	Explain the life cycle phases of the *ERP* implementation with illustrations.	08
		OR	
7	a	Evaluate the critical factors involved in successful implementation of *ERP* and highlight their impact.	08
	b	Assess the post *ERP* implementation activities with suitable examples. How important are these activities?	08
8	a	Explain the relevance of e-commerce in today's markets. Explain with examples.	08
	b	Differentiate between digital market and traditional market by considering competitive factors.	08
		OR	
9	a	List and explain the components of E-commerce. Illustrate with an example.	08
	b	Explain how different departments participate in an e-commerce project. Is it right from the systems point of view?	08
10	a	Identify the nine essential e-commerce processes required for the successful operation or management of e-commerce activities.	08
	b	Evaluate some of the trends of e-commerce applications with examples.	08
		OR	
11	a	Analyze the benefits and challenges of e-commerce.	08
	b	How do you rate the e-commerce activities from the Buyer's experience? Give suitable examples.	08

12IM6D5

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-17

Industrial Engineering and Management

ENTERPRISE RESOURCE PLANNING AND E-COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	Differentiate between the business function and business process.	02
	1.2	What is subject oriented, integrated, non-volatile & time-variant data warehouse?	02
	1.3	Give one example each for a proprietary & a open source ERP software.	02
	1.4	One of the most critical step in the ERP implementation is the creation of an _____.	01
	1.5	The minimal information system consists of three elements _____, _____ and _____	02
	1.6	List the objective of ERP implementation.	02
	1.7	Give one example each for $B2C$ & $B2E$ e-business transaction models.	02
	1.8	Draw the single block diagram showing the relation between $B2C$ & $B2E$ models.	02
	1.9	Differentiate between $OLAP$ and $OLTP$.	02
	1.10	List the sales related business transaction.	02
	1.11	_____ Module facilitates the process of maintaining the appropriate level of stock in a warehouse.	01

PART B

2	a	"Thinking in terms of business process helps mangers to look at their organizations from a customer's perspective:. Explain with a real life example.	08
	b	List the common myth about the ERP system and demystify them.	08
		OR	
3	a	Summarize the evolution of ERP from $1960s$ to $1990s$.	08
	b	Highlight different reasons for the growth of ERP market.	08

4	a	Explain the integration of ERP, supply chain and customers relationship applications with the help of neat sketch.	10
	b	Why is $OLAP$ is called fast analysis & shared multi-dimensional information.	06
		OR	
5	a	How do you address different security issues involved in ERP implementation?	08
	b	Bring out the difference between data mining and data warehouse system.	08
6	a	Build a business model of a ERP system for an education institution like $RVCE$.	10
	b	List and explain the various skills required by the people who are involved in the ERP implementation.	06
		OR	
7	a	Develop a framework to implement ERP system in a health care organization.	08
	b	Choosing the right ERP vendors goes beyond evaluating the software functionality. Explain.	08
8	a	Explain most widely used client server applications.	08
	b	Compare different e-business transaction models with suitable examples.	08
		OR	
9	a	With the help of a block diagram explain internet administrative structure.	08
	b	What are the important concepts introduced in the IT Act, 2000.	08
10	a	Compare traditional marketing over e-marketing.	08
	b	Summarize the general security issues that e-business must consider.	08
		OR	
11	a	Discuss the different components of m-commerce.	08
	b	Recommend a marketing strategy for m-commerce for a restaurant at your city.	08

12IM6D5

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

ENTERPRISE RESOURCE PLANNING AND E - COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	_____ is the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical, contemporary measures of performance such as cost, quality, service and speed.	01
	1.2	Define $OLAP$.	02
	1.3	Differentiate between business function and process.	02
	1.4	The extraction of hidden predictive information from large database is called as _____.	01
	1.5	What is the subject oriented, integrated, non-volatile and time variant data warehouse?	02
	1.6	Define $OLTP$.	02
	1.7	Classify different E-business transaction models.	02
	1.8	What are the important concepts introduced in the IT act, 2000.	02
	1.9	What is digital signature?	01
	1.10	Establish relation between $B2B$ and $B2C$ models.	02
	1.11	What dynamic IP addressing.	02
	1.12	Define e-supply chain management.	01

PART-B

2	a	Explain with examples why it is important to have interdepartmental communication for the smooth and efficient functioning of an organization.	08
	b	What are the problems in ERP implementation according to survey finding reported in the literature?	08
		OR	
3	a	What are the common myths about ERP implementation?	08
	b	"Thinking in terms of business processes helps managers to look at their organizations from a customer perspective". Explain with example.	08

4	a	Discuss different tools and techniques used in data mining.	08
	b	Define BPR. List different phases involved in BPR implementation. Explain any two phases in detail.	08
		OR	
5	a	Develop a flow for sales and distribution module for a garment company.	08
	b	Explain the integration of ERP, supply chain and customer relationship applications with the help of neat sketch.	08
6	a	Develop a questionnaire containing minimum ten questions to collect the requirements of customers with respect to ERP software.	08
	b	Evaluate the pros and cons of proprietary and open source ERP implementation.	08
		OR	
7	a	Create a test plan to evaluate the effectiveness of ERP implementation. Give two examples showcasing your test cases.	08
	b	Select right implementation strategy for organization having different business processes. Justify your answer.	08
8	a	Plan a survey to evaluate the popularity of e-commerce over traditional commerce.	08
	b	Compare different e-business transaction models with suitable examples.	08
		OR	
9	a	Prioritize the key challenges that exist in the e-commerce industries.	08
	b	What are the processes involved in $B2C$ transaction model?	08
10	a	Explain the client server applications of internet.	08
	b	Recommend a marketing strategy for m-commerce business model.	08
		OR	
11	a	Prepare a list of factors governing E-Security. How these factors can be addressed.	08
	b	Evaluate the effectiveness of internet standards and specifications in Indian context.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-16

Industrial Engineering and Management

ENTERPRISE RESOURCE PLANNING AND E-COMMERCE (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART-A

1	1.1	_____ is a collection of activities that takes one or more kinds of input and creates a output that is of value to the customer.	01
	1.2	The first phase of ERP implementation is _____	01
	1.3	What is subject oriented, integrated, non-volatile and time variant data warehouse?	02
	1.4	_____ are responsible for fixing any problems in software that the implementation team encounters.	01
	1.5	GIS stands for _____.	01
	1.6	What are the characteristics of the information?	02
	1.7	List any four concepts introduced in the IT Act, 2000 of government of India.	02
	1.8	Give one example for $B2C$ transaction model.	01
	1.9	Expand FTP.	01
	1.10	Establish a relationship between $B2B$ and $B2C$ models using a block diagram.	02
	1.11	_____ is the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical, contemporary measures of performance such as cost, quality, service and speed.	01
	1.12	_____ is the process of identifying valid, Novel, potentially useful, and ultimately comprehensive knowledge from database that is used to make crucial business decisions.	01
	1.13	_____ module streamline the management of human resources and human capital.	01
	1.14	Define e-supply chain management.	02
	1.15	What is M-commerce?	01

PART-B

2	a	Explain with examples why it is important to have interdepartmental communication for smooth and efficient functioning of an organization.	08
	b	What is meant by Integrated Data Model? Explain how it is created.	08

OR

3	a	Many enterprises are replacing their legacy business systems with integrated versions. Elaborate on the advantages of installing ERP systems in the enterprises.	08
	b	Provide an overview on the evolution of ERP systems.	08
4	a	Define BPR. List different phases involved in BPR implementation. Explain phase one in detail.	08
	b	Bring out the differences between online analytical and online Transactional processing systems.	08
		OR	
5	a	Explain the functions, subsystems and features of the sales and distribution module in ERP system.	08
	b	What are the functions, subsystems and features of the HR module in ERP systems?	08
6	a	Explain why it is not a good idea to develop an ERP package in-house.	08
	b	Choosing the right ERP vendor goes beyond evaluating the software functionality. Explain.	08
		OR	
7		Discuss with the help of a schematic representation, the different phases of ERP implementation life cycle.	16
8	a	List the key strengths of using the internet for business in India.	08
	b	Compare different e-business transaction models with suitable examples.	08
		OR	
9	a	Explain through a flow chart the processes involved in $B2C$ e-commerce.	08
	b	Prioritize the key opportunities that exist in the e-commerce industries.	08
10	a	List and explain the client-server protocols of internet.	08
	b	Recommend a marketing strategy for M-commerce business model.	08
		OR	
11	a	Prepare a list of factors governing e-security. How these factors can be addressed?	08
	b	Evaluate the effectiveness of internet standards and specification in India context.	08

Management Information Systems

VI Semester SEE Question Papers

Course codes: 12IM6C5

- Apr/May 2018 (12IM6C5)

12IM6C5

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

VI Semester B. E. Examinations April/May-18

Industrial Engineering and Management

MANAGEMENT INFORMATION SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1 Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

2 Answer FIVE full questions from Part B.

PART A

1	1.1	Identify the fundamental roles of *IS* in Business.	02
	1.2	What are the steps involved in developing Information Systems Solutions?	02
	1.3	What are the Information System Resources?	02
	1.4	List different developments in many types of storage media and devices.	02
	1.5	What are the five fundamental database structures?	02
	1.6	What are the major telecommunications trends?	02
	1.7	What are the categories of major financial management systems?	02
	1.8	What are the different levels of managerial decision making that must be supported by information technology in a successful organization?	02
	1.9	What is *OLAP*?	02
	1.10	What is "what if analysis"?	02

PART B

2	a	Explain why knowledge of information systems is important for business professionals. Identify and discuss with neat sketch five areas of information systems knowledge that they need.	08
	b	Discuss with neat sketch several major types of information systems from your experiences that support business operations and managerial decision making at business organizations in the real world.	08
		OR	
3	a	Continued developments in information technology and its business applications have resulted in the evolution of several major types of databases. As an *MIS* manager illustrate several major conceptual categories of databases that may be found in many organizations.	08
	b	As businesses continue to use some open technologies to improve communication with customers and partners, they can gain many competitive advantages along the way in product development, cost savings, marketing, distributions and leveraging their partnerships. Which type of internet technology do you suggest as MIS manager? Discuss with neat sketch how the business will derive the benefits.	08
4	a	Briefly discuss with a neat sketch, how the major functional areas of business are supported by in functional business information systems.	08

	b	Discuss how the major application components of enterprise resource planning demonstrate the cross functional approach of *ERP* systems and also show different flows that support *ERP* systems.	08
		OR	
5	a	Identify the essential processes of an e-commerce system and give examples of how it is implemented in e-commerce applications.	08
	b	What are e-Commerce market places? Identify and discuss the different types of e-commerce market places used by business today.	08
6	a	Enumerate the major differences in the information and decision support capabilities of management information systems and decision support systems.	08
	b	Describe how online analytical processing can meet key information needs of managers.	08
		OR	
7		Briefly discuss how the following emerging areas of decision support systems support decision making in business: a) Neural network b) Expert systems c) Fuzzy logic d) Genetic algorithm.	16
8	a	Betting on new *IT* innovations mean betting the future of the company. Leading edge firms are sometime said to be on the "bleeding edge". Hence, planning for competitive advantage is especially important in today's competitive business arena and complex information technology environment. So, strategic business/*IT* planning involves an evaluation of the potential benefits and risks a company faces when using *IT*-based strategies and technologies for competitive advantage. In such situation, how do you proceed for planning *IT* strategies? Discuss.	08
	b	"Business model" was one of the great buzzwords of the internet boom, routinely invoked, as the writer Michael Lewis puts it, "to invoke all manner of half-baked plans". A good business model however remains essential to every successful organization, whether it's a new venture or an established player. Justify this statements with the questions that illustrate the components of e-business models that can be developed as part of the strategic business/*IT* planning process.	08
		OR	
9	a	Explain how prototyping can be used as an effective technique to improve the process of system development for end users and IS specialists.	08
	b	Describe and give examples to illustrate how you might use each of the steps of the information systems development life cycle to develop and implement a business information system.	08
10	a	Identify and discuss the implementations process activities and timetables for a company installing an intranet-based employee benefits system in its human resource management department.	08
	b	Identify and discuss the implementation activities needed to transform a newly developed information system into an operational system for end users in the implementation process.	08
		OR	
11	a	Briefly discuss the different dimensions of ethical responsibility of business professionals.	08
	b	*IT* managers are increasingly concerned about securing the user end of their network and how they are facing this challenge? Discuss the security management tools and techniques for their networks.	08